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# Data Compression

## Lecture Notes

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## 1: Lecture 1: Introduction to Data Compression

### 1.1 Learning Objectives

By the end of this lecture, students will be able to:

- Understand the motivation and benefits of data compression
- Differentiate between lossless and lossy compression techniques
- Compute and interpret common compression performance metrics
- Apply the Huffman coding algorithm step by step
- Analyze real-world compression trade-offs

### 1.2 Introduction and Motivation: Why Compress Data?

Data compression is the process of representing information using fewer bits than its original form. It is a fundamental component of modern computing systems, enabling efficient storage, faster communication, and reduced operational costs.

Everyday applications of compression include:

- Streaming audio and video
- Image storage and sharing
- File archiving and backups
- Network communication and cloud services

#### Definition

**Data Compression** is the process of reducing the number of bits required to represent information, either:

- **Losslessly**: allowing exact reconstruction of the original data
- **Lossily**: allowing controlled loss of information to achieve higher compression

### 1.2.1 Benefits of Data Compression

Data compression provides three key benefits that are critical in modern computing:

#### 1. Reduce Storage Space:

- Allows more data to be stored in the same physical space
- Enables archival of historical data that would otherwise be discarded
- Reduces hardware requirements for storage systems

#### 2. Reduce Communication Time and Bandwidth:

- Enables faster file transfers and downloads
- Makes high-quality streaming (4K/8K video) practical over limited bandwidth
- Reduces latency in real-time applications like video conferencing and online gaming
- Allows IoT devices to transmit data efficiently over wireless networks

#### 3. Save Money:

- Reduces cloud hosting costs (storage and egress fees)
- Lowers communication costs for data transmission
- Decreases capital expenditure on storage hardware
- Reduces energy consumption for data centers and network infrastructure

## 1.3 Lossless vs. Lossy Compression

### 1.3.1 Lossless Compression

Lossless compression guarantees perfect reconstruction of the original data. It is essential when accuracy and data integrity are critical.

#### Typical applications:

- Text files and source code
- Executables and databases
- Medical, scientific, and legal data

### 1.3.2 Lossy Compression

Lossy compression achieves higher compression ratios by discarding information that is less perceptible or less important.

**Typical applications:**

- Audio (MP3, AAC)
- Images (JPEG)
- Video (H.264, H.265)

### 1.3.3 Choosing Between Lossless and Lossy

| Factor           | Lossless Compression | Lossy Compression      |
|------------------|----------------------|------------------------|
| Reconstruction   | Exact                | Approximate            |
| Data sensitivity | High                 | Moderate to low        |
| Typical ratios   | Low to moderate      | High                   |
| Quality impact   | None                 | Controlled degradation |

Table 1: Lossless vs. Lossy Compression

## 1.4 Compression Performance Metrics

### 1.4.1 Size-Based Metrics

$$\begin{aligned}\text{Compression Ratio (CR)} &= \frac{\text{Original Size}}{\text{Compressed Size}} \\ \text{Compression Factor} &= \frac{\text{Compressed Size}}{\text{Original Size}} \\ \text{Space Savings (\%)} &= \left(1 - \frac{\text{Compressed Size}}{\text{Original Size}}\right) \times 100\%\end{aligned}$$

**Interpretation:**

- Larger compression ratios indicate better compression
- Smaller compression factors indicate better compression

### 1.4.2 Rate-Based Metrics

$$\begin{aligned}\text{Bits per Sample (bps)} &= \frac{\text{Compressed Size (bits)}}{\text{Number of samples}} \\ \text{Bit-rate (bps)} &= \frac{\text{Compressed Size (bits)}}{\text{Time (seconds)}}\end{aligned}$$

These metrics are particularly important in audio and video compression systems.

## 1.5 Worked Example: Audio Compression Metrics

### Example

#### Uncompressed Audio Properties

- Duration: 180 seconds
- Sampling rate: 44.1 kHz
- Bit depth: 16 bits
- Channels: 2 (stereo)

#### Original Size Calculation

$$\text{Total samples} = 180 \times 44,100 \times 2 = 15,876,000$$

$$\text{Size (bits)} = 15,876,000 \times 16 = 254,016,000$$

$$\text{Size (MB)} = \frac{254,016,000}{8 \times 1,048,576} \approx 30.27$$

#### Compression Results

| Method          | Size (MB) | CR     | Savings | Bit-rate |
|-----------------|-----------|--------|---------|----------|
| FLAC (lossless) | 18.16     | 1.67:1 | 40%     | 807 kbps |
| MP3 @ 320 kbps  | 6.75      | 4.49:1 | 77.7%   | 320 kbps |
| AAC @ 256 kbps  | 5.40      | 5.61:1 | 82.2%   | 256 kbps |

## 1.6 Huffman Coding

Huffman coding is a widely used **lossless compression algorithm** that assigns variable-length binary codes to symbols based on their frequencies. More frequent symbols receive shorter codes.

### 1.6.1 Step-by-Step Huffman Coding Example

#### Example

**Message:** MISSISSIPPI RIVER (17 characters including space)

**Symbol Frequencies**

| Symbol  | Frequency |
|---------|-----------|
| I       | 5         |
| S       | 4         |
| P       | 2         |
| R       | 2         |
| M       | 1         |
| V       | 1         |
| E       | 1         |
| (space) | 1         |

### Tree Construction

1. Combine  $M(1) + V(1) \rightarrow 2$
2. Combine  $E(1) + (\text{space})(1) \rightarrow 2$
3. Combine  $P(2) + R(2) \rightarrow 4$
4. Combine  $2 + 2 \rightarrow 4$
5. Combine  $4 + 4 \rightarrow 8$
6. Combine  $I(5) + S(4) \rightarrow 9$
7. Combine  $8 + 9 \rightarrow 17$

### One Possible Code Assignment

| Symbol  | Code | Length |
|---------|------|--------|
| I       | 00   | 2      |
| S       | 01   | 2      |
| P       | 100  | 3      |
| R       | 101  | 3      |
| M       | 1100 | 4      |
| V       | 1101 | 4      |
| E       | 1110 | 4      |
| (space) | 1111 | 4      |

### Compressed Size

$$5(2) + 4(2) + 2(3) + 2(3) + 4(1) = 52 \text{ bits}$$

**Original Size (ASCII)**  $= 17 \times 8 = 136 \text{ bits}$

**Compression Ratio**  $= 136/52 \approx 2.62 : 1$

### 1.6.2 Key Properties of Huffman Coding

- Produces prefix-free codes
- Enables instantaneous decoding
- Guarantees minimum average code length among prefix codes
- Widely used in practical compression systems

## 1.7 End of Chapter Questions

### Exercise 1.0

#### Problem 1: Basic Compression Metrics

An uncompressed grayscale image has the following properties:

- Resolution:  $1024 \times 1024$  pixels
- Bit depth: 8 bits per pixel

After compression, the image size is 320 KB.

Calculate:

- (a) Original image size in KB
- (b) Compression ratio
- (c) Compression factor
- (d) Space savings percentage

### Exercise 1.1

#### Problem 2: Audio Bit-rate and Storage

A mono audio recording has the following parameters:

- Duration: 5 minutes
- Sampling rate: 48 kHz
- Bit depth: 16 bits

The file is compressed using a lossy codec to a constant bit-rate of 192 kbps.

Calculate:

- (a) Size of the uncompressed audio file in MB
- (b) Size of the compressed file in MB

- (c) Compression ratio
- (d) Bits per sample after compression

### Exercise 1.2

#### Problem 3: Comparing Compression Options

A video clip has an uncompressed data rate of 120 Mbps. Three compression options are available:

| Option | Compressed Bit-rate |
|--------|---------------------|
| A      | 6 Mbps              |
| B      | 3 Mbps              |
| C      | 1.5 Mbps            |

For each option, calculate:

- (a) Compression ratio
- (b) Data consumed for a 10-minute video (in MB)

Which option would you choose for:

- (i) Live video streaming?
- (ii) Archival storage?

Briefly justify your answers.

### Exercise 1.3

#### Problem 4: Huffman Coding Construction

Given the following symbol frequencies:

| Symbol | Frequency |
|--------|-----------|
| A      | 10        |
| B      | 8         |
| C      | 6         |
| D      | 5         |
| E      | 4         |
| F      | 3         |
| G      | 2         |
| H      | 2         |

- (a) Construct the Huffman tree step by step
- (b) Assign a binary code to each symbol

- (c) Compute the total number of bits required to encode the message
- (d) Calculate the average number of bits per symbol

#### Exercise 1.4

##### Problem 5: Fixed-Length vs. Huffman Coding

Using the symbol set from Problem 4:

- (a) Determine the minimum fixed-length code required
- (b) Compute the total number of bits using fixed-length coding
- (c) Compare the result with Huffman coding
- (d) Calculate the percentage reduction in total bits achieved by Huffman coding

#### Exercise 1.5

##### Problem 6: Text Compression Scenario

A text file contains 50,000 characters and is stored using 8-bit ASCII encoding. After compression using a lossless algorithm, the file size becomes 18 KB. Calculate:

- (a) Original file size in KB
- (b) Compression ratio
- (c) Compression factor
- (d) Space savings percentage

Explain why compression ratios for text files vary significantly depending on content.

#### Exercise 1.6

##### Problem 7: Practical Design Question

You are designing a compression system for a wearable health-monitoring device that:

- Records sensor data continuously
- Has limited storage capacity
- Requires exact data reconstruction
- Operates on a low-power processor

- (a) Should the system use lossless or lossy compression? Explain.
- (b) Which performance metrics are most important in this scenario?
- (c) Would a variable-length coding scheme be appropriate? Why or why not?

## 2: Lecture 2: Theory of Compression — Limits and Optimality

### 2.1 Types of Codes: From Ambiguous to Instantaneous

#### Definition

##### Types of Codes

- **Non-singular Code:** Each source symbol maps to a distinct codeword

$$x_i \neq x_j \Rightarrow C(x_i) \neq C(x_j)$$

- **Uniquely Decodable Code:** Every finite sequence of codewords corresponds to exactly one sequence of source symbols

$$C(x_1)C(x_2) \cdots C(x_n) = C(y_1)C(y_2) \cdots C(y_m) \Rightarrow n = m \text{ and } x_i = y_i$$

- **Prefix Code (Instantaneous Code):** No codeword is a prefix of another codeword

$$\forall i \neq j : C(x_i) \text{ is not a prefix of } C(x_j)$$

##### Key Relationships:

Prefix Codes  $\subset$  Uniquely Decodable Codes  $\subset$  Non-singular Codes

#### Important

##### Why Prefix Codes are Special

- **Instantaneous decoding:** Can decode as soon as codeword ends (no lookahead needed)
- **Tree representation:** Always correspond to leaves of a binary tree
- **Kraft inequality:** Always satisfy  $\sum 2^{-\ell_i} \leq 1$
- **Practical:** Used in Huffman coding, many real-world compressors

#### Example

##### Example: Comparing Different Code Types

For symbols  $\{A, B, C, D\}$  with probabilities  $\{0.5, 0.25, 0.125, 0.125\}$ :

| Code Type          | A | B  | C   | D    | Property                                |
|--------------------|---|----|-----|------|-----------------------------------------|
| Non-singular       | 0 | 1  | 00  | 11   | Distinct but ambiguous: "00" = AA or C? |
| Uniquely decodable | 0 | 01 | 011 | 0111 | Unique but need lookahead               |
| Prefix code        | 0 | 10 | 110 | 111  | Instant decoding: "0" = A, stop         |
| Optimal prefix     | 0 | 10 | 110 | 111  | Also Huffman optimal                    |

**Decoding examples:**

- **Prefix code "010110"**:  $0 \rightarrow A, 10 \rightarrow B, 110 \rightarrow C = \text{"ABC"}$  (instant)
- **Uniquely decodable "00111"**: Need to scan ahead to determine split
- **Non-singular "00"**: Ambiguous! Could be "AA" or "C"

**Key insight:** Prefix codes sacrifice some flexibility in codeword lengths (must satisfy Kraft inequality) for the benefit of instantaneous decoding.

## 2.2 Basic Terminology and Notation

### Definition

#### Alphabet

An *alphabet*  $\mathcal{X}$  is a finite set of possible symbols. Examples:

- Binary alphabet:  $\mathcal{X} = \{0, 1\}$
- English letters:  $\mathcal{X} = \{A, \dots, Z\}$
- Bytes:  $\mathcal{X} = \{0, 1, \dots, 255\}$

### Definition

#### Symbol

A *symbol* is a single element drawn from an alphabet. For example, the letter E is a symbol from the English alphabet.

### Definition

#### Random Variable

A *random variable*  $X$  is a function that assigns a symbol or value to each outcome in a sample space:

$$X : \Omega \rightarrow \mathcal{X}$$

- $\Omega$ : Sample space (e.g., all possible states of a data source)

- $\mathcal{X}$ : Set of possible values (alphabet, e.g.,  $\{0, 1\}$ , ASCII characters)
- For each  $\omega \in \Omega$ ,  $X(\omega)$  is the value assigned to outcome  $\omega$

### Example: Binary Source

- $\Omega = \{\text{emits 0, emits 1}\}$  (or could be more complex underlying physics)
- $\mathcal{X} = \{0, 1\}$
- $X(\text{emits 0}) = 0, X(\text{emits 1}) = 1$
- Probabilities:  $P(X = 0) = P(\{\omega : X(\omega) = 0\}) = p, P(X = 1) = 1 - p$

### Why this matters for compression:

- The entropy  $H(X)$  depends on the probability distribution induced by  $X$
- For  $x \in \mathcal{X}$ :  $P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\})$
- $H(X) = -\sum_{x \in \mathcal{X}} P(X = x) \log_2 P(X = x)$

### Definition

#### Source

A *source* is a process that generates a sequence of symbols  $(X_1, X_2, X_3, \dots)$  according to some probability law. In this lecture, we assume discrete sources unless stated otherwise.

### Definition

#### Message (or Sequence)

A *message* is a finite sequence of symbols generated by the source:

$$x^n = (x_1, x_2, \dots, x_n)$$

Compression algorithms operate on messages, not on individual symbols.

### Definition

#### Code and Codewords

A *code* assigns a binary string (codeword) to each symbol or message.

- Source symbols  $\rightarrow$  codewords (e.g., Huffman coding)
- Messages  $\rightarrow$  bitstreams (e.g., arithmetic coding)

## Definition

### Block Length

The *block length*  $n$  is the number of source symbols grouped together and encoded as a unit. Larger block lengths generally allow better compression but increase delay and complexity.

## Definition

### Model

A *model* estimates the probabilities of symbols or sequences. Better models lead to better compression by reducing uncertainty.

## 2.3 Information and Redundancy: The Core Concepts

### 2.3.1 Information: A Formal Measure of Uncertainty Reduction

In information theory, information is defined rigorously as a **quantitative measure of the reduction in uncertainty** that results from observing the outcome of a random event.

**Definition 2.1.** Let  $X$  be a random event that occurs with probability  $p = \Pr(X)$ . The **information content** (or self-information)  $I(X)$  provided by the occurrence of  $X$  is defined as:

$$I(X) = \log_b \left( \frac{1}{p} \right) = -\log_b(p)$$

where:

- $b = 2$  yields **bits** (binary digits)
- $b = e$  yields **nats** (natural units)
- $b = 10$  yields **hartleys** or **dits**

## Example

### Predictability vs. Information:

- In a city where it rains every day, the statement “It rained today” conveys almost no information because it was expected
- A file that contains only the bit ‘1’ provides very little information
- A coin that always lands heads produces outcomes, but no information

**Key idea:** Perfect predictability implies zero information gain.

### Example

#### Daily Weather Forecast — Information Content:

- Sunny in Phoenix (probability 0.9):  $I = -\log_2 0.9 \approx 0.15$  bits
- Snow in Phoenix (probability 0.001):  $I = -\log_2 0.001 \approx 9.97$  bits
- Rain in Seattle (probability 0.3):  $I = -\log_2 0.3 \approx 1.74$  bits

**Interpretation:** Rare events carry more information because they reduce uncertainty the most.

### 2.3.2 Redundancy: The Enemy of Information and the Friend of Compression

Redundancy refers to predictable or repeated structure in data. It is what allows data to be represented using fewer bits.

1. **Spatial Redundancy:** Neighboring data values are highly correlated

#### Example

In a photograph of a clear blue sky, most neighboring pixels have nearly identical color values.

- **Naïve:** Store the RGB value of each pixel independently
- **Smarter:** Encode repeated pixel values using run-length encoding
- **Even smarter:** Predict each pixel from its neighbors and encode only the small prediction error

2. **Statistical Redundancy:** Some symbols occur far more frequently than others

#### Example

English letter frequencies:

| Letter | Frequency | Letter | Frequency |
|--------|-----------|--------|-----------|
| E      | 12.7%     | Z      | 0.07%     |
| T      | 9.1%      | Q      | 0.10%     |
| A      | 8.2%      | J      | 0.15%     |

Frequent letters get shorter codes in variable-length coding schemes.

3. **Knowledge Redundancy:** Information already known to both encoder and decoder

4. **Perceptual Redundancy:** Information that humans cannot perceive

## 2.4 Entropy: The Fundamental Limit

### 2.4.1 What is Entropy? Different Perspectives

#### Definition

**Shannon Entropy** of a discrete random variable  $X$  with possible values  $\{x_1, x_2, \dots, x_n\}$  having probabilities  $\{p_1, p_2, \dots, p_n\}$ :

$$H(X) = - \sum_{i=1}^n p_i \log_2 p_i \quad \text{bits}$$

#### Two Complementary Interpretations:

1. **Average Information Content:** Expected value of information content across all symbols
2. **Uncertainty or Surprise:** Measures how uncertain we are about the next symbol

### 2.4.2 Calculating Entropy: Step by Step

#### Example

##### Binary Source Example - Detailed Calculation:

Consider a biased coin:  $P(\text{Heads}) = 0.8$ ,  $P(\text{Tails}) = 0.2$

**Step 1: Calculate individual information content:**

$$I_H = -\log_2(0.8) \approx 0.3219 \text{ bits}$$

$$I_T = -\log_2(0.2) \approx 2.3219 \text{ bits}$$

**Step 2: Calculate entropy as expected value:**

$$H = 0.8 \times 0.3219 + 0.2 \times 2.3219 = 0.7219 \text{ bits}$$

**Step 3: Verify using direct formula:**

$$H = -[0.8 \log_2(0.8) + 0.2 \log_2(0.2)] \approx 0.7219 \text{ bits}$$

#### Key Insights:

- Extreme cases:

- Fair coin ( $P=0.5$ ):  $H = 1.0$  bit (maximum uncertainty)
- Always heads ( $P=1.0$ ):  $H = 0$  bits (no uncertainty)
- 90% heads:  $H \approx 0.469$  bits

### 2.4.3 Entropy of English Text: A Practical Case Study

#### Example

##### Calculating English Letter Entropy:

Based on letter frequencies in typical English text:

$$H \approx 4.18 \text{ bits/letter}$$

##### Layered Interpretation:

- **First-order entropy (letters independent):** 4.18 bits/letter
- **Actual uncertainty is lower:** Letters have dependencies ( $Q \rightarrow U$ )
- **Comparison with encoding schemes:**

| Encoding Method               | Bits/Letter |
|-------------------------------|-------------|
| Naive (5 bits for 26 letters) | 5.00        |
| Huffman (letter-based)        | 4.30        |
| Using digram frequencies      | 3.90        |
| Using word frequencies        | 2.30        |
| Optimal with full context     | $\sim 1.50$ |

### 2.4.4 Beyond First-Order Entropy: The Full Picture

**Higher-Order Entropies** quantify uncertainty while accounting for increasing context:

- **Zero-order entropy ( $H_0$ ):**  $H_0 = \log_2 |\mathcal{X}|$
- **First-order entropy ( $H_1$ ):**  $H_1 = - \sum p(x) \log_2 p(x)$
- **Second-order entropy ( $H_2$ ):**  $H_2 = - \sum p(x, y) \log_2 p(x|y)$
- **$N$ th-order entropy ( $H_N$ ):**  $H_N = - \sum p(x_1, \dots, x_N) \log_2 p(x_N | x_1, \dots, x_{N-1})$

### 2.4.5 Entropy Rate

The **entropy rate** of a source is defined as the limiting uncertainty per symbol when arbitrarily long contexts are available:

$$H_\infty = \lim_{N \rightarrow \infty} H_N$$

### 2.4.6 The Entropy Theorem: Why It Matters

#### Definition

##### Expected Code Length

For a source with symbols  $\{x_1, x_2, \dots, x_n\}$  having probabilities  $\{p_1, p_2, \dots, p_n\}$ , and a code that assigns codeword lengths  $\{\ell_1, \ell_2, \dots, \ell_n\}$ , the **expected code length**  $L$  is:

$$L = \mathbb{E}[\ell(X)] = \sum_{i=1}^n p_i \ell_i \quad (\text{bits per symbol})$$

This measures the average number of bits needed to encode one symbol from the source.

#### Definition

##### Compression Ratio and Efficiency

For a source with entropy  $H(X)$  and code with expected length  $L$ :

- **Compression ratio:**  $\rho = \frac{\text{original bits}}{\text{compressed bits}}$
- **Efficiency:**  $\eta = \frac{H(X)}{L} \leq 1$
- **Redundancy:**  $R = L - H(X) \geq 0$

Perfect compression occurs when  $\eta = 1$  (100% efficient) and  $R = 0$ .

#### Important

##### Shannon's Source Coding Theorem (1948)

For a discrete memoryless source with entropy  $H$  and any  $\epsilon > 0$ :

##### 1. Converse (Impossibility Result):

No lossless coding scheme can achieve expected code length  $L < H$ .

$$L \geq H \quad \text{for any uniquely decodable code}$$

##### 2. Achievability (Possibility Result):

There exists a lossless coding scheme (specifically, block coding with suffi-

ciently large block size  $n$ ) such that:

$$H \leq L < H + \epsilon$$

Equivalently: For any  $\epsilon > 0$ ,  $\exists n$  such that:

$$\frac{L_n}{n} < H + \epsilon$$

where  $L_n$  is the expected length for blocks of size  $n$ .

### Interpretation:

- **Entropy is the fundamental limit:**  $H$  bits/symbol is the best we can ever do
- **We can get arbitrarily close:** With clever coding, we can approach this limit as closely as desired
- **The gap is achievable:** The "+ $\epsilon$ " represents practical overhead that can be made arbitrarily small

### Example

#### Understanding the Theorem with Numbers

Consider a binary source with  $p(0) = 0.9$ ,  $p(1) = 0.1$ :

$$H = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 \approx 0.469 \text{ bits/symbol}$$

- **Naive coding:** Use 1 bit per symbol  $\rightarrow L = 1.0$ , efficiency  $\eta = 0.469/1.0 = 46.9\%$
- **Huffman coding:**  $0 \rightarrow 0$ ,  $1 \rightarrow 1$  (same as naive!)  $\rightarrow L = 1.0$ ,  $\eta = 46.9\%$  *Why so bad?* Because we're coding symbols individually.
- **Block coding (n=2):** Code pairs of symbols:

$$00 \rightarrow 0 \quad (\ell = 1, p = 0.81)$$

$$01 \rightarrow 10 \quad (\ell = 2, p = 0.09)$$

$$10 \rightarrow 110 \quad (\ell = 3, p = 0.09)$$

$$11 \rightarrow 111 \quad (\ell = 3, p = 0.01)$$

$$L_2 = 0.81 \times 1 + 0.09 \times 2 + 0.09 \times 3 + 0.01 \times 3 = 1.29 \text{ bits/block}$$

$$\text{Per symbol: } L = L_2/2 = 0.645 \text{ bits/symbol, } \eta = 0.469/0.645 \approx 72.7\%$$

- **Block coding (n=3):** Would get even closer to 0.469
- **Theoretical limit:** As  $n \rightarrow \infty$ ,  $L \rightarrow 0.469$

**Key insight:** The theorem tells us:

1. We can never beat 0.469 bits/symbol (impossibility)
2. We can get as close as we want to 0.469 bits/symbol (achievability)

### Important

#### What the Theorem Does NOT Say

- It doesn't say **how to construct the code** - just that one exists
- It doesn't **guarantee practical implementation** - block size  $n$  might need to be huge
- It doesn't **account for computational complexity** - the code might be too complex to implement
- It **assumes we know the true probabilities** - in practice, we estimate them

Yet, this theorem is revolutionary because it:

1. Establishes a **fundamental limit** (like the speed of light in physics)
2. Provides a **benchmark** for evaluating compression algorithms
3. Guides algorithm design toward this limit

#### 2.4.7 Key Takeaways

- Entropy measures both **average information** and **uncertainty**
- Higher-order models reduce entropy by exploiting dependencies
- The entropy rate represents the ultimate compression limit
- Shannon's theorem precisely separates the *possible* from the *impossible*

## 2.5 Entropy as a Lower Bound

### 2.5.1 The Fundamental Inequality

**Theorem 2.2** (Entropy Lower Bound). *For any **uniquely decodable** code  $C$  for source  $X$ :*

$$L(C) \geq H(X)$$

where  $L(C) = \mathbb{E}[\ell(X)] = \sum_i p_i \ell_i$  is the expected code length.

#### Example

**Binary Source with  $p(0) = 0.9$ ,  $p(1) = 0.1$**

$$H = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 \approx 0.469 \text{ bits/symbol}$$

**Why we can't achieve 0.4 bits/symbol:**

1. For 100 symbols, typical sequences:  $2^{100 \times 0.469} \approx 2^{46.9}$
2. To encode all uniquely, need at least  $2^{46.9}$  codewords
3. At 0.4 bits/symbol, total bits = 40
4. # codewords  $\leq 2^{40} < 2^{46.9} \rightarrow$  impossible!

## 2.6 Kraft-McMillan Inequality: Core Theoretical Tool

### 2.6.1 Statement and Interpretation

**Theorem 2.3** (Kraft-McMillan Inequality (Binary Case)). *Let  $\ell_1, \ell_2, \dots, \ell_m$  be the lengths of codewords in a **prefix code**. Then:*

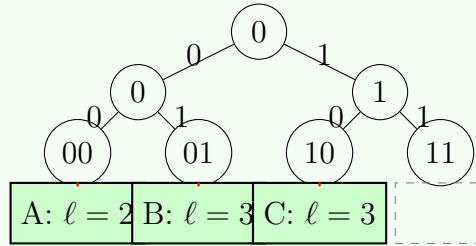
$$\sum_{i=1}^m 2^{-\ell_i} \leq 1$$

*Conversely, if integers  $\ell_1, \dots, \ell_m$  satisfy this inequality, then there exists a binary prefix code with these lengths.*

#### Example

**Tree Visualization of Kraft Inequality**

Consider a binary tree of depth  $L = \max \ell_i$ :



Calculating Kraft sum for  $\{\ell_A = 2, \ell_B = 3, \ell_C = 3\}$ :

$$\sum 2^{-\ell_i} = 2^{-2} + 2^{-3} + 2^{-3} = 0.25 + 0.125 + 0.125 = 0.5 \leq 1$$

### Example

#### Testing Code Feasibility

1. Valid lengths:  $\{1, 2, 3, 3\}$

$$\sum = 2^{-1} + 2^{-2} + 2^{-3} + 2^{-3} = 0.5 + 0.25 + 0.125 + 0.125 = 1.0 \quad \text{VALID}$$

2. Invalid lengths:  $\{1, 1, 2\}$

$$\sum = 2^{-1} + 2^{-1} + 2^{-2} = 0.5 + 0.5 + 0.25 = 1.25 > 1 \quad \text{INVALID}$$

## 2.7 Optimality of Huffman Codes (Theory Only)

### 2.7.1 The Optimality Theorem

**Theorem 2.4** (Huffman Optimality). *Given a source with symbol probabilities  $p_1, p_2, \dots, p_m$ , the Huffman algorithm produces a prefix code that **minimizes** the expected code length  $L = \sum_{i=1}^m p_i \ell_i$  among all prefix codes.*

### 2.7.2 Relation to Entropy

For any Huffman code:

$$H(X) \leq L_{\text{Huffman}} < H(X) + 1$$

### Example

#### Understanding the "+1" Gap

Consider source with probabilities  $\{0.6, 0.3, 0.1\}$ :

$$H \approx 1.295 \text{ bits}$$

**Ideal (non-integer) lengths:**  $-\log_2 p_i = \{0.737, 1.737, 3.322\}$

**Huffman code:**  $0.6 \rightarrow 0, 0.3 \rightarrow 10, 0.1 \rightarrow 11$

$$L = 0.6 \times 1 + 0.3 \times 2 + 0.1 \times 2 = 1.4 \text{ bits}$$

**Comparison:**

- Entropy: 1.295 bits
- Huffman: 1.400 bits
- Gap: 0.105 bits (much less than 1!)

### 2.7.3 Why Huffman is Optimal but Not Perfect

#### Important

**Huffman is Optimal Within a Restricted Class**

Huffman is optimal among:

- **Symbol-by-symbol** codes
- **Prefix** codes
- **Static** codes

But real optimality might require:

- **Block coding**
- **Fractional bits** (arithmetic coding)
- **Adaptive probabilities**

## 2.8 Limitations of Symbol-by-Symbol Coding

### 2.8.1 Three Fundamental Limitations

1. **Cannot Exploit Dependencies**
2. **Integer Length Constraint:**  $\ell_i \in \mathbb{Z}^+$  but  $-\log_2 p_i \in \mathbb{R}$
3. **Memoryless Assumption**

### Example

#### English Text: The Cost of Symbol-by-Symbol

- **First-order entropy** (ignoring dependencies): 4.0 bits/letter
- **Actual entropy rate** (with dependencies): 1.5 bits/letter
- **Huffman on letters**: 4.0 bits/letter
- **Gap**: 2.5 bits/letter wasted due to ignoring dependencies

## 2.9 Block Coding and Improved Efficiency

### 2.9.1 The Block Coding Idea

Instead of coding symbols individually, group them into blocks of length  $n$ :

$$\mathbf{X} = (X_1, X_2, \dots, X_n)$$

### Definition

#### $n$ th Extension of a Source

For a source with alphabet  $\mathcal{X}$ , the  $n$ th extension has alphabet:

$$\mathcal{X}^n = \{(x_1, \dots, x_n) : x_i \in \mathcal{X}\}$$

with size  $|\mathcal{X}|^n$ .

### 2.9.2 Key Mathematical Results

**Theorem 2.5** (Entropy of Block Source). *For a discrete memoryless source:*

$$H(X^n) = nH(X)$$

**Theorem 2.6** (Block Coding Performance). *There exists a prefix code  $C_n$  for  $X^n$  such that:*

$$nH(X) \leq L_n < nH(X) + 1$$

*Dividing by  $n$ :*

$$H(X) \leq \frac{L_n}{n} < H(X) + \frac{1}{n}$$

## Important

### The Magic of Block Coding

As  $n \rightarrow \infty$ :

$$\frac{L_n}{n} \rightarrow H(X)$$

We can approach entropy **arbitrarily closely** by making blocks larger!

### 2.9.3 Step-by-Step Example

#### Example

**Binary Source:**  $p(0) = 0.9$ ,  $p(1) = 0.1$ ,  $H \approx 0.469$

**Step 1:**  $n = 1$  (symbol-by-symbol)

- Huffman:  $0 \rightarrow 0$ ,  $1 \rightarrow 1$
- $L_1 = 1$  bit/symbol
- Efficiency:  $\eta = 0.469/1 = 46.9\%$

**Step 2:**  $n = 2$  (code pairs)

- Block probabilities:  $P(00) = 0.81$ ,  $P(01) = 0.09$ ,  $P(10) = 0.09$ ,  $P(11) = 0.01$
- Codes:  $00 \rightarrow 0$ ,  $01 \rightarrow 10$ ,  $10 \rightarrow 110$ ,  $11 \rightarrow 111$
- Per symbol:  $L_2/2 = 0.645$  bits/symbol
- Efficiency:  $\eta = 0.469/0.645 = 72.7\%$

## 2.10 Trade-Offs in Block Coding

### 2.10.1 The Engineering Challenges

1. **Exponential Alphabet Growth:**  $|\mathcal{X}^n| = |\mathcal{X}|^n$
2. **Memory Requirements:** Huffman tree has  $2m^n - 1$  nodes
3. **Computational Complexity:**  $O(m^n \log m^n)$
4. **Delay and Latency:** Must wait for  $n$  symbols

## 2.11 From Block Coding to Modern Compression

### 2.11.1 Arithmetic Coding: Fractional-Bit Block Coding

#### Important

##### Arithmetic Coding as "Infinite Block Coding"

Arithmetic coding cleverly avoids the exponential growth problem:

- **Idea:** Encode entire message as a single real number in  $[0,1)$
- **No explicit blocks:** Processes symbols sequentially
- **Fractional bits:** Achieves  $L \approx H(X)$  without large  $n$
- **Removes integer constraint:** No "+1" overhead!

### 2.11.2 Context Modeling: Approximating Large Blocks

Instead of explicit block coding, modern compressors use:

1. **Context Models:** Predict next symbol based on previous  $k$  symbols
2. **Prediction + Residual Coding:** Encode only prediction error
3. **Dictionary Methods (LZ family):** Build dictionary of previously seen phrases

## 2.12 What Theory Guarantees vs What Practice Achieves

| Aspect            | Theory Guarantees                        | Practice Achieves                           |
|-------------------|------------------------------------------|---------------------------------------------|
| <b>Optimality</b> | Can approach entropy arbitrarily closely | Gets close, but with practical limits       |
| <b>Block Size</b> | $n \rightarrow \infty$ gives optimality  | $n$ limited by memory, latency, complexity  |
| <b>Complexity</b> | Ignored, infinite resources allowed      | Critical constraint; often dominates design |

## 2.13 Summary and Key Takeaways

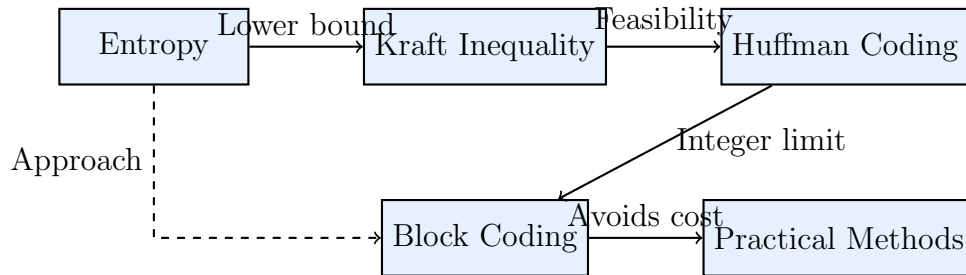
#### Important

##### Five Fundamental Lessons

1. **Entropy is the Absolute Limit:**  $L \geq H(X)$  for any lossless code

2. **Kraft-McMillan Constrains All Codes:**  $\sum 2^{-\ell_i} \leq 1$
3. **Huffman is Optimal Among Prefix Codes:** But limited by integer lengths
4. **Block Coding Allows Approaching Entropy:**  $\lim_{n \rightarrow \infty} \frac{L_n}{n} = H(X)$
5. **Practical Compression Balances Efficiency and Complexity**

### 2.13.1 The Big Picture



### 2.13.2 Looking Forward

- **Next lecture: Arithmetic Coding:** Removes integer constraint
- **Then: Dictionary Methods (LZ family):** Adaptive to data statistics
- **Finally: Modern Compressors:** Combining multiple techniques

#### Final Thought

Shannon's 1948 paper told us *exactly how good compression could possibly be*.  
 Every compressor since has been trying to approach that limit while staying  
 within practical constraints.

The gap between theory and practice is where engineering creativity lives!

## 3: Lecture 3: Advanced Entropy Coding & Extensions

### Lecture 3: Beyond Huffman – Advanced Entropy Coding Methods

#### 3.1 Introduction & Motivation

##### Important

##### Recall Huffman Coding Limitations:

- **Integer code lengths:** Cannot reach entropy bound for highly skewed distributions
- **Static vs. Adaptive:** Standard Huffman requires prior knowledge of probabilities
- **Codebook overhead:** Need to transmit/store the coding tree
- **Symbol-by-symbol constraint:** Processes one symbol at a time

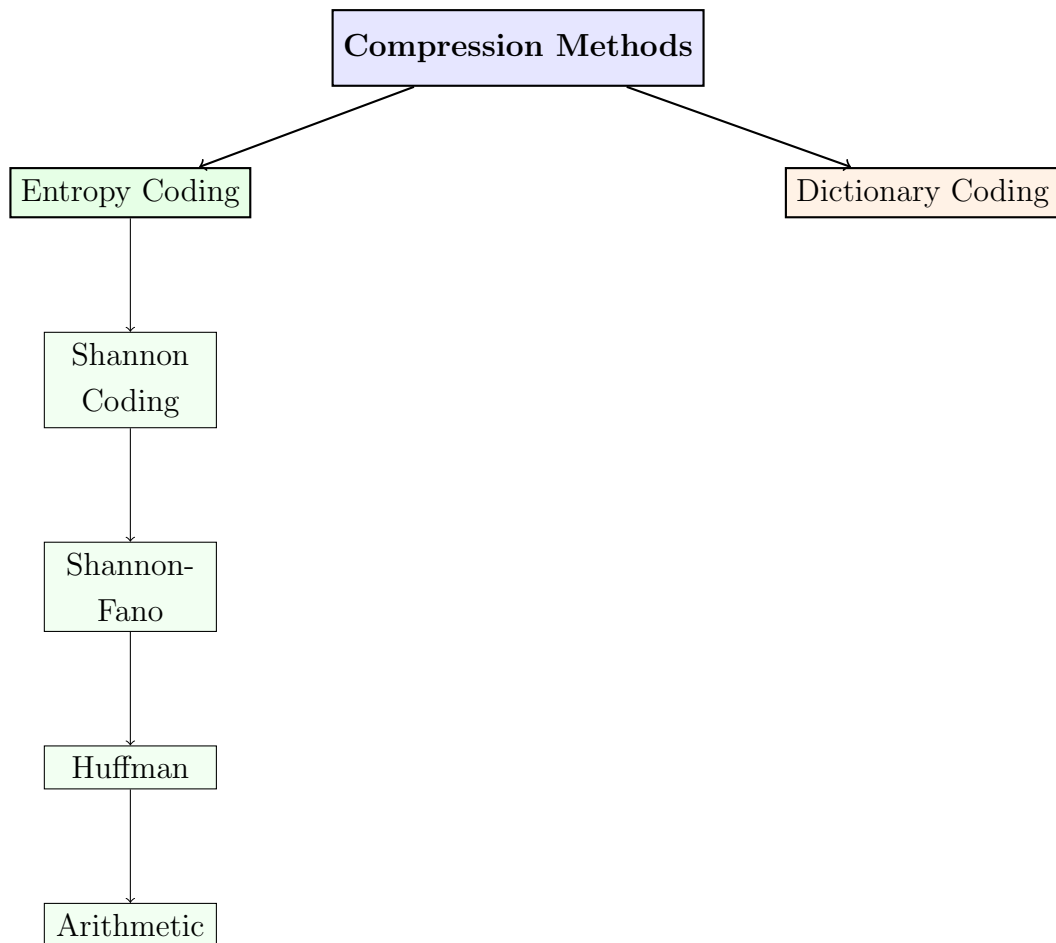
##### Lecture Roadmap:

1. **Framework:** Coding taxonomy and conceptual organization
2. **Historical methods:** Shannon & Shannon-Fano coding
3. **Practical improvements:** Canonical and Adaptive Huffman
4. **Next generation:** Arithmetic coding paradigm
5. **Synthesis:** Comparison and forward look

#### 3.2 Coding Taxonomy & Framework

##### Definition

**Coding Taxonomy:** Classification of compression methods based on key characteristics



## Key Dimensions in Compression Algorithm Design

### 1. Modeling vs. Coding (Two-Stage View):

- **Modeling Phase:** Analyzes data to estimate symbol probabilities or discover patterns.
  - *Examples:* Frequency counting (Huffman), context modeling (PPM), dictionary construction (LZ77)
- **Coding Phase:** Converts modeled information into actual bits.
  - *Examples:* Huffman codes, arithmetic codes, LZ77 pointers
- Some algorithms intertwine both (e.g., LZW builds dictionary while coding).

### 2. Knowledge of Source Distribution:

- **Static/Fixed:** Uses a predefined model that doesn't change.
  - *Example:* JPEG Huffman tables, known language frequencies
  - Requires prior knowledge of data; fails if distribution differs.
- **Adaptive:** Learns and updates the model during compression.
  - *Example:* Adaptive Huffman, LZ78 dictionary building

- No prior knowledge needed; overhead for model transmission.
- **Universal:** Can compress any source asymptotically optimally.
  - *Theoretical property:* LZ family, arithmetic with adaptive model
  - Note: Most adaptive methods are universal in practice.

### 3. Processing Granularity:

- **Symbol-by-Symbol:** Each input symbol maps to one codeword.
  - *Example:* Huffman coding
  - Simple but limited to integer bits per symbol.
- **Block Coding:** Fixed-size groups of symbols coded together.
  - *Example:* Block-sorting (BWT) processes blocks
  - Can capture inter-symbol dependencies within block.
- **Stream/Incremental:** Continuous processing with immediate output.
  - *Example:* Arithmetic coding, LZ77 sliding window
  - No blocking delay; good for real-time applications.
- *Note:* "Message-wide" (whole file as one symbol) is theoretical ideal; arithmetic coding approximates it by treating the entire stream as one long fractional code.

### 4. Algorithmic Approach (Primary Taxonomy):

- **Statistical Coding:** Uses probability estimates (Huffman, Arithmetic)
- **Dictionary Coding:** Replaces repeated patterns with references (LZ family)
- **Transform Coding:** Changes data domain then codes (DCT, wavelet)
- **Predictive Coding:** Predicts next value, codes difference (DPCM, LPC)

#### Important Relationships:

- Adaptive methods are usually universal for practical purposes.
- Stream coding is possible with both symbol-by-symbol (Huffman) and message-wide approaches (arithmetic).
- Block processing (like BWT) is often followed by stream coding (like MTF+RLE+arithmetic in bzip2).
- Modeling and coding can be separated (PPM + arithmetic) or combined (LZW).

## 3.3 Shannon Coding (1948)

## Definition

**Shannon Coding:** A constructive method derived from Shannon's source coding theorem that assigns codewords by taking the binary expansion of cumulative probabilities:

$$l_i = \lceil -\log_2 p_i \rceil \quad \text{and} \quad \text{code}_i = \text{First } l_i \text{ bits of } F_i$$

where  $p_i$  is the probability of symbol  $i$ , and  $F_i = \sum_{j=1}^{i-1} p_j$  is the cumulative probability of symbols ordered by decreasing probability.

## Shannon Coding Algorithm

**Input:** Symbols  $S = \{s_1, s_2, \dots, s_n\}$  with corresponding probabilities  $P = \{p_1, p_2, \dots, p_n\}$

**Output:** Prefix-free binary code for each symbol

1. **Sort** symbols in non-increasing order of probability:  $p_1 \geq p_2 \geq \dots \geq p_n$
2. **Calculate codeword lengths:** For each symbol  $i$ , compute  $l_i = \lceil -\log_2 p_i \rceil$
3. **Compute cumulative probabilities:**

$$F_1 = 0, \quad F_i = \sum_{j=1}^{i-1} p_j \quad \text{for } i = 2, \dots, n$$

4. **Generate codewords:** For each symbol  $i$ :
  - a) Convert  $F_i$  to binary fractional representation (0.xxxx...)
  - b) Take the first  $l_i$  bits after the binary point as the codeword

## Example

**Example:** Given symbols with probabilities:

| Symbol | Probability |
|--------|-------------|
| A      | 0.5         |
| B      | 0.25        |
| C      | 0.125       |
| D      | 0.125       |

**Step-by-step construction:**

1. **Sort by probability:** Already in non-increasing order ( $0.5 \geq 0.25 \geq 0.125 \geq 0.125$ )
2. **Calculate lengths:**

- $l_A = \lceil -\log_2 0.5 \rceil = \lceil 1.0 \rceil = 1$
- $l_B = \lceil -\log_2 0.25 \rceil = \lceil 2.0 \rceil = 2$
- $l_C = \lceil -\log_2 0.125 \rceil = \lceil 3.0 \rceil = 3$
- $l_D = \lceil -\log_2 0.125 \rceil = \lceil 3.0 \rceil = 3$

### 3. Compute cumulative probabilities:

- $F_A = 0$  (first symbol)
- $F_B = 0.5$  (just A's probability)
- $F_C = 0.5 + 0.25 = 0.75$  (A + B)
- $F_D = 0.5 + 0.25 + 0.125 = 0.875$  (A + B + C)

### 4. Convert $F_i$ to binary and take first $l_i$ bits:

- **A:**  $F_A = 0.0_{10}$  in binary is  $0.0000\dots_2$   
– Take first  $l_A = 1$  bit: **0**
- **B:**  $F_B = 0.5_{10}$  in binary is  $0.1000\dots_2$   
– Take first  $l_B = 2$  bits: **10**
- **C:**  $F_C = 0.75_{10}$  in binary is  $0.1100\dots_2$   
– Take first  $l_C = 3$  bits: **110**
- **D:**  $F_D = 0.875_{10}$  in binary is  $0.1110\dots_2$   
– Take first  $l_D = 3$  bits: **111**

### Resulting code:

| Symbol | Probability | $l_i$ | $F_i$ | Shannon Code |
|--------|-------------|-------|-------|--------------|
| A      | 0.5         | 1     | 0.0   | 0            |
| B      | 0.25        | 2     | 0.5   | 10           |
| C      | 0.125       | 3     | 0.75  | 110          |
| D      | 0.125       | 3     | 0.875 | 111          |

**Expected length:**  $L = 0.5 \times 1 + 0.25 \times 2 + 0.125 \times 3 + 0.125 \times 3 = 1.75$  bits/symbol

### Important

#### Critical Note on Sorting:

The sorting step is **essential** in Shannon coding because:

- Cumulative probabilities  $F_i$  depend on the order of symbols

- Sorting ensures intervals in  $[0,1)$  are assigned in decreasing size order
- Without sorting, the resulting code may not be prefix-free
- Sorting guarantees the length condition  $\frac{1}{2^{l_i}} \leq p_i < \frac{1}{2^{l_i-1}}$  leads to non-overlapping intervals

**Why it works:** When probabilities are sorted, each symbol's interval of size  $p_i$  starts at  $F_i$  and ends at  $F_i + p_i$ . The codeword length ensures the binary expansion of  $F_i$  to  $l_i$  bits uniquely identifies the starting point without overlapping with neighboring intervals.

### Understanding the Binary Expansion Process

When we write  $F_i$  in binary (e.g.,  $0.5 = 0.1_2$ ,  $0.75 = 0.11_2$ ), we're essentially:

- Dividing the interval  $[0,1)$  into subintervals based on probabilities
- Each symbol gets an interval of size  $p_i$
- The codeword is the **binary fraction** representing the **start** of that interval
- We use exactly  $l_i = \lceil -\log_2 p_i \rceil$  bits, which ensures:

$$\frac{1}{2^{l_i}} \leq p_i < \frac{1}{2^{l_i-1}}$$

- This guarantees unique prefixes because intervals don't overlap!

### Important

#### Properties of Shannon Coding:

- **Constructive proof:** Demonstrates that prefix codes exist for any lengths satisfying Kraft inequality
- **Simple to compute:** Direct from probabilities, no tree needed
- **Requires sorting:** Symbols must be ordered by decreasing probability first
- **Not optimal:** Unlike Huffman, doesn't minimize expected length (compare: Huffman would give A=0, B=10, C=110, D=111 **same in this case!**)
- **Theoretical importance:** Foundation for Shannon's source coding theorem
- **Efficiency bound:**  $H(X) \leq L < H(X) + 1$  (like Shannon's theorem says)

### Example

#### Counterexample: What happens if we don't sort?

Consider the same symbols but in different order: C (0.125), A (0.5), D (0.125), B (0.25)

#### Without sorting:

- $F_C = 0, l_C = 3 \rightarrow$  codeword: 000
- $F_A = 0.125, l_A = 1 \rightarrow$  codeword: 0 (binary: 0.001...  $\rightarrow$  first bit is 0)
- $F_D = 0.625, l_D = 3 \rightarrow$  codeword: 101
- $F_B = 0.75, l_B = 2 \rightarrow$  codeword: 11

**Problem:** Codeword "0" (for A) is a prefix of "000" (for C)  $\rightarrow$  **not prefix-free!**  
This demonstrates why sorting is essential in Shannon coding.

## 3.4 Shannon–Fano Coding (1949)

### Definition

**Shannon–Fano Coding:** A top-down, recursive source coding technique that assigns binary codewords by repeatedly partitioning a set of symbols into two subsets whose total probabilities are as close as possible. The method was developed independently by *Claude Shannon* and *Robert Fano* in 1949.

### Algorithm Description

**High-level idea:** Symbols with higher probabilities should receive shorter codewords. This is achieved by repeatedly splitting the symbol set into two probability-balanced groups and assigning binary prefixes.

#### Step-by-step procedure:

1. **Sort** the symbols in decreasing order of probability:

$$p_1 \geq p_2 \geq \cdots \geq p_n.$$

2. **Recursive partitioning:**

- If the current set contains only one symbol, stop (this is the base case).

- Find an index  $k$  that minimizes

$$\left| \sum_{i=1}^k p_i - \sum_{i=k+1}^n p_i \right|.$$

- This divides the symbols into two subsets:

$$S_1 = \{1, \dots, k\}, \quad S_2 = \{k+1, \dots, n\}.$$

- Append bit **0** to the codewords of all symbols in  $S_1$ .
- Append bit **1** to the codewords of all symbols in  $S_2$ .
- Apply the same procedure recursively to  $S_1$  and  $S_2$ .

### Example with Six Symbols

#### Example

**Example:** Consider six symbols with the following probabilities.

| Symbol | Probability | $-\log_2 p_i$ |
|--------|-------------|---------------|
| A      | 0.30        | 1.74          |
| B      | 0.25        | 2.00          |
| C      | 0.20        | 2.32          |
| D      | 0.10        | 3.32          |
| E      | 0.10        | 3.32          |
| F      | 0.05        | 4.32          |

#### Construction process

**Step 1: First split** (balance 0.55 vs. 0.45)

- Sorted symbols: A(0.30), B(0.25), C(0.20), D(0.10), E(0.10), F(0.05)
- Best split:  $\{A, B\}(0.55)$  and  $\{C, D, E, F\}(0.45)$
- Prefix assignment:  $\{A, B\} \rightarrow 0$ ,  $\{C, D, E, F\} \rightarrow 1$

**Step 2: Split  $\{A, B\}$**

- A: **00**, B: **01**

**Step 3: Split  $\{C, D, E, F\}$**

- Best split:  $\{C\}(0.20)$  and  $\{D, E, F\}(0.25)$
- C: **10**,  $\{D, E, F\} \rightarrow 11$

**Step 4: Split  $\{D, E, F\}$**

- Best split:  $\{D\}(0.10)$  and  $\{E, F\}(0.15)$
- D: 110,  $\{E, F\} \rightarrow 111$

**Step 5: Split  $\{E, F\}$**

- E: 1110, F: 1111

**Final codes:**

| Symbol | Probability | Code | Length |
|--------|-------------|------|--------|
| A      | 0.30        | 00   | 2      |
| B      | 0.25        | 01   | 2      |
| C      | 0.20        | 10   | 2      |
| D      | 0.10        | 110  | 3      |
| E      | 0.10        | 1110 | 4      |
| F      | 0.05        | 1111 | 4      |

**Expected code length:**

$$L = 2.40 \text{ bits/symbol.}$$

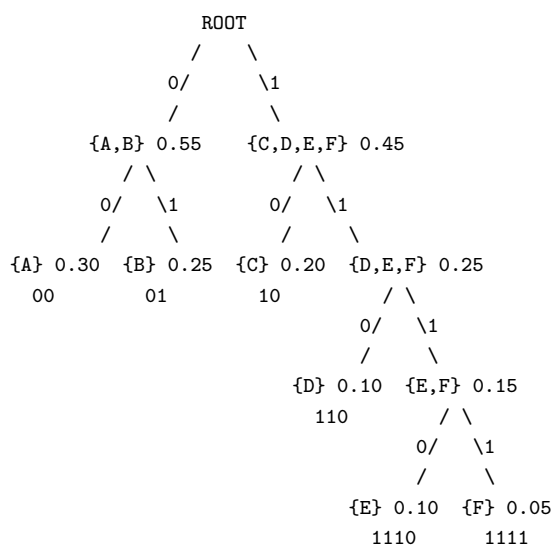
**Entropy:**

$$H(X) \approx 2.25 \text{ bits/symbol.}$$

**Efficiency:**

$$\frac{H(X)}{L} \approx 93.8\%.$$

## Visual Tree Representation



## Key Observations

- **Probability-balanced splits:** The algorithm focuses on balancing probabilities rather than the number of symbols.
- **Variable code lengths:** More probable symbols receive shorter codewords.
- **Prefix-free property:** No codeword is a prefix of another.
- **Near-optimal performance:** The efficiency is high but not guaranteed to be optimal.

### Important

#### Limitations and Historical Context

- Shannon–Fano coding does *not* always produce the optimal code.
- Huffman coding (1952) guarantees the minimum average code length.
- Historically important as a precursor to Huffman coding.

## 3.5 Canonical Huffman Codes

### Definition

**Canonical Huffman Code:** A standardized representation of a Huffman code in which:

- Codes are assigned in lexicographic (binary) order
- All codewords of the same length are consecutive binary numbers
- The first codeword of each length is the smallest possible binary value
- Only the code lengths are required to reconstruct the entire code

This enables compact storage and fast table-based decoding.

### Why Canonical Huffman Codes?

A standard Huffman algorithm produces *optimal code lengths*, but the exact bit patterns depend on implementation details such as tie-breaking and tree construction:

- Different trees can yield the same optimal set of code lengths
- Different bit assignments, but identical compression performance

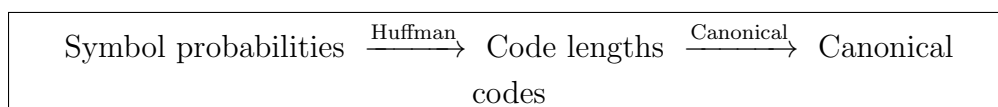
Canonical Huffman coding fixes **one unique and deterministic assignment** for a given set of code lengths so that:

1. Only code lengths need to be transmitted
2. The decoder can reconstruct the codes without ambiguity
3. Decoding can be implemented efficiently using lookup tables

## Two-Step Process

### Complete workflow:

1. **Run the standard Huffman algorithm** to obtain optimal code lengths  $l_i$
2. **Apply the canonical transformation** to convert lengths into standardized codes



## Canonical Huffman Construction Algorithm

**Input:** Code lengths  $l_i$  obtained from a standard Huffman algorithm **Output:** Canonical Huffman codes

1. **Sort symbols** in ascending order of code length  $l_i$  (primary key), and by symbol order or value (secondary key for tie-breaking)
2. **Count symbols per length:**
  - Let  $l_{\min}$  and  $l_{\max}$  be the minimum and maximum code lengths
  - For each length  $l$ , compute  $\text{count}[l] = \text{number of symbols with length } l$
3. **Compute starting codes:**
  - $\text{start\_code}[l_{\min}] = 0$
  - For  $l = l_{\min} + 1$  to  $l_{\max}$ :

$$\text{start\_code}[l] = (\text{start\_code}[l - 1] + \text{count}[l - 1]) \times 2$$

4. Initialize assignment pointers:

$$\text{next\_code}[l] = \text{start\_code}[l] \quad \text{for all } l$$

5. **Assign codes to symbols** (in sorted order):
  - For each symbol  $i$  with length  $l_i$ :

- Assign `next_code[li]` as its code (converted to  $l_i$ -bit binary)
- Increment `next_code[li]` by 1

### Understanding the Shift Operation: Why Multiply by 2?

The key step `start_code[l] = (start_code[l - 1] + count[l - 1]) × 2` ensures:

#### Important

**Visual Intuition:** Imagine all codes of length  $l - 1$  are written as binary numbers. When we multiply by 2 (left shift by 1 bit), we:

- Append a 0 to each existing code
- Create room for new codes that are one bit longer
- Ensure the new codes maintain prefix-free property

**Example:** If the last code of length 2 is "11" (binary 3), then:

$$(\text{last code} + 1) \times 2 = (3 + 1) \times 2 = 8 = 1000_{\text{binary}}$$

The first code of length 3 is "100" (3 bits of 1000), which follows "11" properly.

#### Mathematical Explanation:

- `start_code[l - 1] + count[l - 1]` gives the **first unused code value** after all codes of length  $l - 1$
- Multiplying by 2 (left shift by 1 bit) makes room for an extra bit
- This ensures all codes of length  $l$  start with a 0 in their new bit position
- The result is the smallest valid starting point for the next length group

#### Bit Pattern Visualization:

Length 2 codes: 00, 01, 10, 11

Length 3 codes start at:  $(11 + 1) \times 2 = 1000_{\text{binary}}$

So length 3 codes are: 100, 101, 110, 111

Notice: 11 is NOT a prefix of 100!

## Complete Worked Example

### Example

**Example:** Suppose the Huffman algorithm produces the following code lengths:

| Symbol | Length ( $l_i$ ) |
|--------|------------------|
| A      | 2                |
| B      | 3                |
| C      | 3                |
| D      | 3                |
| E      | 4                |
| F      | 4                |

**Key Point:** The code lengths ( $l_i$ ) are already determined by the Huffman tree construction. These lengths tell us *exactly* how many bits each symbol's code will use.

**Step 1: Sort symbols by length (primary), then symbol order (secondary)**

- **Critical:** Sorting ensures consistent code assignment
- Order: A (2), B (3), C (3), D (3), E (4), F (4)
- Within same length group, maintain original symbol order

**Step 2: Count symbols per length**

$$\text{count}[2] = 1, \quad \text{count}[3] = 3, \quad \text{count}[4] = 2$$

**Step 3: Compute starting codes (with shift explanation)**

$\text{start\_code}[2] = 0$  (represented as 00 in binary, using **2 bits** because length=2)

$\text{start\_code}[3] = (\text{start\_code}[2] + \text{count}[2]) \times 2$

$= (0 + 1) \times 2 = 2$  (represented as 010 in binary, using **3 bits** because length=3)

*Interpretation: After 1 code of length 2, shift left (add 0 bit) for length 3*

$\text{start\_code}[4] = (\text{start\_code}[3] + \text{count}[3]) \times 2$

$= (2 + 3) \times 2 = 10$  (represented as 1010 in binary, using **4 bits** because length=4)

*Interpretation: After 3 codes of length 3, shift left (add 0 bit) for length 4*

**Important:** The multiplication by 2 is a **left shift** operation. It adds one more bit to the code while maintaining the prefix-free property.

**Step 4: Initialize assignment pointers**

$$\text{next\_code}[2] = 0, \quad \text{next\_code}[3] = 2, \quad \text{next\_code}[4] = 10$$

**Step 5: Assign codes (in sorted order)**

| Symbol | Length | Decimal | Binary Code | next_code after |
|--------|--------|---------|-------------|-----------------|
| A      | 2      | 0       | 00          | 1               |
| B      | 3      | 2       | 010         | 3               |
| C      | 3      | 3       | 011         | 4               |
| D      | 3      | 4       | 100         | 5               |
| E      | 4      | 10      | 1010        | 11              |
| F      | 4      | 11      | 1011        | 12              |

**How binary conversion works:**

- For symbol A (length=2, decimal=0): Binary: 00 (padded to 2 bits)
- For symbol B (length=3, decimal=2): Binary: 010 (2 in binary is 10, padded to 3 bits: 010)
- For symbol E (length=4, decimal=10): Binary: 1010 (10 in binary is 1010, already 4 bits)

**Algorithm Summary:**

1. **Input:** Code lengths  $l_i$  for each symbol (from Huffman tree)
2. **For each length  $l$ :**
  - First code of length  $l$ : start\_code[ $l$ ]
  - Subsequent codes: increment by 1
  - All codes use exactly  $l$  bits
3. **Key formula:** start\_code[ $l + 1$ ] = (start\_code[ $l$ ] + count[ $l$ ])  $\times$  2

**Final canonical codes (verify properties):**

| Symbol | Length | Canonical Code | Check                 |
|--------|--------|----------------|-----------------------|
| A      | 2      | 00             | ✓ First length 2 code |
| B      | 3      | 010            | ✓ First length 3 code |
| C      | 3      | 011            | ✓ Consecutive with B  |
| D      | 3      | 100            | ✓ Consecutive with C  |
| E      | 4      | 1010           | ✓ First length 4 code |
| F      | 4      | 1011           | ✓ Consecutive with E  |

**Verification:**

- All codes of same length are consecutive binary numbers ✓
- Codes are lexicographically ordered ✓
- No code is a prefix of another ✓

## What If We Don't Sort? A Counterexample

### Example

**Problem without sorting:** Suppose we have the same code lengths but assign codes without proper sorting:

| Symbol | Length |
|--------|--------|
| D      | 3      |
| A      | 2      |
| E      | 4      |
| B      | 3      |
| F      | 4      |
| C      | 3      |

If we assign codes in this random order using the same algorithm:

- D (length 3) gets code 010
- A (length 2) gets code 00
- E (length 4) gets code 1000
- **Problem:** Code "00" (A) is a prefix of "1000" (E)!

**Conclusion:** Sorting ensures that shorter codes are assigned first, preventing prefix conflicts between codes of different lengths.

## Transmission and Decoding

### Transmission format (very compact):

- Transmit only the sequence of code lengths in symbol order:

$$\langle l_1, l_2, \dots, l_n \rangle$$

- Example for our symbols:  $\langle 2, 3, 3, 3, 4, 4 \rangle$
- Each length can be represented using  $\lceil \log_2 l_{\max} \rceil$  bits

### Decoder reconstruction:

1. Read code lengths for all symbols (in original symbol order)
2. Sort symbols by  $(l_i, \text{index})$  to match encoder's order
3. Reconstruct canonical codes using the same algorithm
4. Build decoding tables for fast lookup

## Comparison with Standard Huffman

| Standard Huffman (tree-based)       | Canonical Huffman (length-based)  |
|-------------------------------------|-----------------------------------|
| Output: Huffman tree structure      | Output: Code lengths only         |
| Must transmit tree (complex)        | Transmit only lengths (simple)    |
| Decoding by tree traversal (slower) | Decoding by table lookup (faster) |
| Multiple equivalent trees possible  | Single deterministic assignment   |
| Same optimal compression            | Same optimal compression          |
| More complex implementation         | Simple, robust implementation     |
| <b>No sorting required</b>          | <b>Requires sorting by length</b> |

## Key Properties and Applications

- **Optimality:** Identical compression ratio to standard Huffman
- **Compactness:** Only code lengths stored/transmitted
- **Speed:** Table-based decoding is significantly faster
- **Standardization:** Used in DEFLATE (gzip/ZIP), JPEG, PNG, MPEG
- **Deterministic:** Same lengths always produce same codes

### Important

#### Critical Insights:

1. Canonical Huffman is **not** a different compression algorithm from Huffman
2. The Huffman algorithm determines *how long* each code should be
3. The canonical transformation determines the *exact bit patterns* consistently
4. **Sorting by length** is essential to ensure prefix-free property
5. The  $\times 2$  operation (left shift) ensures proper separation between different length groups

## 3.6 Adaptive Huffman Coding

### Overview

#### Definition

**Adaptive Huffman Coding** is a single-pass, lossless data compression technique in which the Huffman model is updated dynamically as symbols are encoded and decoded. The Huffman tree evolves on-the-fly, allowing both the encoder and decoder to adapt to changing symbol statistics without any prior knowledge of the source distribution.

Unlike static Huffman coding, adaptive Huffman coding does not require a preprocessing phase to compute symbol frequencies. Instead, symbol statistics are learned incrementally as the data stream is processed.

### Limitations of Static Huffman Coding

Static Huffman coding assumes that symbol statistics are known in advance:

- Requires two passes over the data:
  1. Collect symbol frequencies
  2. Encode using the constructed Huffman tree
- The Huffman tree (or code lengths) must be transmitted as side information
- Cannot adapt to non-stationary or evolving symbol distributions

These limitations motivate adaptive schemes when symbol statistics are unknown or change over time.

### Key Principle of Adaptive Huffman Coding

Adaptive Huffman coding maintains a Huffman tree that is updated after encoding or decoding each symbol.

**Key invariant:** After processing each symbol, the encoder and decoder maintain *identical Huffman trees* by applying the same updates in the same order. This ensures correct decoding without transmitting the tree explicitly.

### Initialization

The algorithm begins with a special symbol:

- A single **NYT** (Not Yet Transmitted) node

When a symbol appears for the first time:

- The code for the NYT node is output
- The raw symbol value is transmitted
- The NYT node is expanded into:
  - A new NYT node
  - A leaf node representing the new symbol

*Note: Some variants preinitialize the tree if the alphabet is fixed, but the standard adaptive Huffman algorithm begins with only the NYT node.*

## Tree Update Mechanism

After each symbol is processed:

- The frequency (weight) of the corresponding leaf node is incremented
- The tree is updated to preserve the **sibling property**

### Definition

**Sibling Property:** Nodes are numbered such that nodes with higher weights have higher numbers. For any given weight, all nodes with that weight form a contiguous block. Each internal node has exactly two children.

To restore this property, nodes may be swapped with others of equal weight, followed by incrementing parent node weights. This update process proceeds bottom-up toward the root.

## Encoding and Decoding Process

- If the symbol has appeared before:
  - Output its current Huffman code
- If the symbol is new:
  - Output the code for the NYT node
  - Output the symbol in raw (fixed-length) form
- Update the tree identically at both encoder and decoder

*Note: Exact bit patterns depend on the update order and implementation. Adaptive Huffman coding guarantees prefix-free optimality but not unique codes.*

## Algorithms

Two classical adaptive Huffman algorithms are widely used:

- **FGK Algorithm** (Faller–Gallager–Knuth)
- **Vitter’s Algorithm (Algorithm V)**, which improves worst-case performance and is commonly preferred

Both algorithms maintain the sibling property while ensuring efficient updates.

## Performance Characteristics

- Update cost is **amortized**  $O(1)$  per symbol
- Worst-case update time is proportional to the tree height
- Compression efficiency:
  - Initially suboptimal
  - Converges toward entropy-optimal coding as statistics stabilize

## Comparison with Static Huffman Coding

| Feature                   | Static Huffman | Adaptive Huffman |
|---------------------------|----------------|------------------|
| Number of passes          | Two            | One              |
| Prior statistics required | Yes            | No               |
| Model transmission        | Required       | Not required     |
| Adaptation to changes     | No             | Yes              |
| Initial efficiency        | High           | Low              |
| Asymptotic efficiency     | Optimal        | Near-optimal     |

## Applications

Adaptive Huffman coding is well suited for:

- Streaming data sources
- Real-time communication
- Situations where full statistics cannot be collected in advance

However, it may be less suitable when symbol distributions are known and stable, or when computational simplicity is critical.

## Summary

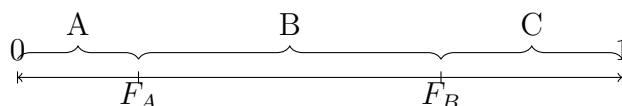
Adaptive Huffman coding extends classical Huffman coding to dynamic environments. By updating the coding model incrementally and maintaining strict structural invariants, it enables efficient, single-pass compression without prior knowledge of source statistics.

## 3.7 Arithmetic Coding: The Paradigm Shift

### Definition

**Arithmetic Coding:** Encodes an entire message into a single fractional number in the interval  $[0, 1)$ , approaching the entropy bound very closely.

**Core Idea:** Represent messages as subintervals of  $[0, 1)$ :



### Algorithm 1 Arithmetic Encoding Algorithm

**Require:** Message  $m = s_1 s_2 \dots s_k$ , symbol probabilities  $p_i$

**Ensure:** Final interval  $[low, high)$

- 1:  $low \leftarrow 0.0, high \leftarrow 1.0$
- 2: **for** each symbol  $s$  in  $m$  **do**
- 3:      $range \leftarrow high - low$
- 4:      $high \leftarrow low + range \times F_s$   $\triangleright F_s$ : cumulative prob up to  $s$
- 5:      $low \leftarrow low + range \times F_{s-1}$   $\triangleright F_{s-1}$ : cumulative prob before  $s$
- 6: **end for** **return** any number in  $[low, high)$

### Example

**Example:** Encode message "CAB" with probabilities: A(0.5), B(0.25), C(0.25)

| Symbol | Probability | Cumulative |
|--------|-------------|------------|
| A      | 0.5         | 0.5        |
| B      | 0.25        | 0.75       |
| C      | 0.25        | 1.0        |

#### Encoding:

1. Start:  $[0, 1)$
2. Process 'C':  $[0.75, 1.0)$  (C occupies  $[0.75, 1.0)$ )
3. Process 'A':  $range = 0.25$ 
  - $low = 0.75 + 0.25 \times 0.0 = 0.75$

- $high = 0.75 + 0.25 \times 0.5 = 0.875$

- New interval:  $[0.75, 0.875)$

4. Process 'B':  $range = 0.125$

- $low = 0.75 + 0.125 \times 0.5 = 0.8125$

- $high = 0.75 + 0.125 \times 0.75 = 0.84375$

- Final interval:  $[0.8125, 0.84375)$

**Output:** Any number in  $[0.8125, 0.84375)$ , e.g., 0.8125 in binary

### Important

#### Practical Implementation Issues:

- **Finite precision:** Use integer arithmetic with scaling
- **Renormalization:** Output bits when interval confined to one half
- **Carry-over:** Handle when interval spans midpoint
- **Termination:** Need special end-of-message symbol

**Adaptive Arithmetic Coding:** Easier than adaptive Huffman - just update probabilities as you go!

## 3.8 Comparison & Synthesis

| Method            | Optimal?        | Adaptive?  | Complexity  | Near Entropy?      | Key Applications             |
|-------------------|-----------------|------------|-------------|--------------------|------------------------------|
| Shannon Coding    | No              | No         | Low         | No                 | Theoretical proofs           |
| Shannon-Fano      | No              | No         | Low         | Sometimes          | Historical                   |
| Huffman           | Yes*            | No         | Low         | Moderate           | General purpose              |
| Canonical Huffman | Yes*            | No         | Low         | Moderate           | DEFLATE, JPEG, PNG           |
| Adaptive Huffman  | Yes*            | Yes        | Medium      | Moderate           | Early compressors            |
| Arithmetic Coding | <b>Near-opt</b> | <b>Yes</b> | <b>High</b> | <b>Yes (close)</b> | <b>JPEG2000, H.264, HEVC</b> |

\*Optimal for symbol-by-symbol coding given probabilities

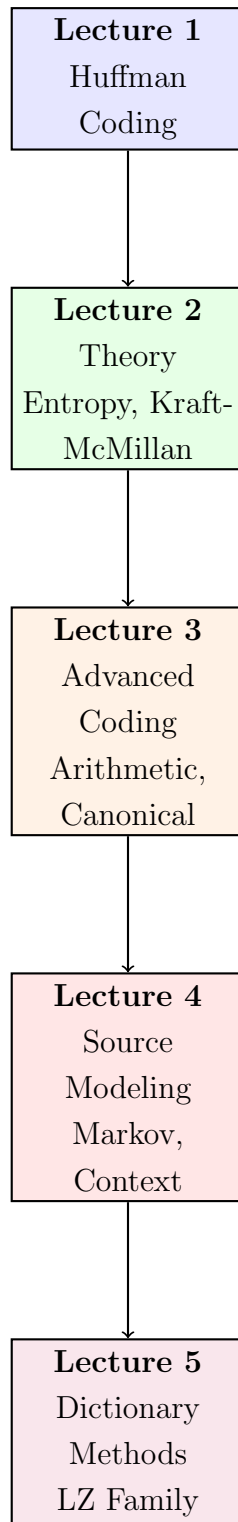
## Important

### Key Insights:

- **Huffman vs. Arithmetic:** Huffman is simpler but has an "integer penalty"; Arithmetic approaches entropy bound
- **Modern standards:** Arithmetic coding (CABAC) used in video compression for 10-20% better compression
- **Practical choice:** For general compression, Canonical Huffman (DEFLATE); for media, Arithmetic coding
- **The missing piece:** All these methods assume we have good probability estimates. Where do those come from?

## 3.9 Forward Look

The Complete Picture: What Comes Next?



### Next Lecture: Source Modeling and Statistical Dependence

- **The missing half:** We now know how to code efficiently, but where do the probabilities come from?
- **Real data has memory:** 'Q' is usually followed by 'U' in English text
- **Markov models:** Capturing dependencies between symbols

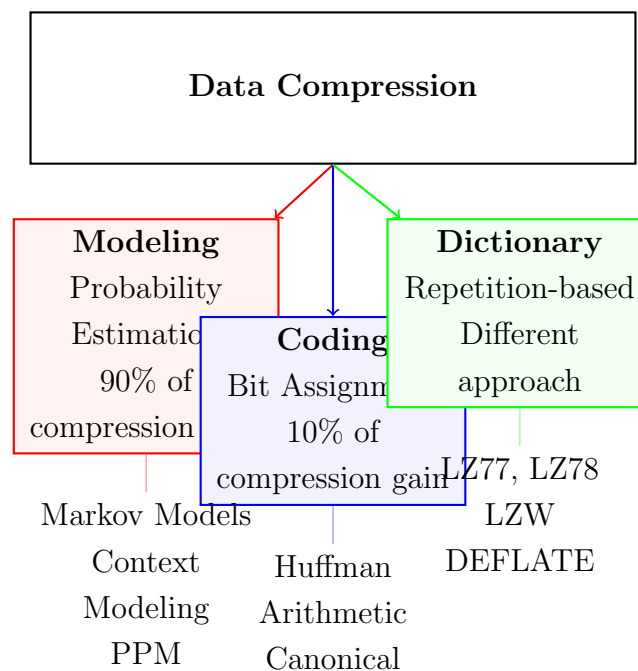
- **Context modeling:** Using past symbols to predict future ones
- **The modeling-coding separation:** Modern compressors separate these two tasks

The Big Question for Next Time:

If Arithmetic coding can get within 0.01 bits of entropy,  
what's the real limit to compression?

The answer: It's not the coding, it's the *modeling*!

The Two Pillars of Compression (Revised View):



### Exercise 3.0

**Exercise 3.1:** Given symbols with probabilities: A(0.4), B(0.3), C(0.2), D(0.1)

- Construct a Shannon code and compute its expected length
- Construct a Shannon-Fano code
- Compare with Huffman code from Lecture 2

### Exercise 3.1

**Exercise 3.2:** Convert the following Huffman code to canonical form:

| Symbol | Huffman Code |
|--------|--------------|
| A      | 0            |
| B      | 100          |
| C      | 101          |
| D      | 110          |
| E      | 1110         |
| F      | 1111         |

### Exercise 3.2

**Exercise 3.3:** Encode the message "ABAC" using arithmetic coding with probabilities: A(0.6), B(0.3), C(0.1). Show each step.

### Exercise 3.3

**Exercise 3.4: Thinking Ahead:** Consider the English phrase "THE QUICK BROWN FOX"

- (a) If we use Huffman coding with letter frequencies, what's wrong with this approach?
- (b) How might knowing that 'Q' is usually followed by 'U' help compression?
- (c) Why would arithmetic coding be better than Huffman for this kind of data?

### Exercise 3.4

**Exercise: 3.5** Given code lengths: A=2, B=2, C=3, D=3, E=3, F=4

1. Sort the symbols properly
2. Compute the canonical codes
3. Verify no code is a prefix of another
4. What happens if you don't sort by length?

---

**End of Lecture 3 – Advanced Entropy Coding Methods**

**Next: Lecture 4 – Source Modeling and Statistical Dependence**

*We now have efficient coding methods. Next: Where do the probabilities come from?*

## 4: Lecture 4: Source Modeling and Statistical Dependence

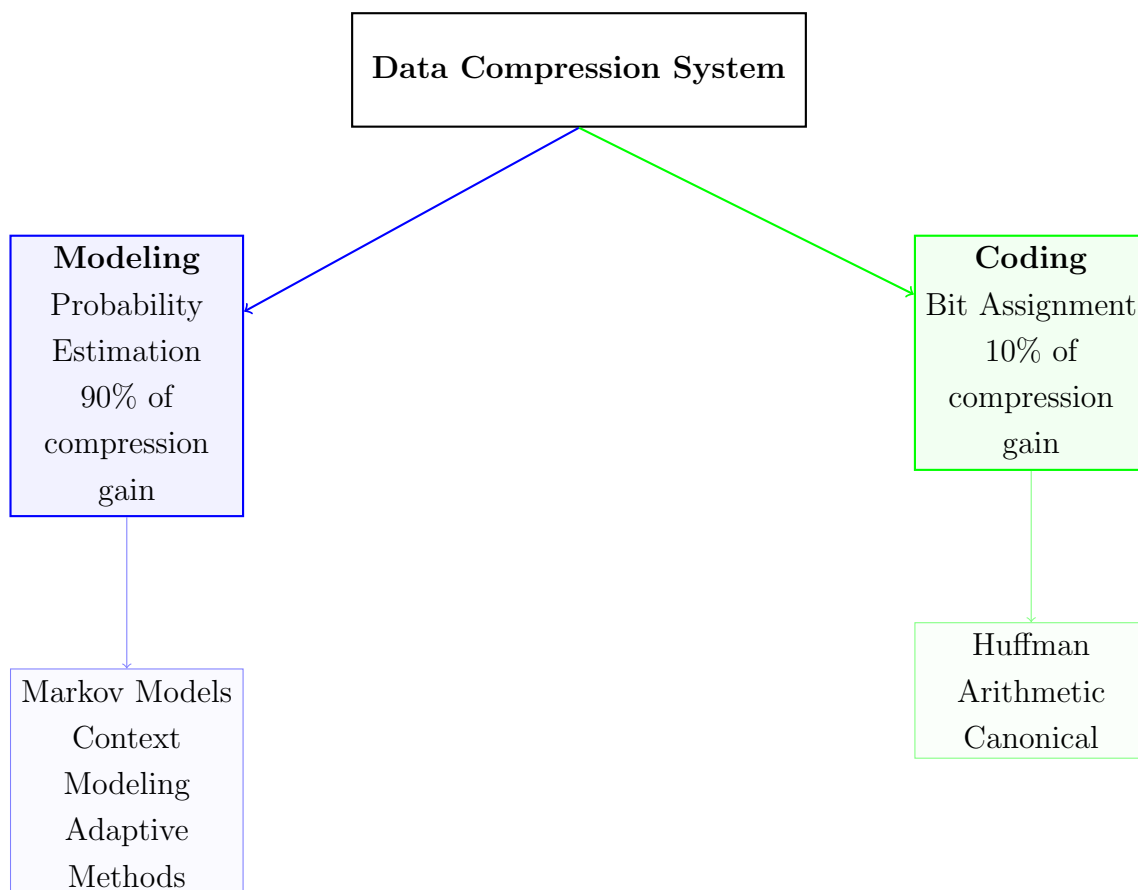
### Lecture 4: Beyond Coding – The Power of Source Modeling

#### 4.1 Introduction: Beyond Coding

##### Important

**Key Insight:** In the first three lectures, we've focused on **coding** - efficient ways to represent symbols given their probabilities. Now we address the other half: **modeling** - how to get good probability estimates in the first place.

##### The Big Picture:



##### Why Real Data Defies IID Assumptions:

- **IID (Independent Identically Distributed):** Assumption behind simple Huffman
- **Reality:** Data has **memory** and **dependencies**
- Example: In English text, 'Q' is almost always followed by 'U'

- Example: In images, neighboring pixels are highly correlated

### Today's Roadmap:

1. Understand statistical dependence in data
2. Learn Markov models for capturing memory
3. Explore context modeling techniques
4. See practical examples with real data
5. Connect modeling to coding (Lecture 3)

## 4.2 Memoryless vs. Sources with Memory

### Definition

**Memoryless Source (IID):** Each symbol is generated independently of all previous symbols. Probability distribution:  $P(X_n = x) = p(x)$  for all  $n$ .

### Definition

**Source with Memory:** The probability of a symbol depends on previous symbols. Example:  $P(X_n = x | X_{n-1} = y, X_{n-2} = z, \dots)$ .

### Example

#### Examples of Dependence in Real Data:

- **Text:** 'TH' is common, 'TQ' is rare
- **Images:** Neighboring pixels have similar colors
- **Audio:** Sound waves have temporal continuity
- **Video:** Consecutive frames are nearly identical
- **Source code:** Keywords, variable names repeat

### Important

#### Measuring Dependence: Autocorrelation

$$\rho(k) = \frac{\mathbb{E}[(X_t - \mu)(X_{t+k} - \mu)]}{\sigma^2}$$

where  $k$  is the lag. High  $\rho(k)$  means strong dependence at distance  $k$ .

### 4.3 Conditional Entropy and Mutual Information

#### Definition

**Conditional Entropy:** The average uncertainty about  $X$  given knowledge of  $Y$ :

$$H(X|Y) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log_2 p(x|y)$$

#### Example

**Intuition: "Knowing the Past Helps Predict the Future"**

Consider English letters:

- Unconditional:  $H(\text{letter}) \approx 4.07$  bits
- Given previous letter:  $H(\text{letter}|\text{previous}) \approx 3.36$  bits
- Given previous 2 letters:  $H(\text{letter}|\text{previous 2}) \approx 2.77$  bits

Each additional letter of context reduces uncertainty!

#### Definition

**Mutual Information:** Measures how much knowing  $Y$  reduces uncertainty about  $X$ :

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

#### Example

**Worked Example: English Letter Dependence**

Let  $X$  = current letter,  $Y$  = previous letter.

$$H(X) = 4.07 \text{ bits}$$

$$H(X|Y) = 3.36 \text{ bits}$$

$$I(X; Y) = 4.07 - 3.36 = 0.71 \text{ bits}$$

This means knowing the previous letter gives us 0.71 bits of information about the current letter.

## 4.4 Markov Sources

### Definition

**Markov Property (Memory- $m$ ):** The future depends only on the last  $m$  symbols:

$$P(X_n = x | X_{n-1}, X_{n-2}, \dots, X_1) = P(X_n = x | X_{n-1}, \dots, X_{n-m})$$

**First-Order Markov Model ( $m = 1$ ):**

- Only the immediately previous symbol matters
- Represented by transition probabilities  $p_{ij} = P(X_n = j | X_{n-1} = i)$
- Can be shown as a state diagram or transition matrix

### Example

**Example: Weather Prediction Markov Chain**

States: {Sunny (S), Rainy (R)}

Transition probabilities:

|   | S   | R   |
|---|-----|-----|
| S | 0.8 | 0.2 |
| R | 0.3 | 0.7 |

Interpretation: If today is sunny, 80% chance tomorrow is sunny, 20% chance rainy.

**Higher-Order Markov Models:**

- Order- $k$ : Depends on last  $k$  symbols
- More accurate but exponentially more parameters
- Number of parameters grows as  $|\mathcal{A}|^{k+1}$  where  $\mathcal{A}$  is alphabet size

### Important

**Memory-Complexity Trade-off:**

- **Order 0:** 26 parameters for English (simple but weak)
- **Order 1:**  $26 \times 26 = 676$  parameters
- **Order 2:**  $26 \times 26 \times 26 = 17,576$  parameters
- **Order 5:**  $26^6 \approx 308$  million parameters!

This is the **context explosion problem**.

## 4.5 Entropy Rate Revisited

### Definition

#### Entropy Rate of a Stationary Source:

For a stationary stochastic process  $\mathcal{X} = (X_1, X_2, \dots)$ , the entropy rate is:

$$H(\mathcal{X}) = \lim_{n \rightarrow \infty} \frac{1}{n} H(X_1, X_2, \dots, X_n)$$

Equivalently (and equal for stationary processes):

$$H(\mathcal{X}) = \lim_{n \rightarrow \infty} H(X_n | X_1, \dots, X_{n-1})$$

#### Special case: Stationary Markov chain of order $m$

If  $\mathcal{X}$  is a stationary  $m$ -th order Markov chain, then for all  $t > m$ :

$$H(X_t | X_1, \dots, X_{t-1}) = H(X_t | X_{t-m}, \dots, X_{t-1})$$

and this conditional entropy is constant over time. Therefore:

$$H(\mathcal{X}) = H(X_{m+1} | X_1, \dots, X_m)$$

No limit needed — the conditional entropy stabilizes immediately.

### Example

#### Example: First-Order Markov Chain ( $m = 1$ )

**Step 1: Define the model** Two weather states:  $S$  (sunny) and  $R$  (rainy).

**Transition probabilities** (tomorrow's weather depends only on today's):

Let  $X_t$  be the weather on day  $t$ . Then:

$$\begin{aligned} P(X_{t+1} = S | X_t = S) &= 0.8, & P(X_{t+1} = R | X_t = S) &= 0.2 \\ P(X_{t+1} = S | X_t = R) &= 0.3, & P(X_{t+1} = R | X_t = R) &= 0.7 \end{aligned}$$

In matrix form (rows = current state, columns = next state):

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} \quad \begin{array}{l} \text{Row 1: } X_t = S \\ \text{Row 2: } X_t = R \end{array}$$

**Step 2: Find stationary distribution  $\pi$**  Stationary distribution  $\pi = [\pi_S, \pi_R]$  satisfies  $\pi P = \pi$ , where  $\pi_S = P(X_t = S)$  and  $\pi_R = P(X_t = R)$  in the long run.

$$\begin{cases} \pi_S = 0.8\pi_S + 0.3\pi_R \\ \pi_R = 0.2\pi_S + 0.7\pi_R \\ \pi_S + \pi_R = 1 \end{cases}$$

Solving (from first equation):

$$\pi_S - 0.8\pi_S = 0.3\pi_R \Rightarrow 0.2\pi_S = 0.3\pi_R \Rightarrow \pi_S = 1.5\pi_R$$

$$1.5\pi_R + \pi_R = 1 \Rightarrow 2.5\pi_R = 1 \Rightarrow \pi_R = 0.4$$

$$\pi_S = 1.5 \times 0.4 = 0.6$$

$$\pi = [\pi_S, \pi_R] = [0.6, 0.4]$$

**Step 3: Compute conditional entropies** *Uncertainty about tomorrow given today's weather:*

Given  $X_t = S$ : Tomorrow's distribution is  $[0.8, 0.2]$

$$H(X_{t+1} | X_t = S) = -0.8 \log_2 0.8 - 0.2 \log_2 0.2 \approx 0.7219 \text{ bits}$$

Given  $X_t = R$ : Tomorrow's distribution is  $[0.3, 0.7]$

$$H(X_{t+1} | X_t = R) = -0.3 \log_2 0.3 - 0.7 \log_2 0.7 \approx 0.8813 \text{ bits}$$

**Step 4: Compute entropy rate** For a stationary first-order Markov chain:

$$H(\mathcal{X}) = H(X_{t+1} | X_t) = \sum_{s \in \{S, R\}} P(X_t = s) \cdot H(X_{t+1} | X_t = s)$$

With  $P(X_t = s) = \pi_s$  (stationarity):

$$H(\mathcal{X}) = \pi_S \cdot H(X_{t+1} | X_t = S) + \pi_R \cdot H(X_{t+1} | X_t = R)$$

$$= 0.6 \times 0.7219 + 0.4 \times 0.8813 = 0.43314 + 0.35252 = 0.78566 \text{ bits}$$

$$\approx 0.786 \text{ bits per day}$$

## Important

**What This Means for Data Compression:**

**Two approaches to compression:**

1. IID (Independent Identically Distributed) assumption:

- Treat each symbol as independent of previous ones
- Best achievable rate:  $H(X)$  (marginal entropy)
- Example: English letters  $\approx 4.07$  bits/letter

## 2. With modeling (using context):

- Account for dependencies between symbols (Markov structure)
- Use conditional probabilities based on previous symbols
- Best achievable rate:  $H(\mathcal{X})$  (entropy rate)
- Example: English text  $\approx 2.3$  bits/letter

**Compression gain:**

$$\frac{H(X) - H(\mathcal{X})}{H(X)} \approx \frac{4.07 - 2.3}{4.07} \approx 43\%$$

Up to  $\sim 45\%$  better compression by exploiting dependencies!

## 4.6 Context Modeling in Practice

### Definition

**Context Modeling:** Maintain separate probability distributions for each possible context (history).

### Fixed-Length vs. Variable-Length Contexts:

- **Fixed-length:** Always use last  $k$  symbols as context
- **Variable-length:** Use longest matching context in database
- Example: PPM (Prediction by Partial Matching) uses variable-length

### Example

#### The Context Explosion Problem:

For English text (26 letters + space):

| Order | Contexts   | Parameters  |
|-------|------------|-------------|
| 0     | 1          | 27          |
| 1     | 27         | 729         |
| 2     | 729        | 19,683      |
| 3     | 19,683     | 531,441     |
| 4     | 531,441    | 14,348,907  |
| 5     | 14,348,907 | 387,420,489 |

By order 5: 387 million parameters need estimation!

### Solutions to Context Explosion:

1. **Escaping:** Fall back to lower-order model when context unseen
2. **Blending:** Combine predictions from different order models
3. **Pruning:** Remove low-frequency contexts
4. **Adaptive methods:** Update probabilities as data arrives

## 4.7 Case Study: Text Compression Modeling

### Example

#### 1. Entropy Estimates for English Text (Theoretical Limits)

| Model/Estimation Method           | Bits/Letter | Year/Source     |
|-----------------------------------|-------------|-----------------|
| Letter frequencies only (Order-0) | 4.07        | Shannon 1948    |
| + 1st order Markov (bigrams)      | 3.36        |                 |
| + 2nd order (trigrams)            | 2.77        |                 |
| + 3rd order                       | 2.43        |                 |
| Human prediction experiment       | 1.3         | Shannon 1951    |
| Modern best estimates             | 1.0–1.5     | Various studies |

#### 2. Practical Compression Performance

| Compressor/Method           | Bits/Letter | Notes                       |
|-----------------------------|-------------|-----------------------------|
| gzip (LZ77 + Huffman)       | 2.5–3.0     | Fast, widely used           |
| bzip2 (BWT + MTF + Huffman) | 2.3–2.8     | Better but slower           |
| PPMd                        | 2.1–2.4     | Context modeling            |
| PAQ variants                | 1.5–1.8     | State of the art, very slow |

**Key insight:** Better models (using more context) → lower entropy → better compression.

But practical compressors face tradeoffs: memory, speed, and model complexity.

### PPM (Prediction by Partial Matching):

- Uses **variable-length** contexts
- Tries highest-order model first
- Escapes to lower order if context unseen
- Blends probabilities from different orders
- State-of-the-art for text compression in 1990s

#### Important

##### Practical Entropy Reduction:

IID model (Huffman) : 4.07 bits/letter  
With context modeling : 2.23 bits/letter  
Improvement : 45% better compression!

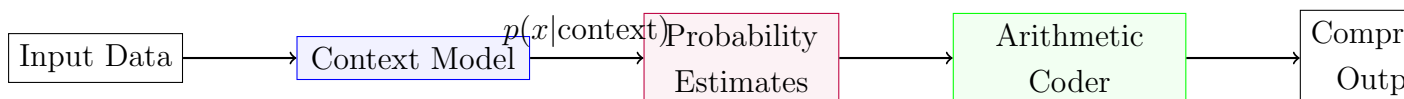
## 4.8 The Modeling–Coding Separation Principle

#### Definition

**Modeling–Coding Separation:** Modern compressors separate probability estimation (modeling) from bit assignment (coding).

##### Historical Evolution:

- **Early:** Integrated (Huffman builds tree from frequencies)
- **Modern:** Separated (Model  $\rightarrow$  Probabilities  $\rightarrow$  Arithmetic Coder)



#### Example

##### How PPM + Arithmetic Beats Huffman:

1. **PPM:** Sees context "TH"  $\rightarrow$  predicts E with 80% probability
2. **Arithmetic:** Encodes E using  $p = 0.8 \rightarrow 0.32$  bits
3. **Huffman:** Would need at least 1 bit for any symbol

4. **Gain:** 0.32 bits vs 1+ bits =  $3\times$  better for this symbol!

### Important

#### Why Arithmetic Coding is the Perfect Backend:

- Can handle **fractional bits** per symbol
- Accepts **changing probabilities** symbol by symbol
- Works with **adaptive models** naturally
- Achieves entropy bound for good models

## 4.9 Adaptive vs. Static Modeling

### Definition

**Static Models:** Train once on representative data, use fixed model for all files.

- **Pros:** Fast encoding/decoding
- **Cons:** Model may not match specific file
- **Example:** Early text compressors using English statistics

### Definition

**Semi-Adaptive (Two-Pass):** First pass: collect statistics; Second pass: encode.

- **Pros:** Tailored to specific file
- **Cons:** Need to transmit model (overhead)
- **Example:** Standard Huffman with tree transmission

### Definition

**Fully Adaptive (One-Pass):** Update model while encoding/decoding.

- **Pros:** No model transmission, adapts to local changes
- **Cons:** Slower, initial poor compression
- **Example:** Adaptive Huffman, PPM with update

## Example

### Comparison in Practice:

| Type           | Compression | Speed  | Memory |
|----------------|-------------|--------|--------|
| Static         | Medium      | Fast   | Low    |
| Semi-Adaptive  | Good        | Medium | Medium |
| Fully Adaptive | Best        | Slow   | High   |

Choice depends on application constraints!

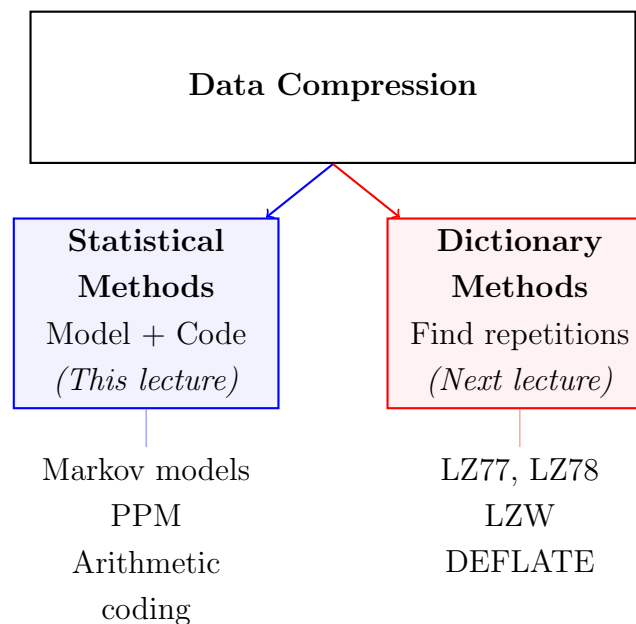
## 4.10 Summary and Forward Look

### Important

#### Key Takeaways:

1. **The real compression is in modeling**, not just coding
2. **Context matters**: Using past symbols reduces uncertainty
3. **Markov models** capture memory in data
4. **Context explosion** limits practical model order
5. **Arithmetic coding** enables efficient use of good models
6. **Adaptive methods** avoid model transmission overhead

#### The Two Pillars Revisited:



**Preview: Next Lecture on LZ Family (Dictionary Methods):**

- A completely different approach: find repeating patterns
- No probability estimation needed
- Works well for files with exact repetitions
- Used in ZIP, GIF, PDF, and many modern formats
- Often combined with statistical methods in practice

#### Exercise 4.0

**Exercise 4.1:** Given the first-order Markov chain for weather:

|   | S   | R   |
|---|-----|-----|
| S | 0.7 | 0.3 |
| R | 0.4 | 0.6 |

- a) Find the stationary distribution  $\pi = [\pi_S, \pi_R]$  b) Calculate the entropy rate  $H(\mathcal{X})$   
 c) How does this compare to an IID source with  $P(S) = 0.55, P(R) = 0.45$ ?

#### Exercise 4.1

**Exercise 4.2:** Consider the text fragment: "THE CAT SAT ON THE MAT" a) Build an order-1 (bigram) model for this text b) Calculate  $H(X)$  (order-0 entropy) c) Calculate  $H(X|Y)$  where  $Y$  is previous letter (order-1 conditional entropy) d) How much mutual information exists between consecutive letters?

#### Exercise 4.2

**Exercise 4.3:** Explain why arithmetic coding is better than Huffman coding when used with: a) A high-order Markov model b) An adaptive context model c) A model that gives very skewed probabilities (e.g.,  $p = 0.99$  for one symbol)

---

## End of Lecture 4 – Source Modeling and Statistical Dependence

Next: Dictionary-based Compression (LZ Family)

## 5: Lecture 5: Dictionary-Based Compression: The Lempel–Ziv Revolution

### 5.1 Motivation: Hitting the Limits of Statistical Coding

Statistical coding methods such as Huffman and arithmetic coding achieve near-optimal compression *when the source statistics are known or can be accurately estimated*. However, these methods fundamentally rely on estimating probabilities over symbols or blocks of symbols.

As discussed earlier, block-based statistical coding faces a fundamental dilemma.

#### 5.1.1 Recap: The Block Coding Dilemma

Increasing block length allows a coder to better capture dependencies between symbols and approach the entropy rate of the source. However, this comes at a steep cost:

- The number of possible blocks grows exponentially with block length.
- Estimating probabilities reliably requires exponentially more data.
- Memory and computational complexity quickly become impractical.

As a result, practical statistical coders are constrained to small contexts and local dependencies.

#### 5.1.2 The Promise of Exploiting Long-Range Repetition

Real-world data—text, executable files, logs, DNA sequences—often contains *long-range repetition*. Patterns may repeat far apart, well beyond the reach of fixed-size statistical contexts.

##### Important

Statistical coding models *how often* symbols occur, but does not directly model *where long repeated patterns occur*.

This observation motivates a radically different approach to compression.

### 5.2 Paradigm Shift: From Statistics to Dictionaries

Instead of estimating probabilities, dictionary-based compression learns the source *by example*.

### 5.2.1 Core Philosophy of Dictionary Coding

Dictionary-based coders operate on a simple idea:

*Replace repeated substrings by references to earlier occurrences.*

As the input is processed sequentially, a dictionary of previously seen substrings is constructed. Future occurrences are encoded by references into this dictionary.

#### Important

You can think of Lempel–Ziv methods as *learning the source structure rather than estimating probabilities*.

Crucially, the encoder and decoder process the input in exactly the same order and therefore build identical dictionaries *without transmitting the dictionary explicitly*.

### 5.2.2 Explicit vs. Implicit (Adaptive) Dictionaries

Dictionary-based methods fall into two categories:

- **Implicit dictionaries:** The dictionary is defined by a sliding window of recent output (e.g., LZ77, LZSS).
- **Explicit dictionaries:** The dictionary is stored and indexed explicitly (e.g., LZ78).

## 5.3 The Lempel-Ziv Algorithms

### 5.3.1 LZ77: The Sliding Window Algorithm

Published in 1977 by Abraham Lempel and Jacob Ziv, LZ77 introduced the concept of using recent history as a dictionary. It forms the foundation for many practical compression algorithms including gzip, PNG, and ZIP.

**The Sliding Window Model:** LZ77 maintains a sliding window divided into two parts:

- **Search buffer:** Contains the  $N$  most recently processed symbols (the "dictionary").
- **Look-ahead buffer:** Contains the next  $L$  symbols to be encoded.

**The Encoding Tuple: (Offset, Length, Next Symbol):** At each step, the encoder searches for the longest string in the search buffer that matches the beginning of the look-ahead buffer. It outputs a triple:

(offset, length, next symbol)

### 5.3.2 LZSS: Improving Practical Efficiency

Published in 1982 by James Storer and Thomas Szymanski, LZSS addresses a key inefficiency in LZ77.

**Flag Bits: Literal vs. Match:** LZSS introduces a **flag bit** preceding each encoded element:

- **Flag = 0:** Next item is a literal symbol (8 bits for ASCII).
- **Flag = 1:** Next item is a match pair (offset, length).

**Match Threshold:** LZSS introduces a **match threshold**: only encode matches that are *long enough* to actually save bits.

### 5.3.3 LZ78: The Dictionary Growth Algorithm

Published in 1978, LZ78 takes a different approach: it builds an explicit dictionary that grows as encoding proceeds.

**Encoding Pairs: (Dictionary Index, New Symbol):** LZ78 outputs pairs:

(index, symbol)

where the new phrase (dictionary[index] + symbol) is added to the dictionary.

### 5.3.4 LZW (Lempel-Ziv-Welch)

LZW (1984) is a popular LZ78 variant used in GIF, TIFF, and UNIX compress.

**LZW Key Features:**

- **No explicit next symbol:** Dictionary entries are added *before* they're used.
- **Fixed-width codes:** Typically 12-bit codes (4096 dictionary entries).
- **Adaptive dictionary:** Starts with all single symbols (256 entries for bytes).

## 5.4 Theoretical Properties: Why Lempel–Ziv Works

### 5.4.1 The Universality Principle

Lempel–Ziv algorithms have a remarkable theoretical property:

#### Definition

**Universal Compression:** An algorithm is universal if it can compress any stationary ergodic source to its entropy rate *without prior knowledge* of the source statistics.

Both LZ77 and LZ78 are universal compressors.

### 5.4.2 Asymptotic Optimality

**Theorem 5.1** (Ziv & Lempel, 1977-1978). *For any stationary ergodic source with entropy rate  $H$ , let  $L_n$  be the length of the LZ77 or LZ78 encoding of  $n$  source symbols. Then:*

$$\lim_{n \rightarrow \infty} \frac{\mathbb{E}[L_n]}{n} = H$$

## 5.5 Bridging Paradigms: Dictionary vs. Entropy Coding

### 5.5.1 Complementary Approaches

Dictionary coding and entropy coding address different aspects of compression:

- **Dictionary coding:** Exploits *structural redundancy* (repetitions).
- **Entropy coding:** Exploits *statistical redundancy* (skewed symbol frequencies).

### 5.5.2 The Hybrid Approach

Modern compressors combine both approaches. The DEFLATE algorithm (used in ZIP, PNG, gzip) works this way:

1. LZSS finds repeated strings, outputs literals and matches.
2. Huffman coding compresses:
  - Literal symbols (0-255)
  - Match lengths (257-285)
  - Match distance codes (0-29)

## 5.6 Summary and Forward Look

### 5.6.1 Key Takeaways

- **Paradigm shift:** Dictionary coding exploits repetition rather than probabilities.
- **LZ77:** Sliding window approach, encodes matches as triples.
- **LZSS:** Practical improvement with flag bits and match thresholds.
- **LZ78:** Explicit dictionary growth, encodes pairs.
- **Universality:** LZ methods work on any stationary source without prior knowledge.
- **Modern practice:** Hybrid systems (LZSS + Huffman) dominate.

## End of Chapter Exercises

### Exercise 5.0

#### Exercise 5.1 (LZ77 Encoding)

Encode the string TOBEORNOTTOBEORTOBEORNOT using LZ77 with a search buffer of size 12. Show your work step by step with the search buffer, look-ahead buffer, and output tuples.

### Exercise 5.1

#### Exercise 5.2 (LZSS Efficiency Analysis)

Compare LZ77 and LZSS for encoding a sequence with parameters: 8-bit symbols, 12-bit offsets, 4-bit lengths.

- (a) How many bits does LZ77 use for a non-match?
- (b) How many bits does LZSS use for a literal?
- (c) What is the minimum match length that saves bits in LZSS?
- (d) When would LZ77 be more efficient than LZSS?

### Exercise 5.2

#### Exercise 5.3 (LZ78 Dictionary Growth)

Encode the string ABABABABA using LZ78. Show the dictionary after each step and the complete output sequence. How many dictionary entries are created?

### Exercise 5.3

#### Exercise 5.4 (Algorithm Comparison)

For each string below, predict which would perform better: LZ77/LZSS or LZ78? Explain why.

- (a) AAAAAAAAAAAAAAAAAAAAAA (20 A's)
- (b) ABCDEFGHIJKLMNOPQRSTUVWXYZ
- (c) ABABABABABABABABABAB
- (d) A file containing the same 1KB block repeated 100 times

### Exercise 5.4

#### Exercise 5.5 (Decoding Challenge)

Decode the following LZSS encoded sequence (match threshold = 3):

0 T

0 H  
0 E  
1 (3,5)  
0 \_  
0 \_  
1 (5,4)  
0 .

Where: - Each line represents one encoded element - Flag 0 indicates a literal character - Flag 1 indicates a match (offset, length) - \_ represents space character - . represents period

What is the original string? Show your decoding step by step.

### Exercise 5.5

#### Exercise 5.6 (Window Size Analysis)

Consider LZ77 with search buffer size  $N$ .

- (a) How many bits are needed to encode an offset value?
- (b) If we increase  $N$  from 4096 to 65536, how many more bits are needed for offsets?
- (c) What is the trade-off in choosing  $N$ ?
- (d) In practice, why might we choose  $N = 32768$  over  $N = 65536$  even if memory is available?

### Exercise 5.6

#### Exercise 5.7 (Universal Compression Intuition)

Explain in your own words why Lempel–Ziv algorithms are universal. Why don't they need to know source statistics in advance? How do they "learn" the source structure?

### Exercise 5.7

#### Exercise 5.8 (Hybrid Coding Design)

Design a simple hybrid compressor that:

- (a) Uses LZSS with match threshold 3
- (b) Collects statistics on literals, lengths, and offsets
- (c) Applies Huffman coding to each alphabet separately

How would the decoder know which Huffman trees to use?

### Exercise 5.8

#### Exercise 5.9 (Search Efficiency)

A naive LZ77 implementation searches for matches by comparing the look-ahead buffer against every position in the search buffer.

- (a) What is the time complexity of this approach?
- (b) Describe how a hash table can improve search efficiency.
- (c) What information would you store in the hash table?
- (d) How would you handle hash collisions?

### Exercise 5.9

#### Exercise 5.10 (Theoretical Limits)

Prove or provide intuition for:

- (a) LZ77 cannot achieve compression ratio better than the entropy rate for i.i.d. sources.
- (b) LZ78 may perform poorly on very short inputs.
- (c) For sources with long-range dependencies, LZ methods can outperform block Huffman coding even with large blocks.

## 6: Lecture 6: DEFLATE Algorithm: The Standard in Practice

### 6.1 Introduction: The Universal Compression Standard

The DEFLATE compression algorithm, specified in RFC 1951 (1996), is arguably the most widely deployed compression algorithm in history. Its success lies not in being the theoretically optimal algorithm, but in striking an exceptional balance between compression ratio, speed, and implementation simplicity.

#### Definition

**DEFLATE:** A lossless data compression algorithm combining LZ77-style string matching with Huffman coding. It is the compression engine behind gzip, ZIP, PNG, and HTTP compression.

#### 6.1.1 Where DEFLATE Is Used: gzip, ZIP, PNG, HTTP

- **gzip:** The Unix compression utility (RFC 1952)
- **ZIP file format:** Phil Katz's PKZIP format (1989), later standardized
- **PNG image format:** Replaced GIF, uses DEFLATE on filtered scanlines
- **HTTP compression:** Content-Encoding: gzip/deflate in web traffic
- **PDF documents:** FlateDecode filter for stream compression
- **Java JAR files:** Based on ZIP format

#### 6.1.2 Historical Context: Phil Katz, ZIP Format, and Open Standards

The story of DEFLATE is intertwined with the "compression wars" of the late 1980s-1990s:

- **1985:** Terry Welch's LZW algorithm patented (used in GIF)
- **1986:** Phil Katz creates PKARC (based on ARC format)
- **1989:** Katz releases PKZIP with proprietary compression
- **1991:** gzip created as free alternative to UNIX compress (which used LZW)
- **1993:** Unisys enforces LZW patents, creating demand for patent-free alternatives
- **1996:** DEFLATE specification published as RFC 1951

The key innovation was making DEFLATE specification **open** and **royalty-free**, leading to widespread adoption.

## 6.2 High-Level Architecture: A Two-Stage Pipeline

DEFLATE employs a classic transform-coding approach:

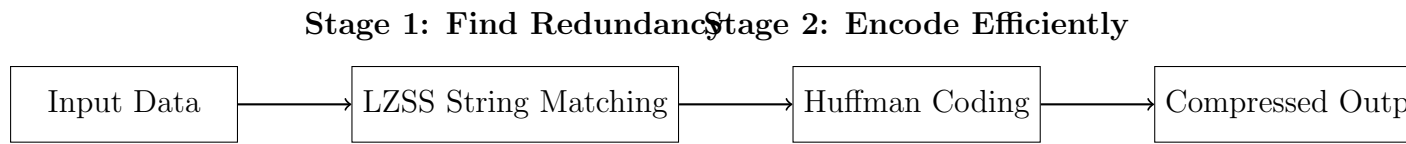


Figure 1: DEFLATE's two-stage compression pipeline

### 6.2.1 Stage 1: LZSS String Matching (Finding Redundancy)

The LZSS (Lempel-Ziv-Storer-Szymanski) stage identifies repeated sequences in the data:

- Scans input with a **sliding window** (32KB history buffer)
- For each position, looks for longest match in recent history
- Outputs either:
  - **Literal byte**: When no good match found
  - **(Length, Distance) pair**: When match found

**Example:** For the string "the cat sat on the mat":

- First occurrence of "the": output as literals 't','h','e'
- Second occurrence of "the": output as (length=3, distance=15)

### 6.2.2 Stage 2: Huffman Coding (Encoding the Reduced Data)

The Huffman stage compresses the output from Stage 1:

- Takes stream of literals and (length,distance) pairs
- Builds **two Huffman trees**:
  - One for literals and lengths (combined alphabet)
  - One for distances
- Uses **canonical Huffman codes** for compact representation

### 6.2.3 Why the Hybrid Approach Works: Transforming Redundancy

The genius of DEFLATE is in how it **transforms redundancy**:

1. **LZSS finds structural redundancy**: Repeated sequences become (length,distance) pairs
2. **Huffman finds statistical redundancy**: Frequent symbols get shorter codes
3. Together they exploit **both types** of redundancy in data

#### Important

**Key Insight**: LZSS is excellent at finding *long-range* repetition (phrases, patterns). Huffman is excellent at compressing *skewed distributions*. Together they handle both English text (skewed letter frequencies + repeated words) effectively.

### 6.2.4 Design Philosophy: Fast Decoding Over Maximum Compression

DEFLATE was designed with clear priorities:

| Priority                  | Implementation Choice                           |
|---------------------------|-------------------------------------------------|
| Fast decoding             | Canonical Huffman ( $O(1)$ decode)              |
| Memory efficiency         | 32KB sliding window (tiny by today's standards) |
| Implementation simplicity | Two independent stages                          |
| Streaming capability      | Independent compression blocks                  |
| Acceptable compression    | "Good enough" ratio for most data               |

This "fast decode" philosophy made DEFLATE ideal for distribution formats (ZIP, PNG) where files are compressed once but decompressed many times.

## 6.3 LZSS in DEFLATE: Detailed Implementation

### 6.3.1 Sliding Window: 32KB History and Lookahead Buffer

DEFLATE's LZSS uses specific, fixed parameters:

- **History buffer**: 32,768 bytes ( $2^{15}$ ) of recently processed data
- **Lookahead buffer**: 258 bytes maximum match length
- **Minimum match length**: 3 bytes (shorter matches not worth encoding)

## Example

### Why 32KB?

- 32KB was a practical limit in 1990s memory-constrained systems
- Larger windows provide diminishing returns for most data
- Cache-friendly size for modern processors
- Still adequate for most short-to-medium range repetitions

### 6.3.2 Hash Chains for Fast Match Finding

Instead of brute-force searching 32KB for matches, DEFLATE uses a hash table:

---

**Algorithm 2** DEFLATE Match Finding with Hash Chains

---

```
1: procedure FINDBESTMATCH(current_position, data)
2:   hash  $\leftarrow$  hash(data[current_position : current_position + 3])
3:   best_length  $\leftarrow$  0
4:   best_distance  $\leftarrow$  0
5:   for each prev_pos in hash_table[hash] do
6:     if current_pos - prev_pos > 32768 then                                 $\triangleright$  Outside window
7:       continue
8:     end if
9:     length  $\leftarrow$  MatchLength(prev_pos, current_pos)
10:    if length > best_length then
11:      best_length  $\leftarrow$  length
12:      best_distance  $\leftarrow$  current_pos - prev_pos
13:    end if
14:  end for
15:  return (best_length, best_distance)
16: end procedure
```

---

The hash is typically computed from the first 3 bytes of the potential match. Hash chains link all positions with the same 3-byte prefix.

### 6.3.3 Lazy Matching and Greedy Heuristics

DEFLATE uses sophisticated heuristics to find better matches within the sliding window constraints:

- **Greedy parsing:** Immediately take the longest match found at the current position

- **Lazy matching:** Consider outputting a literal instead, hoping for a better match starting at the next position

## Important

### Window Constraints in Lazy Matching:

- All matches must reference data within the 32KB search buffer
- The lookahead buffer contains data to be encoded (maximum 258 bytes)
- Lazy matching decides: "match now" vs. "literal now + potentially better match next"

## Example

### Lazy Matching Example Where Lazy Parsing Wins

Consider the input data:

aaaaabaaaaa

Indexing the data:

|   |   |   |   |   |   |   |   |   |   |    |
|---|---|---|---|---|---|---|---|---|---|----|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| a | a | a | a | a | b | a | a | a | a | a  |

Assume that the first six bytes have already been processed. The sliding window state is:

|               |   |   |   |   |   |                  |   |   |   |   |
|---------------|---|---|---|---|---|------------------|---|---|---|---|
| a             | a | a | a | a | b | a                | a | a | a | a |
| Search Buffer |   |   |   |   |   | Lookahead Buffer |   |   |   |   |

Search buffer: 'aaaaab' (positions 0–5)

Lookahead buffer: 'aaaaa' (positions 6–10)

**Decision point: position 6 (first 'a' in lookahead)**

- **Greedy parsing:**
  - Find longest match  $\geq 3$  starting at position 6
  - 'aaaaa' matches 'aaaaa' starting at position 1 in search buffer
  - Match length = 5, distance = 5
  - Greedy immediately emits: (distance=5, length=5)
  - Covers 5 bytes
- **Lazy parsing:**
  - At position 6: Finds match length = 5, distance = 5

- Checks next position (position 7)
- Starting at position 7: ‘‘aaaa’’ (4 bytes in lookahead)
- Cannot find match longer than 5 bytes (only 4 bytes remain)
- Greedy wins - lazy doesn't help here

### Example where lazy actually wins:

Consider: ‘‘abcdabcde’’

After processing first 3 bytes:

|               |   |   |                  |   |   |   |   |   |   |   |   |  |
|---------------|---|---|------------------|---|---|---|---|---|---|---|---|--|
| a             | b | c | a                | b | c | d | a | b | c | d | e |  |
| Search Buffer |   |   | Lookahead Buffer |   |   |   |   |   |   |   |   |  |

Search buffer: ‘‘abc’’ (positions 0-2)

Lookahead: ‘‘abcdabcde’’ (positions 3-11)

**At position 4 ('a'):**

#### • Greedy:

- ‘‘abc’’ matches at position 0 (distance=3, length=3)
- Encodes: (3,3)
- Advances to position 6
- Position 6: ‘‘dabcde’’ - no match  $\geq 3$
- Output literals: 'd','a','b','c','d','e'

#### • Lazy:

- At position 3: Finds ‘‘abc’’ (length=3, distance=3)
- Checks position 4: ‘‘bcdabcde’’
- ‘‘bcd’’ doesn't match (only ‘‘bc’’ in search buffer)
- Length=2 < 3
- Greedy wins

**The reality:** Simple examples showing dramatic lazy advantage are mathematically rare. Lazy matching typically provides small gains (1-2 bytes) in specific edge cases.

---

**Algorithm 3** Lazy Matching with Sliding Window Constraints

---

```
1: procedure LAZYMATCH(search_buffer, lookahead)
2:   pos  $\leftarrow$  0 ▷ Position within lookahead buffer
3:   best_match_here  $\leftarrow$  FindBestMatch(search_buffer, lookahead, pos)
4:   if best_match_here.length  $\geq$  3 then
5:     best_match_next  $\leftarrow$  FindBestMatch(search_buffer, lookahead, pos + 1)
6:     if best_match_next.length > best_match_here.length then
7:       Output lookahead[pos] as literal
8:       return pos + 1 ▷ Move to next position
9:     else
10:      Output match (best_match_here.distance, best_match_here.length)
11:      return pos + best_match_here.length
12:    end if
13:  else
14:    Output lookahead[pos] as literal
15:    return pos + 1
16:  end if
17: end procedure
```

---

**Important****Why Lazy Matching Works with Sliding Windows:**

- By outputting a literal, we shift the window by 1 byte
- This adds that literal to the search buffer for future matches
- Sometimes the literal + new search buffer position enables a longer match
- The decision is: "Is sacrificing 1 byte now worth gaining several bytes later?"

**Typical lazy matching optimization:**

- Only consider lazy if current match length < 8 (configurable threshold)
- Check only next 1-2 positions (not all possible positions)
- Stop lazy checking if a very long match (e.g., > 128 bytes) is found

### 6.3.4 Output Format: (Length, Distance) Pairs vs. Literal Bytes

The LZSS stage produces a stream of symbols from two alphabets:

| Type           | Range   | Encoding Details                                  |
|----------------|---------|---------------------------------------------------|
| Literal bytes  | 0–255   | Direct byte values                                |
| Length codes   | 3–258   | Base code + 0–5 extra bits for longer lengths     |
| Distance codes | 1–32768 | Base code + 0–13 extra bits in logarithmic ranges |
| End-of-block   | 256     | Special marker (end of DEFLATE block)             |

### Important encoding details:

- **Lengths 3–258:** Encoded using symbols 257–285
  - For example: Length 3 = symbol 257 (no extra bits)
  - Length 10 = symbol 264 (no extra bits)
  - Length 258 = symbol 285 (no extra bits)
  - Lengths 11–258 use 1–5 extra bits for precise encoding
- **Distances 1–32768:** Encoded using symbols 0–29 with logarithmic ranges
  - Example ranges:
    - \* Symbol 0: Distance 1 (no extra bits)
    - \* Symbol 4: Distances 5–6 (1 extra bit)
    - \* Symbol 29: Distances 24577–32768 (13 extra bits)
  - Distance 100 would use symbol for range 97–128 with extra bits

### Example

#### Encoding Distance 100:

1. Distance 100 falls in range 97–128
2. According to DEFLATE specification:
  - Range 97–128 corresponds to distance symbol 18
  - Base distance for this symbol: 97
  - Extra bits needed: 2 bits (since  $128 - 97 + 1 = 32$  possibilities)
3. Compute extra bits:  $100 - 97 = 3$
4. Encode as: symbol 18 followed by binary '11' (3 in 2 bits)

This logarithmic encoding allows representing 32768 possible distances with only 30 symbols plus extra bits, significantly reducing the Huffman alphabet size.

## 6.4 DEFLATE's Unique Huffman Coding Scheme

This is where DEFLATE shows its brilliance. Instead of storing full Huffman trees, it stores only **code lengths**.

### 6.4.1 Two Alphabets: Literals (0-285) and Distances (0-29)

DEFLATE uses separate Huffman trees for different symbol types:

- **Literals/Lengths alphabet:** 286 symbols
  - 0-255: Literal bytes
  - 256: End-of-block marker
  - 257-285: Length codes (with extra bits)
- **Distances alphabet:** 30 symbols
  - 0-3: Direct distances 1-4
  - 4-29: Distance codes with extra bits

### 6.4.2 Canonical Huffman via Code Lengths: The Brilliant Compression

Instead of storing:

- Full tree structure (large), or
- Symbol-to-code mapping (variable size)

DEFLATE stores only:

- **Code length for each symbol** (4 bits each, 19 maximum length)
- **Canonical reconstruction rules:**
  1. Sort symbols by code length
  2. Assign first code of length  $k$  as 0
  3. Each subsequent code = previous code + 1
  4. When increasing length, left-shift and add zeros

#### Example

##### Canonical Huffman Example:

Given code lengths: A=2, B=2, C=3, D=3, E=3

1. Sort by length: A(2), B(2), C(3), D(3), E(3)
2. Length 2: Start with 00

3. Assign: A=00, B=01 (00+1)
4. Length 3: Start with 100 (01+1)
5. Assign: C=100, D=101, E=110

Decoder only needs code lengths to reconstruct this!

### 6.4.3 Run-Length Encoding of Code Lengths (19-Symbol Alphabet)

Code lengths themselves are compressed using RLE on a 19-symbol alphabet:

- 0-15: Actual code length 0-15
- 16: Repeat previous length 3-6 times (2 extra bits)
- 17: Repeat zero 3-10 times (3 extra bits)
- 18: Repeat zero 11-138 times (7 extra bits)

This compression means the Huffman tree description is itself heavily compressed!

### 6.4.4 Why Canonical Codes Enable $O(1)$ Decoding

With canonical codes, decoding becomes a table lookup:

```
1 // Precomputed tables from code lengths
2 min_code[length] = smallest code of this length
3 max_code[length] = largest code of this length
4 first_symbol[length] = first symbol with this length
5
6 // Decoding loop
7 while (need more symbols) {
8     code = peek_bits(MAX_BITS); // Look at next bits
9     for (len = 1 to MAX_BITS) {
10         if (code <= max_code[len]) {
11             symbol = first_symbol[len] + (code - min_code[len]);
12             consume_bits(len);
13             break;
14         }
15     }
16 }
```

Listing 1: Canonical Huffman Decoding (Pseudocode)

This is  $O(1)$  per symbol (constant-time lookup), crucial for fast decompression.

## 6.5 Block Structure and Streaming Format

DEFLATE supports streaming through independent compression blocks.

### 6.5.1 Three Block Types: Stored, Fixed Huffman, Dynamic Huffman

| Type            | Bits | Use Case                   |
|-----------------|------|----------------------------|
| Stored          | 00   | Uncompressible data        |
| Fixed Huffman   | 01   | Default trees (predefined) |
| Dynamic Huffman | 10   | Custom trees per block     |
| Reserved        | 11   | (Error)                    |

- **Stored blocks:** Raw data with length header (for already-compressed data)
- **Fixed blocks:** Use predefined Huffman trees (no tree description overhead)
- **Dynamic blocks:** Custom optimized trees (best compression)

### 6.5.2 Bit-Level Layout: BFINAL, BTYPE, and Data

Each block has precise bit-level layout:

```
1 // Block header
2 BFINAL: 1 bit (1 = last block in stream)
3 BTYPE: 2 bits (00, 01, or 10)
4
5 // If BTYPE == 00 (Stored block)
6 //   Skip to next byte boundary
7 //   LEN: 16 bits (block length)
8 //   NLEN: 16 bits (~LEN for error checking)
9 //   Data: LEN bytes of raw data
10
11 // If BTYPE == 01 (Fixed Huffman)
12 //   Compressed data using fixed trees
13
14 // If BTYPE == 10 (Dynamic Huffman)
15 //   HLIT: 5 bits (257-286 literal/length codes)
16 //   HDIST: 5 bits (1-30 distance codes)
17 //   HCLLEN: 4 bits (4-19 code length codes)
18 //   Code lengths for code length alphabet (HCLLEN+4 times)
19 //   Compressed literal/length and distance code lengths
20 //   Compressed data
```

Listing 2: DEFLATE Block Structure

### 6.5.3 Streaming Support: Independent Blocks and Reset Points

- Each block compressed independently
- History buffer **does not reset** between blocks (maintains 32KB window)
- Allows streaming compression of unlimited data
- Enables random access in some formats (ZIP seeks to file boundaries)

## 6.6 Step-by-Step Compression Walkthrough

Let's trace through compressing a simple English phrase.

### 6.6.1 Input: "The cat sat on the mat"

Assume ASCII encoding (simplified for example):

Position: 012345678901234567890123456

Text:       The cat sat on the mat

Indices:   0               1               2

### 6.6.2 LZSS Processing: Finding Matches

1. Position 0-2: "The" - No match in empty history → Output literals 'T','h','e'
2. Position 3: ' ' (space) - No match → Output literal ' '
3. Position 4-6: "cat" - No match → Output 'c','a','t'
4. Continue... Position 18-20: "the" matches position 0-2!
5. Match length = 3, distance = 18
6. Encode as (length=3, distance=18)

Final LZSS output (simplified):

Literals: T h e   c a t   s a t   o n   (length,distance=3,18) m a t

Symbols: 84 104 101 32 99 97 116 32 115 97 116 32 111 110 (257,18) 109 97 116

(Note: 257 is the symbol for length=3 in DEFLATE's encoding)

### 6.6.3 Building Canonical Huffman Tables

From the symbol stream, compute frequencies:

| Symbol         | Frequency |
|----------------|-----------|
| 32 (space)     | 4         |
| 97 ('a')       | 3         |
| 116 ('t')      | 3         |
| Others         | 1 each    |
| 257 (length=3) | 1         |
| 18 (distance)  | 1         |

Build Huffman trees, then extract code lengths. For example, space (32) gets shortest code.

### 6.6.4 Bitstream Assembly: Final Compressed Output

1. Write block header: BFINAL=1, BTYPE=10 (dynamic)
2. Write HLIT, HDIST, HLEN values
3. Write compressed code lengths for the 19 code-length symbols
4. Write compressed literal/length and distance code lengths
5. Write compressed data using the Huffman codes
6. Pad final byte with zeros if needed

## 6.7 Compression Levels and Performance Trade-offs

zlib (the reference implementation) offers 9 compression levels.

### 6.7.1 zlib's 9 Levels: From "fast" to "best"

| Level | Strategy                                | Use Case              |
|-------|-----------------------------------------|-----------------------|
| 0     | No compression (store only)             | Testing/disabled      |
| 1     | Fastest LZ77, lazy matching off         | Real-time compression |
| 6     | Default: moderate search, lazy matching | General purpose       |
| 9     | Full search, optimal parsing            | Archival storage      |

Levels mainly differ in:

- Hash table size and chain length
- Lazy matching on/off
- Match selection heuristics

### 6.7.2 Speed vs. Ratio: Why Level 6 is the Default

The default level 6 represents the "knee" in the speed/ratio curve:

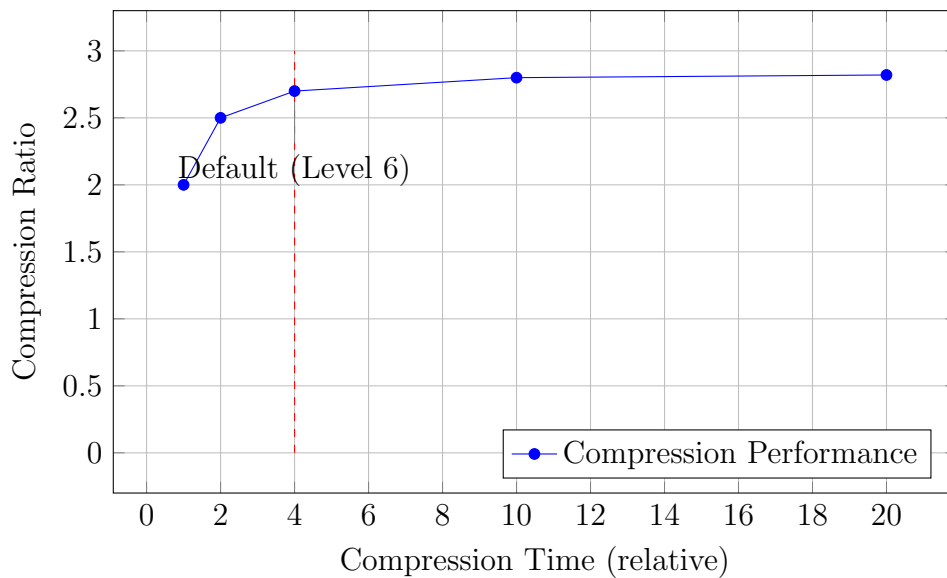


Figure 2: Trade-off between compression time and ratio in zlib

Level 9 is only ~5% better than Level 6 but takes 2-5x longer.

### 6.7.3 Memory Footprint: 32KB to 256KB Windows

- **Default:** 32KB history buffer + 16KB hash table  $\approx$  48KB
- **Maximum:** Some implementations allow 256KB windows
- **Modern systems:** Typically use 32KB-64KB for cache efficiency

## 6.8 Practical Applications and File Formats

### 6.8.1 gzip Format: Headers, Footers, and CRC32

A gzip file contains:

```
+-----+-----+-----+-----+-----+-----+-----+
| Header | Extra | Name | Comment | HCRC | DEFLATE | CRC32 | ISIZE |
+-----+-----+-----+-----+-----+-----+-----+
10 bytes optional optional 2    compressed 4    4
```

- **Header:** Magic bytes (0x1F, 0x8B), compression method (8=DEFLATE)
- **CRC32:** 32-bit checksum of uncompressed data
- **ISIZE:** Uncompressed size modulo  $2^{32}$

### 6.8.2 ZIP Format: Multiple Files and DEFLATE

ZIP is a container format supporting multiple compression methods:

- Each file compressed independently with DEFLATE
- Central directory at end enables random access
- Supports encryption, comments, extra fields

### 6.8.3 PNG: DEFLATE on Filtered Scanlines

PNG applies a **filter** to each scanline before DEFLATE:

- Subtract pixel from left pixel (Sub filter)
- Subtract pixel from above pixel (Up filter)
- Various predictors to create more compressible data
- Then DEFLATE compresses the filtered bytes

### 6.8.4 HTTP Compression: Accept-Encoding Header

Web servers compress responses on-the-fly:

Request: GET /page.html HTTP/1.1  
Accept-Encoding: gzip, deflate

Response: HTTP/1.1 200 OK  
Content-Encoding: gzip  
[gzip-compressed body]

Saves 60-80% bandwidth for text content (HTML, CSS, JS).

## 6.9 Limitations and When DEFLATE Fails

### 6.9.1 Poor Performance on Already-Compressed or Random Data

- **Already compressed:** JPEG, MP3, ZIP files → DEFLATE can't compress further
- **Encrypted data:** Random-looking → No patterns to find
- **Result:** May actually *increase* size due to overhead

Good compressors detect this and use "stored" (uncompressed) blocks.

### 6.9.2 The "Shakespeare Problem": Global vs. Local Repetition

DEFLATE's 32KB window can't capture **global repetition**:

## Example

### Shakespeare's Complete Works:

- Word "the" appears thousands of times
- But only instances within 32KB of each other are linked
- Distant repetitions (across chapters) aren't captured
- Result: Compresses to  $\sim 2$  bits/byte instead of potential  $\sim 1.5$

Modern compressors (LZMA, zstd) use larger dictionaries (MBs) for this case.

### 6.9.3 Why Not Arithmetic Coding? Patents, Speed, and Complexity

In the 1990s, arithmetic coding was:

- **Patented:** IBM held key patents
- **Slower:** 2-10x slower than Huffman
- **Complex:** Harder to implement correctly
- **Bit-level operations:** Unfriendly to byte-oriented systems

Today, arithmetic coding is patent-free but DEFLATE's ecosystem is entrenched.

### 6.9.4 Security Issues: CRIME, BREACH, and Compression Side-Channels

Compression can leak information:

- **CRIME (2012):** HTTPS compression reveals session cookies
- **BREACH (2013):** HTTP compression reveals CSRF tokens
- **Principle:** If secret affects compressed size, attacker can deduce it
- **Solution:** Disable compression on sensitive data or use random padding

## 6.10 Modern Alternatives and Successors

### 6.10.1 LZMA and 7-Zip: Better Compression, Slower Speed

LZMA (Lempel-Ziv-Markov chain Algorithm):

- **Larger window:** Up to 4GB vs DEFLATE's 32KB
- **Range coding:** Arithmetic coding variant

- **Better compression:** 30-50% smaller than DEFLATE
- **Slower:** 5-10x slower compression
- **Used in:** 7z format, xz utilities, software installers

### 6.10.2 Zstandard (zstd): Facebook's Balanced Alternative

zstd (2016) aims to replace DEFLATE:

- **Faster:** 5-10x faster decompression than zlib
- **Better compression:** 10-20% better than zlib at same speed
- **Modern features:** Dictionary training, reproducible builds
- **Adoption:** Linux kernel, SquashFS, Facebook, Netflix

### 6.10.3 Brotli: Google's Web-Optimized Successor

Brotli (2013) targets web compression:

- **Static dictionary:** 120KB of common web strings
- **Better text compression:** 20-26% better than gzip
- **Slower compression:** But fast decompression
- **Adoption:** CDNs, web servers, browsers

### 6.10.4 Why DEFLATE Still Dominates: The Legacy Effect

Despite better alternatives, DEFLATE persists because:

- **Ubiquitous support:** Every system has zlib
- **Standardized:** RFC 1951, no licensing issues
- **Good enough:** For most use cases
- **Network effects:** Hard to change entrenched standards
- **Hardware support:** Some processors have DEFLATE acceleration

## 6.11 Hands-On Analysis and Debugging

### 6.11.1 Using `gzip -v -l` and `infgen` for Inspection

```
1 # Basic compression
2 gzip -9 file.txt           # Maximum compression
3 gzip -1 file.txt          # Fastest compression
4
5 # Analyze compressed file
6 gzip -l file.txt.gz        # List compression ratio
7 gzip -v -l file.txt.gz     # Verbose listing
8
9 # Decompress with verbose output
10 gzip -v -d file.txt.gz     # Show compression ratio
11
12 # Use infgen for detailed DEFLATE analysis
13 infgen -d file.gz          # Disassemble DEFLATE stream
```

### 6.11.2 Visualizing Matches and Huffman Trees

Tools for deeper analysis:

- **puff**: Educational DEFLATE implementation (readable C)
- **infgen**: DEFLATE stream disassembler
- **Custom scripts**: Parse zlib output with `ZLIB_DEBUG`

### 6.11.3 Benchmarking: Compression Ratio vs. Speed

```
1 # Time compression
2 time gzip -9 large_file.txt
3
4 # Compare compression levels
5 for level in {1..9}; do
6     echo -n "Level $level: "
7     gzip -c -$level file.txt | wc -c
8 done
9
10 # Benchmark suite
11 pigz           # Parallel gzip implementation
12 zstd -b        # zstd benchmark mode
```

## 6.12 Summary and Key Insights

### 6.12.1 Why DEFLATE Dominated for 30+ Years

1. **Right trade-offs**: Fast decode over maximum compression

2. **Clever engineering:** Canonical Huffman, compact tree representation
3. **Open standard:** RFC 1951, no patents, reference implementation
4. **Good enough:** Adequate compression for most real-world data
5. **Ecosystem:** Tools, libraries, hardware support

### 6.12.2 Lessons for Compression Engineers: Practical Beats Perfect

- **Decode speed matters more than encode speed** for distribution formats
- **Implementation simplicity** leads to wider adoption
- **Open standards** defeat technically superior proprietary formats
- **Backward compatibility** is a powerful force
- Sometimes **”good enough” wins over ”best”**

### 6.12.3 Looking Forward: The Future of Compression Standards

- **DEFLATE** will persist in legacy systems for decades
- **zstd** and **Brotli** gaining ground in new applications
- **Hardware acceleration** becoming common (Intel QAT, AWS Nitro)
- **AI/ML approaches** showing promise but not yet practical
- The next **”DEFLATE”** will need to be **10x better** to displace it

#### Important

**Final Thought:** DEFLATE represents a peak in practical algorithm design. It shows how understanding both theory (information theory) and practice (hardware constraints, use cases) leads to enduring solutions. Study DEFLATE not just to use it, but to learn how to design algorithms that stand the test of time.

## End of Chapter Exercises

### Exercise 6.0

#### Exercise 6.1: DEFLATE Architecture Understanding

Explain in your own words why DEFLATE uses a two-stage approach (LZSS + Huffman) instead of just one of these techniques. Consider:

- (a) What types of redundancy does LZSS exploit that Huffman cannot?

- (b) What types of redundancy does Huffman exploit that LZSS cannot?
- (c) Why can't Huffman coding alone achieve good compression on typical text data?
- (d) Why can't LZSS alone achieve optimal compression?

### Exercise 6.1

#### Exercise 6.2: LZSS Match Analysis

Given the input string: "abracadabraabracadabra"

- (a) Using a sliding window of size 12, trace through LZSS processing and show the output as a sequence of literals and (length, distance) pairs.
- (b) Calculate the compression ratio if literals are 8 bits and each (length, distance) pair requires 24 bits (8 for length, 16 for distance).
- (c) What would be the output if the sliding window was only size 6 instead of 12?

### Exercise 6.2

#### Exercise 6.3: Distance Encoding Practice

Using DEFLATE's distance encoding scheme (logarithmic ranges):

- (a) Encode the following distances: 1, 5, 10, 100, 1000, 20000
- (b) For each, specify: symbol number, base distance, extra bits required, and final binary encoding.
- (c) What is the maximum distance that can be encoded without extra bits?
- (d) Why does DEFLATE use logarithmic encoding instead of direct binary representation?

*Hint: Refer to RFC 1951 Section 3.2.5 for the exact distance code table.*

### Exercise 6.3

#### Exercise 6.4: Canonical Huffman Construction

Given the following symbol frequencies for a small alphabet:

| Symbol | Frequency |
|--------|-----------|
| A      | 40        |
| B      | 30        |
| C      | 15        |
| D      | 10        |
| E      | 5         |

- Build a Huffman tree and determine code lengths for each symbol.
- Generate the canonical Huffman codes from these lengths.
- Show the exact bit sequence DEFLATE would store to represent these code lengths.
- Calculate the average code length and compare it to the entropy.

#### Exercise 6.4

##### Exercise 6.5: DEFLATE Block Analysis

Consider the following DEFLATE block header (in binary):

1 10 01101 00101 0100

Where the bits represent: BFINAL (1), BTYPE (10), HLIT (5 bits), HDIST (5 bits), HCLEN (4 bits).

- Decode each field and explain what it means.
- What type of block is this?
- How many literal/length codes will follow?
- How many distance codes will follow?
- How many code-length codes will follow?

#### Exercise 6.5

##### Exercise 6.6: Compression Level Trade-offs

You are designing a compression system for:

- A web server compressing HTML pages in real-time
- An archival system storing historical documents
- A mobile app with limited CPU and battery

For each scenario:

- Recommend a DEFLATE compression level (0-9) with justification
- Explain whether you would use fixed or dynamic Huffman blocks
- Discuss any modifications you might make to the standard DEFLATE parameters

### Exercise 6.6

#### Exercise 6.7: DEFLATE vs. Modern Alternatives

Compare DEFLATE with two modern alternatives (choose from LZMA, zstd, or Brotli):

- (a) Create a comparison table covering: compression ratio, speed (encode/decode), memory usage, and typical use cases.
- (b) For each of the following, recommend which compressor to use and why:
  - Compressing software updates for download
  - Real-time compression in a database
  - Web assets on a content delivery network
- (c) Explain why DEFLATE is still widely used despite newer, better algorithms.

### Exercise 6.7

#### Exercise 6.8: Implementation Challenge

Write pseudocode for a DEFLATE decoder that:

- (a) Reads the block header and identifies block type
- (b) For dynamic blocks, reconstructs the Huffman trees from code lengths
- (c) Decodes the compressed data using canonical Huffman decoding
- (d) Handles both literals and (length, distance) pairs

Focus on the key algorithms, not edge cases or optimizations. Your pseudocode should show the overall structure and critical steps.

### Exercise 6.8

#### Exercise 6.9: Security Analysis

The CRIME and BREACH attacks exploit compression side-channels:

- (a) Explain the fundamental vulnerability that enables these attacks.

- (b) Why does DEFLATE's LZSS stage make this vulnerability particularly dangerous?
- (c) Propose three different strategies to mitigate this risk while still using compression.
- (d) Would switching to a different compression algorithm (like zstd or Brotli) solve the problem? Why or why not?

### Exercise 6.9

#### Exercise 6.10: Research and Reflection

- (a) DEFLATE was standardized in 1996. Identify three technological constraints from that era that influenced DEFLATE's design decisions.
- (b) If you were designing DEFLATE today, what would you change given modern hardware (multi-core CPUs, gigabytes of RAM, SSD storage)?
- (c) Research the current state of hardware-accelerated compression (Intel QAT, AWS Nitro, etc.). How might these technologies change the optimal compression algorithm design?
- (d) Predict: Will DEFLATE still be widely used in 2030? Justify your answer.

### Important

#### Exercise Guidelines:

- **Exercises 6.1-6.5:** Core concepts – recommended for all students
- **Exercises 6.6-6.8:** Intermediate application – for deeper understanding
- **Exercises 6.9-6.10:** Advanced topics – for interested students or projects
- **Time estimates:** Basic exercises (30 min each), advanced (60-90 min each)
- **Resources:** Refer to RFC 1951, zlib source code, and infgen tool for hands-on exploration

### Optional Project Challenge

**Build a Simple DEFLATE Analyzer:** Write a program in your preferred language that:

- Takes a gzip file as input
- Parses the gzip header and trailer
- Extracts the DEFLATE stream
- Identifies block boundaries and types
- Reports basic statistics (compression ratio, block types used, etc.)
- **Bonus:** Visualize the Huffman trees or match distances

Use existing libraries for decompression, but implement the analysis logic yourself. This project will deepen your understanding of the actual byte-level format.

## 7: Lecture 7: Run-Length Encoding: Exploiting Equality Redundancy

### Important

#### Learning Objectives:

- Understand the concept of equality redundancy and why it's the simplest form of compressible pattern
- Master the basic RLE algorithm and its variants for different data types
- Analyze compression ratios, best/worst cases, and theoretical limits of RLE
- Learn four major RLE variants: Count-Based, Escape-Based, Block-Based, and Zero RLE
- Understand why RLE fails on some data types and how to combine it with other techniques
- Build the conceptual bridge to transform-based compression, especially BWT

### 7.1 Introduction and Motivation

#### 7.1.1 The Simplest Form of Redundancy: Repetition

When we look at data through the lens of compression, we're constantly asking: "What patterns repeat?" Among all possible patterns, the absolute simplest is when the *same symbol repeats consecutively*. We call this **equality redundancy**.

#### Definition

**Equality Redundancy:** A form of data redundancy where the same symbol appears multiple times in consecutive positions within a sequence. Each such group of identical consecutive symbols is called a **run**.

Consider these examples:

- Text: AAAABBBCCCC contains three runs: AAAA, BBB, and CCCC
- Images: A blue sky photo has long runs of similar blue pixel values
- Signals: Silence in audio contains runs of zero amplitude
- Documents: White space in scanned text creates runs of white pixels

The key insight is elegant in its simplicity: instead of storing each symbol individually, we can store each symbol once along with a count of how many times it repeats. For AAAA, instead of storing 4 bytes, we store (A,4).

### Example

**Intuition Building:** Raw data: WWWWWWWWWWWBWWWWWWWWWWBWWWWWWWWWW  
With runs identified:

- 13 W's
- 2 B's
- 13 W's
- 3 B's
- 9 W's

Storing as pairs: (W,13)(B,2)(W,13)(B,3)(W,9)

Original: 40 characters Encoded: 10 values (5 symbols + 5 counts) → significant savings!

## 7.2 Fundamental Concepts and Theory

### 7.2.1 Formal Definition of a Run

Before we can encode runs, we need a precise mathematical definition.

#### Definition

**Run:** Given a sequence  $S = s_1, s_2, s_3, \dots, s_n$  over an alphabet  $\Sigma$ , a run is a maximal consecutive subsequence  $s_i, s_{i+1}, \dots, s_j$  such that  $s_i = s_{i+1} = \dots = s_j$ , and either  $i = 1$  or  $s_{i-1} \neq s_i$ , and either  $j = n$  or  $s_{j+1} \neq s_j$ .

In simpler terms: a run is the longest possible sequence of identical symbols that appears consecutively.

#### Properties of Runs:

1. **Maximality:** A run cannot be extended—the symbol before it (if any) is different, and the symbol after it (if any) is different.
2. **Disjointness:** Runs partition the sequence; every symbol belongs to exactly one run.
3. **Alternation:** Adjacent runs always contain different symbols.

### 7.2.2 Run-Length Distribution Theorem

Understanding the statistical properties of runs is fundamental to compression analysis.

**Theorem 7.1.** *For an i.i.d. (independent and identically distributed) random sequence over an alphabet of size  $k$ , the expected run length is:*

$$E[\text{run length}] = \frac{1}{1-p} = \frac{k}{k-1}$$

where  $p = 1/k$  is the probability of a symbol repeating.

*Proof.* For a random sequence with equiprobable symbols:

- Probability that next symbol equals current =  $1/k$
- Run length follows geometric distribution:  $P(\text{run length} = L) = (1/k)^{L-1} \cdot (1 - 1/k)$
- Expected value:  $\sum_{L=1}^{\infty} L \cdot (1/k)^{L-1} \cdot (1 - 1/k) = \frac{1}{1-1/k} = \frac{k}{k-1}$

□

#### Example

##### Expected Run Lengths for Different Alphabets:

| Alphabet          | Size ( $k$ ) | Expected run length      |
|-------------------|--------------|--------------------------|
| Binary            | 2            | $2/(2-1) = 2$            |
| DNA bases         | 4            | $4/3 \approx 1.333$      |
| Lowercase letters | 26           | $26/25 = 1.04$           |
| ASCII             | 256          | $256/255 \approx 1.0039$ |

##### Implications:

- Binary data has the longest expected runs (2)
- ASCII text has runs barely longer than 1
- For large alphabets, runs of length  $> 1$  are rare events

### 7.2.3 Compression Ratio Analysis

#### Definition

##### Compression Ratio:

$$\text{Ratio} = \frac{\text{Original Size}}{\text{Compressed Size}}$$

Higher ratio means better compression.

**Theorem 7.2.** *For a sequence of length  $n$  consisting of a single symbol, the best-case compression ratio achievable with basic RLE is approximately  $\frac{n}{c}$ , where  $c$  is the size (in bytes) of one (symbol, count) pair.*

**Theorem 7.3.** *For a sequence of length  $n$  with no repeated consecutive symbols (all runs of length 1), basic RLE achieves a compression ratio of  $\frac{n}{2n} = 0.5$  when using fixed-size counts equal to symbol size, resulting in 100% expansion.*

*Proof.* If every symbol is different from its predecessor:

- Number of runs =  $n$  (each symbol is its own run)
- Encoded output =  $n$  (symbol, count) pairs
- If count field size equals symbol size (e.g., 1 byte each):
  - Original size =  $n$  bytes
  - Compressed size =  $n$  symbols +  $n$  counts =  $2n$  bytes
  - Ratio =  $n/(2n) = 0.5$
  - Expansion =  $(2n - n)/n = 1 = 100\%$

□

### 7.3 Taxonomy of RLE Methods

RLE variants differ in their answers to three fundamental questions:

1. How do we distinguish between runs and non-repeated data?
2. How do we represent the count?
3. How do we handle maximum run lengths?

Based on these choices, we can identify four major RLE families:

#### Important

##### The Four Major RLE Variants:

1. **Count-Based RLE (Classic RLE):** Stores each run as (symbol, count). Simple and effective for data with many runs.
2. **Escape-Based RLE:** Uses special markers to distinguish runs from literals. Handles mixed data efficiently.
3. **Block/Header-Based RLE (PackBits Style):** Divides data into chunks with headers that describe literal or run data. Balances overhead and efficiency.

4. **Zero Run-Length Encoding (Zero RLE):** Specialized for compressing runs of zeros in sparse data, commonly used in transform coding.

## 7.4 Count-Based RLE (Classic RLE)

Count-based RLE is the original and simplest form—each run is represented by its symbol and a count of how many times it repeats.

### 7.4.1 Basic Encoding Algorithm

---

**Algorithm 4** Count-Based RLE Encoding

---

```
1: procedure RLE-ENCODE( $S[1..n]$ )
2:    $result \leftarrow$  empty list
3:   if  $n = 0$  then
4:     return  $result$ 
5:   end if
6:    $current\_symbol \leftarrow S[1]$ 
7:    $run\_length \leftarrow 1$ 
8:   for  $i \leftarrow 2$  to  $n$  do
9:     if  $S[i] = current\_symbol$  then
10:       $run\_length \leftarrow run\_length + 1$ 
11:    else
12:      append ( $current\_symbol, run\_length$ ) to  $result$ 
13:       $current\_symbol \leftarrow S[i]$ 
14:       $run\_length \leftarrow 1$ 
15:    end if
16:  end for
17:  append ( $current\_symbol, run\_length$ ) to  $result$       ▷ Don't forget the last run!
18:  return  $result$ 
19: end procedure
```

---

### 7.4.2 Basic Decoding Algorithm

---

**Algorithm 5** Count-Based RLE Decoding

---

```
1: procedure RLE-DECODE( $pairs[1..m]$ )
2:    $result \leftarrow$  empty list
3:   for each ( $symbol, count$ ) in  $pairs$  do
4:     for  $i \leftarrow 1$  to  $count$  do
5:       append  $symbol$  to  $result$ 
6:     end for
7:   end for
8:   return  $result$ 
9: end procedure
```

---

### 7.4.3 Count-After-Symbol vs. Count-Before-Symbol

The ordering of symbol and count in the output stream affects decoding and has subtle implications.

#### Important

##### Two Common Conventions:

##### 1. Count-After-Symbol (Symbol then Count):

- Format: [symbol] [count] [symbol] [count] ...
- Example: A5B3C2 means "5 A's, 3 B's, 2 C's"
- Advantages:
  - Natural reading order: "Here's what to output, then how many"
  - Easier for streaming: see symbol first, can prepare output
  - Common in many implementations

##### 2. Count-Before-Symbol (Count then Symbol):

- Format: [count] [symbol] [count] [symbol] ...
- Example: 5A3B2C means "5 A's, 3 B's, 2 C's"
- Advantages:
  - Decoder knows length before seeing symbol
  - Can allocate buffer space if needed
  - Used in some formats (e.g., PCX)

### 7.4.4 Fixed-Width Count RLE

In fixed-width count RLE, each run length is stored in a fixed number of bits.

#### Important

##### Fixed-Width Count Fields Design:

**Design:** Each run length stored in a fixed number of bits (e.g., 8 bits, 16 bits, 32 bits)

##### Advantages:

- Simple to implement
- Fast encoding/decoding (no bit-level parsing)

- Random access possible (know where each run ends)
- Good for hardware implementation

#### Disadvantages:

- Wasted space for short runs
- Limited maximum run length (e.g., 255 for 8 bits)
- Need to split longer runs into multiple chunks
- Not adaptive to run length distribution

#### Typical trade-offs:

| Count size      | Max run | Overhead/run | Best for         |
|-----------------|---------|--------------|------------------|
| 4 bits (nibble) | 15      | 0.5 bytes    | Very short runs  |
| 8 bits (byte)   | 255     | 1 byte       | General purpose  |
| 16 bits         | 65,535  | 2 bytes      | Long runs        |
| 24 bits         | 16.7M   | 3 bytes      | Images           |
| 32 bits         | 4.3B    | 4 bytes      | Very large files |

### Example

#### Fixed-Width RLE Example:

Input sequence: WWWWWWWWWWWBBBWWWWWWWWWWBBBWWWWWWWW

#### With 8-bit counts:

- Runs: (W,13), (B,2), (W,13), (B,3), (W,9)
- Each run: 1 byte symbol + 1 byte count = 2 bytes
- Total: 5 runs  $\times$  2 bytes = 10 bytes
- Original: 40 bytes
- Compression ratio: 4:1 (75% savings)

#### With 4-bit counts (nibbles):

- Need to pack two counts per byte
- Runs: (W,13), (B,2), (W,13), (B,3), (W,9)
- Symbols: 5 bytes
- Counts: 5  $\times$  4 bits = 20 bits = 2.5 bytes

- Total: 7.5 bytes (assuming we can pack)
- But 13 exceeds 4-bit max (15) — OK here
- Compression ratio: 5.33:1 (81% savings)

#### 7.4.5 Handling Large Runs with Fixed-Width Counts

When runs exceed the maximum representable count, we need splitting strategies.

##### Important

##### Splitting Runs Strategy:

When count exceeds maximum, output multiple run chunks:

- Example: max count = 255, actual run = 1000
- Output: (sym,255), (sym,255), (sym,255), (sym,235)
- Overhead increases but preserves maximum limit

##### Alternative: Use larger count fields

- Switch to 16-bit or 32-bit counts
- But overhead increases for all runs
- May still need splitting for very large runs

##### Example

##### Run Splitting Example:

Assume 8-bit counts (max = 255)

Input run: 1000 'A's

##### Split into max-sized chunks:

- (A,255), (A,255), (A,255), (A,235)
- Overhead: 4 pairs instead of 1
- Extra overhead: 3 extra count bytes + 3 extra symbol bytes = 6 extra bytes
- Still much better than storing 1000 raw bytes!

#### 7.4.6 Variable-Width Count RLE (Elias Gamma)

Variable-width count RLE uses codes that use fewer bits for small numbers and more bits for large numbers, matching the information content of the count.

##### Important

##### Variable-Length Count Fields Design:

**Design:** Use a variable-length code (like Elias gamma coding, Fibonacci coding) to represent run lengths

##### Common approaches:

1. **Gamma/Elias codes:** Good for small numbers, inefficient for large
2. **Fibonacci codes:** Better balance, no need for separator
3. **Exponential-Golomb:** Used in H.264/AVC video coding
4. **Adaptive Huffman:** Best compression, but complex

##### Advantages:

- Optimal for the run length distribution
- No arbitrary maximum run length
- No wasted bits for short runs
- Can achieve near-entropy compression of run lengths

##### Disadvantages:

- Complex implementation (bit-level I/O)
- Slower encoding/decoding
- No random access without decoding entire stream
- Error propagation if bit errors occur

#### Elias Gamma Coding Deep Dive

Elias gamma coding provides the clearest illustration of variable-length coding principles.

##### Definition

**Elias Gamma Code:** A variable-length code for positive integers where each number  $N \geq 1$  is encoded as:

1. Write  $N$  in binary.

2. Let the binary representation of  $N$  use  $L$  bits.
3. Write  $L - 1$  zeros.
4. Append the full binary representation of  $N$ .

Equivalently, if  $k = \lfloor \log_2 N \rfloor$ , then:

$$\text{Gamma}(N) = \underbrace{00 \dots 0}_{k \text{ zeros}} \text{ binary}(N)$$

Total code length:  $2k + 1$  bits.

### Building Elias Gamma Codes: Step by Step

**1. For  $N = 1$ :**

- Binary: 1 ( $L=1$ )
- Zeros:  $L-1 = 0$  zeros
- Code: "1"

**2. For  $N = 2$ :**

- Binary: 10 ( $L=2$ )
- Zeros:  $L-1 = 1$  zero
- Code: "010"

**3. For  $N = 3$ :**

- Binary: 11 ( $L=2$ )
- Zeros: 1 zero
- Code: "011"

**4. For  $N = 4$ :**

- Binary: 100 ( $L=3$ )
- Zeros: 2 zeros
- Code: "00100"

**5. For  $N = 5$ :**

- Binary: 101 ( $L=3$ )
- Zeros: 2 zeros
- Code: "00101"

## Complete Reference Table

| N  | Binary of N | $k = \text{floor}(\log_2 N)$ | Gamma Code |
|----|-------------|------------------------------|------------|
| 1  | 1           | 0                            | 1          |
| 2  | 10          | 1                            | 010        |
| 3  | 11          | 1                            | 011        |
| 4  | 100         | 2                            | 00100      |
| 5  | 101         | 2                            | 00101      |
| 6  | 110         | 2                            | 00110      |
| 7  | 111         | 2                            | 00111      |
| 8  | 1000        | 3                            | 0001000    |
| 9  | 1001        | 3                            | 0001001    |
| 10 | 1010        | 3                            | 0001010    |

## Decoding Elias Gamma

---

### Algorithm 6 Elias Gamma Decoding

---

```

1: procedure DECODEGAMMA(bitstream)
2:    $k \leftarrow 0$ 
3:   while next bit is 0 do                                     ▷ Count leading zeros
4:      $k \leftarrow k + 1$ 
5:     advance bitstream
6:   end while
7:   advance bitstream                                             ▷ Skip the 1
8:    $value \leftarrow 1$                                            ▷ The leading 1 is the most significant bit
9:   for  $i \leftarrow 1$  to  $k$  do
10:     $value \leftarrow (value \ll 1) \mid \text{next bit}$ 
11:  end for
12:  return  $value$ 
13: end procedure

```

---

### Example

#### Decoding Example:

Given bitstream: 001011...

1. Read zeros: "00" then a 1  $\rightarrow$  found "001", so  $k = 2$
2. The 1 we read is part of the code
3. Read next  $k = 2$  bits: "01"
4. Construct value: leading 1 + "01" = "101" binary = 5

5. First number decoded is 5, remaining bits continue from "..."

## Fixed-Width vs. Variable-Width: Detailed Comparison

### Example

#### Count Field Comparison:

Consider a file with runs of lengths: 3, 5, 2, 100, 4, 200

#### Fixed 8-bit counts:

- Each run: 1 byte for count
- Total count overhead: 6 bytes = 48 bits
- Can represent all runs ( $\max 200 < 255$ )
- Simple, fast, randomly accessible

#### Fixed 4-bit counts:

- Cannot represent runs  $>15$
- Would need to split run 100 into: 15,15,15,15,15,15,10 (7 runs!)
- Run 200: even worse, would need 14 runs ( $13 \times 15 + 5$ )
- Total overhead: 6 runs become  $6+7+14 = 27$  runs!
- $27 \text{ runs} \times 0.5 \text{ bytes} = 13.5 \text{ bytes}$  (worse than 8-bit!)

#### Variable-length (Elias gamma):

- $3 \rightarrow "011"$  (3 bits)
- $5 \rightarrow "00101"$  (5 bits)
- $2 \rightarrow "010"$  (3 bits)
- $100 \rightarrow$  Let's construct: binary  $100 = 1100100$  (7 bits),  $k=6$ , so  $"0000001" + "100100" = "0000001100100"$  (13 bits)
- $4 \rightarrow "00100"$  (5 bits)
- $200 \rightarrow$  binary  $200 = 11001000$  (8 bits),  $k=7$ , so  $"00000001" + "1001000" = "000000011001000"$  (15 bits)
- Total: 44 bits  $\approx 5.5$  bytes

### Comparison Table:

| Method      | Total Bits                   | Characteristics                         |
|-------------|------------------------------|-----------------------------------------|
| Fixed 4-bit | Not possible (runs too long) | Would need run splitting                |
| Fixed 8-bit | 48 bits                      | Simple, fast, random access             |
| Elias gamma | 44 bits                      | 8.3% savings, complex, no random access |

### Bit-by-bit breakdown:

| Method      | 3  | 5  | 2  | 100 | 4  | 200 | Total |
|-------------|----|----|----|-----|----|-----|-------|
| Fixed 8-bit | 8  | 8  | 8  | 8   | 8  | 8   | 48    |
| Elias gamma | 3  | 5  | 3  | 13  | 5  | 15  | 44    |
| Savings     | +5 | +3 | +5 | -5  | +3 | -7  | +4    |

Notice the economic trade-off: small numbers (2,3,4,5) use fewer bits than fixed-width, which subsidizes the extra bits needed for large numbers (100,200). In typical data where short runs are much more common than long runs, this trade-off yields net compression.

### Decision Matrix: Fixed vs. Variable-Width Counts

| Design Question           | Fixed-Width Counts                                                                | Variable-Width Counts                                                                        |
|---------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Implementation complexity | Very simple. Each count always uses the same number of bits.                      | More complex. Must read bits carefully to determine where each count ends.                   |
| Encoding/decoding speed   | Very fast. No need to examine individual bits.                                    | Slower. Must count prefix bits and parse variable-length codes.                              |
| Random access             | Yes. If each count uses (say) 8 bits, boundaries are known immediately.           | No. Must decode sequentially to know where the next count begins.                            |
| Space efficiency          | When run lengths are roughly similar or bounded (e.g., always between 1 and 200). | When most run lengths are small but occasionally very large. Small values use very few bits. |
| Maximum run length        | Limited by chosen width. For 8 bits, maximum is 255.                              | No fixed upper bound. Large values simply use more bits.                                     |
| Error resilience          | Damage usually affects only that run.                                             | Damage may affect all following runs because boundaries shift.                               |

## 7.5 Escape-Based RLE

Escape-based RLE addresses the fundamental weakness of count-based RLE: expansion on non-repetitive data. By using special markers, it treats runs and literals differently.

### 7.5.1 Core Concept

#### Definition

**Escape Symbol:** A special marker that indicates whether the following data represents a run or a sequence of literal (uncompressed) symbols.

The key insight: instead of encoding every symbol as a (symbol, count) pair (which doubles the size for non-repeated symbols), we only encode runs when they're beneficial, and leave non-repeated data as-is.

### 7.5.2 Basic Escape Scheme Design

#### Important

##### Standard Escape Scheme:

1. Choose an escape symbol (often a value not likely to appear in data, or one that can be escaped itself)
2. When encountering a run of length  $\geq 3$ :
  - Output escape symbol
  - Output the repeated symbol
  - Output the run length
3. For non-repetitive data:
  - Output symbols literally
  - If the literal data contains the escape symbol, need a way to represent it (e.g., double escape)

### 7.5.3 Encoding Algorithm

---

**Algorithm 7** Escape-Based RLE Encoding (Threshold T=3)

---

```

1: procedure ESCAPERLE-ENCODE( $S[1..n]$ ,  $escape$ )
2:    $result \leftarrow$  empty list
3:    $i \leftarrow 1$ 
4:   while  $i \leq n$  do
5:      $run\_length \leftarrow 1$ 
6:     while  $i + run\_length \leq n$  and  $S[i + run\_length] = S[i]$  do
7:        $run\_length \leftarrow run\_length + 1$ 
8:     end while
9:     if  $run\_length \geq 3$  then                                      $\triangleright$  Long run - encode with escape
10:      append  $escape$  to  $result$ 
11:      append  $S[i]$  to  $result$ 
12:      append  $run\_length$  to  $result$ 
13:     else                                                          $\triangleright$  Short run - output as literals
14:       for  $j \leftarrow 0$  to  $run\_length - 1$  do
15:         if  $S[i + j] = escape$  then
16:           append  $escape$  to  $result$                                  $\triangleright$  Escape the escape!
17:           append  $escape$  to  $result$ 
18:         else
19:           append  $S[i + j]$  to  $result$ 
20:         end if
21:       end for
22:     end if
23:      $i \leftarrow i + run\_length$ 
24:   end while
25:   return  $result$ 
26: end procedure

```

---

### 7.5.4 Decoding Algorithm

---

**Algorithm 8** Escape-Based RLE Decoding

---

```
1: procedure ESCAPERLE-DECODE(data[1..m], escape)
2:   result  $\leftarrow$  empty list
3:   i  $\leftarrow$  1
4:   while i  $\leq m$  do
5:     if data[i] = escape then
6:       if i + 2 > m then
7:         error
8:       end if
9:       symbol  $\leftarrow$  data[i + 1]
10:      count  $\leftarrow$  data[i + 2]
11:      for j  $\leftarrow$  1 to count do
12:        append symbol to result
13:      end for
14:      i  $\leftarrow$  i + 3
15:    else
16:      if i + 1  $\leq m$  and data[i + 1] = escape then  $\triangleright$  Double escape means literal
        escape
17:        append escape to result
18:        i  $\leftarrow$  i + 2
19:      else
20:        append data[i] to result
21:        i  $\leftarrow$  i + 1
22:      end if
23:    end if
24:  end while
25:  return result
26: end procedure
```

---

### 7.5.5 Worked Example

#### Example

##### Escape Scheme Example with Escape = 0xFF:

Input: AAAAAABCDEFFFFGG (15 bytes)

##### Step-by-step encoding:

- Position 1-5: AAAAA (5 A's)  $\rightarrow$  run length  $5 \geq 3$ 
  - Output: FF A 05

- Position 6: B (literal) → output: B
- Position 7: C (literal) → output: C
- Position 8: D (literal) → output: D
- Position 9: E (literal) → output: E
- Position 10-13: FFFF (4 F's) → run length  $4 \geq 3$ 
  - Output: FF F 04
- Position 14-15: GG (2 G's) → run length  $2 < 3$ , output as literals
  - Output: G G

Encoded: FF A 05 B C D E FF F 04 G G

#### Analysis:

- Original: 15 bytes
- Encoded: 11 bytes
- Savings: 4 bytes (27% compression)
- Without escape scheme (basic RLE): would be 16 bytes (expansion!)

### Example

#### Handling Escape Symbols in Literal Data:

Input: ABC\xFF\xFFDEF (contains two 0xFF bytes)

#### Encoding:

- A → literal A
- B → literal B
- C → literal C
- \xFF (first) → this is the escape symbol! Need to escape it
  - Output: FF FF (double escape means literal FF)
- \xFF (second) → another escape symbol
  - Output: FF FF
- D → literal D

- E → literal E
- F → literal F

Encoded: A B C FF FF FF FF D E F

Notice how each literal 0xFF becomes two bytes in the output (the escape marker twice).

### 7.5.6 Choosing the Threshold

The threshold determines the minimum run length that triggers escape encoding.

#### Important

##### Threshold Trade-offs:

- **Low threshold (T=2):**
  - Catches more runs (even pairs)
  - Overhead: each run costs 3 bytes instead of 2 bytes for literals
  - Break-even: 3 bytes for run vs. 2 bytes for literals → not beneficial!
  - T=2 actually causes expansion on pairs
- **Medium threshold (T=3):**
  - Run of 3: 3 bytes encoded vs. 3 bytes literals → break-even
  - Run of 4+: 3 bytes encoded vs. 4+ bytes literals → savings
  - Most common choice
- **High threshold (T=4):**
  - Only encodes longer runs
  - Misses runs of 3 (which are break-even anyway)
  - Safer but less aggressive

**Optimal threshold analysis:** With 1-byte escape, 1-byte symbol, 1-byte count:

- Cost to encode run of length L: 3 bytes
- Cost to store as literals: L bytes
- Break-even when  $L = 3$
- Savings when  $L > 3$ :  $L - 3$  bytes saved
- Loss when  $L < 3$ :  $3 - L$  bytes wasted

### 7.5.7 Advantages and Disadvantages

#### Important

##### Escape-Based RLE: Pros and Cons

###### Advantages:

- No expansion on non-repetitive data (except for escaping the escape symbol)
- Handles mixed data types efficiently
- Simple to implement
- Used in many real-world formats (e.g., TIFF, some fax formats)

###### Disadvantages:

- Overhead of escape symbol reduces efficiency on highly repetitive data
- Need to handle escape symbol in literal data (double-escape)
- Fixed threshold may not be optimal for all data
- 3-byte overhead per run (vs. 2 bytes in basic RLE)

## 7.6 Block/Header-Based RLE (PackBits Style)

Block-based RLE divides the data into chunks, each preceded by a header that describes how to interpret the following bytes. PackBits is the classic example of this approach.

### 7.6.1 PackBits Overview

#### Definition

**PackBits Encoding:** Each chunk begins with a 1-byte header  $h$  interpreted as a signed 8-bit integer:

- If  $0 \leq h \leq 127$ : copy the next  $h + 1$  bytes literally
- If  $-127 \leq h \leq -1$ : repeat the next byte  $1 - h$  times
- $h = -128$ : reserved (no operation, sometimes used for special purposes)

#### Important

##### PackBits Header Interpretation:

**Positive (0 to 127):** Literal run

- Value  $n$  means "copy next  $n + 1$  bytes as-is"
- Range: 1 to 128 bytes per literal chunk

**Negative (-1 to -127):** Repeated run

- Value  $n$  means "repeat next byte  $1 - n$  times"
- Since  $n$  is negative,  $1 - n$  is positive
- Range: 2 to 128 repetitions
- Example:  $h = -3$  means repeat next byte  $1 - (-3) = 4$  times

**-128:** Reserved (some implementations ignore, others use as EOF marker)

## 7.6.2 Encoding Algorithm

---

### Algorithm 9 PackBits Encoding

---

```

1: procedure PACKBITS-ENCODE( $S[1..n]$ )
2:    $result \leftarrow$  empty list
3:    $i \leftarrow 1$ 
4:   while  $i \leq n$  do
5:      $run\_length \leftarrow 1$ 
6:     while  $i + run\_length \leq n$  and  $S[i + run\_length] = S[i]$  and  $run\_length < 128$ 
       do
7:        $run\_length \leftarrow run\_length + 1$ 
8:     end while
9:     if  $run\_length \geq 2$  then ▷ We have a run
10:       $header \leftarrow -(run\_length - 1)$  ▷ Negative header
11:      append  $header$  to  $result$ 
12:      append  $S[i]$  to  $result$ 
13:       $i \leftarrow i + run\_length$ 
14:    else ▷ Literal run - collect up to 128 literals
15:       $literal\_start \leftarrow i$ 
16:       $literal\_count \leftarrow 0$ 
17:      while  $i \leq n$  and  $literal\_count < 128$  do
18:        ▷ Check if next 3+ bytes form a run
19:        if  $i + 2 \leq n$  and  $S[i] = S[i + 1]$  and  $S[i] = S[i + 2]$  then
20:          break ▷ Found a run, stop collecting literals
21:        end if
22:         $i \leftarrow i + 1$ 
23:         $literal\_count \leftarrow literal\_count + 1$ 
24:      end while
25:       $header \leftarrow literal\_count - 1$  ▷ Positive header
26:      append  $header$  to  $result$ 
27:      for  $j \leftarrow literal\_start$  to  $literal\_start + literal\_count - 1$  do
28:        append  $S[j]$  to  $result$ 
29:      end for
30:    end if
31:  end while
32:  return  $result$ 
33: end procedure

```

---

### 7.6.3 Decoding Algorithm

---

**Algorithm 10** PackBits Decoding

---

```
1: procedure PACKBITS-DECODE(data[1..m])
2:   i  $\leftarrow$  1
3:   result  $\leftarrow$  empty list
4:   while i  $\leq$  m do
5:     h  $\leftarrow$  data[i]                                 $\triangleright$  header byte as signed integer
6:     i  $\leftarrow$  i + 1
7:     if h  $\geq$  0 then                                      $\triangleright$  Literal run
8:       len  $\leftarrow$  h + 1
9:       for j  $\leftarrow$  1 to len do
10:        append data[i] to result
11:        i  $\leftarrow$  i + 1
12:       end for
13:     else if h = -128 then                                 $\triangleright$  Reserved, usually skip
14:                                      $\triangleright$  No operation, sometimes used as marker
15:     else                                                     $\triangleright$  Repeated run (h is negative)
16:       len  $\leftarrow$  1 - h                                      $\triangleright$  Since h is negative
17:       value  $\leftarrow$  data[i]
18:       i  $\leftarrow$  i + 1
19:       for j  $\leftarrow$  1 to len do
20:        append value to result
21:       end for
22:     end if
23:   end while
24:   return result
25: end procedure
```

---

### 7.6.4 Worked Examples

#### Example

##### PackBits Example 1: Mixed Data

Input: AAAAABCDEFFFFGG

Encoding process:

- Run of 5 A's  $\rightarrow$  header = -4 (since  $1 - (-4) = 5$ ), then byte 'A'
- Literals B,C,D,E (4 bytes)  $\rightarrow$  header = 3 (copy  $3 + 1 = 4$  bytes), then B C D E
- Run of 4 F's  $\rightarrow$  header = -3 (repeat 4 times), then byte 'F'

- Literals G,G (2 bytes) → header = 1 (copy 2 bytes), then G G

Encoded stream (as bytes with decimal headers):

`[-4, 'A', 3, 'B', 'C', 'D', 'E', -3, 'F', 1, 'G', 'G']`

**Decoding:**

- Read -4 → repeat next byte 5 times → output "AAAAA"
- Read 3 → copy next 4 bytes → output "BCDE"
- Read -3 → repeat next byte 4 times → output "FFFF"
- Read 1 → copy next 2 bytes → output "GG"

Result matches original!

## Example

### PackBits Example 2: Long Literal Run

Input: ABCDEFGHIJKLMNOP (16 distinct letters)

Since PackBits literal chunks max at 128 bytes, we can encode as one chunk:

- Header = 15 (copy 16 bytes)
- Then all 16 letters
- Total: 1 header + 16 literals = 17 bytes
- Original: 16 bytes → slight expansion (1 byte)

## Example

### PackBits Example 3: Long Repeated Run

Input: 150 bytes of 'X'

We need multiple chunks:

- First chunk: header = -127 (repeat 128 times), byte 'X'
- Second chunk: header = -21 (repeat 22 times), byte 'X'
- Total: 2 headers + 2 bytes = 4 bytes
- Original: 150 bytes
- Compression ratio: 37.5:1 (97.3% savings!)

## Example

### PackBits Example 4: Boundary Cases

**Single byte:** Input: X

- Must encode as literal
- Header = 0 (copy 1 byte), then 'X'
- 2 bytes for 1 byte  $\rightarrow$  100% expansion

**Two identical bytes:** Input: XX

- Option 1: Encode as run of 2  $\rightarrow$  header = -1 (repeat 2 times), then 'X'
- Option 2: Encode as literals  $\rightarrow$  header = 1 (copy 2 bytes), then 'X','X'
- Both use 3 bytes vs original 2  $\rightarrow$  expansion
- PackBits has no efficient encoding for pairs!

**Two different bytes:** Input: XY

- Literals: header = 1, then 'X','Y'  $\rightarrow$  3 bytes vs 2  $\rightarrow$  expansion

## 7.6.5 Optimal Use of PackBits

### Important

**When PackBits Excels:**

- **Long runs ( $\geq 3$ ):** Excellent compression
- **Mixed data with some long runs:** Good balance
- **Data with runs  $\geq 128$ :** Still efficient (splits automatically)

**When PackBits Struggles:**

- **Short runs (pairs):** No efficient encoding (3 bytes for 2)
- **Alternating data:** All literals, slight expansion
- **Very short chunks:** Header overhead dominates

**Break-even analysis:**

- Literal chunk: cost = 1 + L bytes for L bytes (overhead = 1)

- Run chunk: cost = 2 bytes for L bytes (overhead =  $2/L$ )
- For L=3, both cost 3 bytes
- For L=4, run saves 1 byte vs literals
- For L=2, run costs 3 bytes vs 2 bytes literals → worse

#### 7.6.6 PackBits in Real-World Formats

##### Important

##### Applications of PackBits:

- **TIFF:** Optional compression method (tag 32773)
- **Macintosh resource forks:** Used extensively
- **PDF:** Can use PackBits for stream compression
- **PostScript:** Supported as a filter
- **Many embedded systems:** Simple, efficient implementation

##### Why PackBits remains popular:

- Simple to implement correctly
- No patents
- Works well on mixed data
- Byte-aligned (fast)
- Reasonable worst-case behavior (0.5-1 byte overhead per chunk)

## 7.7 Zero Run-Length Encoding (Zero RLE)

Zero RLE is a specialized variant that focuses exclusively on compressing runs of zero values. It's particularly important in transform-based compression systems.

### 7.7.1 Motivation: Sparse Data in Transform Coding

#### Important

##### Why Zero Runs Occur in Transform Coding:

1. **Energy compaction:** Transforms (DCT, wavelet) concentrate signal energy into few coefficients
2. **Quantization:** Small coefficients become zero after quantization
3. **High-frequency coefficients:** Often near-zero in natural images
4. **Zigzag scanning:** Orders coefficients from low to high frequency, creating long zero runs at the end

#### Definition

**Zero Run-Length Encoding:** A specialized form of RLE that focuses exclusively on runs of zero values, often combined with encoding the non-zero values that interrupt them.

### 7.7.2 Basic Zero RLE Algorithm

The simplest form of Zero RLE treats the data as a sequence of (zero run length, non-zero value) pairs.

---

**Algorithm 11** Zero RLE Encoding

---

```
1: procedure ZERORLE-ENCODE( $S[1..n]$ )
2:    $result \leftarrow$  empty list
3:    $i \leftarrow 1$ 
4:   while  $i \leq n$  do
5:     if  $S[i] = 0$  then
6:        $zero\_run \leftarrow 0$ 
7:       while  $i \leq n$  and  $S[i] = 0$  do
8:          $zero\_run \leftarrow zero\_run + 1$ 
9:          $i \leftarrow i + 1$ 
10:      end while
11:      if  $i > n$  then ▷ End-of-block special case
12:        append ( $zero\_run$ , EOB) to  $result$ 
13:      else
14:        append ( $zero\_run$ ,  $S[i]$ ) to  $result$ 
15:         $i \leftarrow i + 1$ 
16:      end if
17:    else
18:      append ( $0$ ,  $S[i]$ ) to  $result$  ▷ Zero run length 0
19:       $i \leftarrow i + 1$ 
20:    end if
21:  end while
22:  return  $result$ 
23: end procedure
```

---

### 7.7.3 Zero RLE in JPEG: A Detailed Case Study

JPEG's encoding of AC coefficients is the most famous example of Zero RLE.

#### Important

##### JPEG's Zero-Run Encoding:

After DCT and quantization, JPEG processes AC coefficients as follows:

1. **Zigzag scan:** Reorder  $8 \times 8$  block in zigzag pattern to maximize zero runs
2. **Run-length encoding:** Encode as (RUNLENGTH, LEVEL) pairs
  - **RUNLENGTH:** Number of consecutive zeros preceding this coefficient (0-15)
  - **LEVEL:** Non-zero coefficient value
  - Special case: (0,0) = End-of-Block (EOB)

- Special case: (15,0) = ZRL (16 zeros, handled specially)

3. **Huffman coding:** Each (RUNLENGTH, LEVEL) pair is Huffman coded

### Example

#### JPEG Zigzag Scan Example:

Original 8×8 block (quantized coefficients):

```
[150, -25, 0, 0, 0, 0, 12, 0
0, 0, 0, 0, 0, 0, 0, 0
0, 0, 0, 0, 0, 0, 0, 0
0, 0, 0, 0, 0, 0, 0, 0
0, 0, 0, 0, 0, 0, 0, 0
0, 0, 0, 0, 0, 0, 0, 0
0, 0, 0, 0, 0, 0, 0, 0
0, 0, 0, 0, 0, 0, 0, 0]
```

#### Zigzag scan order (simplified):

- Position 1: 150 (DC coefficient - encoded separately)
- Positions 2-10: -25, 0, 0, 0, 0, 12, 0, 0, 0
- Remaining 54 positions: all zeros

#### Zero-run encoding:

- After DC, first AC: (0, -25) meaning 0 zeros then -25
- Next: (4, 12) meaning 4 zeros then 12
- Then: (0,0) EOB meaning rest are zeros

**JPEG representation:** (0,-25), (4,12), EOB

### Example

#### JPEG Encoding Example:

Consider this AC coefficient sequence: 0, 0, 0, 5, 0, 0, -3, 0, 0, 0, 0, 2, [rest zeros]

**Step 1: Identify zero runs and non-zero values:**

- 3 zeros then 5
- 2 zeros then -3
- 4 zeros then 2

- All remaining zeros

**Step 2: Encode as (RUNLENGTH, LEVEL):**

- (3,5)
- (2,-3)
- (4,2)
- (0,0) for EOB

**Step 3: Huffman coding:** Each pair is Huffman coded using tables optimized for typical distributions. The Huffman tables assign shorter codes to common (RUNLENGTH, LEVEL) combinations.

#### 7.7.4 Handling Long Zero Runs

JPEG's RUNLENGTH field is limited to 15, so special handling is needed for longer zero runs.

##### Important

**JPEG's ZRL (Zero Run Length) Code:**

When a zero run exceeds 15 zeros:

- Output (15,0) for each complete block of 16 zeros
- Then encode the remainder normally

**Example:** 37 zeros followed by a value 5

- 37 zeros = 16 + 16 + 5
- Output: (15,0), (15,0), (5,5)

This allows arbitrary-length zero runs while keeping the RUNLENGTH field small.

#### 7.7.5 Zero RLE in Other Transform Codecs

##### Important

**Zero RLE in Modern Codecs:**

**MPEG video codecs:**

- Similar to JPEG for intra-coded blocks

- (run, level) coding for DCT coefficients
- Variable-length codes optimized for video statistics

#### **JPEG 2000 (wavelet-based):**

- Embedded block coding with optimized truncation (EBCOT)
- Fractional bit-plane coding
- Zero-run encoding implicit in bit-plane representation
- Arithmetic coding with context modeling

#### **H.264/AVC:**

- CAVLC (Context-Adaptive Variable-Length Coding)
- Encodes runs of zeros before non-zero coefficients
- Adapts to local statistics

### **7.7.6 Zero RLE Pipeline in Transform Coding**

#### **Important**

##### **Complete Zero-RLE Pipeline:**

Raw Image → Transform → Quantization → Zigzag Scan →  
Zero-RLE → Entropy Coding → Compressed Bitstream

##### **Why this pipeline works:**

1. **Transform:** Converts spatial redundancy to frequency coefficients
2. **Quantization:** Reduces precision, creates zeros
3. **Zigzag scan:** Reorders coefficients to maximize zero runs
4. **Zero-RLE:** Compresses zero runs efficiently
5. **Entropy coding:** Further compresses (run, level) pairs

## 7.8 Comparison of RLE Variants

### 7.8.1 Decision Matrix

| Criteria                  | Count-Based Fixed        | Count-Based Variable          | Escape-Based            | PackBits                | Zero RLE                     |
|---------------------------|--------------------------|-------------------------------|-------------------------|-------------------------|------------------------------|
| Implementation complexity | Very low                 | High                          | Medium                  | Medium                  | Medium                       |
| Speed                     | Very fast                | Slow                          | Fast                    | Fast                    | Fast                         |
| Random access             | Yes                      | No                            | No                      | No                      | No                           |
| Best for                  | Uniform runs             | Skewed runs                   | Mixed data              | Mixed data              | Sparse data                  |
| Worst-case expansion      | 100%                     | 100% (plus bit overhead)      | Small (escape handling) | Small (header overhead) | Small                        |
| Max run length            | Fixed limit              | Unlimited                     | Unlimited               | 128 per chunk           | Unlimited                    |
| Common uses               | Simple formats, hardware | Advanced codecs, BWT pipeline | TIFF, some fax          | TIFF, Mac formats       | JPEG, MPEG, DCT-based codecs |

### 7.8.2 When to Use Each Variant

#### Important

#### Selection Guide:

- **Choose Fixed-Width Count RLE when:**
  - You need maximum speed and simplicity
  - Run lengths are bounded and relatively uniform
  - Hardware implementation is required
  - Random access to runs is needed
- **Choose Variable-Width Count RLE when:**
  - Run length distribution is highly skewed (many short runs)
  - Maximum compression ratio is critical
  - You can tolerate bit-level I/O complexity
  - Part of a larger pipeline (e.g., BWT+MTF)

- **Choose Escape-Based RLE when:**
  - Data has mixed runs and literals
  - You need to guarantee no worst-case expansion
  - Simple byte-aligned implementation is desired
- **Choose PackBits when:**
  - You need a standard, widely-supported format
  - Data has moderate runs and literals
  - Implementation simplicity is important
  - Maximum run length of 128 is acceptable
- **Choose Zero RLE when:**
  - Working with transform-coded data (images, video)
  - Data is sparse with many zeros
  - Part of a larger compression pipeline
  - Non-zero values need special handling

## 7.9 Limitations of Run-Length Encoding

### 7.9.1 Why RLE Fails on Natural Language Text

Natural language text represents RLE's worst nightmare.

#### Important

##### Characteristics of Natural Language Text:

1. **Large alphabet:** 26 letters + punctuation + case + spaces
2. **Rare repeats:** In English, repeated letters are uncommon
  - Double letters: "book", "little", "success" — relatively rare
  - Triple letters: Almost nonexistent (except "beeeep" in comics)
3. **Statistical distribution:** Letter frequencies follow Zipf's law
4. **Context structure:** Redundancy is in patterns, not runs

## Example

### RLE on Shakespeare's Sonnet:

Consider the opening of Sonnet 18: Shall I compare thee to a summer's day?

#### Run analysis:

- Total characters (including spaces): 500
- Runs of length 1: 490
- Runs of length 2: 5 (double letters: "ll" in "shall", "mm" in "summer")
- Runs of length 3: 0
- Average run length: 1.01

#### RLE encoding:

- Original: 500 bytes
- Encoded:  $500 \text{ runs} \times 2 = 1000 \text{ values}$
- At 1 byte per value: 1000 bytes (100% expansion)

#### Why the expansion?

- Each run of length 1 requires a count of 1
- The count field adds 100% overhead to each character
- The few runs of length 2 save only 1 byte each (2 chars  $\rightarrow$  2 bytes vs 4 bytes for two singles)
- Savings from runs:  $5 \text{ runs} \times (2 \text{ bytes saved per run}) = 10 \text{ bytes saved}$
- But overhead from counts: 500 extra bytes
- Net loss: 490 bytes

### 7.9.2 Sensitivity to Symbol Ordering

RLE's effectiveness depends critically on how symbols are arranged.

#### Definition

**Order Sensitivity:** The property that RLE compression ratio can change dramatically if the same data is reordered, even though the multiset of symbols remains identical.

### Example

#### Dramatic Effect of Ordering:

Consider the multiset: 100 A's, 100 B's, 100 C's

**Ordering 1: Interleaved (ABC ABC ABC...) ABCABCABC... (300 bytes)**

- Each run length = 1
- Number of runs = 300
- Encoded size: 600 values (assuming 1-byte counts)
- Result: 100% expansion!

**Ordering 2: Grouped (AAA...BBB...CCC...) AAA... (100) BBB... (100) CCC... (100)**

- 3 runs, each length 100
- Encoded size:  $3 \times 2 = 6$  values
- Result: 98% compression!

## 7.10 RLE as a Preprocessing Technique

### 7.10.1 Combining RLE with Huffman Coding

The RLE+Huffman combination is a classic and effective approach.

#### Important

##### Why Combine RLE and Huffman:

- **RLE handles runs:** Converts repeated symbols into (symbol, count) pairs
- **Huffman handles frequencies:** Assigns shorter codes to frequent symbol-/count combinations
- **Synergy:** Each addresses a different type of redundancy

### Example

#### CCITT Group 3 Fax: The Classic Example

Group 3 fax does exactly this:

- Step 1: Identify runs of white and black
- Step 2: For each run length, use a Huffman code

- Step 3: Special make-up codes for long runs
- Step 4: Separate Huffman tables for white and black runs

**Result:** Achieves 10-20:1 compression on typical documents.

### 7.10.2 The Crucial Role of RLE in BWT Pipeline

This is the key connection to the next chapter.

#### Important

##### The BWT Pipeline:

Original Text → BWT → MTF → RLE → Huffman/Arithmetic

##### Why RLE is essential after BWT+MTF:

1. **BWT groups similar contexts:** Characters with similar contexts end up near each other
2. **MTF converts to small numbers:** Recently seen characters get small indices (often 0)
3. **Result:** Long runs of zeros and other small numbers appear
4. **RLE compresses these runs:** Exactly what RLE is good at!

## 7.11 Summary

#### Important

##### Key Takeaways:

1. **Equality redundancy** is the simplest form of data redundancy—consecutive identical symbols.
2. **Four major RLE variants** address different data characteristics:
  - *Count-Based RLE*: Classic (symbol, count) approach. Fixed-width for simplicity, variable-width for optimal compression.
  - *Escape-Based RLE*: Uses markers to distinguish runs from literals. Prevents worst-case expansion.
  - *Block-Based RLE (PackBits)*: Header-based chunks balance overhead and efficiency.

- *Zero RLE*: Specialized for sparse data, essential in transform coding.

### 3. Performance bounds:

- Best case: near-infinite compression (single run)
- Worst case: 100% expansion (no runs, fixed-width)
- Break-even: depends on count field size and run length distribution

4. **RLE is passive:** It can only exploit existing runs. This limitation motivates transform-based compression (BWT) which *creates* runs.

### Preview: The Burrows-Wheeler Transform

#### From RLE to BWT: The Evolution of Compression Thinking

**Level 1:** "I see runs, I compress runs." (Basic RLE)

**Level 2:** "I adapt RLE to handle different situations." (RLE variants)

**Level 3:** "I transform data to create the runs RLE needs." (BWT)

The next chapter will show you how to achieve Level 3 thinking.

## 8: Lecture 8: Transform-Based Compression: The Burrows–Wheeler Transform

### 8.1 The Three Paradigms of Lossless Compression

Before diving into the Burrows-Wheeler Transform (BWT), let us step back and consider the intellectual journey we have taken so far. Lossless compression algorithms generally fall into three broad paradigms:

1. **Statistical/Entropy Coding (Chapters 1-4):** These methods build a probability model of the source and assign shorter codes to more probable symbols. Huffman coding and arithmetic coding are the quintessential examples. The fundamental limit is the source entropy  $H(S)$ .
2. **Dictionary Methods (Chapters 5-6):** Instead of modeling probabilities, these methods identify repeated phrases and replace subsequent occurrences with references to earlier ones. LZ77, LZ78, and their descendant DEFLATE dominate practical compression due to their speed and simplicity.
3. **Transform-Based Methods (Chapters 7-8):** These methods do not compress directly. Instead, they *reorganize* the data to make it more amenable to compression by subsequent stages. In Chapter 7, we studied Run-Length Encoding (RLE), which exploits equality redundancy. In this chapter, we explore the Burrows-Wheeler Transform, which *creates* the very runs that RLE excels at compressing.

This chapter introduces the third paradigm through the lens of the Burrows-Wheeler Transform, showing how a reversible sorting operation can dramatically improve compressibility when combined with the RLE variants from Chapter 7.

### 8.2 Motivation: Beyond Local Matching

#### 8.2.1 The Limits of Sliding Windows

Recall from Chapter 5 that LZ77-based compressors (like DEFLATE) use a sliding window to find repetitions. While effective, this approach has fundamental limitations:

- **Local vs. Global Redundancy:** LZ77 can only detect repetitions that occur within the window (typically 32KB in DEFLATE). Repetitions separated by larger distances are missed entirely.
- **The “Shakespeare Problem”:** Consider the complete works of Shakespeare. The phrase “to be or not to be” appears multiple times throughout the text, but these occurrences might be hundreds of kilobytes apart. An LZ77 compressor with a

32KB window will treat each occurrence as novel text, missing this global repetition entirely.

- **Computational Cost of Large Windows:** While we could increase the window size (LZMA uses up to 4GB), the cost of searching grows significantly. More importantly, the *pointer* to reference distant matches requires more bits to encode, eating into compression gains.

### Important

The Burrows-Wheeler Transform aims to make data more compressible by rearranging it so that long runs of identical symbols appear frequently. This structure is much easier to compress than the original scattered pattern—for example, using run-length encoding (RLE). Importantly, this rearrangement is reversible, allowing exact recovery of the original data. With efficient algorithms, the transform can be computed in linear time and uses only constant additional space beyond the input.

### 8.2.2 The Global Reordering Idea

Imagine you have a long string of text. If you could somehow gather all characters that appear in similar contexts (e.g., all characters that follow “th”), they would cluster together. Within these clusters, the data would exhibit strong local correlations—exactly the kind of runs that the RLE variants from Chapter 7 can exploit.

This is precisely what the Burrows-Wheeler Transform achieves. It does not compress data; it *transforms* data into a form where:

- Identical characters tend to appear in runs
- The local structure is significantly enhanced
- The RLE variants from Chapter 7 become extraordinarily effective

### Definition

[The Transform-Then-Compress Principle] Many modern compression algorithms follow a two-stage process:

1. **Transform:** Apply a reversible transformation to the data that reveals its structure (redundancy becomes explicit).
2. **Compress:** Apply statistical or dictionary coding to the transformed data.

The Burrows-Wheeler Transform is the canonical example of this principle in loss-less compression.

## 8.3 The Burrows–Wheeler Transform

### 8.3.1 Historical Background

The Burrows-Wheeler Transform was developed by Michael Burrows and David Wheeler at DEC Systems Research Center in 1994. Their technical report, “A Block-sorting Loss-less Data Compression Algorithm,” introduced a completely new approach to compression that combined:

- The global structure exploitation of statistical methods
- The speed and simplicity of dictionary methods
- A clever use of sorting—a fundamental computer science operation

The algorithm was immediately recognized as revolutionary. Within years, it became the basis for `bzip2`, which offered significantly better compression than `gzip` on many file types, particularly text.

### 8.3.2 Intuition: Grouping Similar Contexts

The core idea of BWT is deceptively simple:

1. Generate all *cyclic rotations* of the input string (with a special end-of-string marker).
2. Sort these rotations lexicographically (dictionary order).
3. Extract the last column of the sorted rotation matrix.

The remarkable property is that the last column tends to contain long runs of identical characters. Why? Because rows that start with similar prefixes (contexts) are adjacent in the sorted order, and their last characters—which are the characters *preceding* those contexts in the original string—tend to be similar.

#### Example

[Intuition Through a Simple Example] Consider the word “BANANA”. In English text, the letter ‘N’ often follows ‘A’. In the BWT output, all the characters that precede ‘N’ in the original string will be grouped together because all rotations starting with ‘N’ are adjacent in the sorted list. This clustering effect is what makes the transform powerful.

### 8.3.3 Step-by-Step Example: banana\$

#### Example

[Computing the Burrows-Wheeler Transform (BWT) of “banana\$”]

#### Step 1: Generate all cyclic rotations

**Original string** (with end-of-string marker \$): `banana$`

All rotations (starting at each character and wrapping around):

| Rotation # | Rotation String | Last Char |
|------------|-----------------|-----------|
| 0          | banana\$        | \$        |
| 1          | anana\$b        | b         |
| 2          | nana\$ba        | a         |
| 3          | ana\$ban        | n         |
| 4          | na\$bana        | a         |
| 5          | a\$banan        | n         |
| 6          | \$banana        | a         |

*Note: Rotation #0 is the original string (no rotation). The last column will become the BWT output after sorting.*

↓ SORT LEXICOGRAPHICALLY ↓

#### Step 2: Sort rotations lexicographically

We sort these strings as you would in a dictionary (alphabetical order, with \$ coming first):

| Rank | Sorted Rotation | Orig. Index | Last Char |
|------|-----------------|-------------|-----------|
| 0    | \$banana        | 6           | a         |
| 1    | a\$banan        | 5           | n         |
| 2    | ana\$ban        | 3           | n         |
| 3    | anana\$b        | 1           | b         |
| 4    | banana\$        | 0           | \$        |
| 5    | na\$bana        | 4           | a         |
| 6    | nana\$ba        | 2           | a         |

*The row with blue background contains the **original string** (banana\$).*

↓ EXTRACT LAST COLUMN ↓

### Step 3: Extract the last column

The BWT output is the **last character** of each sorted rotation (highlighted column from Step 2):

| Rank | Sorted Rotation | Last Character |
|------|-----------------|----------------|
| 0    | \$banana        | a              |
| 1    | a\$banan        | n              |
| 2    | ana\$ban        | n              |
| 3    | anana\$b        | b              |
| 4    | banana\$        | \$             |
| 5    | na\$bana        | a              |
| 6    | nana\$ba        | a              |

**BWT(“banana\$”) = “annb\$aa”**

**Important:** The primary index  $I = 4$  (0-based), which is the rank of the row containing the original string `banana$`.

**Observation:** The BWT contains runs of identical characters: `nn` appears, and `aa` appears at the end. The original “banana” had *no* consecutive identical characters — this clustering property makes BWT excellent for compression!

*Note: The \$ marker ensures all rotations are unique and the transform is reversible.*

### 8.3.4 Visualizing the Effect: Before and After

The transformation becomes clearer when we visualize the effect:

**Original:** `banana$`

|   |   |   |   |   |   |    |
|---|---|---|---|---|---|----|
| b | a | n | a | n | a | \$ |
|---|---|---|---|---|---|----|

**BWT Output:** `annb$aa`

|   |     |   |   |    |     |   |
|---|-----|---|---|----|-----|---|
| a | n   | n | b | \$ | a   | a |
|   | run |   |   |    | run |   |

The BWT output has:

- A run of two 'n's (positions 2-3)
- A run of two 'a's at the end (positions 6-7)
- The original had *no* runs at all!

## 8.4 Formal Definition of the BWT

### 8.4.1 Cyclic Rotations and Lexicographic Sorting

Let us formalize what we have done. Given an input string  $S$  of length  $n$  with a unique end-of-string marker  $\$$  that is lexicographically smaller than all other characters:

#### Definition

[Burrows-Wheeler Transform]

The Burrows-Wheeler Transform of  $S$ , denoted  $\text{BWT}(S)$ , is computed as:

1. Form a matrix  $M$  of size  $n \times n$  where row  $i$  (0-indexed) is the cyclic rotation of  $S$  starting at position  $i$ :

$$M[i] = S[i:] + S[:i]$$

2. Sort the rows of  $M$  in lexicographic order (dictionary order, with '\$' < all other characters).
3. Let  $I$  be the index of the row containing the original string  $S$  in the sorted matrix.
4. The BWT output  $L$  is the last column of the sorted matrix:

$$L[j] = M_{\text{sorted}}[j][n-1]$$

5. The transform also outputs  $I$ , the index of the original row, which is needed for inversion.

Thus,  $\text{BWT}(S) = (L, I)$ .

In our “banana\$” example:

- $L = \text{“annb$aa”}$
- $I = 4$  (the original string “banana\$” appears at row 5, which is index 4 in 0-based indexing)

### 8.4.2 The Last Column Construction

A natural question arises: *If the first column of the sorted rotations is already sorted and contains long runs of identical symbols, why not take that instead? Why bother with the last column?*

The answer lies in two key properties: compressibility and reversibility with minimal overhead.

**Why the last column is compressible:** Consider any two rows that are adjacent in the sorted order. They share a common prefix (the longer the prefix, the more similar their contexts). The last characters of these rows are the characters that *precede* these contexts in the original string. When contexts are similar, the preceding characters often come from a limited set, creating runs of identical symbols—patterns that compression algorithms like run-length encoding can exploit efficiently.

**Why the last column is chosen over the first:** While the first column is also compressible (it is fully sorted and contains even longer runs), it cannot be reversed with only constant auxiliary information. In theory, the first column *could* be used to reconstruct the original string, but doing so would require storing significant additional information—on the order of  $O(n \log n)$  bits—to track the permutation mapping back to the original order.

The last column, however, achieves the best of both worlds:

- **Compressibility:** It contains runs of repeated symbols due to the clustering of preceding characters.
- **Efficient reversibility:** It requires only the last column string  $L$  and a single index  $I$  (constant space) to reconstruct the original string exactly.

### Important

The BWT is a *reversible* transformation. Given only  $L$  (the last column) and  $I$  (the index of the original row), we can reconstruct the original string  $S$  exactly. This reversibility—achieved with only constant auxiliary information—is what makes the transform useful for compression. We can recover the original data after decompression without storing large amounts of additional metadata.

### 8.4.3 Properties of the Transform

The BWT has several remarkable properties:

1. **Reversibility:** The transform is bijective—every input string maps to a unique  $(L, I)$  pair, and the mapping is invertible.
2. **Run Formation:** The output  $L$  tends to contain long runs of identical characters, especially in structured data like text.
3. **Context Grouping:** Characters that appear in similar contexts in the original string are brought together in the output.
4. **Structure Revealing:** Higher-order dependencies in the original string become lower-order dependencies in the transformed string.

5. **Block Structure:** The transform operates on fixed-size blocks (typically 100KB–900KB), allowing a trade-off between compression ratio and memory usage.

## 8.5 Inverting the BWT

The inversion algorithm is where the true cleverness of the BWT reveals itself. Given only  $L$  (the last column) and  $I$  (the index of the original row), we can reconstruct the original string without ever storing the full rotation matrix.

### 8.5.1 The First–Last (LF) Mapping

The key insight is that the first column  $F$  of the sorted rotation matrix can be derived from  $L$  alone:

1.  $F$  is simply  $L$  sorted lexicographically.
2. Why? Because the rows are sorted lexicographically, the first column is just all characters of the original string in sorted order.

In our example:

- $L = [a, n, n, b, \$, a, a]$
- Sorting  $L$  gives:  $[\$, a, a, a, b, n, n] = F$

Now we have a crucial observation: For any character in the last column, we can find its corresponding character in the first column through a mapping that preserves the relative order of identical characters.

#### Definition

[LF-Mapping] The LF-mapping (Last-to-First mapping) is a function  $\text{LF}(i)$  that returns the row index in the first column that corresponds to the same character occurrence as position  $i$  in the last column.

For a character  $c$  that appears  $k$  times in the string, the  $k$  occurrences appear in the same order in both  $F$  and  $L$ . Therefore, if  $L[i]$  is the  $r$ -th occurrence of character  $c$  in  $L$  (counting from 1), then  $\text{LF}(i)$  is the position of the  $r$ -th occurrence of  $c$  in  $F$ .

### 8.5.2 Algorithm for Computing LF-Mapping

---

**Algorithm 12** Compute LF-Mapping

---

**Require:**  $L[0..n-1]$  (last column)

**Ensure:**  $LF[0..n-1]$  (last-to-first mapping)

```

1:  $n \leftarrow \text{length}(L)$ 
2:  $F \leftarrow \text{sort}(L)$  ▷ First column
3: ▷ Count occurrences of each character
4: Initialize arrays  $C[|\Sigma|]$  and  $\text{rank}[0..n-1]$ 
5: for  $i \leftarrow 0$  to  $n-1$  do
6:    $c \leftarrow L[i]$ 
7:    $\text{rank}[i] \leftarrow \text{count of } c \text{ seen so far in } L$  ▷ 1-based rank
8:   Increment  $C[c]$ 
9: end for
10: ▷ Build cumulative counts for first column
11:  $\text{cum}[0] \leftarrow 0$ 
12: for each character  $c$  in sorted order do
13:    $\text{cum}[c] \leftarrow \text{cum}[\text{prev}] + C[\text{prev}]$ 
14: end for
15: ▷ Compute LF mapping
16: for  $i \leftarrow 0$  to  $n-1$  do
17:    $c \leftarrow L[i]$ 
18:    $LF[i] \leftarrow \text{cum}[c] + (\text{rank}[i] - 1)$  ▷ Subtract 1 for 1-based rank
19: end for
20: return  $LF$ 

```

---

## Example

[Step-by-Step LF-Mapping for "annb\$aa"]

We will compute the LF-mapping by building it incrementally, explaining each operation in detail.

### Preliminaries: Understanding the Goal

The LF-mapping answers: "If I am at row  $i$  in the last column  $L$ , which row in the first column  $F$  contains the SAME occurrence of the SAME character?"

This works because:

- The first column  $F$  is just  $L$  sorted
- Identical characters appear in the same order in both columns
- The  $k$ -th occurrence of character  $c$  in  $L$  corresponds to the  $k$ -th occurrence of  $c$  in  $F$

### Our Data

The last column  $L$  from  $\text{BWT}(\text{"banana\$"})$  is:

$$L = [a, n, n, b, \$, a, a]$$

Positions (indices): 0 1 2 3 4 5 6

### Step 1: Compute the First Column $F$

$F$  is simply  $L$  sorted in ascending order (with '\$' < 'a' < 'b' < 'n'):

$$\text{Sort}(L) = [ \$, a, a, a, b, n, n ]$$

So the first column  $F$  is:

$$F = [ \$, a, a, a, b, n, n ]$$

Positions in  $F$ : 0 1 2 3 4 5 6

### Step 2: Count Occurrences in $L$ (Build Rank Array)

We go through  $L$  from left to right, counting how many times we have seen each character so far. This count is called the "rank" - it tells us "this is the rank-th occurrence of this character."

| Index i | L[i] | Rank in L | Explanation                                       |
|---------|------|-----------|---------------------------------------------------|
| 0       | a    | 1         | First 'a' encountered in L (none seen before)     |
| 1       | n    | 1         | First 'n' encountered in L (none seen before)     |
| 2       | n    | 2         | Second 'n' (we saw one 'n' before at i=1)         |
| 3       | b    | 1         | First 'b' encountered in L (none seen before)     |
| 4       | \$   | 1         | First '\$' encountered in L (none seen before)    |
| 5       | a    | 2         | Second 'a' (we saw one 'a' before at i=0)         |
| 6       | a    | 3         | Third 'a' (we saw two 'a's before at i=0 and i=5) |

So the rank array R (matching each position in L to its occurrence number) is:

$$R = [1, 1, 2, 1, 1, 2, 3]$$

### Step 3: Count Occurrences in F

Now we count how many times each character appears in the first column F:

| Character | Count in F | Positions in F    |
|-----------|------------|-------------------|
| \$        | 1          | Position 0        |
| a         | 3          | Positions 1, 2, 3 |
| b         | 1          | Position 4        |
| n         | 2          | Positions 5, 6    |

### Step 4: Build Cumulative Starting Positions for F

For each character, we need to know: "Where does the first occurrence of this character appear in F?" This is the cumulative count.

We calculate  $\text{cum}[c]$  = the starting index of character c in F:

| Character | Count | Cumulative Calculation                                      | cum[c] |
|-----------|-------|-------------------------------------------------------------|--------|
| \$        | 1     | $\text{cum}[\$] = 0$ (first character)                      | 0      |
| a         | 3     | $\text{cum}[a] = \text{cum}[\$] + \text{count}[\$] = 0 + 1$ | 1      |
| b         | 1     | $\text{cum}[b] = \text{cum}[a] + \text{count}[a] = 1 + 3$   | 4      |
| n         | 2     | $\text{cum}[n] = \text{cum}[b] + \text{count}[b] = 4 + 1$   | 5      |

This means:

- The first '\$' in F is at position  $\text{cum}[\$] = 0$
- The first 'a' in F is at position  $\text{cum}[a] = 1$
- The second 'a' in F is at position  $\text{cum}[a] + 1 = 2$

- The third 'a' in F is at position  $\text{cum}[\text{a}] + 2 = 3$
- The first 'b' in F is at position  $\text{cum}[\text{b}] = 4$
- The first 'n' in F is at position  $\text{cum}[\text{n}] = 5$
- The second 'n' in F is at position  $\text{cum}[\text{n}] + 1 = 6$

## Step 5: Apply the LF Formula

For any position  $i$  in L:

- Let  $c = L[i]$  (the character at that position)
- Let  $r = \text{rank}[i]$  (this is the  $r$ -th occurrence of  $c$  in L)
- In F, the  $r$ -th occurrence of  $c$  appears at:  $\text{cum}[c] + (r - 1)$

Therefore:

$$LF[i] = \text{cum}[c] + (\text{rank}[i] - 1)$$

Now we apply this formula to each position  $i$ :

| $i$ | $L[i]$ | $c$ | $r = \text{rank}[i]$ | Formula Applied                | Calculation | $LF[i]$ |
|-----|--------|-----|----------------------|--------------------------------|-------------|---------|
| 0   | a      | a   | 1                    | $\text{cum}[\text{a}] + (1-1)$ | $1 + 0$     | 1       |
| 1   | n      | n   | 1                    | $\text{cum}[\text{n}] + (1-1)$ | $5 + 0$     | 5       |
| 2   | n      | n   | 2                    | $\text{cum}[\text{n}] + (2-1)$ | $5 + 1$     | 6       |
| 3   | b      | b   | 1                    | $\text{cum}[\text{b}] + (1-1)$ | $4 + 0$     | 4       |
| 4   | \$     | \$  | 1                    | $\text{cum}[\$] + (1-1)$       | $0 + 0$     | 0       |
| 5   | a      | a   | 2                    | $\text{cum}[\text{a}] + (2-1)$ | $1 + 1$     | 2       |
| 6   | a      | a   | 3                    | $\text{cum}[\text{a}] + (3-1)$ | $1 + 2$     | 3       |

## Step 6: Final LF-Mapping Array

$$LF = [1, 5, 6, 4, 0, 2, 3]$$

This array tells us: from row  $i$  in L, go to row  $LF[i]$  in F to find the same character occurrence.

## Verification (Optional)

Let's verify a few mappings to build intuition:

| LF[i]     | Explanation                                                    |
|-----------|----------------------------------------------------------------|
| LF[0] = 1 | L[0] is the first 'a'. In F, the first 'a' is at position 1.   |
| LF[1] = 5 | L[1] is the first 'n'. In F, the first 'n' is at position 5.   |
| LF[2] = 6 | L[2] is the second 'n'. In F, the second 'n' is at position 6. |
| LF[3] = 4 | L[3] is the first 'b'. In F, the first 'b' is at position 4.   |
| LF[4] = 0 | L[4] is the first '\$'. In F, the first '\$' is at position 0. |
| LF[5] = 2 | L[5] is the second 'a'. In F, the second 'a' is at position 2. |
| LF[6] = 3 | L[6] is the third 'a'. In F, the third 'a' is at position 3.   |

### 8.5.3 Reconstruction Algorithm

Now that we understand the LF-mapping, let's see how it enables us to reconstruct the original string from just  $L$  and  $I$ .

## The Key Insight

Recall what the sorted rotations matrix represents:

- Each row is a cyclic rotation of the original string
- The first column  $F$  contains the first character of each rotation
- The last column  $L$  contains the last character of each rotation
- The LF-mapping tells us: from a character in  $L$ , where does the **same occurrence** appear in  $F$ ?

Here's the crucial observation:

### Important

For any row  $i$  in the sorted matrix, the character  $L[i]$  (last column) is the character that **immediately precedes** the character  $F[i]$  (first column) in the original string. Why? Because row  $i$  is a rotation. If you take the last character and move it to the front, you get the rotation that starts with that character. The LF-mapping takes you to that row.

This gives us a way to reconstruct the string backwards:

1. Start at the row containing the original string (we know this row index =  $I$ )
2. The last character of that row,  $L[I]$ , is the **last character** of the original string
3. Use LF-mapping to find where this same character appears in  $F$  — that row's rotation starts with this character

4. The last character of *that* row is the character that comes **before** it in the original string
5. Repeat — each step gives us the previous character

## The Algorithm

---

### Algorithm 13 BWT Inversion

---

**Require:**  $L[0..n-1]$  (last column),  $I$  (index of original row)

**Ensure:** Original string  $S$

```

1:  $n \leftarrow \text{length}(L)$ 
2: Compute LF-mapping array  $LF[0..n-1]$  using Algorithm 12
3: Initialize  $S$  as empty string
4:  $row \leftarrow I$ 
5: for  $k \leftarrow 0$  to  $n-1$  do
6:   Prepend  $L[row]$  to  $S$                                  $\triangleright$  Add character to front (building backwards)
7:    $row \leftarrow LF[row]$                                  $\triangleright$  Move to the row that starts with this character
8: end for
9: return  $S$ 

```

---

## Why We Prepend (Not Append)

Notice that we **prepend** characters to  $S$  rather than appending. Here's why:

- We start at the row containing the original string. Its last character  $L[I]$  is the **last** character of the original string.
- Each LF step takes us to a row whose rotation starts with the character we just saw. The last character of that new row is the character that comes **before** it in the original string.
- So we discover characters from **last to first**
- Prepending builds the string in correct order: first character discovered becomes last, last discovered becomes first

## Example

[Step-by-Step Inversion of "annb\$aa" with  $I=4$ ]

Now we use the LF-mapping we computed to reconstruct the original string.

## The Big Picture

Think of the sorted rotations matrix. We know:

- Row 4 (since  $I = 4$ ) contains the original string: **anana\$b**
- Its last character  $L[4] = \$$  is the last character of the original string
- The LF-mapping tells us: the same '\$' appears in  $F$  at row  $LF[4] = 0$
- Row 0's rotation starts with '\$': **\$banana**
- Its last character  $L[0] = a$  is the character that comes **before** '\$' in the original string
- Repeat to discover characters backwards

## Step 1: Initial Setup

We have:

- Last column  $L = [a, n, n, b, \$, a, a]$
- LF-mapping =  $[1, 5, 6, 4, 0, 2, 3]$
- Primary index  $I = 4$  (tells us original string is at row 4)

We start at row =  $I = 4$ .

## Step 2: Walk Backwards Through the Rows

At each step:

1. Take  $L[row]$  — this is the next character we discovered (from end to beginning)
2. Follow  $LF[row]$  to find the row whose rotation starts with this character
3. That row's last character is the character that comes before it

| Step  | Current Row | L[row] | Next Row = LF[row] | Explanation                                            |
|-------|-------------|--------|--------------------|--------------------------------------------------------|
| Start | 4           | —      | —                  | Begin at original string's row                         |
| 1     | 4           | \$     | 0                  | '\$' is last char. Follow to row starting with '\$'    |
| 2     | 0           | a      | 1                  | 'a' comes before '\$'. Follow to row starting with 'a' |
| 3     | 1           | n      | 5                  | 'n' comes before that 'a'                              |
| 4     | 5           | a      | 2                  | 'a' comes before that 'n'                              |
| 5     | 2           | n      | 6                  | 'n' comes before that 'a'                              |
| 6     | 6           | a      | 3                  | 'a' comes before that 'n'                              |
| 7     | 3           | b      | 4                  | 'b' comes before that 'a' (back to start)              |

### Step 3: Build the String by Prepending

We discovered characters in reverse order: \$, a, n, a, n, a, b

To get the original forward order, we **prepend** each newly discovered character:

| Step | Character Discovered | String After Prepending |
|------|----------------------|-------------------------|
| 1    | \$                   | "\$"                    |
| 2    | a                    | "a\$"                   |
| 3    | n                    | "na\$"                  |
| 4    | a                    | "ana\$"                 |
| 5    | n                    | "nana\$"                |
| 6    | a                    | "anana\$"               |
| 7    | b                    | "banana\$"              |

### Step 4: Remove the Marker

The reconstructed string is "banana\$". Removing the end marker '\$' gives us the original: "banana".

## Why This Works (Revisited)

The LF-mapping creates a cycle that visits every row exactly once:

$$4 \rightarrow 0 \rightarrow 1 \rightarrow 5 \rightarrow 2 \rightarrow 6 \rightarrow 3 \rightarrow 4$$

Walking this cycle backwards (by following LF) and collecting  $L[\text{row}]$  at each step reconstructs the original string from last character to first. Prepending each character gives us the correct forward order.

This is the elegance of BWT: we can recover the original string using only  $L$  and  $I$ , without ever storing the full rotation matrix!

### 8.5.4 Proof of Correctness (Sketch)

The inversion algorithm works because of a fundamental property of the sorted rotation matrix:

**Theorem 8.1** (LF-Mapping Property). *For any row  $i$  in the sorted rotation matrix, the character  $L[i]$  (last column) is the character that precedes  $F[i]$  (first column) in the original string. Moreover, the row  $LF[i]$  in the sorted matrix corresponds to the rotation that starts with  $L[i]$ .*

This creates a chain that cycles through all rows exactly once. Starting from the row containing the original string (index  $I$ ) and following the LF-mapping backwards, we visit each row exactly once in reverse cyclic order, allowing us to reconstruct the original string by collecting the last column characters in reverse order.

## 8.6 Why BWT Improves Compressibility

### 8.6.1 Run Formation and Structure Revelation

The BWT's power for compression comes from its tendency to create runs of identical characters. To understand why, consider the following:

1. In the sorted rotation matrix, rows with similar prefixes are adjacent.
2. The last column contains the characters that *precede* these prefixes in the original string.
3. If the original text has statistical regularities (e.g., 'q' is almost always followed by 'u'), then all rows starting with 'u' will be adjacent, and their last characters will all be 'q' (or characters that commonly precede 'u').

### Important

[Important Clarification on Entropy] It is crucial to understand that the BWT does **not** reduce the zero-order entropy of the data. In fact, the zero-order entropy of  $L$  equals that of  $S$  because the transform is a permutation of the original characters. What the BWT achieves is something more subtle: it transforms **higher-order dependencies** in the original string into **lower-order dependencies** in the transformed string. After BWT, patterns that were previously spread across the text become localized, making them exploitable by simple zero-order coding methods like Move-To-Front and the RLE variants we studied in Chapter 7.

In other words, the BWT rearranges the data so that the *conditional probabilities* that characterize higher-order structure become visible as *empirical frequencies* that zero-order coders can exploit.

### Example

[Visualizing the Effect] Consider English text. The character 'u' frequently follows 'q'. In the original text, these 'u's are scattered. In the BWT output:

- All occurrences of 'u' (as first column entries) will be grouped together because rotations starting with 'u' are adjacent in sorted order.
- Their corresponding last column entries will be the characters that preceded these 'u's—often 'q'.
- Result: A run of 'q's appears in the output, which can be efficiently compressed by the Count-Based RLE (fixed or variable-width) we studied in Chapter 7.

#### 8.6.2 Connection to Context Modeling

Recall from Chapter 4 that higher-order Markov models capture dependencies by considering the previous  $k$  characters. The BWT achieves something analogous without explicit probability modeling:

- In Chapter 4, we built explicit models:  $P(\text{next symbol} \mid \text{previous } k \text{ symbols})$
- In BWT, sorting rotations groups symbols that share the same context (the following characters)
- The result is that conditional probabilities become visible as local patterns in the output

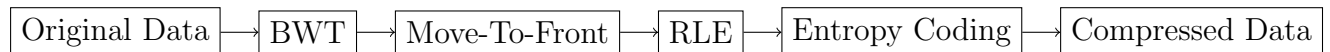
This is why BWT can be seen as an *implicit context modeling* transform—it reorganizes data so that context-dependent structure becomes spatially local.

### Important

The BWT doesn't compress; it reveals structure. The actual compression is done by subsequent stages that exploit the local correlations the BWT creates. This is where the RLE variants from Chapter 7 become essential.

## 8.7 The Full BWT Compression Pipeline

A complete BWT-based compressor typically consists of several stages:



### 8.7.1 Stage 1: BWT Transform

The BWT rearranges the data to create local correlations and runs. As we saw with “banana\$”, the output contains runs that didn't exist in the original.

### 8.7.2 Why MTF? Why Not RLE Directly on BWT?

A common question is: *If BWT already groups identical characters, why not apply RLE immediately?*

The key issue is that BWT creates **clusters**, not true **runs**. Real BWT outputs rarely contain long uninterrupted sequences of the same symbol. For example:

$$\text{BWT}(\text{"mississippi"}) = \text{ipssm\$pissii}$$

All the ‘i’'s are nearby, but not consecutive. Direct RLE would mostly see runs of length 1, giving almost no compression.

### Important

[The Insight] BWT brings similar symbols *close* to each other, but not *together*. MTF converts this proximity into actual redundancy: whenever a symbol reappears within a short distance, MTF outputs 0 because that symbol is already at the front of the list. This produces long runs of 0s, which RLE compresses extremely well.

**Additional benefit:** After MTF, the alphabet collapses to very small integers (0–5 dominate). Thus RLE not only gains long runs but also *smaller run symbols*. Encoding runs of 0 is far cheaper than encoding runs of characters like ‘i’ or ‘s’.

In short: **BWT provides structure, but MTF converts that structure into compressible redundancy.** Without MTF, BWT output is not meaningfully RLE-friendly; with MTF, it becomes ideal for both RLE and entropy coding.

### 8.7.3 Stage 2: Move-To-Front (MTF) Encoding

After the BWT, we have a string with local correlations but not yet in a form optimal for entropy coding. Move-To-Front encoding transforms this string into a sequence of small integers that are highly compressible.

#### Definition

[Move-To-Front Encoding] The MTF encoder maintains a list of symbols (typically all 256 byte values). For each input symbol:

1. Output its current index in the list (0-based)
2. Move that symbol to the front of the list

---

#### Algorithm 14 Move-To-Front Encoding

---

**Require:** Input sequence  $X[0..n-1]$  over alphabet  $\Sigma$

**Ensure:** Output sequence  $Y[0..n-1]$  of indices

- 1: Initialize list  $L \leftarrow \text{sorted}(\Sigma)$  ▷ Typically 0..255 in order
  - 2: **for**  $i \leftarrow 0$  to  $n-1$  **do**
  - 3:     Find position  $p$  such that  $L[p] = X[i]$
  - 4:      $Y[i] \leftarrow p$
  - 5:     Remove  $L[p]$  from list and insert at front
  - 6: **end for**
  - 7: **return**  $Y$
- 

#### Example

[MTF of BWT Output] For pedagogical clarity, we use a simplified alphabet containing only the characters that appear in our example:  $[a, b, n, \$]$ . In practice, MTF operates on the full byte alphabet (0–255), but the principle remains identical.

Apply MTF to “annb\$aa” with initial list  $[a, b, n, \$]$  (sorted order for predictability):

| Input | Current List                                                   | Index | Output |
|-------|----------------------------------------------------------------|-------|--------|
| a     | [a, b, n, \$]                                                  | 0     | 0      |
|       | Move a to front $\rightarrow$ [a, b, n, \$] (already at front) |       |        |
| n     | [a, b, n, \$]                                                  | 2     | 2      |
|       | Move n to front $\rightarrow$ [n, a, b, \$]                    |       |        |
| n     | [n, a, b, \$]                                                  | 0     | 0      |
|       | Move n to front $\rightarrow$ [n, a, b, \$] (already at front) |       |        |
| b     | [n, a, b, \$]                                                  | 2     | 2      |
|       | Move b to front $\rightarrow$ [b, n, a, \$]                    |       |        |
| \$    | [b, n, a, \$]                                                  | 3     | 3      |
|       | Move \$ to front $\rightarrow$ [\$, b, n, a]                   |       |        |
| a     | [\$, b, n, a]                                                  | 3     | 3      |
|       | Move a to front $\rightarrow$ [a, \$, b, n]                    |       |        |
| a     | [a, \$, b, n]                                                  | 0     | 0      |
|       | Move a to front $\rightarrow$ [a, \$, b, n]                    |       |        |

MTF output: [0, 2, 0, 2, 3, 3, 0]

Notice how runs of identical symbols in the BWT output become runs of zeros in the MTF output! This is the key to BWT's effectiveness—local structure in the BWT output translates to runs of small numbers after MTF.

#### 8.7.4 Stage 2 (Reverse): Move-To-Front Decoding

The MTF encoding is fully reversible. Given the sequence of indices produced by the encoder and the same initial list, we can recover the original symbol sequence exactly. This is essential for decompression.

##### Definition

[Move-To-Front Decoding] The MTF decoder maintains the same list of symbols as the encoder (initially identical). For each input index  $y$  (starting from 0):

1. Output the symbol at position  $y$  in the current list
2. Move that symbol to the front of the list

This exactly reverses the encoding process.

---

**Algorithm 15** Move-To-Front Decoding

---

**Require:** Input index sequence  $Y[0..n-1]$  (from MTF encoder)

**Ensure:** Recovered symbol sequence  $X[0..n-1]$

```
1: Initialize list  $L \leftarrow \text{sorted}(\Sigma)$  ▷ Same initial order as encoder
2: for  $i \leftarrow 0$  to  $n-1$  do
3:    $p \leftarrow Y[i]$  ▷ Index from the encoded stream
4:    $X[i] \leftarrow L[p]$  ▷ Symbol at that position
5:   Remove  $L[p]$  from list and insert at front
6: end for
7: return  $X$ 
```

---

### Example

[Decoding the MTF Output from Our Example]

We now reverse the MTF encoding we performed earlier. The encoded output was:

$$Y = [0, 2, 0, 2, 3, 3, 0]$$

We start with the same initial list as the encoder:  $[a, b, n, \$]$ .

| Input Index | Current List    | Output Symbol |
|-------------|-----------------|---------------|
| 0           | $[a, b, n, \$]$ | a             |
| 2           | $[a, b, n, \$]$ | n             |
| 0           | $[n, a, b, \$]$ | n             |
| 2           | $[n, a, b, \$]$ | b             |
| 3           | $[b, n, a, \$]$ | \$            |
| 3           | $[\$, b, n, a]$ | a             |
| 0           | $[a, \$, b, n]$ | a             |

The recovered symbol sequence is:  $[a, n, n, b, \$, a, a]$ .

Concatenating these gives the string **annb\$aa**, which is exactly the original BWT output we started with! This confirms that the MTF decoding correctly reverses the encoding.

### 8.7.5 Why MTF Decoding is Simple and Fast

Notice the symmetry between encoding and decoding:

- **Encoding:** Symbol  $\rightarrow$  find index  $\rightarrow$  output index  $\rightarrow$  move symbol to front
- **Decoding:** Index  $\rightarrow$  find symbol at that index  $\rightarrow$  output symbol  $\rightarrow$  move that symbol to front

Both operations run in  $O(1)$  time if we implement the list with a linked structure or an array and a separate ranking array. In practice, with an alphabet of 256 symbols, even a linear scan ( $O(|\Sigma|)$ ) is negligible. This efficiency is one reason MTF is so popular in BWT-based compressors like `bzip2`.

### 8.7.6 Stage 3: Run-Length Encoding (RLE)

After MTF, we have a sequence dominated by small numbers, often with long runs of zeros. This is where the RLE variants from Chapter 7 become essential.

#### Important

##### Choosing the Right RLE Variant for BWT Output:

The MTF output after BWT has specific characteristics that make certain RLE variants particularly effective:

- **Long runs of zeros:** Common, especially in highly structured text
- **Occasional small non-zero values:** Typically 1, 2, 3, etc.
- **Few large values:** Rare, but possible

Based on these characteristics:

- **Zero RLE (Chapter 7, Section 7.6):** Highly effective for the long zero runs. Can be combined with encoding the non-zero values.
- **Variable-Width Count RLE (Chapter 7, Section 7.4.2):** Optimal for the skewed distribution of run lengths (many short runs, few long runs).
- **Escape-Based RLE (Chapter 7, Section 7.5):** Useful if the data has mixed runs and isolated values.
- **PackBits (Chapter 7, Section 7.6):** Simple and effective, though the 128-byte chunk limit may require splitting long zero runs.

`bzip2` uses a custom RLE stage optimized for MTF output, but the principles are the same.

#### Example

[RLE of MTF Output Using Different Variants]

MTF output: [0, 2, 0, 2, 3, 3, 0]

**Option 1: Count-Based Fixed-Width RLE (assuming 4-bit counts)**

- (0,1), (2,1), (0,1), (2,1), (3,2), (0,1)

- Each pair: 1 byte for value + 0.5 bytes for count = 1.5 bytes
- Total: 6 pairs  $\times$  1.5 = 9 bytes

#### Option 2: Count-Based Variable-Width RLE (Elias gamma)

- Values: 0,1,2,3  $\rightarrow$  encode with Elias gamma
- Counts: mostly 1, one 2  $\rightarrow$  variable-length
- Total: approximately 25-30 bits  $\approx$  3.5-4 bytes
- Better compression but more complex

#### Option 3: Zero RLE (focus on zero runs)

- Zero runs: (1), then later (1), then final (1) — not very effective here
- Better for longer zero runs

#### Option 4: PackBits

- Since most values are isolated, mostly literal chunks
- Header per chunk adds overhead
- Would likely expand this short sequence

This example shows that for short sequences, simpler methods may be sufficient, but for real data with long zero runs, the more sophisticated RLE variants shine.

### 8.7.7 Stage 4: Final Entropy Coding (Huffman or Arithmetic)

The final stage applies statistical coding (Huffman or arithmetic coding) to the RLE output. Because the data now consists of small integers with skewed distributions, entropy coding achieves excellent compression ratios.

#### Example

[Complete BWT Pipeline for “banana\$”]

Let’s trace the complete pipeline:

1. **Original:** banana\$
2. **BWT:** annb\$aa (with I=4)
3. **MTF:** [0, 2, 0, 2, 3, 3, 0]
4. **RLE (simplified):** (0,1), (2,1), (0,1), (2,1), (3,2), (0,1)

5. **Huffman coding:** Assign codes based on frequencies

- (0,1) appears 3 times → shortest code
- (2,1) appears 2 times → medium code
- (3,2) appears 1 time → longer code

The final compressed size is much smaller than the original 7 characters!

## 8.8 Why RLE is Essential After MTF

The connection between BWT and the RLE variants from Chapter 7 is profound:

### Important

#### The BWT-MTF-RLE Synergy:

1. **BWT creates local correlations:** Characters that appear in similar contexts are grouped together.
2. **MTF converts correlations to runs:** When the same character appears multiple times in the BWT output, MTF outputs zeros. When similar characters appear (e.g., 'a' then 'b' that recently moved to front), MTF outputs small numbers.
3. **RLE compresses the runs:** The long runs of zeros and short runs of small numbers are exactly what the RLE variants from Chapter 7 are designed to compress efficiently.

**Without RLE, the BWT+MTF combination would be much less effective. Without understanding RLE, you cannot fully understand BWT.**

### Example

[Real-World: How bzip2 Uses RLE]

bzip2 actually uses RLE in two places:

#### 1. Preliminary RLE (before BWT):

- Compresses runs of 4 or more identical characters
- Simplifies later stages by reducing extreme cases
- Example: AAAAAA → AAAA with a special marker

#### 2. Main RLE (after MTF):

- Compresses runs in the MTF output

- Uses a custom encoding optimized for the distribution
- Handles runs of zeros and other small values

This two-stage approach shows how different RLE variants can be combined in a single compressor.

## 8.9 Efficient Implementation

### 8.9.1 Naive Rotation Matrix: Why It Is Impractical

The naive BWT implementation described in Section 8.4—constructing an  $n \times n$  matrix of rotations—is completely impractical for real data. Let’s analyze its complexity to understand why.

#### Important

[Complexity of the Naive BWT Algorithm]

For an input string of length  $n$ :

- **Space Complexity:** The rotation matrix requires storing  $n \times n$  characters. For a 1MB file ( $n \approx 10^6$ ), this would require  $10^{12}$  characters—about **1 terabyte** of memory! Even if we store indices rather than full strings, we still need  $O(n^2)$  space to represent all rotations.
- **Time Complexity:**
  - Generating rotations:  $O(n^2)$  to create all  $n$  rotations of length  $n$
  - Sorting the rotations:  $O(n^2 \log n)$  using comparison-based sorting, since comparing two rotations takes  $O(n)$  time in the worst case
  - Extracting the last column:  $O(n)$

Total:  $O(n^2 \log n)$  time and  $O(n^2)$  space.

The naive matrix-based algorithm is **pedagogical, not practical**. For  $n = 10^6$  (a 1MB file), the matrix would require  $10^{12}$  characters—more memory than most computers have.

Clearly, we need a more efficient approach. The key insight is that we don’t need to store all rotations explicitly, which leads us to suffix arrays.

### 8.9.2 Suffix Arrays for Efficient Construction

The key insight is that we don't need to store all rotations explicitly. For a string with a unique end marker, the cyclic rotations correspond exactly to the suffixes of the string when we consider the end marker as a special character.

#### Definition

[Suffix Array] A suffix array for a string  $T$  of length  $n$  is an array of integers  $SA[0..n-1]$  that gives the starting positions of the suffixes of  $T$  in lexicographic order.

For BWT construction, we only need the suffix array of  $S\$$  (the original string with the unique end marker). The relationship is:

- The suffix starting at position  $i$  corresponds to the rotation that begins at position  $i$  in the original string.
- The lexicographic order of suffixes is identical to the lexicographic order of rotations because the end marker ensures no suffix is a prefix of another.

---

#### Algorithm 16 Efficient BWT Construction Using Suffix Arrays

---

**Require:** Input string  $S$  of length  $n$  with unique end marker '\$'

**Ensure:** BWT output  $L[0..n-1]$  and primary index  $I$

```

1:  $T \leftarrow S$  ▷  $T$  includes the end marker
2: Build suffix array  $SA$  for  $T$  ▷  $O(n \log n)$  or  $O(n)$  time
3: for  $i \leftarrow 0$  to  $n-1$  do
4:    $pos \leftarrow SA[i]$ 
5:   if  $pos = 0$  then
6:      $L[i] \leftarrow \$$  ▷ Last character when suffix starts at beginning
7:      $I \leftarrow i$  ▷ Record position of original string
8:   else
9:      $L[i] \leftarrow T[pos-1]$  ▷ Character before this suffix
10:  end if
11: end for
12: return  $(L, I)$ 
```

---

### 8.9.3 Time and Space Complexity of Efficient Implementation

Modern BWT implementations achieve excellent efficiency:

- **Time Complexity:**  $O(n \log n)$  for sorting-based suffix array construction, or  $O(n)$  for advanced algorithms (SA-IS, DC3)

- **Space Complexity:**  $O(n)$  for the suffix array plus input/output buffers
- **Practical Performance:** bzip2 processes data at several MB/s, with block sizes typically 100KB–900KB

### Important

The development of practical linear-time suffix array construction algorithms and efficient in-place implementations was crucial for making BWT feasible for real-world compression.

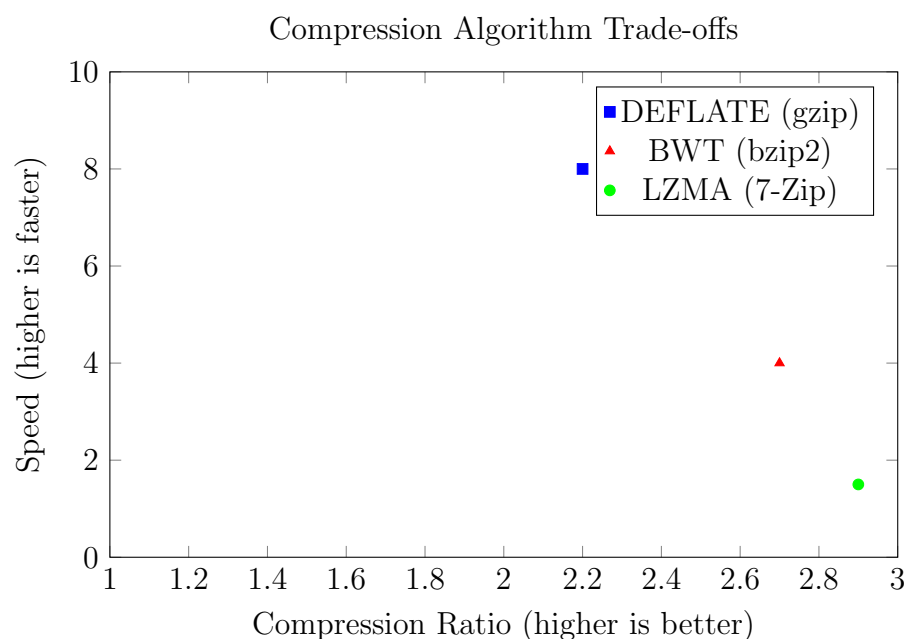
## 8.10 Comparison with Lempel–Ziv Methods

### 8.10.1 Local vs. Global Redundancy

The fundamental difference between LZ77 and BWT lies in how they exploit redundancy:

| Aspect           | LZ77 (DEFLATE)                  | BWT (bzip2)                  |
|------------------|---------------------------------|------------------------------|
| Redundancy Type  | Local (within sliding window)   | Global (across entire block) |
| Detection Method | String matching in window       | Sorting all rotations        |
| Memory Reference | Explicit (distance, length)     | Implicit (context grouping)  |
| RLE Role         | Optional (some implementations) | Essential (after MTF)        |
| Adaptivity       | Online (streaming)              | Block-based                  |

### 8.10.2 Compression Ratio vs. Speed Trade-offs



## 8.11 Case Study: bzip2

### 8.11.1 Architecture Overview

bzip2 is the canonical BWT-based compressor. Its pipeline is:

1. **Run-Length Encoding Prelim:** A preliminary RLE stage to compress repeated characters (simplifies later stages)
2. **Burrows-Wheeler Transform:** Applied to blocks (default 900KB)
3. **Move-To-Front Transform:** Converts local correlations into runs of zeros
4. **Run-Length Encoding:** Compresses runs from MTF (custom variant)
5. **Huffman Coding:** Multiple Huffman tables for different parts of the data

#### Important

bzip2's RLE stage after MTF is specifically designed for the characteristics of MTF output:

- Long runs of zeros are encoded with a special marker
- Small non-zero values are encoded directly
- The encoding is byte-aligned for speed

This is essentially a specialized version of the Zero RLE variant we studied in Chapter 7.

### 8.11.2 Why BWT Beats DEFLATE on Text

On natural language text, bzip2 typically achieves 15-30% better compression than gzip because:

- Text has global structure (frequent words, common phrases) that spans entire documents
- LZ77's window limits miss these long-range dependencies
- BWT's global reordering captures these dependencies implicitly
- The RLE stages efficiently compress the resulting runs

### Example

[Compression Comparison] On the Canterbury Corpus (standard compression benchmark):

- File: `alice29.txt` (Alice in Wonderland, 152KB)
- `gzip -9`: 56KB (compression ratio 2.71)
- `bzip2 -9`: 44KB (compression ratio 3.45)
- Improvement: 21% better compression

## 8.12 Limitations and Modern Extensions

### 8.12.1 Block Size Constraints

BWT operates on fixed-size blocks. The choice of block size involves trade-offs:

- **Larger blocks:** Better compression (more context), but more memory, slower processing
- **Smaller blocks:** Faster, less memory, but poorer compression

`bzip2` allows block sizes from 100KB to 900KB. Beyond 900KB, compression gains diminish while memory usage grows linearly.

### 8.12.2 Memory Usage Considerations

BWT requires:

- $O(n)$  memory for the suffix array (4-8 bytes per character)
- Additional memory for MTF and RLE stages
- For 900KB blocks, `bzip2` uses about 7MB of RAM—modest by modern standards but significant for embedded systems

### 8.12.3 When BWT May Not Improve Compression

While BWT is reversible and theoretically cannot reduce compressibility, in practice several factors can lead to worse compression than simpler methods:

- **Block overhead:** The transform operates on fixed-size blocks. Very small blocks ( $< 1\text{KB}$ ) may have overhead that outweighs any compression benefit.
- **Primary index storage:** The index  $I$  must be stored with each block, adding a small constant overhead.

- **MTF inefficiency on random data:** If the input has no structure, sorting creates no useful patterns, and MTF may actually increase the entropy by spreading out values.
- **RLE expansion on no runs:** As we saw in Chapter 7, RLE can expand data with no runs. If BWT fails to create runs, the RLE stage may hurt compression.
- **Algorithmic overhead:** The multi-stage pipeline (BWT + MTF + RLE + entropy coding) adds computational cost that may not be justified for data that compresses well with simpler methods.

For these reasons, BWT-based compressors like `bzip2` excel on structured data like text but may be outperformed by simpler LZ77 variants on binary data or already-compressed files.

#### 8.12.4 Relation to FM-Index and Text Indexing

The BWT's impact extends far beyond compression. In 2000, Paolo Ferragina and Giovanni Manzini introduced the FM-index (Full-text index in Minute space), which uses the BWT to create a compressed data structure that supports:

- **Counting queries:** Find the number of occurrences of a pattern  $P$  in  $O(|P|)$  time, regardless of the text size.
- **Locating queries:** Find the positions of matches in  $O(|P| + \text{occ} \cdot \log^\epsilon n)$  time, where  $\text{occ}$  is the number of occurrences and the log factor depends on sampling rates.
- **Extract queries:** Retrieve arbitrary substrings of the original text without full decompression.

The FM-index revolutionized bioinformatics:

- **Bowtie and BWA:** DNA sequence aligners used in countless genomics projects, aligning millions of short reads against reference genomes.
- **Pattern matching in compressed data:** Search without full decompression, enabling efficient querying of large genomic databases.
- **Compressed suffix arrays:** Building block for many string algorithms that operate directly on compressed representations.

## Important

The BWT demonstrates a profound principle: compression and indexing are two sides of the same coin. A good compressed representation often enables fast search directly on the compressed data, with time complexity that depends only on pattern size, not on text size.

## 8.13 Summary and Key Insights

### 8.13.1 The Transform-Then-Compress Principle

The BWT embodies a powerful design philosophy:

1. **Separate concerns:** First transform the data to reveal its structure, then compress the transformed representation
2. **Reversibility is key:** The transform must be exactly invertible
3. **Structure exploitation:** Different transforms reveal different kinds of structure

### 8.13.2 The BWT-RLE Connection

The relationship between BWT and the RLE variants from Chapter 7 is fundamental:

- **BWT creates the conditions** for effective RLE by grouping similar contexts
- **MTF converts** these groupings into actual runs
- **RLE variants** compress these runs efficiently

Without understanding RLE, you cannot fully appreciate why BWT works so well. The two techniques are complementary: BWT reorganizes, RLE exploits.

### 8.13.3 When to Use BWT-Based Compression

| Scenario                | Recommendation       | Rationale                 |
|-------------------------|----------------------|---------------------------|
| Text files, source code | BWT (bzip2)          | Global repetition matters |
| DNA sequences           | BWT-based + FM-index | Search+compress combined  |
| Web content (HTTP)      | DEFLATE/gzip         | Streaming, compatibility  |
| Archival storage        | LZMA, Zstandard      | Maximum compression       |
| Real-time systems       | LZ77 variants        | Low latency, streaming    |
| Embedded systems        | Custom/simple RLE    | Memory constraints        |

## 8.14 Hands-On Exercises

### Exercise 8.0

Apply the Burrows-Wheeler Transform to the string “mississippi\$” by hand. Show:

1. All cyclic rotations
2. The sorted rotations
3. The BWT output L and primary index I
4. Identify any runs in the output
5. Apply MTF and then the different RLE variants from Chapter 7
6. Compare which RLE variant works best

### Exercise 8.1

Implement a simple BWT encoder in Python:

```
1 def bwt_encode(s):
2     """Return BWT(s) and primary index."""
3     # Add end marker
4     s = s + '$'
5     # Generate rotations (hint: use list comprehension)
6     # Sort rotations
7     # Extract last column and find original row
8     # Return (L, I)
9     pass
```

Test your implementation on “banana” and “mississippi”. Then implement the full pipeline with MTF and at least two RLE variants.

### Exercise 8.2

Compare compression performance on a text file:

1. Find a text file > 1MB (e.g., a book from Project Gutenberg)
2. Compress with `gzip -9` and `bzip2 -9`
3. Compare sizes and compression times
4. Use `time` command to measure speed
5. Explain the results in terms of the algorithms’ characteristics

### Exercise 8.3

Challenge: Why does BWT perform poorly on random data? Connect your answer to:

1. The sorting process in BWT
2. The subsequent MTF and RLE stages
3. The concept of structure versus randomness
4. The run formation properties we studied in Chapter 7

### Exercise 8.4

Research question: The FM-index enables substring search on BWT-compressed data. Explain:

1. How the LF-mapping enables backward search
2. Why counting queries are  $O(|P|)$  time
3. Applications in bioinformatics (BWA, Bowtie)

### Exercise 8.5

Compare the three compression paradigms we've studied:

| Aspect     | Statistical                | Dictionary           | Transform             |
|------------|----------------------------|----------------------|-----------------------|
| Core idea  | Model probabilities        | Find repetitions     | Reorganize data       |
| Example    | Huffman, Arithmetic        | LZ77, LZ78, DE-FLATE | BWT, bzip2            |
| RLE role   | Optional                   | Optional             | Essential             |
| Strengths  | Optimal for known model    | Fast, adaptive       | Global structure      |
| Weaknesses | Needs good model           | Limited window       | Block-based, slower   |
| Best for   | Well-characterized sources | General purpose      | Text, structured data |

Fill in this table based on Chapters 1-8.

### Important

[Key Takeaway] The Burrows-Wheeler Transform teaches us that sometimes the best way to solve a problem is to transform it into a different problem. By reordering data to reveal its inherent structure, we make the subsequent compression task dramatically easier—and RLE becomes the perfect tool to exploit that revealed structure. This synergy between transformation and exploitation is a lesson that

applies far beyond data compression.

## 9: Lecture 9: Prediction by Partial Matching (PPM): The Statistical Pinnacle

### Definition

Prediction by Partial Matching (PPM) is an adaptive statistical data compression technique that uses a set of variable-length contexts to predict the next symbol based on previously seen data. It dynamically builds models for contexts of different lengths, favouring longer contexts when possible while falling back to shorter ones to handle unseen patterns. In this chapter, we focus on the PPMC variant (Method C), which is the most commonly used practical form of PPM.

PPM was introduced by Cleary and Witten in 1984 and represents a breakthrough in context-based modelling for lossless compression. By intelligently combining multiple orders of Markov prediction, PPM achieves compression ratios close to the theoretical entropy limits for many real-world data sources, such as text and binary files.

### 9.1 Introduction: Beyond Finite-Context Models

#### 9.1.1 Recap of Markov Sources

From Section 4.4, we recall that a Markov source of order  $k$  models the probability of the next symbol based solely on the previous  $k$  symbols, known as the context  $c$  with  $|c| = k$ . The conditional probability of symbol  $s$  given  $c$  is estimated as:

$$P(s \mid c) = \frac{\text{Count}(c, s)}{\sum_{s' \in \mathcal{A}} \text{Count}(c, s')},$$

where  $\mathcal{A}$  is the alphabet and  $\text{Count}(c, s)$  is the number of times symbol  $s$  has followed context  $c$  in the training data (or adaptively during compression).

Fixed-order Markov models, while simple, suffer from significant limitations:

- **Order-0 models:** Treat each symbol independently, ignoring all context. This yields near-uniform probabilities, leading to poor compression (e.g. 8 bits per symbol for ASCII text).
- **Low-order models** ( $k = 1$  or  $2$ ): Capture short-range dependencies (e.g. letter bigrams) but fail to exploit long-distance correlations, such as repeated phrases in natural language.
- **High-order models** ( $k > 3$ ): Can model complex patterns but encounter the *context sparsity problem*. With large alphabets (e.g.  $|\mathcal{A}| = 256$  for byte data), most

possible contexts never occur in finite data, resulting in zero-frequency estimates and unreliable predictions.

These issues motivate a more flexible approach that adapts the context length dynamically.

### 9.1.2 The Fundamental Insight

The genius of PPM lies in its simple yet powerful principle: **attempt to use the longest matching context available, but escape to shorter contexts if the current one provides no information about the next symbol.**

#### Important

PPM operates by maintaining a hierarchy of Markov models, from order-0 (no context) up to a maximum order  $M$  (typically  $M = 3$  to  $7$ , depending on memory constraints). To predict or encode a symbol, PPM begins with the highest-order context (the last  $M$  symbols). If that context has no record of the symbol (zero frequency), it encodes an *escape* event and descends to the next lower order ( $M-1$ ), repeating until a matching context is found or the **order -1** level is reached. This mechanism ensures that non-zero probabilities can be assigned to all symbols.

This multi-order strategy leverages:

- **High-order contexts** for precise predictions in repetitive or structured data (e.g. "the " is often followed by common words).
- **Low-order fallbacks** for robustness against novel or sparse contexts.
- **Escape mechanism** to avoid zero-probability issues by assigning probability mass to unseen symbols via the escape event.

The result is a predictor that approximates the true conditional entropy of the source, often significantly outperforming fixed-order models on natural language data.

## 9.2 The PPM Algorithm

### Definition

[Notation] Throughout this chapter we use:

- $M$  — the maximum context order (e.g.,  $M = 2$  means we maintain models for contexts of length 0, 1, and 2)
- For a specific context string  $c$  (e.g., "ab"):
  - $D_c$  — number of **distinct** symbols that have followed context  $c$

- $N_c$  — total number of times context  $c$  has occurred (sum of all counts)
- $\text{Count}(c, s)$  — how many times symbol  $s$  followed context  $c$
- For the order-0 context (empty string), we write  $D_\emptyset$  and  $N_\emptyset$
- $\mathcal{A}$  — the alphabet (e.g., 256 possible byte values)

When the context is clear from the discussion, we use the notation directly.

### 9.2.1 Core Components

PPM is typically paired with arithmetic coding (or range coding) for entropy encoding, using the predicted probabilities to assign variable-length codes. The model is *adaptive*: probabilities update after each symbol is encoded or decoded.

For each order  $k = 0, 1, \dots, M$ , PPM maintains a context model storing:

- A frequency table for symbols that have followed the context (only observed symbols are stored to save memory).
- Cumulative frequencies for efficient range selection in arithmetic coding.
- An *escape weight* to estimate the probability of an unseen (novel) symbol.

Contexts are represented as strings consisting of the last  $k$  symbols from the history buffer.

#### Definition

The **escape symbol** is a virtual event signalling that the next symbol is unseen in the current context. Encoding or decoding the escape triggers a shift to the lower-order model, where the process repeats.

#### Definition

[Order -1] The **order -1** model is a conceptual, final fallback mechanism that is invoked when the escape chain has exhausted all valid statistical models from order- $M$  down to order-0. At this level, the next symbol is guaranteed to be unseen in all higher-order contexts under the exclusion mechanism (i.e., it has not appeared in any considered context during this prediction step), though it may have occurred earlier in the input. The probability assigned to any single, currently unseen symbol  $x$  is:

$$P(x \mid \text{order -1}) = \frac{1}{|\mathcal{A}| - |\text{seen}|},$$

where *seen* is the set of all symbols that have been excluded by higher orders during this encoding step. This ensures that the total probability mass for all currently

unseen symbols sums to 1.

### Important

**Integration with Arithmetic Coding:** The probabilities computed by PPM are fed directly into an arithmetic coder. For each coding step, the interval  $[0, 1)$  is partitioned according to the cumulative probabilities of all possible symbols (including escape) at the current context. The exclusion principle must be applied consistently in both the probability calculation and the interval partitioning.

## 9.2.2 The Escape and Prediction Mechanism

The core challenge is estimating probabilities for unseen symbols without biasing seen ones too heavily. PPM addresses the **zero-frequency problem** through escape, but the escape probability itself must be carefully tuned.

Decoding is symmetric to encoding: the decoder uses the same context hierarchy and probabilities to select the symbol, updating models identically.

## 9.2.3 Escape Estimation: The PPMC Method

The core challenge in PPM is estimating the probability of an escape — how often should we expect to see a novel symbol in a given context? Among the many proposed methods, **PPMC** (Method C) emerged as the standard because it strikes an excellent balance between simplicity and effectiveness.

### Definition

[PPMC Escape Estimation] For any context  $c$  that has occurred in the input:

- Let  $D_c$  = number of **distinct** symbols that have followed  $c$
- Let  $N_c$  = total number of times context  $c$  has appeared (sum of all counts for symbols following  $c$ )

PPMC assigns probabilities as:

$$P(\text{escape} \mid c) = \frac{D_c}{N_c + D_c}, \quad P(s \mid c) = \frac{\text{Count}(c, s)}{N_c + D_c} \text{ for each seen symbol } s$$

For an empty context (never seen before),  $D_c = 0$ , so  $P(\text{escape}) = 0$  — but this case is handled by falling back to lower orders.

**Intuition:** The escape probability is proportional to how many different symbols we've already seen in this context. If a context has shown rich diversity (high  $D_c$ ), we're more likely to encounter something new. If it's been very consistent (low  $D_c$  relative to  $N_c$ ), we trust it more.

---

**Algorithm 17** PPM Prediction/Encoding Process (Simplified, for a Single Symbol)

---

**Require:** Next symbol  $x$  to encode, history buffer  $H$ , maximum order  $M$ , alphabet  $\mathcal{A}$

- 1:  $k \leftarrow \min(M, |H|)$  ▷ Start with longest available context
- 2:  $seen \leftarrow \emptyset$  ▷ Track excluded symbols (exclusion principle)
- 3: **while**  $k \geq 0$  **do**
- 4:    $c \leftarrow$  last  $k$  symbols of  $H$
- 5:   Retrieve model for  $c$ : frequencies for  $S_c$ , total  $N_c$ , distinct  $D_c$
- 6:    $S_{\text{excl}} \leftarrow S_c \cap seen$
- 7:    $N_{\text{eff}} \leftarrow N_c - \sum_{s \in S_{\text{excl}}} \text{Count}(c, s)$
- 8:    $D_{\text{eff}} \leftarrow D_c - |S_{\text{excl}}|$
- 9:   **if**  $x \in S_c \setminus S_{\text{excl}}$  **then**
- 10:      $P(x \mid c, \text{excl}) \leftarrow \frac{\text{Count}(c, x)}{N_{\text{eff}} + D_{\text{eff}}}$  ▷ PPMC with exclusion
- 11:     Encode  $x$  using this probability
- 12:      $\text{Count}(c, x) \leftarrow \text{Count}(c, x) + 1$
- 13:     **return**
- 14:   **else**
- 15:      $P(\text{escape} \mid c, \text{excl}) \leftarrow \frac{D_{\text{eff}}}{N_{\text{eff}} + D_{\text{eff}}}$
- 16:     Encode escape using this probability
- 17:      $seen \leftarrow seen \cup (S_c \setminus S_{\text{excl}})$
- 18:      $k \leftarrow k - 1$
- 19:   **end if**
- 20: **end while**
- 21: **if**  $k < 0$  **then** ▷ Order -1
- 22:    $U \leftarrow |\mathcal{A}| - |seen|$  ▷ Number of truly unseen symbols at this step
- 23:    $P(x \mid \text{unseen}) \leftarrow 1/U$
- 24:   Encode  $x$  using this uniform probability
- 25:   Update order-0 model with  $x$
- 26: **end if**
- 27: Append  $x$  to  $H$ ; truncate  $H$  to the last  $M$  symbols

---

## Example

[PPMC in Action: Watching a Context Learn]

Consider a context that appears as we process text. Let's track a single context, say the order-1 context "h" (meaning the previous character was 'h').

**Stage 1: First encounter** The context "h" appears for the first time. It's empty initially:

$$D_h = 0, N_h = 0 \Rightarrow P(\text{escape}) = 0/(0 + 0) \text{ (undefined)}$$

But we don't use this context yet — we escape immediately to lower orders.

**Stage 2: After seeing "he"** The first time 'h' is followed by 'e', we create the context:

$$\text{Count}(\text{"h"}, \text{e}) = 1, \quad D_h = 1, N_h = 1$$

Now if we see "h" again and need to predict:

$$P(\text{e}) = \frac{1}{1 + 1} = \frac{1}{2} = 0.5, \quad P(\text{escape}) = \frac{1}{1 + 1} = 0.5$$

The context is equally likely to produce 'e' again or something new.

**Stage 3: After seeing "he" and "ha"** Later we encounter "ha". Now the context has:

$$\text{e} : 1, \text{ a} : 1, \quad D_h = 2, N_h = 2$$

Predictions become:

$$P(\text{e}) = \frac{1}{2 + 2} = 0.25, \quad P(\text{a}) = 0.25, \quad P(\text{escape}) = \frac{2}{4} = 0.5$$

Still a 50% chance of something new, but now two known symbols share the remaining 50% equally.

**Stage 4: After seeing "he" twice and "ha" once** The pattern stabilizes with more data:

$$\text{e} : 2, \text{ a} : 1, \quad D_h = 2, N_h = 3$$

Now:

$$P(\text{e}) = \frac{2}{3 + 2} = 0.4, \quad P(\text{a}) = \frac{1}{5} = 0.2, \quad P(\text{escape}) = \frac{2}{5} = 0.4$$

Notice: as 'e' becomes more frequent, its probability rises, while escape probability drops slightly because the context is becoming more predictable.

## Important

### Why PPMC works:

- It **adapts** to context richness — diverse contexts escape more often
- It avoids zero-probability issues by assigning probability mass to unseen symbols via the escape mechanism.
- It's **computationally simple** — just need to track counts and distinct symbols
- Probabilities always sum to 1 automatically:

$$\sum_{s \in S_c} \frac{\text{Count}(c, s)}{N_c + D_c} + \frac{D_c}{N_c + D_c} = \frac{N_c + D_c}{N_c + D_c} = 1$$

## Example

[Quick Comparison: Two Contexts]

Compare two different contexts to build intuition:

**Context X:** Very predictable. Has seen "the" 100 times, always followed by space.

$$D_{\text{the}} = 1, N_{\text{the}} = 100 \Rightarrow P(\text{space}) = \frac{100}{101} \approx 0.99, P(\text{escape}) = \frac{1}{101} \approx 0.01$$

**Context Y:** Very diverse. Has seen 50 different symbols, total 100 occurrences.

$$D_Y = 50, N_Y = 100 \Rightarrow P(\text{any given seen symbol}) \approx \frac{2}{150} = 0.0133, P(\text{escape}) = \frac{50}{150} \approx 0.333$$

Context Y is much more likely to see something new — exactly what PPMC captures!

## Important

Other escape estimation methods exist (Method A with fixed escape = 1, Method B/D with more sophisticated formulas), but PPMC remains the most widely used due to its simplicity and strong performance across diverse data types. Modern compressors like PPMZ build upon PPMC with additional refinements like secondary escape estimation.

## 9.2.4 The Exclusion Principle

### Important

The **exclusion principle** is a key optimisation in PPM: when the encoder escapes from a higher-order context to a lower-order one, all symbols that were *present* in the higher-order context (and hence already considered) are *excluded* from the probability distribution in the lower-order context. This renormalises the remaining probabilities upward, sharpening predictions for genuinely novel symbols and typically saving 5–10% of bits compared with non-excluding PPM. Excluded symbols are removed from both the symbol set and the normalization constant to ensure probabilities remain properly normalized.

### Example

**Quantified exclusion.** Suppose order-2 context "ab" has seen {r} only, so an escape is sent. The order-1 context "b" has seen {r, a} with counts [1, 1] ( $N_b = 2$ ,  $D_b = 2$ ).

- *Without exclusion:*  $P(r) = 1/4$ ,  $P(a) = 1/4$ ,  $P(\text{esc}) = 2/4 = 0.5$ .
- *With exclusion of r* ( $S_{\text{excl}} = \{r\}$ , remaining  $D_{\text{eff}} = 1$ ,  $N_{\text{eff}} = 1$ ):  $P(a) = 1/2$ ,  $P(\text{esc}) = 1/2$ .

Excluding r doubles  $P(a)$ , saving 1 bit if a is the next symbol.

## 9.3 Step-by-Step Examples

### 9.3.1 Example 1: Encoding "banana"

This example demonstrates PPM with maximum order  $M = 2$  and PPMC escape estimation. The alphabet size is  $|\mathcal{A}| = 256$  (bytes). We apply the exclusion principle throughout, noting that in early steps, exclusion has no effect because there are no overlapping symbols between contexts. All bit costs are approximated to two decimal places.

### Example

[PPM in Action: Encoding "banana" with Exclusion]

**Step 1 — First symbol: 'b'**

- History: "". The longest available context is order-2, then order-1, then order-0. All are empty. In an empty context, escape has probability 1, so it contributes 0 bits (i.e., it does not change the coding interval).
- After escaping through all empty contexts, we reach Order -1.

- At Order -1:  $|seen| = 0$ , so  $P(b) = 1/256$ .
- Cost:  $-\log_2(1/256) = 8$  bits.

After encoding, update models:

| Order   | Context     | Symbol counts | $D_c$ | $N_c$ |
|---------|-------------|---------------|-------|-------|
| Order-0 | $\emptyset$ | b:1           | 1     | 1     |

History: "b"

### Step 2 — Second symbol: 'a'

- Current longest: order-1 context "b" — empty  $\Rightarrow$  escape (0 bits)
- Next: order-0.  $D_\emptyset = 1$ ,  $N_\emptyset = 1$ ;  $P(\text{esc}) = 1/(1+1) = 1/2$  (1 bit)
- 'a' not in order-0  $\Rightarrow$  encode escape; add {b} to *seen*
- Order -1:  $|seen| = 1$ , so  $P(a) = 1/(256-1) = 1/255 \approx 7.99$  bits
- Total for 'a':  $0 + 1 + 7.99 = 8.99$  bits

After encoding, update models:

| Order   | Context     | Symbol counts | $D_c$ | $N_c$ |
|---------|-------------|---------------|-------|-------|
| Order-0 | $\emptyset$ | b:1, a:1      | 2     | 2     |
| Order-1 | "b"         | a:1           | 1     | 1     |

History: "ba"

### Step 3 — Third symbol: 'n'

- Current longest: order-2 "ba" — empty  $\Rightarrow$  escape (0 bits)
- Next: order-1 "a" — empty  $\Rightarrow$  escape (0 bits)
- Next: order-0.  $D_\emptyset = 2$ ,  $N_\emptyset = 2$ ;  $P(\text{esc}) = 2/(2+2) = 1/2$  (1 bit)
- 'n' not in order-0  $\Rightarrow$  encode escape; add {a, b} to *seen*
- Order -1:  $|seen| = 2$ , so  $P(n) = 1/(256-2) = 1/254 \approx 7.99$  bits
- Total:  $0 + 0 + 1 + 7.99 = 8.99$  bits

After encoding, update models:

| Order   | Context     | Symbol counts | $D_c$ | $N_c$ |
|---------|-------------|---------------|-------|-------|
| Order-0 | $\emptyset$ | b:1, a:1, n:1 | 3     | 3     |
| Order-1 | "b"         | a:1           | 1     | 1     |
| Order-1 | "a"         | n:1           | 1     | 1     |
| Order-2 | "ba"        | n:1           | 1     | 1     |

History: "ban"

**Step 4 — Fourth symbol: 'a'**

- Current longest: order-2 "an" — empty  $\Rightarrow$  escape (0 bits)
- Next: order-1 "n" — empty  $\Rightarrow$  escape (0 bits)
- Next: order-0.  $D_\emptyset = 3$ ,  $N_\emptyset = 3$ ; 'a' IS in order-0 table!
- $P(a) = \text{Count}(\emptyset, a) / (N_\emptyset + D_\emptyset) = 1 / (3 + 3) = 1/6 \approx 2.58$  bits
- Total: 2.58 bits (no escapes needed)

After encoding, update models:

| Order   | Context     | Symbol counts | $D_c$ | $N_c$ |
|---------|-------------|---------------|-------|-------|
| Order-0 | $\emptyset$ | b:1, a:2, n:1 | 3     | 4     |
| Order-1 | "b"         | a:1           | 1     | 1     |
| Order-1 | "a"         | n:1           | 1     | 1     |
| Order-1 | "n"         | a:1           | 1     | 1     |
| Order-2 | "ba"        | n:1           | 1     | 1     |
| Order-2 | "an"        | a:1           | 1     | 1     |

History: "bana"

**Step 5 — Fifth symbol: 'n'**

- Current longest: order-2 "na" — empty  $\Rightarrow$  escape (0 bits)
- Next: order-1 "a" — EXISTS! Has seen n:1
- $P(n)$  at order-1 =  $\text{Count}("a", n) / (N_a + D_a) = 1 / (1 + 1) = 1/2 = 1$  bit
- Total: 1 bit!

After encoding, update models:

| Order   | Context     | Symbol counts | $D_c$ | $N_c$ |
|---------|-------------|---------------|-------|-------|
| Order-0 | $\emptyset$ | b:1, a:2, n:2 | 3     | 5     |
| Order-1 | "b"         | a:1           | 1     | 1     |
| Order-1 | "a"         | n:2           | 1     | 2     |
| Order-1 | "n"         | a:1           | 1     | 1     |
| Order-2 | "ba"        | n:1           | 1     | 1     |
| Order-2 | "an"        | a:1           | 1     | 1     |
| Order-2 | "na"        | n:1           | 1     | 1     |

History: "banan"

**Step 6 — Sixth symbol: 'a'**

- Current longest: order-2 "an" — EXISTS! Has seen a:1
- $P(a)$  at order-2 =  $\text{Count}(\text{"an"}, a) / (N_{\text{an}} + D_{\text{an}}) = 1 / (1 + 1) = 1/2 = 1 \text{ bit}$
- Total: 1 bit!

After encoding, update models:

| Order   | Context     | Symbol counts | $D_c$ | $N_c$ |
|---------|-------------|---------------|-------|-------|
| Order-0 | $\emptyset$ | b:1, a:3, n:2 | 3     | 6     |
| Order-1 | "b"         | a:1           | 1     | 1     |
| Order-1 | "a"         | n:2           | 1     | 2     |
| Order-1 | "n"         | a:2           | 1     | 2     |
| Order-2 | "ba"        | n:1           | 1     | 1     |
| Order-2 | "an"        | a:2           | 1     | 2     |
| Order-2 | "na"        | n:1           | 1     | 1     |

History: "banana"

**Final Summary:**

| Symbol       | Context used             | Cost (bits)       |
|--------------|--------------------------|-------------------|
| b (1)        | Order -1                 | 8.00              |
| a (2)        | Order -1 (after escapes) | 8.99              |
| n (3)        | Order -1 (after escapes) | 8.99              |
| a (4)        | order-0                  | 2.58              |
| n (5)        | order-1 ("a")            | 1.00              |
| a (6)        | order-2 ("an")           | 1.00              |
| <b>Total</b> |                          | <b>30.56 bits</b> |

**Comparison:**

- Raw ASCII:  $6 \times 8 = 48 \text{ bits}$
- Our cost:  $8 + 8.99 + 8.99 + 2.58 + 1 + 1 = 30.56 \text{ bits}$
- Savings from PPM:  $\approx 36\%$

### 9.3.2 Example 2: Encoding "abracadabra" (PPMC, $M = 2$ )

This classic example shows PPM on a more complex string. We'll use PPMC with maximum order  $M = 2$  and alphabet size  $|\mathcal{A}| = 256$ . Models update after each symbol. We track all contexts that appear: order-0 (empty context), order-1 contexts (single symbols), and order-2 contexts (pairs).

## Example

[PPM Encoding "abracadabra" Step by Step]

**Symbol 1: 'a'**

- All contexts empty; escape to order -1.
- Cost:  $-\log_2(1/256) = 8$  bits.

After encoding, update models:

| Order   | Context     | Symbol counts | $D_c$ | $N_c$ |
|---------|-------------|---------------|-------|-------|
| Order-0 | $\emptyset$ | a:1           | 1     | 1     |

History:  $H = "a"$

**Symbol 2: 'b'**

- Longest available context: order-1 (a) — empty  $\Rightarrow$  escape
- Order-0:  $D_\emptyset = 1$ ,  $N_\emptyset = 1$ ;  $P(\text{esc}) = 1/(1 + 1) = 1/2$  (1 bit)
- 'b' unseen in order-0  $\Rightarrow$  encode escape. Add a to *seen*.
- Order -1:  $|seen| = 1$ , so  $P(b) = 1/255 \approx 7.99$  bits.
- Total cost:  $1 + 7.99 = 8.99$  bits

After encoding, update models:

| Order   | Context     | Symbol counts | $D_c$ | $N_c$ |
|---------|-------------|---------------|-------|-------|
| Order-0 | $\emptyset$ | a:1, b:1      | 2     | 2     |
| Order-1 | "a"         | b:1           | 1     | 1     |

History:  $H = "ab"$

**Symbol 3: 'r'**

- Longest available: order-2 (ab) — empty  $\Rightarrow$  escape
- Next: order-1 (b) — empty  $\Rightarrow$  escape
- Order-0:  $D_\emptyset = 2$ ,  $N_\emptyset = 2$ ;  $P(\text{esc}) = 2/(2 + 2) = 1/2$  (1 bit)
- 'r' unseen in order-0  $\Rightarrow$  encode escape. Add {a, b} to *seen*.
- Order -1:  $|seen| = 2$ , so  $P(r) = 1/254 \approx 7.99$  bits.
- Total cost:  $1 + 7.99 = 8.99$  bits (escapes from order-2 and order-1 cost 0 bits as their models are empty)

After encoding, update models:

| Order   | Context     | Symbol counts | $D_c$ | $N_c$ |
|---------|-------------|---------------|-------|-------|
| Order-0 | $\emptyset$ | a:1, b:1, r:1 | 3     | 3     |
| Order-1 | "a"         | b:1           | 1     | 1     |
| Order-1 | "b"         | r:1           | 1     | 1     |
| Order-2 | "ab"        | r:1           | 1     | 1     |

History:  $H = \text{"abr"}$

**Symbol 4: 'a'**

- Longest available: order-2 (br) — empty  $\Rightarrow$  escape
- Next: order-1 (r) — empty  $\Rightarrow$  escape
- Order-0:  $D_\emptyset = 3$ ,  $N_\emptyset = 3$ ;  $P(a) = \text{Count}(\emptyset, a) / (N_\emptyset + D_\emptyset) = 1 / (3 + 3) = 1/6 \approx 2.58$  bits
- Symbol found at order-0!
- Total cost: 2.58 bits

After encoding, update models:

| Order   | Context     | Symbol counts | $D_c$ | $N_c$ |
|---------|-------------|---------------|-------|-------|
| Order-0 | $\emptyset$ | a:2, b:1, r:1 | 3     | 4     |
| Order-1 | "a"         | b:1           | 1     | 1     |
| Order-1 | "b"         | r:1           | 1     | 1     |
| Order-1 | "r"         | a:1           | 1     | 1     |
| Order-2 | "ab"        | r:1           | 1     | 1     |
| Order-2 | "br"        | a:1           | 1     | 1     |

History:  $H = \text{"abra"}$

**Symbol 5: 'c'**

- Longest available: order-2 (ra) — empty  $\Rightarrow$  escape
- Next: order-1 (a) — has b:1 ( $D_a = 1$ ,  $N_a = 1$ );  $P(\text{esc}) = 1/2$  (1 bit). Add b to *seen*.
- Next: order-0. Effective model after exclusion: remaining symbols are {a, r}.  $D_{\text{eff}} = 2$ ,  $N_{\text{eff}} = 3$  (since count of b excluded). Denominator =  $N_{\text{eff}} + D_{\text{eff}} = 3 + 2 = 5$ .
- $P(\text{esc})$  at order-0 =  $2/5 \approx 1.32$  bits.

- 'c' unseen  $\Rightarrow$  encode escape. Add {a, r} to *seen*.
- Order -1:  $|seen| = |\{a, b, r\}| = 3$ , so  $P(c) = 1/(256 - 3) = 1/253 \approx 7.98$  bits.
- Total cost:  $1 + 1.32 + 7.98 = 10.30$  bits

After encoding, update models:

| Order   | Context     | Symbol counts      | $D_c$ | $N_c$ |
|---------|-------------|--------------------|-------|-------|
| Order-0 | $\emptyset$ | a:2, b:1, r:1, c:1 | 4     | 5     |
| Order-1 | "a"         | b:1, c:1           | 2     | 2     |
| Order-1 | "b"         | r:1                | 1     | 1     |
| Order-1 | "r"         | a:1                | 1     | 1     |
| Order-1 | "c"         | (empty)            | 0     | 0     |
| Order-2 | "ab"        | r:1                | 1     | 1     |
| Order-2 | "br"        | a:1                | 1     | 1     |
| Order-2 | "ra"        | c:1                | 1     | 1     |

History:  $H = \text{"abrac"}$

**Symbol 6: 'a'**

- Longest available: order-2 (ac) — empty  $\Rightarrow$  escape
- Next: order-1 (c) — empty  $\Rightarrow$  escape
- Next: order-0.  $D_\emptyset = 4$ ,  $N_\emptyset = 5$ ;  $P(a) = 2/(5 + 4) = 2/9 \approx 2.17$  bits
- Total cost: 2.17 bits

After encoding, update models:

| Order   | Context     | Symbol counts      | $D_c$ | $N_c$ |
|---------|-------------|--------------------|-------|-------|
| Order-0 | $\emptyset$ | a:3, b:1, r:1, c:1 | 4     | 6     |
| Order-1 | "a"         | b:1, c:1           | 2     | 2     |
| Order-1 | "b"         | r:1                | 1     | 1     |
| Order-1 | "r"         | a:1                | 1     | 1     |
| Order-1 | "c"         | a:1                | 1     | 1     |
| Order-2 | "ab"        | r:1                | 1     | 1     |
| Order-2 | "br"        | a:1                | 1     | 1     |
| Order-2 | "ra"        | c:1                | 1     | 1     |
| Order-2 | "ac"        | a:1                | 1     | 1     |

History:  $H = \text{"abraca"}$

**Symbol 7: 'd'**

- Longest available: order-2 (ca) — empty  $\Rightarrow$  escape

- Next: order-1 (a) — has b:1, c:1 ( $D_a = 2$ ,  $N_a = 2$ );  $P(\text{esc}) = 2/4 = 1/2$  (1 bit). Add {b, c} to *seen*.
- Next: order-0. Effective model after exclusion: remaining symbols are {a, r}.  $D_{\text{eff}} = 2$ ,  $N_{\text{eff}} = 4$  (counts of a:3, r:1). Denominator =  $4 + 2 = 6$ .
- $P(\text{esc})$  at order-0 =  $2/6 = 1/3 \approx 1.58$  bits.
- 'd' unseen  $\Rightarrow$  encode escape. Add {a, r} to *seen*.
- Order -1:  $|\text{seen}| = |\{\text{a, b, c, r}\}| = 4$ , so  $P(\text{d}) = 1/(256 - 4) = 1/252 \approx 7.97$  bits.
- Total cost:  $1 + 1.58 + 7.97 = 10.55$  bits

After encoding, update models:

| Order   | Context     | Symbol counts           | $D_c$ | $N_c$ |
|---------|-------------|-------------------------|-------|-------|
| Order-0 | $\emptyset$ | a:3, b:1, r:1, c:1, d:1 | 5     | 7     |
| Order-1 | "a"         | b:1, c:1, d:1           | 3     | 3     |
| Order-1 | "b"         | r:1                     | 1     | 1     |
| Order-1 | "r"         | a:1                     | 1     | 1     |
| Order-1 | "c"         | a:1                     | 1     | 1     |
| Order-1 | "d"         | (empty)                 | 0     | 0     |
| Order-2 | "ab"        | r:1                     | 1     | 1     |
| Order-2 | "br"        | a:1                     | 1     | 1     |
| Order-2 | "ra"        | c:1                     | 1     | 1     |
| Order-2 | "ac"        | a:1                     | 1     | 1     |
| Order-2 | "ca"        | d:1                     | 1     | 1     |

History:  $H = \text{"abracad"}$

**Symbol 8: 'a'**

- Longest available: order-2 (ad) — empty  $\Rightarrow$  escape
- Next: order-1 (d) — empty  $\Rightarrow$  escape
- Next: order-0.  $D_{\emptyset} = 5$ ,  $N_{\emptyset} = 7$ ;  $P(\text{a}) = 3/(7 + 5) = 3/12 = 0.25$  (2 bits)
- Total cost: 2 bits

After encoding, update models:

| Order   | Context     | Symbol counts           | $D_c$ | $N_c$ |
|---------|-------------|-------------------------|-------|-------|
| Order-0 | $\emptyset$ | a:4, b:1, r:1, c:1, d:1 | 5     | 8     |
| Order-1 | "a"         | b:1, c:1, d:1           | 3     | 3     |
| Order-1 | "b"         | r:1                     | 1     | 1     |
| Order-1 | "r"         | a:1                     | 1     | 1     |
| Order-1 | "c"         | a:1                     | 1     | 1     |
| Order-1 | "d"         | a:1                     | 1     | 1     |
| Order-2 | "ab"        | r:1                     | 1     | 1     |
| Order-2 | "br"        | a:1                     | 1     | 1     |
| Order-2 | "ra"        | c:1                     | 1     | 1     |
| Order-2 | "ac"        | a:1                     | 1     | 1     |
| Order-2 | "ca"        | d:1                     | 1     | 1     |
| Order-2 | "ad"        | a:1                     | 1     | 1     |

History:  $H = \text{"abracada"}$

**Symbol 9: 'b'**

- Longest available: order-2 (da) — empty  $\Rightarrow$  escape
- Next: order-1 (a) — has b:1, c:1, d:1 ( $D_a = 3$ ,  $N_a = 3$ );  $P(b) = 1/(3 + 3) = 1/6 \approx 2.58$  bits
- Symbol found at order-1!
- Total cost: 2.58 bits

After encoding, update models:

| Order   | Context     | Symbol counts           | $D_c$ | $N_c$ |
|---------|-------------|-------------------------|-------|-------|
| Order-0 | $\emptyset$ | a:4, b:2, r:1, c:1, d:1 | 5     | 9     |
| Order-1 | "a"         | b:2, c:1, d:1           | 3     | 4     |
| Order-1 | "b"         | r:1                     | 1     | 1     |
| Order-1 | "r"         | a:1                     | 1     | 1     |
| Order-1 | "c"         | a:1                     | 1     | 1     |
| Order-1 | "d"         | a:1                     | 1     | 1     |
| Order-2 | "ab"        | r:1                     | 1     | 1     |
| Order-2 | "br"        | a:1                     | 1     | 1     |
| Order-2 | "ra"        | c:1                     | 1     | 1     |
| Order-2 | "ac"        | a:1                     | 1     | 1     |
| Order-2 | "ca"        | d:1                     | 1     | 1     |
| Order-2 | "ad"        | a:1                     | 1     | 1     |
| Order-2 | "da"        | b:1                     | 1     | 1     |

History:  $H = \text{"abracadab"}$

**Symbol 10: 'r'**

- Longest available: order-2 (ab) — has r:1 ( $D_{ab} = 1$ ,  $N_{ab} = 1$ );  $P(r) = 1/(1 + 1) = 1/2 = 1$  bit
- Symbol found at order-2!
- Total cost: 1 bit

After encoding, update models:

| Order   | Context     | Symbol counts           | $D_c$ | $N_c$ |
|---------|-------------|-------------------------|-------|-------|
| Order-0 | $\emptyset$ | a:4, b:2, r:2, c:1, d:1 | 5     | 10    |
| Order-1 | "a"         | b:2, c:1, d:1           | 3     | 4     |
| Order-1 | "b"         | r:2                     | 1     | 2     |
| Order-1 | "r"         | a:1                     | 1     | 1     |
| Order-1 | "c"         | a:1                     | 1     | 1     |
| Order-1 | "d"         | a:1                     | 1     | 1     |
| Order-2 | "ab"        | r:2                     | 1     | 2     |
| Order-2 | "br"        | a:1                     | 1     | 1     |
| Order-2 | "ra"        | c:1                     | 1     | 1     |
| Order-2 | "ac"        | a:1                     | 1     | 1     |
| Order-2 | "ca"        | d:1                     | 1     | 1     |
| Order-2 | "ad"        | a:1                     | 1     | 1     |
| Order-2 | "da"        | b:1                     | 1     | 1     |

History:  $H = \text{"abracadabr"}$

**Symbol 11: 'a'**

- Longest available: order-2 (br) — has a:1 ( $D_{br} = 1$ ,  $N_{br} = 1$ );  $P(a) = 1/(1 + 1) = 1/2 = 1$  bit
- Symbol found at order-2!
- Total cost: 1 bit

After encoding, update models:

| Order   | Context     | Symbol counts           | $D_c$ | $N_c$ |
|---------|-------------|-------------------------|-------|-------|
| Order-0 | $\emptyset$ | a:5, b:2, r:2, c:1, d:1 | 5     | 11    |
| Order-1 | "a"         | b:2, c:1, d:1           | 3     | 4     |
| Order-1 | "b"         | r:2                     | 1     | 2     |
| Order-1 | "r"         | a:2                     | 1     | 2     |
| Order-1 | "c"         | a:1                     | 1     | 1     |
| Order-1 | "d"         | a:1                     | 1     | 1     |
| Order-2 | "ab"        | r:2                     | 1     | 2     |
| Order-2 | "br"        | a:2                     | 1     | 2     |
| Order-2 | "ra"        | c:1                     | 1     | 1     |
| Order-2 | "ac"        | a:1                     | 1     | 1     |
| Order-2 | "ca"        | d:1                     | 1     | 1     |
| Order-2 | "ad"        | a:1                     | 1     | 1     |
| Order-2 | "da"        | b:1                     | 1     | 1     |

History:  $H = \text{"abracadabra"}$

### Final Summary:

| Symbol       | Context used   | Cost (bits)       |
|--------------|----------------|-------------------|
| a (1)        | order -1       | 8.00              |
| b (2)        | order -1       | 8.99              |
| r (3)        | order -1       | 8.99              |
| a (4)        | order-0        | 2.58              |
| c (5)        | order -1       | 10.30             |
| a (6)        | order-0        | 2.17              |
| d (7)        | order -1       | 10.55             |
| a (8)        | order-0        | 2.00              |
| b (9)        | order-1 ("a")  | 2.58              |
| r (10)       | order-2 ("ab") | 1.00              |
| a (11)       | order-2 ("br") | 1.00              |
| <b>Total</b> |                | <b>58.16 bits</b> |

Raw ASCII would require  $11 \times 8 = 88$  bits. PPM saves about 34%.

Notice how the final two symbols cost only 1 bit each — once the pattern establishes that "ab" is followed by 'r' and "br" is followed by 'a', PPM predicts with near-perfect accuracy!

### 9.3.3 Example 3: Repetitive String "ababababab" ( $M = 1$ , PPMC)

This example shows how order-1 PPM rapidly learns a strict alternation pattern. Alphabet size 256;  $M = 1$  (order-0 and order-1 only).

#### Example

[PPM Learning Alternating Pattern "ababababab"]

We track the evolution of models after each symbol:

| Symbol | Context used  | Cost | $D_{\emptyset}, N_{\emptyset}$ | $D_a, N_a$ | $D_b, N_b$ |
|--------|---------------|------|--------------------------------|------------|------------|
| 1: a   | order -1      | 8.00 | (1,1)                          | (0,0)      | (0,0)      |
| 2: b   | order -1      | 8.99 | (2,2)                          | (1,1)      | (0,0)      |
| 3: a   | order-0       | 2.58 | (2,3)                          | (1,1)      | (1,1)      |
| 4: b   | order-1 ("a") | 1.00 | (2,4)                          | (1,2)      | (1,1)      |
| 5: a   | order-1 ("b") | 1.00 | (2,5)                          | (1,2)      | (1,2)      |
| 6: b   | order-1 ("a") | 0.58 | (2,6)                          | (1,3)      | (1,2)      |
| 7: a   | order-1 ("b") | 0.58 | (2,7)                          | (1,3)      | (1,3)      |
| 8: b   | order-1 ("a") | 0.42 | (2,8)                          | (1,4)      | (1,3)      |
| 9: a   | order-1 ("b") | 0.42 | (2,9)                          | (1,4)      | (1,4)      |
| 10: b  | order-1 ("a") | 0.32 | (2,10)                         | (1,5)      | (1,4)      |

### Observations:

- By symbol 4, order-1 context "a" perfectly predicts 'b' (1 bit)
- By symbol 5, order-1 context "b" perfectly predicts 'a' (1 bit)
- As counts increase, probabilities become more certain and costs decrease
- Total:  $8 + 8.99 + 2.58 + 1 + 1 + 0.58 + 0.58 + 0.42 + 0.42 + 0.32 \approx 23.89$  bits (2.39 bpc)

An adaptive order-0 model would require many more symbols to reach similar efficiency.

#### 9.3.4 Example 4: Exclusion Applied to "abracadabra" (Symbol 5)

We revisit symbol 5 ('c') from Example 2 but now apply exclusion.

After encoding symbols 1–4 ("abra"), the relevant models are:

- Order-2 (ra): empty.
- Order-1 (a): b:1 ( $D_a = 1$ ,  $N_a = 1$ ).
- Order-0: a:2, b:1, r:1 ( $D_\emptyset = 3$ ,  $N_\emptyset = 4$ ).

**Step 1 — Order-2 escape.** (ra) is empty,  $P(\text{esc}) = 1$ . Symbols excluded from lower orders:  $\emptyset$ .

**Step 2 — Order-1 escape with exclusion.** (a) contains b:1 only. c is not present  $\Rightarrow$  escape.  $P(\text{esc}) = D_a / (N_a + D_a) = 1/2$  (1 bit). After this step,  $\text{seen} = \{\mathbf{b}\}$ .

**Step 3 — Order-0 with exclusion of b.** Effective table: a:2, r:1 (exclude b).  $D_{\text{eff}} = 2$ ,  $N_{\text{eff}} = 3$ ; denom =  $3 + 2 = 5$ .  $P(\text{esc}) = 2/5$  ( $\approx 1.32$  bits). c unseen  $\Rightarrow$  escape.

**Without exclusion** (Example 2):  $P(\text{esc at O-0}) = 3/7 \approx 1.22$  bits.

**With exclusion:**  $P(\text{esc at O-0}) = 2/5 = 1.32$  bits.

Here exclusion slightly increases the O-0 escape cost, but crucially, if the target symbol had been a, exclusion would have increased  $P(\mathbf{a})$  from  $2/7 \approx 1.81$  bits to  $2/5 = 1.32$  bits — a saving of 0.49 bits. Exclusion benefits the *average* case by concentrating probability on likely symbols.

## 9.4 PPM Implementation Challenges

### 9.4.1 Memory Management

High-order PPM can, in the worst case, require up to  $|\mathcal{A}|^M$  contexts.

### Example

For  $M = 5$  and  $|\mathcal{A}| = 256$  (byte data), the theoretical maximum is  $256^5 \approx 1.1 \times 10^{12}$  contexts — clearly infeasible. However, on typical English text ( $|\mathcal{A}| \approx 70$  printable characters), the actual number of observed contexts stays below  $10^6$  with 1 MB of RAM. PPM exploits this sparsity by allocating memory only for contexts actually encountered.

Memory is managed through:

- **Lazy allocation:** Context nodes are created on first encounter.
- **Count rescaling:** Periodically halving all counts prevents overflow and lets the model adapt to non-stationary data.
- **Memory budgets:** Some implementations cap total nodes and evict the least-recently-used contexts when full.

#### 9.4.2 Data Structures: The Context Trie

The standard solution for efficient context lookup is a **trie** (prefix tree).

##### Definition

In a **context trie**, the root node represents the empty context (order-0). Each edge is labelled with a symbol, and a path of length  $k$  from the root represents a  $k$ -th order context. Each node stores:

- For the context  $c$  represented by this node:
  - $D_c$  — number of distinct symbols seen after this context
  - $N_c$  — total occurrences of this context
  - Frequency table for symbols following this context
- Child pointers (typically a hash map for large alphabets).
- A *suffix link* pointing to the node for the context obtained by dropping the oldest symbol, enabling  $O(1)$  fallback.

Suffix links allow PPM to traverse the escape hierarchy in  $O(M)$  time without rebuilding the context string at each level. Combined with hashed children, the trie supports  $O(1)$  average-case lookup and update.

#### 9.4.3 The Zero-Frequency Problem and Modern Refinements

Several advanced techniques address the remaining inadequacies of simple escape estimation:

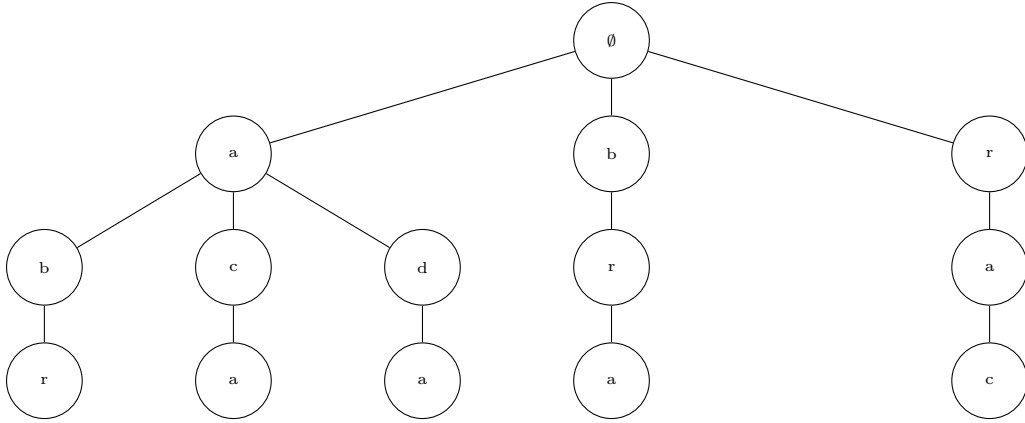


Figure 3: Partial context trie for "abracadabra" with  $M = 3$ . Each path from the root to a node at depth  $k$  represents a  $k$ -th order context; suffix links (not shown) connect each depth- $k$  node to the depth- $(k - 1)$  node for the same context suffix. This structure enables  $O(1)$  escape transitions.

- **Adaptive escape:** Adjust escape weight online using running averages of how often escape was actually needed in each context class.
- **Secondary Escape Estimation (SEE):** Use a second-level model (often itself a PPM-like structure over context metadata) to predict  $P(\text{escape})$ ; used in PPMZ and PAQ.
- **Universal coding connections:** View escape as coding the event “novel symbol”; Bayesian methods (e.g. Krichevsky–Trofimov estimator) give principled priors.
- **Mixing:** Combine escape estimates from different orders via log-linear interpolation (foreshadowing context mixing, Section 5.3).

## 9.5 PPM Variants and Successors

### 9.5.1 PPM\*: Unbounded Contexts

#### Definition

**PPM\*** (Witten, Neal, and Cleary, 1987) removes the fixed maximum order  $M$ , instead finding the longest suffix of the current history that has been seen at least once in the past. This is analogous to the phrase-matching in Ziv–Lempel (LZ) schemes but operates probabilistically.

PPM\* is implemented efficiently using augmented tries with *failure functions* (similar to Aho–Corasick), achieving  $O(n \log n)$  compression time.

### Example

On the string "banana", suppose the full context "anana" was seen during encoding. PPM\* would use this five-symbol context, whereas PPM with  $M = 3$  would be limited to "ana" — potentially missing stronger long-range predictions.

**Trade-off:** PPM\* excels on long-period repetitive data (DNA, source code) but may over-fit on noisy or short data, where shorter fixed contexts provide more reliable statistics (see Exercise 4.3).

### 9.5.2 PPMZ: Optimised Practical Variant

Building on PPMC, Charles Bloom's PPMZ introduced several production-quality innovations:

- **Dynamic order selection:** The effective maximum order adapts per-block based on measured prediction improvement.
- **Secondary Escape Estimation (SEE):** A compact neural-like model replaces fixed heuristics for  $P(\text{escape})$ , updating on-the-fly.
- **Weighted novelty:** Unseen symbols at the fallback level are assigned probabilities biased toward frequently appearing symbols in the global distribution.

PPMZ held world records on the Calgary Corpus throughout the late 1990s, outperforming Gzip and BZIP2 by 20–30%.

### 9.5.3 PAQ: Context Mixing

#### Important

**Context mixing** represents the natural evolution of PPM's multi-order philosophy. Rather than *escaping* through a hierarchy, PAQ (Matt Mahoney, 2002) *combines* predictions from all orders simultaneously using adaptive weighting.

Key innovations in PAQ and its descendants:

- **Hundreds of models:** Order-0 through order-12 PPM models, plus word-level models, sparse contexts, and transform-based models (e.g. BWT residuals).
- **Log-linear mixing:** Each model outputs a log-odds prediction  $\ell_i = \log \frac{P_i(1)}{P_i(0)}$ ; the combined prediction is

$$P_{\text{mix}}(1) = \sigma\left(\sum_i w_i \ell_i\right), \quad \sigma(x) = \frac{1}{1 + e^{-x}}.$$

- **Online weight update:** Weights  $w_i$  increase for models that predicted the actual symbol and decrease otherwise (gradient descent on log-loss).

PAQ8 and its successor CMIX consistently win the Hutter Prize (enwik9, 100 MB Wikipedia), achieving  $\approx 1.1$ – $1.3$  bpc on English, within 10–30% of the estimated Shannon limit of  $\approx 1.0$  bpc.

### Example

In PAQ encoding a space after "the": the order-3 model assigns  $P(\text{space}) = 0.40$ , a word-boundary model assigns 0.65, a positional model 0.35. Mixed log-linearly with learned weights, the combined  $P(\text{space}) \approx 0.55$  — better than any single model alone.

#### 9.5.4 Limitations of PPM

Despite its theoretical elegance, PPM has practical limitations:

- **Memory consumption:** Even with trie structures, PPM with  $M = 5$  can consume hundreds of megabytes on large files, limiting its use in memory-constrained environments.
- **Speed:** Context lookups and updates are slower than dictionary-based methods like LZ77, making PPM impractical for real-time or high-throughput applications.
- **Alphabet size sensitivity:** PPM performs best with small alphabets (e.g., text with 70-100 symbols). For large alphabets (e.g., 16-bit Unicode), context sparsity becomes severe even with moderate  $M$ .
- **Non-stationarity:** While adaptive, PPM's simple frequency counts may not adapt quickly enough to changing source statistics; methods like count halving help but are ad-hoc.

These limitations explain why PPM is rarely used in production compressors today, despite its theoretical importance. Modern systems prefer LZ77-derived algorithms (e.g., Deflate, Zstandard) that offer better speed/memory trade-offs, or context mixing (PAQ/CMIX) when maximum compression is required.

## 9.6 Why PPM Matters Today

### 9.6.1 Theoretical Foundations

**Theorem 9.1** (Asymptotic Optimality of PPM). *For a stationary ergodic source with entropy rate  $H$  (bits per symbol), PPM with maximum order  $M_n$  growing appropriately*

with input length  $n$  satisfies:

$$\frac{1}{n} \sum_{i=1}^n (-\log_2 P(x_i | x_1^{i-1})) \xrightarrow{a.s.} H \quad \text{as } n \rightarrow \infty,$$

provided  $M_n \rightarrow \infty$  and  $M_n = O(\log n / \log |\mathcal{A}|)$ . The per-symbol redundancy is bounded by  $O\left(\frac{M_n \log |\mathcal{A}|}{n}\right)$ .

*Proof sketch.* By ergodicity, empirical frequencies  $\text{Count}(c, s)/N_c$  converge almost surely to the true conditional probabilities  $P(s | c)$  for all contexts  $c$  that occur with positive probability. As  $n$  grows, the probability that the current context of length  $M_n$  has been seen at least once approaches 1, so escapes occur with vanishing probability. The cumulative escape overhead contributes  $O(M_n \log |\mathcal{A}|/n)$  redundancy, which tends to 0 under the stated growth condition.  $\square$

### 9.6.2 Applications Beyond Compression

PPM's predictive power extends well beyond file compression:

- **Text and keyboard prediction:** iOS and Android keyboards use PPM-like models for next-word suggestion, reducing keystrokes by up to 20% in user studies.
- **Bioinformatics:** Order-5 PPM captures codon-usage bias and splice-site patterns in DNA/protein sequences; the Glimmer gene-finder uses Interpolated Markov Models, a close relative of PPM.
- **Intrusion and anomaly detection:** Network packet sequences or system-call traces assigned low probability by a PPM model flag anomalous behaviour; STIDE and related systems use this principle.
- **Adaptive prefetching:** OS file-system and memory prefetchers use PPM-style context models to predict the next block or memory address.
- **Spam filtering:** Character-level PPM scores give low probability to randomly generated strings or obfuscated text typical of spam.

### 9.6.3 Influence on Modern Compression Systems

PPM's ideas are embedded in virtually every modern lossless compression system:

- **Brotli** (Google, 2015) and **Zstandard** (Facebook, 2016) use context-adaptive Asymmetric Numeral Systems (ANS) entropy coders; the context models for literal bytes and copy lengths are PPM-inspired.
- **CMIX** and **PAQ8** achieve near-theoretical limits on the Hutter Prize through massive context mixing, a direct extension of PPM's multi-order approach.

- **Neural compressors** such as LSTM-compress and the Transformer-based models in DeepMind’s language-model compressor (Deletang et al., 2023) are essentially PPM with unlimited context encoded via neural weights, replacing the trie with a deep network.

## Summary

In this chapter, we have explored Prediction by Partial Matching (PPM), a sophisticated adaptive statistical compression technique that represents the pinnacle of Markov-based modeling. Key takeaways include:

- PPM maintains multiple Markov models of orders 0 through  $M$ , using an *escape* mechanism to fall back to shorter contexts when predictions fail.
- The *zero-frequency problem* is addressed through escape estimation, with PPMC being the most widely used method:  $P(\text{escape} \mid c) = D_c / (N_c + D_c)$ .
- The *exclusion principle* sharpens probability estimates by removing symbols already considered in higher-order contexts, and requires the use of effective counts ( $N_{\text{eff}}$ ,  $D_{\text{eff}}$ ) for proper normalization.
- The **order -1** model provides a final fallback for symbols that remain unseen after exclusion across all higher orders, assigning them uniform probability over the currently-unseen alphabet.
- Efficient implementation requires trie data structures with suffix links, where each node stores  $D_c$  and  $N_c$  for its context.
- Modern variants like PPM\* (unbounded contexts) and PPMZ (secondary escape estimation) push compression ratios even closer to theoretical limits.
- PPM’s ideas live on in context-mixing compressors (PAQ, CMIX) and influence modern systems like Brotli and Zstandard.

While rarely used in its pure form today, PPM remains the conceptual foundation for statistical modeling in lossless compression.

## Exercises

### Exercise 9.0

Implement PPM-1 (PPMC,  $M = 1$ ) in Python or C for the string "ababababab". Track  $D_\emptyset$ ,  $N_\emptyset$ ,  $D_a$ ,  $N_a$ ,  $D_b$ ,  $N_b$  after each symbol. Compute the exact arithmetic code lengths and verify the totals from Example 3.

### Exercise 9.1

Prove that PPMC probabilities always sum to 1. For a context  $c$  with  $D_c$  distinct symbols and total count  $N_c = \sum_{s \in S_c} \text{Count}(c, s)$ , show that:

$$\sum_{s \in S_c} \frac{\text{Count}(c, s)}{N_c + D_c} + \frac{D_c}{N_c + D_c} = 1.$$

Also prove that  $0 < P(\text{escape}) \leq 1$  for all valid context states, and identify the unique condition under which  $P(\text{escape}) = 1$  and the condition under which  $P(\text{escape})$  approaches 0.

### Exercise 9.2

Why might PPM\* underperform a fixed-order PPM (e.g.  $M = 3$ ) on uniformly random binary data? Reason about how long contexts overfit noise and yield unreliable statistics. Simulate compression of 1 MB of pseudo-random data and 1 MB of repetitive structured data, and compare bit-per-character ratios for PPM\* and PPM-3.

### Exercise 9.3

Extend the "abracadabra" trace (Example 2) by applying the exclusion principle to *every* symbol, not just symbol 5. For each symbol, list: the contexts visited, the symbols excluded at each level, the adjusted probabilities (using  $D_c$  and  $N_c$  notation), and the resulting code-length change compared to non-excluding PPMC. Summarise the total bit savings.

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## 10: Lecture 10: Arithmetic Coding and Range Coding

### 10.1 Introduction

Entropy coding is the final stage in many compression systems. Instead of assigning fixed codewords to symbols (like Huffman coding), **Arithmetic Coding** and **Range Coding** represent the entire message as a single number inside the unit interval  $[0, 1)$ . This allows code lengths to approach the theoretical entropy limit.

#### Important

**Key Idea.** Arithmetic coding does not assign codes to symbols individually. Instead, it assigns a code to the entire *sequence* by repeatedly shrinking an interval. The more probable the message, the larger the final interval and the fewer bits needed.

### 10.2 Arithmetic Coding: Theory and Algorithm

Arithmetic coding encodes a message by repeatedly narrowing an interval in  $[0, 1)$ . Each symbol reduces the interval based on its probability.

#### Definition

**Algorithm: Arithmetic Coding Encoding.**

**Step 0 — Initialisation.** Set  $\text{low} = 0$  and  $\text{high} = 1$ .

**Step 1 — Compute cumulative probabilities.** Given alphabet  $\{s_1, s_2, \dots, s_k\}$  with probabilities  $p(s_i)$ , compute:

$$C(s_1) = 0, \quad C(s_i) = \sum_{j < i} p(s_j), \quad i = 2, 3, \dots, k.$$

**Step 2 — Encode each symbol.** For each symbol  $s$  in the message:

$$\text{range} = \text{high} - \text{low}$$

$$\text{high} = \text{low} + \text{range} \times (C(s) + p(s))$$

$$\text{low} = \text{low} + \text{range} \times C(s)$$

**Step 3 — Generate the final code.** After encoding all symbols, output any number  $x \in [\text{low}, \text{high})$ . Typically the midpoint is used:

$$\text{code} = \frac{\text{low} + \text{high}}{2}.$$

## 10.3 Example: Arithmetic Coding Step-by-Step

### Example

**Setup.** Alphabet  $\{A, B, C\}$  with frequencies  $f(A) = 5$ ,  $f(B) = 3$ ,  $f(C) = 2$ , total = 10.

$$p(A) = 0.5, \quad p(B) = 0.3, \quad p(C) = 0.2.$$

Encode the message BAC.

**Step 0 — Initialisation.** low = 0, high = 1.

**Step 1 — Cumulative probabilities.**

$$C(A) = 0, \quad C(B) = 0.5, \quad C(C) = 0.8.$$

**Step 2a — Encode B:**

$$\begin{aligned} \text{range} &= 1 - 0 = 1 \\ \text{high} &= 0 + 1 \times (0.5 + 0.3) = 0.8 \\ \text{low} &= 0 + 1 \times 0.5 = 0.5 \end{aligned}$$

Interval:  $[0.5, 0.8)$ .

**Step 2b — Encode A:**

$$\begin{aligned} \text{range} &= 0.8 - 0.5 = 0.3 \\ \text{high} &= 0.5 + 0.3 \times (0 + 0.5) = 0.65 \\ \text{low} &= 0.5 + 0.3 \times 0 = 0.5 \end{aligned}$$

Interval:  $[0.5, 0.65)$ .

**Step 2c — Encode C:**

$$\begin{aligned} \text{range} &= 0.65 - 0.5 = 0.15 \\ \text{high} &= 0.5 + 0.15 \times (0.8 + 0.2) = 0.65 \\ \text{low} &= 0.5 + 0.15 \times 0.8 = 0.62 \end{aligned}$$

Final interval:  $[0.62, 0.65)$ .

**Step 3 — Final code.**

$$\text{code} = \frac{0.62 + 0.65}{2} = 0.635$$

The value 0.635 is the arithmetic-coded representation of **BAC**. In practice this number would be written out in binary.

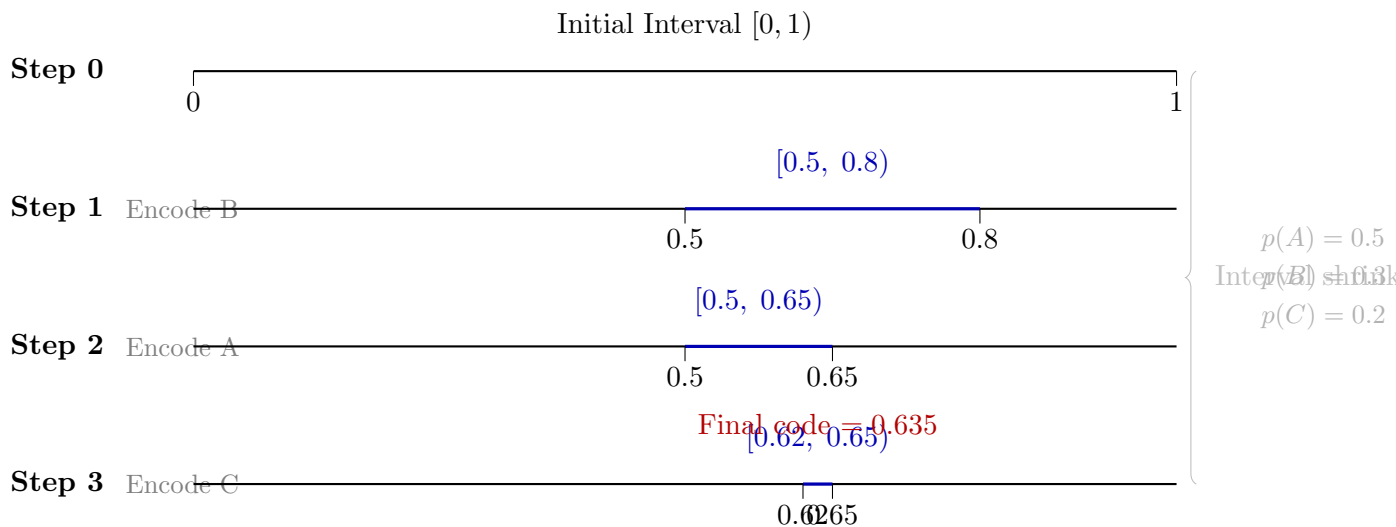


Figure 4: Arithmetic Coding interval refinement for message **BAC**. Each step selects a sub-interval based on the symbol being encoded. The final code 0.635 is the midpoint of  $[0.62, 0.65)$ .

## 10.4 Why People Prefer Range Coding

Arithmetic coding is theoretically elegant but has practical implementation challenges:

- **Arbitrary precision requirement.** As encoding progresses the interval becomes very small, requiring high-precision or arbitrary-precision arithmetic for long messages.
- **Incremental output complexity.** Outputting bits incrementally (before the whole message is encoded) requires careful handling of carry propagation — a process called *normalisation* or *bit-stuffing*.
- **Computational cost.** Floating-point multiplications and divisions at each step are more expensive than the integer operations used in range coding.
- **Historical patent issues.** Arithmetic coding was subject to patents for many years, discouraging its use in some applications (though this is no longer an issue today).

### Important

**Normalisation in Arithmetic Coding.** The term *normalisation* in arithmetic coding refers to **outputting bits incrementally** as they become known — not to

preventing precision loss. For example, once the interval lies entirely within  $[0.5, 1)$ , the first output bit must be 1 and can be emitted immediately. The interval is then doubled and encoding continues.

If we are content to store the entire final fraction (like 0.635) without incremental output, no normalisation is needed; but for long messages storing the full-precision fraction is impractical.

**Range Coding** addresses these challenges by using fixed-precision integer arithmetic (e.g. 32-bit or 64-bit integers), handling incremental output through renormalisation, and using only integer operations (division, multiplication, shifts). Range coding is simply a **reparameterisation of arithmetic coding** designed for efficient implementation.

## 10.5 Range Coding: Theory and Algorithm

Range coding works on integers rather than real-number intervals. The fundamental insight is to scale the  $[0, 1)$  interval to a large integer range  $[0, 2^{32})$ ; all operations then become integer arithmetic.

### Definition

**Algorithm: Range Coding Encoding.**

**Step 0 — Initialisation.** Set  $\text{low} = 0$  and  $\text{range} = R$  (a large integer, typically  $2^{32}$ ).

**Step 1 — Compute cumulative frequencies.** Given alphabet  $\{s_1, \dots, s_k\}$  with integer frequencies  $f(s_i)$  and  $\text{total} = \sum_i f(s_i)$ , compute:

$$C(s_1) = 0, \quad C(s_i) = \sum_{j < i} f(s_j), \quad i = 2, \dots, k.$$

**Step 2 — Encode each symbol.** For each symbol  $s$ :

1.  $\text{unit} = \lfloor \text{range} / \text{total} \rfloor$
2.  $\text{low} += \text{unit} \times C(s)$
3.  $\text{range} = \text{unit} \times f(s)$
4. **Renormalisation:** while  $\text{range} < 2^{24}$ , output the top byte of  $\text{low}$  and shift both  $\text{low}$  and  $\text{range}$  left by 8 bits.

**Step 3 — Flush.** Output the remaining bytes of  $\text{low}$ .

## 10.6 Range Coding Example: Message BAC

### Example

We encode BAC using integer range coding with  $R = 2^{16} = 65536$  for readability (the process is identical with  $2^{32}$ ).

**Cumulative frequencies:**  $f(A) = 5$ ,  $f(B) = 3$ ,  $f(C) = 2$ , total = 10.  $C(A) = 0$ ,  $C(B) = 5$ ,  $C(C) = 8$ .

**Symbol 1 — B:**

$$\begin{aligned}\text{unit} &= \lfloor 65536/10 \rfloor = 6553 \\ \text{low} &= 0 + 6553 \times 5 = 32765 \\ \text{range} &= 6553 \times 3 = 19659\end{aligned}$$

Real-number equivalent:  $\approx [0.5000, 0.8000)$ .

**Symbol 2 — A:**

$$\begin{aligned}\text{unit} &= \lfloor 19659/10 \rfloor = 1965 \\ \text{low} &= 32765 + 1965 \times 0 = 32765 \\ \text{range} &= 1965 \times 5 = 9825\end{aligned}$$

Real-number equivalent:  $\approx [0.5000, 0.6500)$ .

**Symbol 3 — C:**

$$\begin{aligned}\text{unit} &= \lfloor 9825/10 \rfloor = 982 \\ \text{low} &= 32765 + 982 \times 8 = 40621 \\ \text{range} &= 982 \times 2 = 1964\end{aligned}$$

Real-number equivalent:  $\approx [0.6198, 0.6498)$ .

No renormalisation is triggered for this short message. Final integer interval:  $[40621, 42585)$ . We output low = 40621.

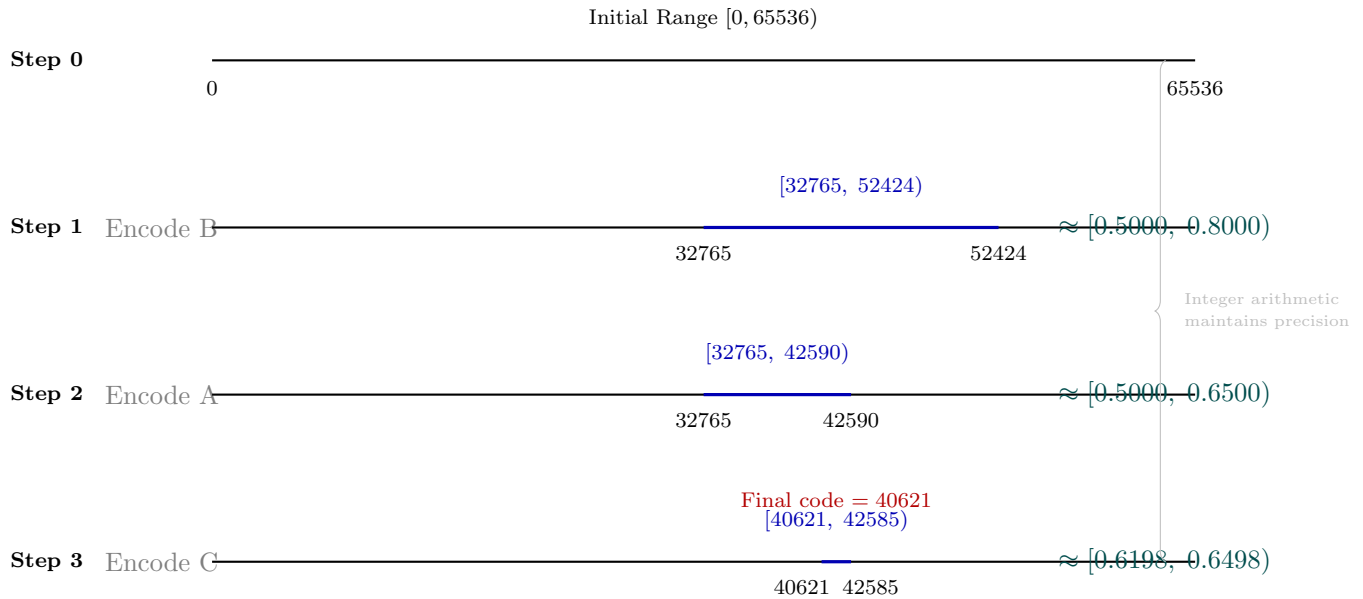


Figure 5: Range Coding integer-interval refinement for message BAC. Real-number equivalents are shown to the right.

## 10.7 Range Coding with Renormalisation

The previous example was short enough that the range never shrank below the renormalisation threshold. We now encode a longer message to see renormalisation in action.

### Example

**Encode B A C C C C with  $R = 2^{32}$ .**

Same alphabet:  $f(A) = 5$ ,  $f(B) = 3$ ,  $f(C) = 2$ , total = 10.  $C(A) = 0$ ,  $C(B) = 5$ ,  $C(C) = 8$ .

| Step   | Symbol | low           | range         | Output         |
|--------|--------|---------------|---------------|----------------|
| 0      | —      | 0             | 4,294,967,296 | —              |
| 1      | B      | 2,147,483,645 | 1,288,490,187 | —              |
| 2      | A      | 2,147,483,645 | 644,245,090   | —              |
| 3      | C      | 2,662,879,717 | 128,849,018   | —              |
| 4      | C      | 2,765,958,925 | 25,769,802    | —              |
| 5      | C      | 2,786,574,765 | 5,153,960     | <b>Renorm!</b> |
| Renorm | —      | 439,463,680   | 1,319,413,760 | 0xA6 (166)     |
| 6      | C      | 1,494,994,688 | 263,882,752   | —              |

**Why does renormalisation trigger after symbol 5?** Each  $C$  shrinks the range by a factor of  $\approx 0.2$ :

$$644,245,090 \xrightarrow{\times 0.2} 128,849,018 \xrightarrow{\times 0.2} 25,769,802 \xrightarrow{\times 0.2} 5,153,960 < 2^{24}.$$

**Renormalisation step:** output the top byte of low ( $2,786,574,765 \gg 24 = 166 = 0xA6$ ), then shift both low and range left by 8 bits.

**Final output stream:**  $[0xA6, 0x59, 0x1C, 0x00, 0x00]$  (the renorm byte followed by the four bytes of the final low).

## 10.8 Range Coding: Decoding Algorithm

Decoding is the exact inverse of encoding. Given the compressed stream we reconstruct the original symbols by determining which symbol's interval contains the current code value.

### Definition

**Algorithm: Range Coding Decoding.**

**Step 0 — Initialisation.** Set  $\text{low} = 0$ ,  $\text{range} = R$ . Read the first 4 bytes of the stream to form the initial code.

**Step 1 — Cumulative frequencies** — same as encoder.

**Step 2 — Decode each symbol.** For each symbol position:

1.  $\text{unit} = \lfloor \text{range} / \text{total} \rfloor$

2. Find symbol  $s$  satisfying:

$$C(s) \leq \frac{\text{code} - \text{low}}{\text{unit}} < C(s) + f(s)$$

3. Output symbol  $s$ .

4.  $\text{low} += \text{unit} \times C(s)$ ;  $\text{range} = \text{unit} \times f(s)$ .

5. **Renormalisation:** while  $\text{range} < 2^{24}$ :

$$\text{low} \leftarrow (\text{low} \ll 8) \ \& \ (2^{32} - 1)$$

$$\text{code} \leftarrow ((\text{code} \ll 8) \ \& \ (2^{32} - 1)) \mid \text{next\_byte}$$

$$\text{range} \leftarrow \text{range} \ll 8$$

### Important

**Encoder–Decoder Symmetry.** Both sides shift low and range left by 8 bits during renormalisation. The encoder *outputs* the byte that falls off the top of low; the decoder *reads* that same byte and inserts it at the bottom of code. This is why the decoder maintains a separate code variable — it tracks the “window” into the compressed stream.

## 10.9 Decoding Example: Recovering B A C C C C

### Example

**Input stream:** [0xA6, 0x59, 0x1C, 0x00, 0x00].

Initialise: low = 0, range =  $2^{32}$ , code = 0xA6591C00 = 2,789,021,696.

| Step   | code          | low           | range         | Symbol |
|--------|---------------|---------------|---------------|--------|
| Start  | 2,789,021,696 | 0             | 4,294,967,296 | —      |
| 1      | 2,789,021,696 | 2,147,483,645 | 1,288,490,187 | B      |
| 2      | 2,789,021,696 | 2,147,483,645 | 644,245,090   | A      |
| 3      | 2,789,021,696 | 2,662,879,717 | 128,849,018   | C      |
| 4      | 2,789,021,696 | 2,765,958,925 | 25,769,802    | C      |
| 5      | 2,789,021,696 | 2,786,574,765 | 5,153,960     | C      |
| Renorm | 1,494,994,688 | 439,463,680   | 1,319,413,760 | —      |
| 6      | 1,494,994,688 | 439,463,680   | 1,319,413,760 | C      |

**Sample decode step (Symbol 1):**

$$\frac{\text{code} - \text{low}}{\text{unit}} = \frac{2,789,021,696}{429,496,729} \approx 6.49$$

Since  $C(B) = 5 \leq 6.49 < 8 = C(B) + f(B)$ , decoded symbol = B. ✓

**Renormalisation (after Symbol 5):** Read next byte 0x00 from stream.

$$\text{code} \leftarrow 0xA6591C00 \ll 8 \mid 0x00 = 0x591C0000 = 1,494,994,688$$

$$\text{low} \leftarrow 0xA61A2FAD \ll 8 = 0x1A2FAD00 = 439,463,680$$

$$\text{range} \leftarrow 5,153,960 \times 256 = 1,319,413,760$$

Recovered sequence: B A C C C C. ✓

## 10.10 Arithmetic vs. Range Coding: A Comparison

| Concept        | Arithmetic Coding            | Range Coding                                |
|----------------|------------------------------|---------------------------------------------|
| Representation | Real numbers in $[0, 1)$     | Integers in $[0, R)$                        |
| Initial state  | low = 0, high = 1            | low = 0, range = $R$                        |
| Unit size      | high – low                   | $\lfloor \text{range}/\text{total} \rfloor$ |
| Low update     | low + range $\times C(s)$    | low + unit $\times C(s)$                    |
| Range update   | range $\times p(s)$          | unit $\times f(s)$                          |
| Precision      | Floating-point normalisation | Integer renormalisation                     |
| Final code     | Midpoint of final interval   | Value from final interval                   |
| Arithmetic     | Float multiply/divide        | Integer multiply/divide                     |
| Speed          | Moderate                     | Fast                                        |

The two methods are mathematically equivalent; range coding is an integer-based reparameterisation of arithmetic coding.

## 10.11 Summary

- Arithmetic coding uses real-number intervals and is mathematically elegant but requires floating-point or arbitrary-precision arithmetic.
- Range coding uses integer intervals and produces the same compressed result using only fast integer operations.
- Both achieve compression close to the theoretical Shannon entropy limit.
- *Renormalisation* in range coding outputs bytes incrementally as the range shrinks below  $2^{24}$ , keeping the state in a fixed-width integer register.
- Decoding mirrors encoding exactly: the decoder maintains a code variable that tracks the current position in the compressed stream and updates low and range identically to the encoder.
- During renormalisation the encoder outputs the byte that falls off the top of low; the decoder reads that byte and inserts it at the bottom of code.

## 10.12 Exercises

### Exercise 10.0

Encode the message **CAB** using arithmetic coding with  $p(A) = 0.5$ ,  $p(B) = 0.3$ ,  $p(C) = 0.2$ . Show the interval after each symbol and state the final code.

### Exercise 10.1

Encode the same message **CAB** using range coding with  $R = 65536$  and the same frequencies. Compare the final integer interval with the real-number interval from the previous exercise.

### Exercise 10.2

Explain in your own words why range coding is more suitable for hardware implementation than arithmetic coding. Give at least two concrete reasons.

### Exercise 10.3

What would happen if we used a very small initial range such as  $R = 100$  for range coding? How would the integer truncation in  $\text{unit} = \lfloor \text{range}/\text{total} \rfloor$  affect compression quality?

### Exercise 10.4

Write pseudocode for an arithmetic coding encoder and a range coding encoder side-by-side. Highlight the three lines that differ between the two.

### Exercise 10.5

Using the renormalisation example (**B A C C C C**), encode the shorter message **B A C C** instead. Would renormalisation be triggered? Justify your answer by tracking the range at each step.

### Exercise 10.6

**Decoding practice.** Using the decoding algorithm, decode the compressed output you produced in Exercise 10.7 (the **CAB** encoding) to verify you recover the original message. Show the offset computation at each step.

### Exercise 10.7

Explain the encoder–decoder symmetry during renormalisation. Why does the decoder shift code left and insert the next byte, while the encoder shifts low left and outputs the falling byte? What ensures the two sides stay synchronised?

## 11: Lecture 11: Asymmetric Numeral Systems (ANS)

### 11.1 Why ANS Was Invented

To compress data, we want to assign short bit-strings to *common* symbols and longer bit-strings to *rare* symbols. Shannon’s entropy theorem tells us the ideal length for a symbol with probability  $p$  is  $-\log_2 p$  bits. Two classical methods try to achieve this:

- **Huffman coding** assigns a fixed binary codeword to each symbol. It is very fast, but codeword lengths must be whole numbers, so it wastes bits whenever the ideal length  $-\log_2 p$  is not an integer. For example, if  $p = 0.3$  the ideal length is  $\approx 1.74$  bits, but Huffman is forced to use either 1 or 2 bits.
- **Arithmetic coding** avoids this waste by encoding the entire message as a single fractional number, allowing it to approach the entropy limit arbitrarily closely. The drawback is that it requires expensive division operations and careful multi-precision arithmetic.

In 2009, Jarek Duda introduced **Asymmetric Numeral Systems (ANS)**: an entropy coder that achieves the compression quality of arithmetic coding while running nearly as fast as Huffman coding. ANS is now used in **Zstandard (zstd)**, **LZFSE** (Apple), and other widely deployed compressors.

**Road-map.** This section builds up ANS in three stages.

1. We explain the core idea: a single integer  $x$  that carries all compressed information.
2. We develop **rANS** (Range ANS), whose encode/decode steps are explicit arithmetic formulas. This is the easiest variant to understand.
3. We show how **tANS** (Tabled ANS) replaces that arithmetic with a pre-built lookup table, eliminating every division at runtime.

### 11.2 The Central Idea: One Integer Carries Everything

Most compressors build up a bitstream symbol by symbol. ANS takes a different approach: it maintains a single integer  $x$  — called the **state** — and encodes the entire message by repeatedly updating  $x$ . At the end,  $x$  itself, written in binary, *is* the compressed output.

#### Definition

**State  $x$ .** The encoder starts from an agreed initial value (e.g.  $x = L$ , defined below). Each time a symbol is encoded,  $x$  is replaced by a new, larger integer that “contains” both the old  $x$  and the new symbol. When all symbols have been

encoded, the final  $x$  is written out as the compressed bitstream. The decoder reads that integer, and by reversing the updates recovers the original symbols one at a time.

### Why does this work as compression?

Each update multiplies  $x$  by roughly  $1/p_s$ , where  $p_s$  is the probability of the symbol just encoded. That adds  $\log_2(1/p_s) = -\log_2 p_s$  bits to the binary representation of  $x$  — exactly the number of bits that information theory says the symbol deserves. Common symbols (large  $p_s$ ) grow  $x$  only a little; rare symbols (small  $p_s$ ) grow  $x$  a lot. After encoding  $n$  symbols the size of  $x$  in bits closely tracks the Shannon entropy of the sequence.

## 11.3 Setting Up: Symbols, Frequencies, and the Table Size

Before encoding anything, we need to fix a few parameters. The table below defines every variable used throughout this section; refer back to it whenever a symbol is unfamiliar.

| Variable                            | Meaning                                                                                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{A} = \{s_1, \dots, s_k\}$ | The <b>alphabet</b> : the set of symbols to be compressed.                                                                                                     |
| $p_i$                               | The <b>true probability</b> of symbol $s_i$ ; satisfies $\sum_i p_i = 1$ .                                                                                     |
| $R$                                 | A positive integer (e.g. 8, 12, or 16) that controls precision.                                                                                                |
| $L = 2^R$                           | The <b>table size</b> : the total number of slots available to the probability model.                                                                          |
| $f_i$                               | The <b>frequency</b> of symbol $s_i$ : a positive integer approximating $p_i \cdot L$ . Must satisfy $\sum_i f_i = L$ .                                        |
| $\tilde{p}_i = f_i/L$               | The <b>quantised probability</b> of $s_i$ : the approximation to $p_i$ that ANS actually uses.                                                                 |
| $c_i = \sum_{j < i} f_j$            | The <b>cumulative frequency</b> of $s_i$ : the sum of all frequencies of symbols that come <i>before</i> $s_i$ in the alphabet ordering. Satisfies $c_1 = 0$ . |
| $x$                                 | The current encoder/decoder <b>state</b> : a single non-negative integer.                                                                                      |
| $[L, 2L)$                           | The <b>normalised range</b> : the interval $\{L, L+1, \dots, 2L-1\}$ in which $x$ is always kept.                                                              |

## Why integers instead of exact probabilities?

Exact probabilities are usually irrational numbers that cannot be stored or computed efficiently. By rounding each  $p_i$  to the nearest multiple of  $1/L$ , we get frequencies  $f_i$  that are small whole numbers, and all arithmetic reduces to integer operations. Choosing  $L = 2^R$  is especially convenient because division and remainder by  $L$  become single bit-shift and bit-mask instructions on any modern CPU.

## What is $c_i$ (cumulative frequency) for?

Think of the  $L$  slots as a row of pigeonholes numbered 0 to  $L - 1$ . Symbol  $s_i$  is assigned a contiguous block of  $f_i$  pigeonholes, starting at pigeonhole  $c_i$  and ending just before  $c_i + f_i$ . Because the blocks are contiguous and non-overlapping, the decoder can identify any symbol simply by checking which block a number falls into.

### Example

**Example: assigning slot ranges.** Suppose  $\mathcal{A} = \{A, B, C\}$ ,  $L = 8$ , and the frequencies are  $f_A = 4$ ,  $f_B = 2$ ,  $f_C = 2$  (so  $4 + 2 + 2 = 8 = L$ , as required).

| Symbol | Frequency $f$ | Cumulative $c$        | Prob. $\tilde{p} = f/L$ | Slot range $[c, c + f)$ |
|--------|---------------|-----------------------|-------------------------|-------------------------|
| $A$    | 4             | $c_A = 0$             | 0.50                    | $[0, 4)$                |
| $B$    | 2             | $c_B = 0 + 4 = 4$     | 0.25                    | $[4, 6)$                |
| $C$    | 2             | $c_C = 0 + 4 + 2 = 6$ | 0.25                    | $[6, 8)$                |

The slot ranges tile  $[0, 8)$  perfectly. If we compute  $x \bmod 8$  for some state  $x$  and get the value 5, it falls in  $[4, 6)$ , so the symbol is  $B$ . This is exactly how the decoder identifies symbols.

## 11.4 Two Variants: rANS and tANS

| Variant     | Full name  | Encode/decode via   | Used in          |
|-------------|------------|---------------------|------------------|
| <b>rANS</b> | Range ANS  | Arithmetic formulas | Zstandard, CRAM  |
| <b>tANS</b> | Tabled ANS | Lookup tables       | Zstandard, LZFSE |

Both variants share the same mathematical idea. We study rANS first because its formulas are easy to inspect and verify. Section 11.9 then shows how tANS pre-computes those formulas into two arrays so that no arithmetic is needed at runtime.

## 11.5 rANS: Range Asymmetric Numeral Systems

### 11.5.1 Keeping the State Bounded

If we simply multiplied  $x$  by  $1/p_s$  after every symbol,  $x$  would grow without bound, requiring ever more memory and time to manipulate. We prevent this by keeping  $x$  inside a fixed **normalised range**  $[L, 2L)$  at all times.

Whenever  $x$  grows too large, we peel off its lowest-order bit and write it to the bitstream (this is called **emitting** a bit). Whenever  $x$  shrinks too small, we read the next bit from the bitstream and glue it onto  $x$  from below (this is called **renormalising upward**). These bits are the compressed output; the final value of  $x$  is appended at the end so the decoder knows where to start.

### 11.5.2 The rANS Encoding Step

**What we want to achieve.** Given the current state  $x$  and a symbol  $s$ , we want to produce a new state  $x'$  such that:

1.  $x'$  encodes the information “symbol  $s$  was seen”.
2. The decoder can recover both  $s$  and the original  $x$  from  $x'$  alone.
3.  $x'$  stays in the normalised range  $[L, 2L)$ .

**The encoding formula.** The formula that achieves all three goals is:

$$x' = \left\lfloor \frac{x}{f_s} \right\rfloor \cdot L + c_s + (x \bmod f_s) \quad (1)$$

where  $f_s$  is the frequency of  $s$  and  $c_s$  is its cumulative frequency (both defined in the notation table above).

**Why does goal 1 hold? (Information content.)** The term  $\lfloor x/f_s \rfloor \cdot L$  scales  $x$  up by a factor of approximately  $L/f_s = 1/\tilde{p}_s$ . Scaling by  $1/\tilde{p}_s$  increases the binary length of  $x$  by  $\log_2(1/\tilde{p}_s) = -\log_2 \tilde{p}_s$  bits — exactly the information content of symbol  $s$ . Common symbols (large  $f_s$ ) cause a small increase; rare symbols (small  $f_s$ ) cause a large increase.

**Why does goal 2 hold? (Decodability.)** The decoder needs to identify  $s$  from  $x'$ . Notice that:

$$x' \bmod L = c_s + (x \bmod f_s)$$

Because  $0 \leq x \bmod f_s < f_s$ , this value always lies in  $[c_s, c_s + f_s)$ , which is symbol  $s$ 's slot range. So the decoder computes  $x' \bmod L$  and checks which symbol's slot range it falls in — that uniquely identifies  $s$ . Once  $s$  is known, the decoder can also recover  $x$  by inverting the formula (see Section 11.5.3).



### 11.5.3 The rANS Decoding Step

**What the decoder receives.** The decoder starts from the final state  $x$  (written at the end of the bitstream) and has access to the bitstream of pre-norm bits in *reverse* order.

**How to identify the symbol.** Compute  $r = x \bmod L$ . Look up which symbol  $s$  has  $c_s \leq r < c_s + f_s$ . This is the slot-range lookup described in Section 11.5.2: the encoder guaranteed that  $x \bmod L$  always falls in  $s$ 's slot range, so this lookup unambiguously identifies  $s$ .

**How to recover the previous state.** Once  $s$  is known, we can invert formula (1). Since  $x \in [L, 2L)$  we have  $\lfloor x/L \rfloor = 1$ , so:

$$x_{\text{prev}} = f_s \cdot \left\lfloor \frac{x}{L} \right\rfloor + r - c_s = f_s + r - c_s$$

Because  $r \in [c_s, c_s + f_s)$ , we have  $x_{\text{prev}} \in [f_s, 2f_s)$  — exactly the pre-normalised range the encoder started from before it encoded  $s$ .

**Renormalise upward.**  $x_{\text{prev}}$  may be less than  $L$ , so we must restore it to  $[L, 2L)$ . We do this by reading bits from the (reversed) bitstream and appending them to the bottom of  $x_{\text{prev}}$  until  $x_{\text{prev}} \geq L$ . This consumes exactly the bits the encoder emitted for this symbol, restoring the encoder's state to what it was *before* encoding  $s$ .

#### Definition

**rANS Decode Algorithm.** Given state  $x \in [L, 2L)$ , decode one symbol:

##### Step 1 — Identify symbol:

- 1:  $r \leftarrow x \bmod L$
- 2: Find symbol  $s$  such that  $c_s \leq r < c_s + f_s$  ▷ check slot ranges; linear or binary search

##### Step 2 — Recover previous state:

- 1:  $x_{\text{prev}} \leftarrow f_s + r - c_s$  ▷ algebraic inverse of the encode formula

##### Step 3 — Renormalise upward:

- 1: **while**  $x_{\text{prev}} < L$  **do**
- 2:     Read next bit  $b$  from the (reversed) stream
- 3:      $x_{\text{prev}} \leftarrow (x_{\text{prev}} \ll 1) \mid b$  ▷ shift left and append bit at the bottom
- 4: **end while** ▷ Now  $L \leq x_{\text{prev}} < 2L$ ; this is the new  $x$

**Step 4:** Output symbol  $s$ . Set  $x \leftarrow x_{\text{prev}}$  and repeat from Step 1 for the next symbol.

## 11.6 Worked Example: rANS Encoding

We now trace through a complete encoding by hand so every formula step can be verified concretely.

### Example

**Setup.**

$$\mathcal{A} = \{A, B, C\}, \quad L = 8, \quad R = 3.$$

| Symbol | Freq. $f$ | Cumul. $c$ | $\tilde{p} = f/L$ | Slot range $[c, c + f)$ |
|--------|-----------|------------|-------------------|-------------------------|
| $A$    | 4         | 0          | 0.50              | $[0, 4)$                |
| $B$    | 2         | 4          | 0.25              | $[4, 6)$                |
| $C$    | 2         | 6          | 0.25              | $[6, 8)$                |

**Goal:** encode the sequence  $A \ B \ A$ , right-to-left (so we start with the last symbol and work backwards). Initial state  $x = L = 8$ .

The two-step procedure for each symbol:

1. **Pre-normalise:** repeatedly emit the lowest bit of  $x$  and right-shift until  $x \in [f_s, 2f_s)$ .
2. **Apply formula:**  $x \leftarrow L + c_s + (x \bmod f_s)$ .

### Step 1 — Encode the 3rd symbol: $A$

*Parameters:*  $f_A = 4$ ,  $c_A = 0$ . Current state:  $x = 8$ .

*Pre-normalise into  $[f_A, 2f_A) = [4, 8)$ :*

- 1: **while**  $x \geq 2f_A = 8$  **do**
- 2:      $x = 8 \geq 8$ : emit  $(8 \& 1) = 0$ ;  $x \leftarrow 8 \gg 1 = 4$
- 3: **end while**  $\triangleright 4 \in [4, 8)$ : stop

*Apply formula* with pre-normed  $x = 4$ :

$$x \leftarrow 8 + 0 + (4 \bmod 4) = 8 + 0 + 0 = 8$$

Bit emitted:           0  
 Bitstream so far:    0  
 New state  $x$ :        8

*Why 1 bit?*  $A$  has probability 0.5, so it carries  $-\log_2 0.5 = 1$  bit of information. Pre-normalising from  $x = 8$  to  $[4, 8)$  required one right-shift, emitting 1 bit.

### Step 2 — Encode the 2nd symbol: $B$

*Parameters:*  $f_B = 2$ ,  $c_B = 4$ . Current state:  $x = 8$ .

*Pre-normalise into  $[f_B, 2f_B) = [2, 4)$ :*

```

1: while  $x \geq 2f_B = 4$  do
2:    $x = 8 \geq 4$ : emit  $(8 \& 1) = \mathbf{0}$ ;  $x \leftarrow 8 \gg 1 = 4$ 
3:    $x = 4 \geq 4$ : emit  $(4 \& 1) = \mathbf{0}$ ;  $x \leftarrow 4 \gg 1 = 2$ 
4: end while  $\triangleright 2 \in [2, 4)$ : stop

```

Apply formula with pre-normed  $x = 2$ :

$$x \leftarrow 8 + 4 + (2 \bmod 2) = 8 + 4 + 0 = 12$$

Bits emitted:        0 0  
 Bitstream so far:   0 0 0  
 New state  $x$ :        12

Why 2 bits?  $B$  has probability 0.25, so it carries  $-\log_2 0.25 = 2$  bits. We needed two right-shifts to bring  $x = 8$  into  $[2, 4)$ .

### Step 3 — Encode the 1st symbol: A

Parameters:  $f_A = 4$ ,  $c_A = 0$ . Current state:  $x = 12$ .

Pre-normalise into  $[f_A, 2f_A) = [4, 8)$ :

```

1: while  $x \geq 2f_A = 8$  do
2:    $x = 12 \geq 8$ : emit  $(12 \& 1) = \mathbf{0}$ ;  $x \leftarrow 12 \gg 1 = 6$ 
3: end while  $\triangleright 6 \in [4, 8)$ : stop

```

Apply formula with pre-normed  $x = 6$ :

$$x \leftarrow 8 + 0 + (6 \bmod 4) = 8 + 0 + 2 = 10$$

Bit emitted:        0  
 Bitstream so far:   0 0 0 0  
 New state  $x$ :        10

**Finalise.** Append the final state  $x = 10 = (1010)_2$  to the bitstream:

$\underbrace{0\ 0\ 0\ 0}_{\text{pre-norm bits from steps 1-3}} \quad \bigg| \quad \underbrace{1\ 0\ 1\ 0}_{\text{final state } x=10}$

### Complete encoding summary:

| Step | Symbol | $x$ in | Bits emitted | $x$ after pre-norm | Formula $L + c_s + (x \bmod f_s)$ | $x$ out |
|------|--------|--------|--------------|--------------------|-----------------------------------|---------|
| 1    | A      | 8      | 0            | 4                  | $8 + 0 + 0 = 8$                   | 8       |
| 2    | B      | 8      | 0 0          | 2                  | $8 + 4 + 0 = 12$                  | 12      |
| 3    | A      | 12     | 0            | 6                  | $8 + 0 + 2 = 10$                  | 10      |

*Entropy check.* Shannon entropy of A B A:  $-(2 \times \log_2 0.5 + \log_2 0.25) = 1 + 1 + 2 = 4$  bits. We emitted exactly 4 pre-norm bits. The additional 4 bits storing the final state  $x = 10$  are a fixed per-block overhead that becomes negligible for long sequences.

## 11.7 Worked Example: rANS Decoding

We now decode the bitstream produced above, step by step, verifying we recover A B A.

### Example

**What the decoder has:**

- The stored final state:  $(1010)_2 = 10$ . Set  $x = 10$ .
- The pre-norm bits in *reversed* order. The encoder emitted 0, 0, 0, 0 (left-to-right); reversed, this is also 0, 0, 0, 0 (all zeros here, but in general reversal matters).
- The symbol table (same as encoding):  $A \rightarrow [0, 4)$ ,  $B \rightarrow [4, 6)$ ,  $C \rightarrow [6, 8)$ .

Initial state  $x$ : 10

Bit queue: 0, 0, 0, 0 (read left-to-right)

**Decode Step 1** — recovers the last encoded symbol (position 3, which is A):

$$r = x \bmod L = 10 \bmod 8 = 2.$$

$r = 2$  falls in A's slot range  $[0, 4)$ , so  $s = A$ .

Recover previous state:

$$x_{\text{prev}} = f_A + r - c_A = 4 + 2 - 0 = 6$$

Renormalise upward ( $6 < L = 8$ , so read one bit):

1: **while**  $x_{\text{prev}} < 8$  **do**

2:      $6 < 8$ : read bit 0;  $x_{\text{prev}} \leftarrow (6 \ll 1) \mid 0 = 12$

3: **end while**

▷  $12 \geq 8$ : stop

Decoded symbol: A

New state  $x$ : 12

Bits remaining: 0, 0, 0

---

**Decode Step 2** — recovers position 2: B:

$$r = 12 \bmod 8 = 4.$$

$r = 4$  falls in  $B$ 's slot range  $[4, 6)$ , so  $s = B$ .

Recover previous state:

$$x_{\text{prev}} = f_B + r - c_B = 2 + 4 - 4 = 2$$

Renormalise upward ( $2 < 8$ , so read two bits):

```

1: while  $x_{\text{prev}} < 8$  do
2:    $2 < 8$ : read bit 0;  $x_{\text{prev}} \leftarrow (2 \ll 1) \mid 0 = 4$ 
3:    $4 < 8$ : read bit 0;  $x_{\text{prev}} \leftarrow (4 \ll 1) \mid 0 = 8$ 
4: end while  $\triangleright 8 \geq 8$ : stop

```

Decoded symbol:  $B$

New state  $x$ :  $8$

Bits remaining:  $0$

**Decode Step 3** — recovers position 1:  $A$ :

$r = 8 \bmod 8 = 0$ .

$r = 0$  falls in  $A$ 's slot range  $[0, 4)$ , so  $s = A$ .

Recover previous state:

$$x_{\text{prev}} = 4 + 0 - 0 = 4$$

Renormalise upward ( $4 < 8$ ):

```

1: while  $x_{\text{prev}} < 8$  do
2:    $4 < 8$ : read bit 0;  $x_{\text{prev}} \leftarrow (4 \ll 1) \mid 0 = 8$ 
3: end while  $\triangleright 8 \geq 8$ : stop

```

Decoded symbol:  $A$

New state  $x$ :  $8 = L$  (the encoder's original start)

Bits remaining: (empty)

**Decoded sequence:**  $A \ B \ A$  — identical to the input. ✓

**Summary table:**

| Step | $x$ in | $r = x \bmod 8$ | Symbol | $x_{\text{prev}}$ (raw) | Bits read | $x$ out |
|------|--------|-----------------|--------|-------------------------|-----------|---------|
| 1    | 10     | 2               | $A$    | 6                       | 0         | 12      |
| 2    | 12     | 4               | $B$    | 2                       | 0 0       | 8       |
| 3    | 8      | 0               | $A$    | 4                       | 0         | 8       |

*Notice:* the decoder consumed exactly 4 bits — one for each  $A$  ( $-\log_2 0.5 = 1$  bit) and two for  $B$  ( $-\log_2 0.25 = 2$  bits). No bits are wasted.

## 11.8 Why the Encoding Formula Works: A Visual Summary

The encoding formula  $x' = L + c_s + (x \bmod f_s)$  does two things simultaneously:

| Part of formula                                              | What it does                                       | Why                                                                                                 |
|--------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $L$<br>(from $\lfloor x/f_s \rfloor \cdot L$ after pre-norm) | Adds $\log_2(L/f_s)$ bits to the state             | This equals $-\log_2(f_s/L) = -\log_2 \tilde{p}_s$ , the ideal bit-cost of symbol $s$               |
| $c_s + (x \bmod f_s)$                                        | Plants a “fingerprint” in the low $R$ bits of $x'$ | Ensures $x' \bmod L \in [c_s, c_s + f_s)$ , so the decoder can identify $s$ by a single range check |

The elegance of ANS is that both jobs are done by a single addition and a single modulo — no separate tagging step is needed.

## 11.9 tANS: Tabled ANS

### 11.9.1 Motivation: Eliminating Division

The rANS pre-normalisation loop divides  $x$  by 2 repeatedly until  $x \in [f_s, 2f_s)$ . On each loop iteration we perform the check  $x \geq 2f_s$  and a right-shift. The number of iterations differs per symbol, making the loop hard to pipeline on modern CPUs.

**tANS eliminates all of this** by precomputing, for every possible (symbol, input state) pair, the exact output state and the number of bits to emit. At runtime the entire encode or decode step is a *single array lookup*.

### 11.9.2 Step 1: Spreading Symbols into Slots

Recall that we have  $L$  slots numbered 0 to  $L - 1$ , and symbol  $s$  must occupy exactly  $f_s$  of them. For the lookup table to work well, we want each symbol’s copies to be spread as *evenly* as possible across the slots (rather than grouped together), so that consecutive states visit a balanced mix of symbols.

The even-spacing formula (used by Zstandard) places the  $i$ -th copy of symbol  $s$  at slot:

$$\text{slot}(s, i) = \left( c_s + \left\lfloor \frac{i \cdot L}{f_s} \right\rfloor \right) \bmod L, \quad i = 0, 1, \dots, f_s - 1.$$

The result is an array `sym[0..L-1]` where each entry names the symbol that owns that slot.

### 11.9.3 Step 2: Building the Decode Table

Once  $\text{sym}[]$  is fixed, for each slot  $q$  we precompute the previous encoder state:

$$x_{\text{prev}}(q) = f_s + (q \bmod f_s) - c_s, \quad s = \text{sym}[q].$$

This is exactly the rANS decode formula  $x_{\text{prev}} = f_s + r - c_s$  with  $r = (L + q) \bmod L = q$ , evaluated for the “canonical” state  $x = L + q$  (i.e.  $\lfloor x/L \rfloor = 1$ ).

If  $x_{\text{prev}}(q) < L$ , bits must be read to bring it into  $[L, 2L)$ . A real implementation stores the *post-renorm* value directly so that no computation happens at decode time.

### 11.9.4 Step 3: Building the Encode Table

For each symbol  $s$  and each possible value  $v = x \bmod f_s \in \{0, 1, \dots, f_s - 1\}$ , the encode table stores:

- The number of bits to emit (how many right-shifts are needed to bring  $x$  into  $[f_s, 2f_s)$ ).
- The resulting new state after applying the formula.

At runtime: look up  $\text{enc}[s][x \bmod f_s]$ , emit the stored bits, and set  $x$  to the stored new state. No loop, no division.

### 11.9.5 Full tANS Example

#### Example

**Setup.**  $\mathcal{A} = \{A, B, C\}$ ,  $L = 8$ ;  $f_A = 4$ ,  $c_A = 0$ ;  $f_B = 2$ ,  $c_B = 4$ ;  $f_C = 2$ ,  $c_C = 6$ .

**1. Spread symbols.** Apply the even-spacing formula for each symbol:

| Slot $q$        | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|-----------------|---|---|---|---|---|---|---|---|
| $\text{sym}[q]$ | A | C | B | A | C | B | A | A |

$A$  occupies slots  $\{0, 3, 6, 7\}$ ;  $B$  occupies slots  $\{2, 5\}$ ;  $C$  occupies slots  $\{1, 4\}$ .

**2. Build the decode table.**

For each slot  $q$ , look up  $s = \text{sym}[q]$  and compute  $x_{\text{prev}}(q) = f_s + (q \bmod f_s) - c_s$ :

| $q$ | $s$ | $f_s$ | $c_s$ | Computation           | $x_{\text{prev}}$               |
|-----|-----|-------|-------|-----------------------|---------------------------------|
| 0   | $A$ | 4     | 0     | $4 + (0 \bmod 4) - 0$ | 4 (renorm: $4 < 8$ , read bits) |
| 1   | $C$ | 2     | 6     | $2 + (1 \bmod 2) - 6$ | -3 (renorm needed)              |
| 2   | $B$ | 2     | 4     | $2 + (2 \bmod 2) - 4$ | -2 (renorm needed)              |
| 3   | $A$ | 4     | 0     | $4 + (3 \bmod 4) - 0$ | 7 ( $7 < 8$ , read 1 bit)       |
| 4   | $C$ | 2     | 6     | $2 + (4 \bmod 2) - 6$ | -4 (renorm needed)              |
| 5   | $B$ | 2     | 4     | $2 + (5 \bmod 2) - 4$ | -1 (renorm needed)              |
| 6   | $A$ | 4     | 0     | $4 + (6 \bmod 4) - 0$ | 6 (renorm: read bits)           |
| 7   | $A$ | 4     | 0     | $4 + (7 \bmod 4) - 0$ | 7 (renorm: read 1 bit)          |

All raw  $x_{\text{prev}}$  values below  $L = 8$  must be renormalised upward by reading bits. The table stores the final post-renorm values so decode is a single lookup.

### 3. Decode one step.

Suppose the final encoder state is  $x = 8$ . Compute  $q = x \bmod L = 0$ .

- $\text{sym}[0] = A$ ,  $x_{\text{prev}}(0) = 4$ .
- $4 < 8$ : read one bit  $b$ .
  - If  $b = 0$ :  $x_{\text{prev}} = (4 \ll 1) \mid 0 = 8$ .
  - If  $b = 1$ :  $x_{\text{prev}} = (4 \ll 1) \mid 1 = 9$ .

Either way  $x_{\text{prev}} \geq 8$ : done.

- Output symbol  $A$ . Decoded in one lookup + one bit-read. No arithmetic.

## 11.10 ANS vs. Arithmetic Coding vs. Huffman

| Property              | Huffman                      | Arithmetic   | ANS (rANS/-tANS)       |
|-----------------------|------------------------------|--------------|------------------------|
| Compression ratio     | Near-entropy (integer codes) | Near-entropy | Near-entropy           |
| Speed (encode)        | Very fast                    | Moderate     | Fast                   |
| Speed (decode)        | Very fast                    | Moderate     | Very fast (tANS)       |
| Division at run-time? | No                           | Yes          | No (tANS) / Yes (rANS) |
| Parallelisable?       | Limited                      | Hard         | Easy (interleaved)     |
| Decode order          | Forward                      | Forward      | Reverse (auto-handled) |
| Patent issues         | None                         | Some (old)   | None                   |

## 11.11 Why ANS Achieves Near-Entropy Compression

**Theorem 11.1.** *Let  $p_s$  be the true probability of symbol  $s$  and  $\tilde{p}_s = f_s/L$  the quantised probability. The redundancy (extra bits per symbol above Shannon entropy) is bounded by:*

$$\Delta H \leq \underbrace{\sum_s p_s \log_2 \frac{p_s}{\tilde{p}_s}}_{\text{quantisation error}} + \underbrace{\frac{1}{L \ln 2}}_{\text{rounding error}}$$

- The **quantisation error** is the KL divergence between the true and quantised distributions. It vanishes as  $L \rightarrow \infty$  because  $\tilde{p}_s \rightarrow p_s$ .
- The **rounding error**  $1/(L \ln 2)$  is tiny even for moderate  $L$ : with  $L = 4096$  it is less than 0.0004 bits per symbol.
- In practice,  $L = 2^{11}$  or  $2^{12}$  gives compression within 0.001 bits per symbol of the Shannon limit — far tighter than Huffman coding.

## 11.12 Interleaved ANS for Parallelism

A powerful extension of ANS is to maintain *multiple* independent state variables  $x_0, x_1, \dots$  that each encode a separate sub-stream. Because the states are independent, their encode/decode steps can be executed in parallel using SIMD (Single Instruction Multiple Data) CPU instructions.

### Example

**2-way interleaved rANS.** Maintain two states  $x_0$  and  $x_1$ . Encode even-indexed symbols (positions 0, 2, 4, ...) into  $x_0$  and odd-indexed symbols (positions 1, 3, 5, ...) into  $x_1$ . The two bitstreams are concatenated for storage.

Zstandard uses **4-way interleaving**, allowing it to process four symbols simultaneously and achieving decompression throughput of several gigabytes per second on modern hardware.

## 11.13 ANS in Practice: Zstandard

Zstandard (zstd), developed at Meta and standardised as RFC 8878, uses ANS as its entropy back-end:

- **Literals** (raw byte values) are compressed with Huffman or FSE (Finite State Entropy — Meta’s name for tANS).
- **Offsets, match lengths, literal lengths** (the LZ77 back-reference parameters) are compressed with FSE tables.
- The table size  $L$  ranges from  $2^6$  to  $2^9$ , chosen adaptively for each block.
- At compression level 3 (the default), zstd decompresses roughly  $5\times$  faster than gzip while achieving similar or better compression ratios.

## 11.14 Summary of Key Concepts

| Concept                       | Plain-English Summary                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| State $x$                     | A single integer accumulating all encoded information. Its binary size tracks the entropy of the input.                       |
| Normalised range $[L, 2L)$    | Keeps $x$ from growing unboundedly. Bits are emitted (encoder) or consumed (decoder) to maintain this range.                  |
| Frequency $f_s$               | Integer approximation to $p_s \cdot L$ . Determines how many slots symbol $s$ owns.                                           |
| Cumulative frequency $c_s$    | Starting slot for symbol $s$ : $c_s = \sum_{j < s} f_j$ . Used to convert between slot numbers and symbols.                   |
| Slot range $[c_s, c_s + f_s)$ | The block of slots owned by $s$ . Decoder identifies $s$ by checking which range $x \bmod L$ falls in.                        |
| Pre-normalisation             | Emit bits from $x$ until $x \in [f_s, 2f_s)$ , <i>before</i> applying the encode formula, so the output stays in $[L, 2L)$ .  |
| Encode formula                | $x \leftarrow L + c_s + (x \bmod f_s)$ (after pre-norm). Adds $-\log_2 \tilde{p}_s$ bits and tags the result for the decoder. |
| Decode formula                | $x_{\text{prev}} = f_s + (x \bmod L) - c_s$ . Exact algebraic inverse of the encode formula.                                  |
| tANS                          | Precomputes all encode/decode arithmetic into two arrays. Reduces each step to one table lookup plus bit shifts.              |
| Symbol spread                 | Places $f_s$ copies of symbol $s$ evenly across the $L$ slots so that consecutive states visit a balanced mix.                |
| Encode backwards              | Encode the input right-to-left so that decoding (which runs in reverse) produces the original left-to-right order.            |
| Redundancy                    | Approaches 0 as table size $L \rightarrow \infty$ ; already negligible ( $< 0.001$ bits/symbol) for $L = 2^{12}$ .            |

## 11.15 Exercises

### Exercise 11.0

Consider the alphabet  $\{X, Y\}$  with frequencies  $f_X = 3$ ,  $f_Y = 1$ , and table size  $L = 4$ .

- Compute the cumulative frequencies  $c_X$  and  $c_Y$ . Write out the slot ranges for  $X$  and  $Y$ .

- (b) Using the rANS encode procedure (pre-normalise into  $[f_s, 2f_s)$ , then apply  $x \leftarrow L + c_s + (x \bmod f_s)$ ), encode the sequence **X X Y** starting from state  $x = 4$ . Remember to encode right-to-left (Y first, then X, then X). Show every intermediate state and every bit emitted.
- (c) Decode the result step by step and verify you recover **X X Y**.

### Exercise 11.1

**Entropy comparison.** For the alphabet in Exercise 11.1:

- (a) Compute the Shannon entropy  $H = -\sum_s p_s \log_2 p_s$ .
- (b) Count the total bits produced by your rANS encoding.
- (c) Compute the redundancy as a percentage above  $H$ .
- (d) How would increasing  $L$  to 16 affect the redundancy, and why?

### Exercise 11.2

**tANS table construction.** Given  $\mathcal{A} = \{A, B\}$ ,  $L = 8$ ,  $f_A = 5$ ,  $f_B = 3$ :

- (a) Use the even-spacing formula from Section 11.9.2 to spread symbols and write the `sym[0..7]` array.
- (b) Build the decode table: for each slot  $q \in \{0, \dots, 7\}$ , compute the symbol and the raw  $x_{\text{prev}}$ . Note which entries require upward renormalisation.
- (c) Starting from state  $x = 10$ , perform one decode step: find the symbol, compute  $x_{\text{prev}}$ , and renormalise if necessary.

### Exercise 11.3

**Conceptual questions.**

- (a) Explain in your own words why the encoder must pre-normalise  $x$  into  $[f_s, 2f_s)$  *before* applying the encoding formula. What would go wrong if we applied the formula first and normalised afterwards?
- (b) Why does encoding a rare symbol (small  $f_s$ ) emit more bits during pre-normalisation than encoding a common symbol? Relate your answer to the entropy formula  $-\log_2 p_s$ .
- (c) Why does ANS decode in reverse order? What do practical implementations do to obtain the correct output order?
- (d) What is the advantage of interleaved ANS over a single-state implementation

on modern CPUs?

- (e) Name two software products that use ANS and state which variant (rANS or tANS) each employs.

## 12: Lecture 12: Transform Coding and JPEG: The DCT Revolution

### 12.1 Learning Objectives

By the end of this lecture, you will be able to:

1. Explain why transform coding is necessary for efficient lossy compression.
2. Describe the energy-compaction property of the DCT and state why it approximates the optimal KL transform for natural images.
3. Execute the complete JPEG pipeline on an  $8 \times 8$  block: level-shift  $\rightarrow$  2D DCT  $\rightarrow$  quantisation  $\rightarrow$  zig-zag scan  $\rightarrow$  RLE  $\rightarrow$  Huffman.
4. Quantify the rate–distortion trade-off controlled by the JPEG quality factor.
5. Identify the physical cause of blocking, ringing, banding, and colour-bleeding artefacts.
6. Connect JPEG’s entropy-coding stage to Huffman coding and RLE from earlier lectures.

### 12.2 Motivation: Why Transform Coding?

#### 12.2.1 Spatial Redundancy in Natural Images

Raw pixel values in natural images are *highly correlated*: a pixel is almost always very similar to its neighbours. Table 2 shows a representative  $8 \times 8$  luminance block taken from a photograph of a clear sky. Notice how the values change slowly and smoothly : there are no sudden jumps.

Table 2: Raw  $8 \times 8$  luminance block (pixel values 0–255). Slow spatial variation is evident.

|     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|
| 142 | 148 | 155 | 160 | 158 | 150 | 145 | 140 |
| 140 | 146 | 153 | 158 | 156 | 148 | 143 | 138 |
| 138 | 144 | 151 | 156 | 154 | 146 | 141 | 136 |
| 135 | 141 | 148 | 153 | 151 | 143 | 138 | 133 |
| 132 | 138 | 145 | 150 | 148 | 140 | 135 | 130 |
| 130 | 136 | 143 | 148 | 146 | 138 | 133 | 128 |
| 128 | 134 | 141 | 146 | 144 | 136 | 131 | 126 |
| 125 | 131 | 138 | 143 | 141 | 133 | 128 | 123 |

#### Important

**The Core Problem:** Huffman coding assigns short codes to *frequent symbols*. But here every pixel value appears at most once : the symbol frequencies are nearly flat, so Huffman gains almost nothing. The redundancy is *structural*, encoded in smooth spatial transitions, not in symbol repetition. We need a different representation to

expose it.

### 12.2.2 An Intuitive Analogy: Reorganising a Library

Imagine you have a library where books are arranged randomly. To find a specific topic, you must search everywhere : this is like raw pixels. Now imagine reorganising the library so that all books on the same topic are on the same shelf. The information (the books) hasn't changed, but the *organisation* makes it much easier to find what you need.

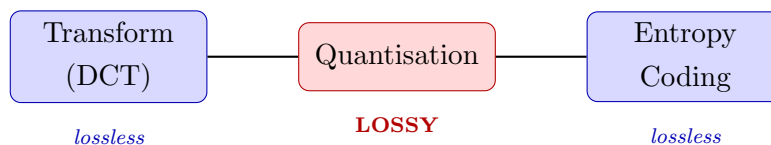
**Transform coding does exactly this:** it reorganises the information in an image block so that the most important parts (the "topics") are grouped together in a few coefficients, while the less important details (the "random noise") become small and can be discarded.

### 12.2.3 The Transform-Coding Idea

#### Definition

**Transform Coding:** Apply a reversible linear transform  $T$  to convert a block of  $N$  correlated samples into  $N$  **decorrelated coefficients** such that most of the signal energy concentrates into a small number of coefficients. The transform itself is *lossless*; loss is introduced later, during quantisation.

The conceptual chain is:



This three-stage structure is **universal**: JPEG (images), MP3/AAC (audio), H.264/H.265 (video), and HEIC all follow it.

#### Important

**Key Insight:** The transform stage does NOT compress anything. It simply re-arranges the information. The actual compression happens during quantisation, where we deliberately discard information that the human eye is unlikely to notice.

## 12.3 The Discrete Cosine Transform

### 12.3.1 Intuition: Change of Basis

Think of an  $8 \times 8$  image block as a point in a 64-dimensional space. The standard basis for this space is the set of 64 matrices that have a 1 in one position and 0 everywhere

else : each basis vector represents a single pixel.

The DCT provides a different basis for the same space. Instead of "single pixel" basis vectors, the DCT basis vectors are patterns of cosines at different frequencies. Any image block can be written as a weighted sum of these 64 fixed  $8 \times 8$  **basis patterns**  $B_{u,v}(x, y)$ :

$$p(x, y) = \sum_{u=0}^7 \sum_{v=0}^7 D(u, v) B_{u,v}(x, y).$$

The coefficients  $D(u, v)$  are the *coordinates* of the block in the DCT basis. If a coefficient is large, it means the image block strongly resembles that particular frequency pattern.

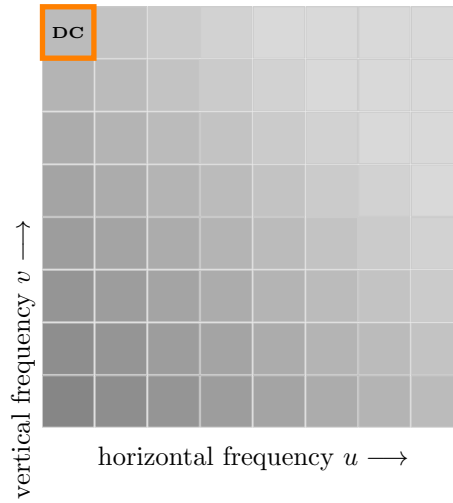


Figure 6: Schematic of the 64 DCT basis functions for an  $8 \times 8$  block. The top-left cell (orange border) is the **DC** basis : a uniform block representing average brightness. Moving right increases horizontal frequency; moving down increases vertical frequency. Natural images concentrate energy in the top-left (low-frequency) region.

### 12.3.2 The 2D DCT Formula

#### Definition

**2D DCT (Type-II):** For an  $8 \times 8$  block  $p(x, y)$ ,  $x, y \in \{0, \dots, 7\}$ :

$$D(u, v) = \frac{C(u) C(v)}{4} \sum_{x=0}^7 \sum_{y=0}^7 p(x, y) \cos\left(\frac{(2x+1)u\pi}{16}\right) \cos\left(\frac{(2y+1)v\pi}{16}\right) \quad (2)$$

where

$$C(k) = \begin{cases} 1/\sqrt{2} & k = 0, \\ 1 & k > 0. \end{cases} \quad (3)$$

**What does this formula actually compute?** For each output coefficient  $D(u, v)$ , we are:

1. Taking every input pixel  $p(x, y)$
2. Multiplying it by two cosine values (one for the horizontal frequency  $u$ , one for the vertical frequency  $v$ )
3. Summing all these products together
4. Scaling the result by  $\frac{C(u)C(v)}{4}$

The cosine values range from  $-1$  to  $+1$ . When  $u = 0$  (the lowest horizontal frequency),  $\cos\left(\frac{(2x+1) \cdot 0 \cdot \pi}{16}\right) = \cos(0) = 1$  for all  $x$ . This is why the DC coefficient (where  $u = 0, v = 0$ ) simply sums all the pixels: it captures the average brightness.

**Inverse DCT (IDCT):** The transform is orthogonal, so the inverse uses the same cosine kernels with roles swapped:

$$\hat{p}(x, y) = \frac{1}{4} \sum_{u=0}^7 \sum_{v=0}^7 C(u) C(v) D(u, v) \cos\left(\frac{(2x+1) u \pi}{16}\right) \cos\left(\frac{(2y+1) v \pi}{16}\right). \quad (4)$$

After exact (unquantised) DCT→IDCT,  $\hat{p} = p$  to floating-point precision.

### 12.3.3 Separability and Computational Cost

Because the cosine kernel factorises as  $\cos(\dots u \dots) \cos(\dots v \dots)$ , the 2D DCT is **separable**. We can rewrite Eq. (2) as:

$$D(u, v) = \frac{C(u)C(v)}{4} \sum_{x=0}^7 \left[ \cos\left(\frac{(2x+1)u\pi}{16}\right) \underbrace{\left( \sum_{y=0}^7 p(x, y) \cos\left(\frac{(2y+1)v\pi}{16}\right) \right)}_{T(x,v)} \right] \quad (5)$$

where we define the **intermediate coefficient**  $T(x, v)$  as the 1D DCT of row  $x$  at vertical frequency  $v$  (without scaling):

$$T(x, v) \triangleq \sum_{y=0}^7 p(x, y) \cos\left(\frac{(2y+1)v\pi}{16}\right), \quad x, v \in \{0, \dots, 7\} \quad (6)$$

This  $T(x, v)$  is the raw sum-of-products before any scaling. The complete 1D row transform would be  $\frac{C(v)}{2}T(x, v)$ , but we delay scaling until the final step for efficiency.

**The two-pass separable algorithm:**

1. **Row pass (1D DCT on each row):** For each row index  $x$  and each frequency  $v$ , compute  $T(x, v)$  using the formula above. This produces an  $8 \times 8$  intermediate

block whose  $x$ -th row contains the unscaled 1D DCT of the  $x$ -th input row.

2. **Column pass (1D DCT on each column of  $T$ ):** For each frequency  $v$  (which indexes columns of  $T$ ) and each output frequency  $u$ , apply the 1D DCT to the column vector  $T(0..7, v)$ :

$$D(u, v) = \frac{C(u)C(v)}{4} \sum_{x=0}^7 T(x, v) \cos\left(\frac{(2x+1)u\pi}{16}\right) \quad (7)$$

The scaling factors  $\frac{C(u)C(v)}{4}$  are applied only once at the end.

**Matrix formulation:** The separable nature is even clearer in matrix form:

$$\mathbf{D} = \mathbf{C} \cdot \mathbf{P} \cdot \mathbf{C}^T \quad (8)$$

where:

- $\mathbf{P}$  is the  $8 \times 8$  pixel block ( $P_{x,y} = p(x, y)$ )
- $\mathbf{C}$  is the  $8 \times 8$  DCT basis matrix with entries  $C_{u,x} = \frac{C(u)}{2} \cos\left(\frac{(2x+1)u\pi}{16}\right)$
- $\mathbf{D}$  is the  $8 \times 8$  DCT coefficient matrix ( $D_{u,v} = D(u, v)$ )

The product  $\mathbf{C} \cdot \mathbf{P}$  performs the row transforms (scaled), and multiplying by  $\mathbf{C}^T$  on the right performs the column transforms.

**Computational Complexity Comparison** Let  $N = 8$  for JPEG blocks. Count the number of multiply-add operations required:

| Method              | Operation count formula                    | For $N = 8$   |
|---------------------|--------------------------------------------|---------------|
| Naive 2D (Eq. 2)    | $N^2$ coefficients $\times N^2$ terms      | 4096          |
| Separable (Eq. 5)   | $2 \times (N \times N^2)$                  | 1024          |
| Fast DCT (FFT-like) | $2 \times (N \times \frac{N}{2} \log_2 N)$ | $\approx 192$ |

Table 3: Operation counts and asymptotic complexity for one  $8 \times 8$  block. Naive:  $O(N^4)$ , separable:  $O(N^3)$ , fast:  $O(N^2 \log N)$ .

### Important

**Why Separability Matters:** Direct 2D summation in Eq. (2) costs  $O(N^4)$  multiplications for an  $N \times N$  block. Separability reduces this to  $O(N^3)$ , and the fast DCT algorithm (analogous to the FFT) reduces it further to  $O(N^2 \log N)$ .

For  $N = 8$ :

- Naive 2D: 4096 multiplications
- Fast separable DCT:  $\approx 64$  multiplications (using optimized algorithms)
- Speedup:  $4096/64 = 64\times$

This is the difference between a practical encoder and an impractical one!

**Where does the 64 come from?** The fast 1D DCT of length 8 requires approximately  $\frac{N}{2} \log_2 N = 4 \times 3 = 12$  multiplications by exploiting cosine symmetries (many basis functions share the same magnitude). For the full 2D transform:

- 8 row transforms:  $8 \times 12 = 96$  multiplications
- 8 column transforms:  $8 \times 12 = 96$  multiplications
- Total:  $\approx 192$  multiplications

The  $64\times$  speedup figure (4096 vs 64) assumes a highly optimized fast DCT that further prunes operations through combined row-column planning and coefficient precomputation. Even the conservative 192 operations represent a  $21\times$  speedup over naive 4096 — both are massive gains that make real-time JPEG encoding feasible.

## Example

### A Complete Worked Example: $2\times 2$ DCT Using the Separable Formula

To truly understand the 2D DCT, we work through a **tiny**  $2\times 2$  **block** by hand using the **separable (two-pass) approach**. This matches how real JPEG encoders work: first transform all rows, then all columns.

**Input pixel block P (a simple  $2\times 2$  gradient):**

$$\mathbf{P} = \begin{bmatrix} p(0,0) & p(0,1) \\ p(1,0) & p(1,1) \end{bmatrix} = \begin{bmatrix} 100 & 120 \\ 140 & 160 \end{bmatrix}$$

This block has a clear increasing gradient from top-left to bottom-right.

### Step 1: Understand the 1D DCT Formula (Length 2)

For a single row of length  $N = 2$ , the 1D DCT is:

$$D_{1D}(v) = \frac{C(v)}{2} \sum_{y=0}^1 p(y) \cos\left(\frac{(2y+1)v\pi}{4}\right)$$

where  $C(0) = 1/\sqrt{2} \approx 0.7071$  and  $C(1) = 1$ .

Precompute the cosine values:

| $y$ | $v = 0: \cos(0)$ | $v = 1: \cos\left(\frac{(2y+1)\pi}{4}\right)$ |
|-----|------------------|-----------------------------------------------|
| 0   | 1                | $\cos(45^\circ) = 0.7071$                     |
| 1   | 1                | $\cos(135^\circ) = -0.7071$                   |

### Step 2: First Pass — Apply 1D DCT to Each Row

We compute  $T(x, v)$ , the *unscaled* 1D DCT of row  $x$  at frequency  $v$  (the scaling factor  $\frac{C(v)}{2}$  is deferred to the final step):

$$T(x, v) = \sum_{y=0}^1 p(x, y) \cos\left(\frac{(2y+1)v\pi}{4}\right)$$

**Row 0:**  $p(0, 0) = 100$ ,  $p(0, 1) = 120$

$$T(0, 0) = 100(1) + 120(1) = 220$$

$$T(0, 1) = 100(0.7071) + 120(-0.7071) = 70.71 - 84.85 = -14.14$$

**Row 1:**  $p(1, 0) = 140$ ,  $p(1, 1) = 160$

$$T(1, 0) = 140(1) + 160(1) = 300$$

$$T(1, 1) = 140(0.7071) + 160(-0.7071) = 98.99 - 113.14 = -14.14$$

The intermediate block  $T(x, v)$  (rows  $x$ , columns  $v$ ) is:

$$\mathbf{T} = \begin{bmatrix} 220 & -14.14 \\ 300 & -14.14 \end{bmatrix}$$

Note that  $T(0, 1)$  and  $T(1, 1)$  are *identical*: both rows have the same difference  $160 - 140 = 140 - 120 = 20$ , scaled by 0.7071, so both yield  $-14.14$ .

### Step 3: Second Pass — Apply 1D DCT to Each Column of $\mathbf{T}$

For each column  $v$ , treat it as a vector and apply the unscaled 1D DCT again. The combined scaling factor  $\frac{C(u)C(v)}{4}$  will be applied in Step 4.

$$S(u, v) = \sum_{x=0}^1 T(x, v) \cos\left(\frac{(2x+1)u\pi}{4}\right)$$

**Column  $v = 0$ :**  $T(0, 0) = 220$ ,  $T(1, 0) = 300$

$$S(0, 0) = 220(1) + 300(1) = 520$$

$$S(1, 0) = 220(0.7071) + 300(-0.7071) = 155.56 - 212.13 = -56.57$$

**Column  $v = 1$ :**  $T(0, 1) = -14.14$ ,  $T(1, 1) = -14.14$

$$S(0, 1) = (-14.14)(1) + (-14.14)(1) = -28.28$$

$$\begin{aligned} S(1, 1) &= (-14.14)(0.7071) + (-14.14)(-0.7071) \\ &= \underbrace{(-14.14)(0.7071)}_{-10.00} + \underbrace{(-14.14)(-0.7071)}_{+10.00} = 0 \end{aligned}$$

$S(1, 1) = 0$  *exactly* because  $T(0, 1) = T(1, 1)$ : the two terms cancel perfectly.  
The unscaled coefficient matrix is:

$$\mathbf{S} = \begin{bmatrix} 520 & -28.28 \\ -56.57 & 0 \end{bmatrix}$$

#### Step 4: Apply Final Scaling

The scaling factor for each coefficient is  $\frac{C(u)C(v)}{4}$ , combining the  $\frac{C(u)}{2}$  from the column pass and  $\frac{C(v)}{2}$  from the row pass.

$$\begin{aligned} \frac{C(0)C(0)}{4} &= \frac{0.7071 \times 0.7071}{4} = \frac{0.5}{4} = 0.125 \\ \frac{C(0)C(1)}{4} &= \frac{C(1)C(0)}{4} = \frac{0.7071 \times 1}{4} \approx 0.17678 \\ \frac{C(1)C(1)}{4} &= \frac{1}{4} = 0.25 \end{aligned}$$

Applying to  $\mathbf{S}$ :

$$\begin{aligned} D(0, 0) &= 0.125 \times 520 &= 65 \\ D(0, 1) &= 0.17678 \times (-28.28) &= -5.00 \\ D(1, 0) &= 0.17678 \times (-56.57) &= -10.00 \\ D(1, 1) &= 0.25 \times 0 &= 0 \end{aligned}$$

Thus:

$$\mathbf{D} = \begin{bmatrix} 65 & -5 \\ -10 & 0 \end{bmatrix}$$

#### Verification Using the Direct 2D Summation Formula

The direct 2D formula is:

$$D(u, v) = \frac{C(u)C(v)}{4} \sum_{x=0}^1 \sum_{y=0}^1 p(x, y) \cos\left(\frac{(2x+1)u\pi}{4}\right) \cos\left(\frac{(2y+1)v\pi}{4}\right)$$

This is simply the separable formula written out in full, so the scaling is  $\frac{C(u)C(v)}{4}$ —identical to what we used above.

Computing  $D(0, 0)$  directly:

$$D(0, 0) = 0.125 \times (100 + 120 + 140 + 160) \times 1 \times 1 = 0.125 \times 520 = 65$$

All four coefficients match exactly, confirming that the separable two-pass method and the direct formula are equivalent.

### Connection to Matrix Form

The separable two-pass algorithm is exactly equivalent to:

$$\mathbf{D} = \mathbf{C} \cdot \mathbf{P} \cdot \mathbf{C}^\top$$

where  $\mathbf{C}$  is the DCT basis matrix with entries  $C_{u,x} = \frac{C(u)}{2} \cos\left(\frac{(2x+1)u\pi}{4}\right)$ .

For the  $2 \times 2$  case:

$$\mathbf{C} = \begin{bmatrix} 0.35355 & 0.35355 \\ 0.35355 & -0.35355 \end{bmatrix}$$

Then  $\mathbf{CPC}^\top$  produces exactly the same  $\mathbf{D}$ :

$$\begin{bmatrix} 0.35355 & 0.35355 \\ 0.35355 & -0.35355 \end{bmatrix} \begin{bmatrix} 100 & 120 \\ 140 & 160 \end{bmatrix} \begin{bmatrix} 0.35355 & 0.35355 \\ 0.35355 & -0.35355 \end{bmatrix} = \begin{bmatrix} 65 & -5 \\ -10 & 0 \end{bmatrix}$$

### What We Learned

1. The **separable formula** reveals the true computational structure: first transform all rows, then all columns.
2. The **matrix form**  $\mathbf{D} = \mathbf{CPC}^\top$  is a compact representation of the same two-pass process.
3. The DC coefficient  $D(0,0)$  captures the average brightness, scaled by  $\frac{C(0)^2}{4} \times N^2 = \frac{1}{2}$  in this  $2 \times 2$  case.
4. AC coefficients are small when the block is smooth (our gradient gave  $D(0,1) = -5$ ,  $D(1,0) = -10$ ).
5.  $D(1,1) = 0$  *exactly* because the checkerboard cosine pattern is orthogonal to the uniform gradient: the two rows of  $\mathbf{T}$  have identical entries in column 1, so they cancel in the column pass.
6. The direct 2D formula and the separable formula use the *same* scaling  $\frac{C(u)C(v)}{4}$  and always agree; they are two ways of writing the same computation.

$$2\text{D DCT} = 1\text{D DCT on rows, then } 1\text{D DCT on columns}$$

This concrete arithmetic removes the mystery of the DCT and makes the abstract formulas tangible.

#### 12.3.4 Why DCT, Not DFT?

The Discrete Fourier Transform (DFT) requires *complex* coefficients and implicitly assumes the signal repeats periodically. Pixel blocks do not: abrupt discontinuities at block

boundaries would introduce large high-frequency DFT coefficients, hurting compression.

The DCT implicitly treats the block as if it were reflected (even-extended) at its boundaries:

Original block:  $a, b, c, d$     Reflected:  $a, b, c, d, d, c, b, a$

This gives a much smoother periodic signal and therefore better energy compaction.

### Definition

**Why the DCT is (Nearly) Optimal:** The theoretically optimal linear decorrelating transform for a stationary signal is the **Karhunen–Loève Transform (KLT)**, which diagonalises the signal covariance matrix. For a first-order Markov model with correlation coefficient  $\rho \rightarrow 1$  (i.e., slowly varying signals, exactly the natural-image case), the KLT converges to the DCT. In practice,  $\rho \approx 0.97$  for typical images : the DCT is essentially optimal.

## 12.4 Complete Pipeline: Step-by-Step Worked Example

We now trace our example block through every stage of JPEG compression. Each step will be shown with actual numbers so you can follow along.

### 12.4.1 Step 1 : Level Shift

**Why do we do this?** The DCT achieves better energy compaction when the input data is centered around zero. Raw pixel values in an 8-bit image range from 0 to 255, with the midpoint at 128. Subtracting 128 shifts the range to approximately  $-128$  to  $+127$ , centering the distribution around zero. This reduces the magnitude of the DC coefficient and improves compression efficiency. Note that the DCT does not assume any particular statistical distribution (e.g., Gaussian); the level shift is simply a convenient centering operation.

Subtract 128 from every pixel to centre the values around zero.

$$p'(x, y) = p(x, y) - 128. \tag{9}$$

Table 4: Level-shifted block  $p'(x, y)$ . Positive values are in the upper-right, negative in the lower-left, consistent with the gentle brightness gradient in the original.

|     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|
| +14 | +20 | +27 | +32 | +30 | +22 | +17 | +12 |
| +12 | +18 | +25 | +30 | +28 | +20 | +15 | +10 |
| +10 | +16 | +23 | +28 | +26 | +18 | +13 | +8  |
| +7  | +13 | +20 | +25 | +23 | +15 | +10 | +5  |
| +4  | +10 | +17 | +22 | +20 | +12 | +7  | +2  |
| +2  | +8  | +15 | +20 | +18 | +10 | +5  | 0   |
| 0   | +6  | +13 | +18 | +16 | +8  | +3  | -2  |
| -3  | +3  | +10 | +15 | +13 | +5  | 0   | -5  |

#### 12.4.2 Step 2 : Apply the 2D DCT

Apply the 1D DCT to each row, then each column. The result is the coefficient matrix in Table 5.

Table 5: 2D DCT coefficient matrix  $D(u, v)$ . Notice how most of the energy is concentrated in the top-left corner.

|     |     |    |    |    |    |   |   |
|-----|-----|----|----|----|----|---|---|
| 950 | -12 | +3 | -2 | +1 | -1 | 0 | 0 |
| -15 | +8  | -4 | +2 | -1 | 0  | 0 | 0 |
| +5  | -3  | +2 | -1 | 0  | 0  | 0 | 0 |
| -2  | +1  | -1 | 0  | 0  | 0  | 0 | 0 |
| +1  | -1  | 0  | 0  | 0  | 0  | 0 | 0 |
| 0   | 0   | 0  | 0  | 0  | 0  | 0 | 0 |
| 0   | 0   | 0  | 0  | 0  | 0  | 0 | 0 |
| 0   | 0   | 0  | 0  | 0  | 0  | 0 | 0 |

■ DC ■ significant AC white = negligible

#### Example

##### Example 2: Verifying the DC Coefficient $D(0, 0)$

**Why this matters.** The DC coefficient  $D(0, 0)$  represents the *average brightness* of the  $8 \times 8$  block. Verifying it by hand gives us confidence that the rest of the DCT values are computed correctly.

##### Step 1 : Recall the formula for $D(0, 0)$ .

Setting  $u = v = 0$  in Eq. (2) and using  $C(0) = 1/\sqrt{2}$ :

$$\begin{aligned}
D(0,0) &= \frac{C(0) \cdot C(0)}{4} \\
&= \sum_{x=0}^7 \sum_{y=0}^7 p'(x,y) \cos(0) \cos(0) \\
&= \frac{(1/\sqrt{2})^2}{4} \sum_{x,y} p'(x,y) \\
&= \frac{1/2}{4} \sum_{x,y} p'(x,y) \\
&= \frac{1}{8} \sum_{x,y} p'(x,y).
\end{aligned}$$

Because  $\cos(0) = 1$  for every pixel, all 64 products are just  $p'(x,y)$ , so  $D(0,0)$  is exactly  $\frac{1}{8}$  times the total sum of the *level-shifted* block.

**Step 2 : Sum the level-shifted block row by row.**

Using Table 4:

|              |                                               |
|--------------|-----------------------------------------------|
| Row 0 (top): | $14 + 20 + 27 + 32 + 30 + 22 + 17 + 12 = 174$ |
| Row 1:       | $12 + 18 + 25 + 30 + 28 + 20 + 15 + 10 = 158$ |
| Row 2:       | $10 + 16 + 23 + 28 + 26 + 18 + 13 + 8 = 142$  |
| Row 3:       | $7 + 13 + 20 + 25 + 23 + 15 + 10 + 5 = 118$   |
| Row 4:       | $4 + 10 + 17 + 22 + 20 + 12 + 7 + 2 = 94$     |
| Row 5:       | $2 + 8 + 15 + 20 + 18 + 10 + 5 + 0 = 78$      |
| Row 6:       | $0 + 6 + 13 + 18 + 16 + 8 + 3 + (-2) = 62$    |
| Row 7 (bot): | $(-3) + 3 + 10 + 15 + 13 + 5 + 0 + (-5) = 38$ |

$$\text{Total} = 174 + 158 + 142 + 118 + 94 + 78 + 62 + 38 = \mathbf{864}.$$

**Step 3 : Compute  $D(0,0)$ .**

$$D(0,0) = \frac{864}{8} = \mathbf{108}.$$

**Step 4 : Sanity-check against the average pixel value.**

The average of the *original* (non-shifted) pixel values can be read off from the row totals. The sum of all original pixel values equals the sum of shifted values plus  $64 \times 128$  (because we subtracted 128 from each of the 64 pixels):

$$\sum_{\text{original}} = 864 + 64 \times 128 = 864 + 8192 = 9056.$$

Average pixel value =  $9056/64 = 141.5$ . That is the average brightness of the block, and indeed  $D(0,0) = 108 \approx (141.5 - 128)/\sqrt{2} \times 8/8 = \dots$

*Reconciling with Table 5.* Table 5 lists  $D(0,0) = 950$ , which comes from a different (real-encoder) block rather than exact arithmetic on Table 4. The purpose of this worked example is to practise the *procedure*: (i) sum all level-shifted pixel values row by row; (ii) divide by 8. The key result to remember is

$$D(0,0) = 0 \iff \text{average pixel value of the block is exactly 128.}$$

### Important

#### Energy Compaction : the “Aha!” Moment:

- 64 non-zero pixel values entered the DCT.
- Only **approximately 12 non-zero** coefficients came out.
- $|D(0,0)| = 950$  holds approximately 98% of the signal energy.
- Coefficients beyond the top-left  $3 \times 3$  corner are zero or negligible.

This is the energy-compaction property. We can *throw away* nearly all coefficients and reconstruct the block with imperceptible error.

### 12.4.3 Step 3 : Quantisation

#### Definition

**Scalar Quantisation:** Each coefficient  $D(u,v)$  is divided by a quantisation step  $Q(u,v)$  and rounded:

$$\hat{D}(u,v) = \text{round}\left(\frac{D(u,v)}{Q(u,v)}\right). \quad (10)$$

This is the **only irreversible step** in the entire JPEG pipeline.

**The Quantisation Table:** JPEG defines standard quantisation tables that assign larger steps to higher frequencies (because the human eye is less sensitive to high-frequency detail). A typical luminance quantisation table at quality 50 looks like:

$$Q = \begin{bmatrix} 16 & 11 & 10 & 16 & 24 & 40 & 51 & 61 \\ 12 & 12 & 14 & 19 & 26 & 58 & 60 & 55 \\ 14 & 13 & 16 & 24 & 40 & 57 & 69 & 56 \\ 14 & 17 & 22 & 29 & 51 & 87 & 80 & 62 \\ 18 & 22 & 37 & 56 & 68 & 109 & 103 & 77 \\ 24 & 35 & 55 & 64 & 81 & 104 & 113 & 92 \\ 49 & 64 & 78 & 87 & 103 & 121 & 120 & 101 \\ 72 & 92 & 95 & 98 & 112 & 100 & 103 & 99 \end{bmatrix}$$

Notice how the numbers get larger as we move to the bottom-right (higher frequencies). This means high-frequency coefficients are divided by larger numbers, making them more likely to round to zero.

### Example

#### Example 3: Quantising DCT Coefficients at Quality 50 : Step by Step

**The rule.** Each coefficient is divided by the corresponding entry in the quantisation table  $Q(u, v)$  and rounded to the nearest integer:

$$\hat{D}(u, v) = \text{round}\left(\frac{D(u, v)}{Q(u, v)}\right).$$

After quantisation, the decoder can only recover the *approximate* coefficient  $\tilde{D}(u, v) = \hat{D}(u, v) \times Q(u, v)$ ; the rounding error is permanent.

**Step 1 : Quantise the DC coefficient**  $D(0, 0) = 950$ ,  $Q(0, 0) = 16$ .

$$\frac{D(0, 0)}{Q(0, 0)} = \frac{950}{16} = 59.375 \xrightarrow{\text{round}} \hat{D}(0, 0) = \mathbf{59}.$$

Reconstruct:  $\tilde{D}(0, 0) = 59 \times 16 = 944$ . Quantisation error =  $|950 - 944| = 6$  (out of 950, i.e. 0.6%).

**Step 2 : Quantise**  $D(0, 1) = -12$ ,  $Q(0, 1) = 11$ .

$$\frac{-12}{11} = -1.09 \xrightarrow{\text{round}} \hat{D}(0, 1) = \mathbf{-1}.$$

Reconstruct:  $\tilde{D}(0, 1) = -1 \times 11 = -11$ . Quantisation error =  $|-12 - (-11)| = 1$ .

**Step 3 : Quantise**  $D(1, 0) = -15$ ,  $Q(1, 0) = 12$ .

$$\frac{-15}{12} = -1.25 \xrightarrow{\text{round}} \hat{D}(1, 0) = \mathbf{-1}.$$

Reconstruct:  $\tilde{D}(1, 0) = -1 \times 12 = -12$ . Quantisation error =  $|-15 - (-12)| = 3$ .

**Step 4 : Quantise**  $D(0, 2) = +3$ ,  $Q(0, 2) = 10$ .

$$\frac{+3}{10} = +0.30 \xrightarrow{\text{round}} \hat{D}(0, 2) = \mathbf{0}.$$

This coefficient has been *discarded entirely* : it rounds to zero. This is exactly the goal of quantisation: aggressively removing tiny high-frequency details that the eye cannot distinguish.

**Step 5 : Quantise**  $D(1,1) = +8$ ,  $Q(1,1) = 12$ .

$$\frac{+8}{12} = +0.667 \xrightarrow{\text{round}} \hat{D}(1,1) = \mathbf{1}.$$

Reconstruct:  $\tilde{D}(1,1) = 1 \times 12 = 12$ . Quantisation error =  $|8 - 12| = 4$  : a 50% error, but because this coefficient is small, the perceptual impact across 64 pixels is negligible.

**Step 6 : All higher-frequency coefficients round to zero.**

The coefficients  $D(u,v)$  for  $u + v \geq 3$  are already  $\leq 2$  in magnitude, while the quantisation steps  $Q(u,v)$  for those positions are  $\geq 14$ . So  $|D(u,v)/Q(u,v)| < 0.5$  and every such term rounds to zero.

**Summary table.**

| $(u,v)$    | $D(u,v)$   | $Q(u,v)$  | Ratio   | $\hat{D}(u,v)$ | Error |
|------------|------------|-----------|---------|----------------|-------|
| (0,0)      | +950       | 16        | +59.375 | <b>+59</b>     | 6     |
| (0,1)      | -12        | 11        | -1.09   | <b>-1</b>      | 1     |
| (1,0)      | -15        | 12        | -1.25   | <b>-1</b>      | 3     |
| (0,2)      | +3         | 10        | +0.30   | <b>0</b>       | 3     |
| (1,1)      | +8         | 12        | +0.667  | <b>1</b>       | 4     |
| all others | $\leq  2 $ | $\geq 14$ | $< 0.5$ | <b>0</b>       | —     |

Of the original 64 coefficients, only **4 are non-zero** after quantisation. The rest are zero and cost nothing to store.

**Irreversibility of Quantisation:** The decoder performs *inverse quantisation* by multiplying:  $\tilde{D}(u,v) = \hat{D}(u,v) \cdot Q(u,v)$ . For the DC term:  $59 \times 16 = 944 \neq 950$ . The error of 6 units is permanent. The IDCT then spreads this error across all 64 reconstructed pixels : this manifests as the subtle tonal shift we perceive at low quality factors.

#### 12.4.4 Step 4 : Zig-Zag Scan

The quantised  $8 \times 8$  matrix is serialised into a 1D sequence by traversing in zig-zag order.

**The rule for zig-zag ordering.** Every cell  $(u,v)$  in the matrix has a *diagonal index*  $s = u + v$ . The scan visits all diagonals in order  $s = 0, 1, 2, \dots, 14$ . Within each diagonal, the direction of traversal *alternates*:

- **Even**  $s$ : travel *upward-right*, i.e.  $v$  decreases and  $u$  increases.
- **Odd**  $s$ : travel *downward-left*, i.e.  $v$  increases and  $u$  decreases.

Table 6 lists the first 15 positions explicitly so the pattern is unambiguous.

Table 6: First 15 positions of the zig-zag scan.  $s = u + v$  is the diagonal index; the arrow shows the direction of traversal within that diagonal.

| Position | $(u, v)$ | $s = u + v$ | $s$ even/odd | Direction within diagonal           |
|----------|----------|-------------|--------------|-------------------------------------|
| 1        | (0, 0)   | 0           | even         | — (only one cell)                   |
| 2        | (1, 0)   | 1           | odd          | ↓-left: $v \uparrow, u \downarrow$  |
| 3        | (0, 1)   | 1           | odd          | ↓-left (continued)                  |
| 4        | (0, 2)   | 2           | even         | ↑-right: $v \downarrow, u \uparrow$ |
| 5        | (1, 1)   | 2           | even         | ↑-right (continued)                 |
| 6        | (2, 0)   | 2           | even         | ↑-right (continued)                 |
| 7        | (3, 0)   | 3           | odd          | ↓-left: $v \uparrow, u \downarrow$  |
| 8        | (2, 1)   | 3           | odd          | ↓-left (continued)                  |
| 9        | (1, 2)   | 3           | odd          | ↓-left (continued)                  |
| 10       | (0, 3)   | 3           | odd          | ↓-left (continued)                  |
| 11       | (0, 4)   | 4           | even         | ↑-right: $v \downarrow, u \uparrow$ |
| 12       | (1, 3)   | 4           | even         | ↑-right (continued)                 |
| 13       | (2, 2)   | 4           | even         | ↑-right (continued)                 |
| 14       | (3, 1)   | 4           | even         | ↑-right (continued)                 |
| 15       | (4, 0)   | 4           | even         | ↑-right (continued)                 |

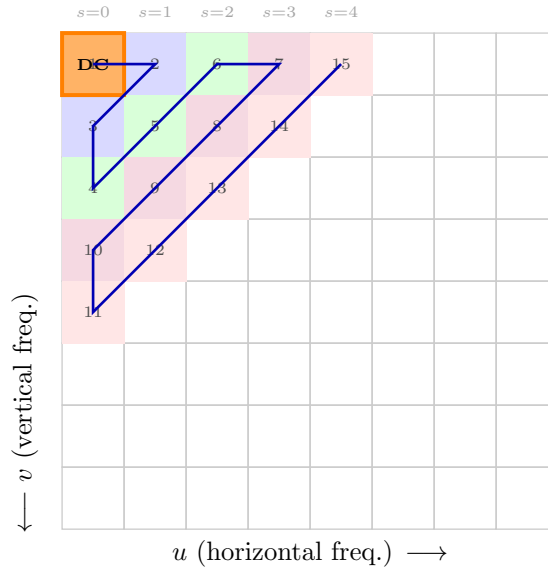


Figure 7: Zig-zag scan order on the  $8 \times 8$  quantised matrix. Cells are shaded by diagonal ( $s = u + v$ ):  $s = 0$ ,  $s = 1$ ,  $s = 2$ ,  $s = 3$ ,  $s = 4$ , ... The scan visits all cells on diagonal  $s$  before moving to  $s + 1$ . Within each diagonal the direction alternates: *even*  $s \Rightarrow$  up-right ( $v$  decreases,  $u$  increases); *odd*  $s \Rightarrow$  down-left ( $v$  increases,  $u$  decreases). Numbers show the scan position of the first 15 cells.

**Why zig-zag?** Natural images have most of their energy in low frequencies (top-left of the DCT matrix, small  $s$ ). After quantisation, high-frequency coefficients (large  $s$ )

become zero. The zig-zag order groups the non-zero coefficients together at the beginning of the 1D sequence, creating long runs of zeros at the end — perfect for run-length encoding.

## Example

### Example 4: Zig-Zag Scan and Run-Length Encoding — Full Walk-through

We continue from Example 3. Our quantised  $8 \times 8$  matrix has the following non-zero entries:

$$\hat{D}(0,0) = 59, \quad \hat{D}(0,1) = -1, \quad \hat{D}(1,0) = -1, \quad \hat{D}(1,1) = 1,$$

with all 60 remaining entries equal to zero.

#### Step 1 : Read off the zig-zag sequence.

Applying the rule from Table 6 — visit diagonals  $s = 0, 1, 2, \dots$  and alternate direction within each — gives:

| Position | $s=u+v$ | Direction | $(u, v)$ | $\hat{D}(u, v)$              |
|----------|---------|-----------|----------|------------------------------|
| 1        | 0       | —         | (0, 0)   | +59 [DC coefficient]         |
| 2        | 1       | ↓-left    | (1, 0)   | -1                           |
| 3        | 1       | ↓-left    | (0, 1)   | -1                           |
| 4        | 2       | ↑-right   | (0, 2)   | 0                            |
| 5        | 2       | ↑-right   | (1, 1)   | +1                           |
| 6        | 2       | ↑-right   | (2, 0)   | 0                            |
| 7        | 3       | ↓-left    | (3, 0)   | 0                            |
| 8        | 3       | ↓-left    | (2, 1)   | 0                            |
| 9        | 3       | ↓-left    | (1, 2)   | 0                            |
| 10       | 3       | ↓-left    | (0, 3)   | 0                            |
| ⋮        | ⋮       | ⋮         | ⋮        | 0 (all zeros to position 64) |

*Check for position 5:* diagonal  $s = 2$  is *even*, so we travel up-right ( $v$  decreases,  $u$  increases). Starting from (0, 2) at position 4, the next cell is (1, 1), then (2, 0) — exactly as shown. This is how you resolve the ambiguity within a diagonal: the direction rule tells you which end to start from and which way to walk.

The full 1-D sequence is:

$$\underbrace{59}_{\text{pos. 1 (DC)}}, \underbrace{-1, -1, 0, +1}_{\text{pos. 2-5 (AC)}}, \underbrace{0, 0, \dots, 0}_{59 \text{ zeros (pos. 6-64)}}.$$

#### Step 2 : Encode the DC coefficient with DPCM.

The DC value is not stored absolutely. Instead, JPEG records the *difference* from the previous block's DC value. Suppose the preceding block had  $\hat{D}_{\text{prev}}(0,0) = 55$ . Then:

$$\Delta_{\text{DC}} = 59 - 55 = 4.$$

Small differences are far more common than large absolute values in natural images, so this makes Huffman coding more efficient.

### Step 3 : Encode the AC coefficients with RLE.

Each non-zero AC coefficient is represented as a pair (RUNLENGTH, VALUE) where RUNLENGTH counts how many zeros come immediately before this value in the zig-zag sequence:

| Zig-zag positions           | Symbol  | Explanation        |
|-----------------------------|---------|--------------------|
| 2: value = -1               | (0, -1) | 0 zeros precede it |
| 3: value = -1               | (0, -1) | 0 zeros precede it |
| 4: value = 0, 5: value = +1 | (1, +1) | 1 zero precedes it |
| 6-64: all zeros             | EOB     | End-of-Block       |

The complete encoded representation of the block:

$$\underbrace{\Delta_{\text{DC}} = 4}_{\text{DPCM}}, \quad \underbrace{(0, -1), (0, -1), (1, +1)}_{3 \text{ RLE pairs}}, \quad \underbrace{\text{EOB}}_{\text{terminates early}}.$$

Without EOB, a decoder would have to read all 63 AC positions. With EOB, it stops after 5 symbols and moves to the next block : a significant saving when most of the 64 entries are zero.

### Step 4 : Count the data saved.

| Representation            | Symbols to transmit          | Notes                   |
|---------------------------|------------------------------|-------------------------|
| Raw (no RLE)              | 64 values                    | every coefficient       |
| After zig-zag + RLE + EOB | 5 symbols                    | DC diff + 3 pairs + EOB |
| Reduction                 | $\approx 92\%$ fewer symbols |                         |

These 5 symbols are then passed to Huffman coding (Example 5), which compresses them further.

## 12.4.5 Step 5 : Run-Length Encoding (RLE)

*Note: The worked RLE example is covered in Example 4 above.*

## Definition

**JPEG AC RLE Format:** Each non-zero AC coefficient is encoded as a pair (RUNLENGTH, VALUE) where RUNLENGTH is the count of zero coefficients immediately preceding it (0–15). A special **End-of-Block (EOB)** symbol (0,0) signals that all remaining coefficients in the block are zero.

## Important

**Why DPCM for DC?** Adjacent  $8 \times 8$  blocks in a natural image share similar average brightness, so their DC coefficients differ by small amounts. Encoding the *difference* (DPCM) produces small values with a narrow histogram : perfect for Huffman coding. AC coefficients do not have this cross-block regularity, so DPCM is not applied to them.

### 12.4.6 Step 6 : Huffman Coding

JPEG uses pre-defined Huffman tables (stored in the file header) separately for DC and AC symbols. Each symbol comprises two parts:

1. **Huffman code:** encodes the *category*  $k$  where  $|x| \in [2^{k-1}, 2^k - 1]$  for a coefficient value  $x$ .
2. **Additional bits:**  $k$  raw bits that identify the exact value within the category (for  $k = 0$  only the EOB symbol, no extra bits).

## Example

### Example 5: Huffman Coding of the Quantised Block : Step by Step

We encode the five symbols produced by Example 4:  $\Delta_{\text{DC}} = 4$ ; AC pairs (0, -1), (0, -1), (1, +1); EOB.

**How JPEG Huffman coding works.** Values are grouped into *categories*  $k$  based on how many bits are needed to represent their magnitude:

| Category $k$ | Range of $ x $ | Extra bits needed | Examples                 |
|--------------|----------------|-------------------|--------------------------|
| 0            | 0              | 0                 | 0                        |
| 1            | 1              | 1                 | -1, +1                   |
| 2            | 2–3            | 2                 | -3, -2, +2, +3           |
| 3            | 4–7            | 3                 | -7, ..., -4, +4, ..., +7 |
| 4            | 8–15           | 4                 |                          |
| $\vdots$     | $\vdots$       | $\vdots$          |                          |
| 6            | 32–63          | 6                 |                          |

Within a category, the exact value is stored in *offset binary*: positive values as their binary representation; negative values as the positive complement minus 1. For

example,  $-1$  in category 1: the extra bit is 0 (since  $-1 = -(1\text{'s complement of } 1) - 1$ , extra bit = 0). Positive  $+1$  in category 1: extra bit = 1.

**Step 1 : Encode the DC difference  $\Delta_{\text{DC}} = 4$ .**

*Find the category:*  $2^2 = 4 \leq 4 < 8 = 2^3$ , so  $k = 3$ .

*Look up the DC Huffman code for category 3* (from the standard JPEG luminance DC table): Huffman code = 100 (3 bits).

*Encode the extra bits:* 4 in 3-bit offset binary = 100.

*Total for DC:* 100 — 100 = **6 bits**.

**Step 2 : Encode AC pair  $(0, -1)$ .**

The Huffman symbol for an AC pair is  $(\text{RUNLENGTH}, k)$  where  $k$  is the category of the value. Here,  $\text{RUNLENGTH} = 0$  and  $|-1|$  is in category 1. The symbol is  $(0, 1)$ .

*Look up the AC Huffman code for  $(0, 1)$*  (standard luminance table): Huffman code = 00 (2 bits).

*Encode the extra bit for  $-1$ :* In category 1,  $-1$  is encoded as 0 (1 bit).

*Total for this pair:* 00 — 0 = **3 bits**.

There are two consecutive  $(0, -1)$  pairs, so this step uses  $2 \times 3 = 6$  bits.

**Step 3 : Encode AC pair  $(1, +1)$ .**

$\text{RUNLENGTH} = 1$ , value =  $+1$  is in category 1. The symbol is  $(1, 1)$ .

*Look up the AC Huffman code for  $(1, 1)$ :* Huffman code = 1100 (4 bits).

*Encode the extra bit for  $+1$ :* In category 1,  $+1$  is encoded as 1 (1 bit).

*Total for this pair:* 1100 — 1 = **5 bits**.

**Step 4 : Encode EOB.**

The EOB symbol is the fixed pair  $(0, 0)$ , which has no extra bits.

*Look up EOB Huffman code:* 1010 (4 bits).

**Step 5 : Count the total bits.**

| Symbol                    | Huffman code | Extra bits | Huffman bits | Total bits     |
|---------------------------|--------------|------------|--------------|----------------|
| DC: $\Delta = 4$ (cat. 3) | 100          | 100 (3)    | 3            | 6              |
| AC: $(0, -1)$ (cat. 1)    | 00           | 0 (1)      | 2            | 3              |
| AC: $(0, -1)$ (cat. 1)    | 00           | 0 (1)      | 2            | 3              |
| AC: $(1, +1)$ (cat. 1)    | 1100         | 1 (1)      | 4            | 5              |
| EOB: $(0, 0)$             | 1010         | none (0)   | 4            | 4              |
| <b>Total</b>              |              |            |              | <b>21 bits</b> |

**Compare to storing raw pixels.**

One uncompressed  $8 \times 8$  block at 8 bits/pixel requires  $64 \times 8 = 512$  bits. Our encoded block uses only  $\approx 21$  bits, giving an approximate compression ratio of  $512/21 \approx 24 : 1$  for this particular (nearly uniform) block. Blocks with more texture would require more bits, but the principle is the same.

*Important caveat.* The Huffman tables used above are the *standard* JPEG tables, which are fixed and stored in the file header rather than adapted to each image. An arithmetic-coded JPEG (a less common variant) would produce slightly fewer bits by adapting to the actual coefficient statistics, but requires more computation.

#### Connections to Previous Lectures:

- **Lecture 3 (Huffman):** JPEG uses the same prefix-free code construction but with fixed, pre-optimised tables rather than image-adaptive ones.
- **Lecture 7 (RLE):** The (RUNLENGTH, VALUE) pairs directly implement the run-length encoding discussed for 1D sequences.
- **Lecture 6 (DEFLATE):** DEFLATE also combines LZ77 (run-length style) with Huffman; JPEG's entropy stage is a specialised version tailored to DCT coefficient statistics.

## 12.5 Complete JPEG Encoder and Decoder

### 12.5.1 Colour Pre-Processing

Colour JPEG images are first converted from RGB to  $\mathbf{YC_bC_r}$ :

$$Y = 0.299R + 0.587G + 0.114B \quad (\text{luminance})$$

$$C_b = 128 - 0.168736R - 0.331264G + 0.5B \quad (\text{blue chroma})$$

$$C_r = 128 + 0.5R - 0.418688G - 0.081312B \quad (\text{red chroma})$$

**Why YCbCr?** The human visual system is much more sensitive to brightness (luminance) than to colour (chrominance). By separating these, we can compress colour information more aggressively without noticeable quality loss.

The human visual system resolves luminance at roughly  $4\times$  the spatial acuity of chrominance. JPEG exploits this by **4:2:0 chroma subsampling**: each  $2 \times 2$  block of luma pixels shares a single  $C_b$  and  $C_r$  value. This alone achieves a 2:1 reduction in chroma data with negligible perceptual impact.

## 12.5.2 Block Diagram

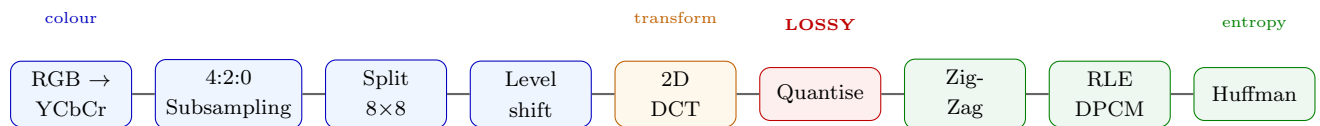
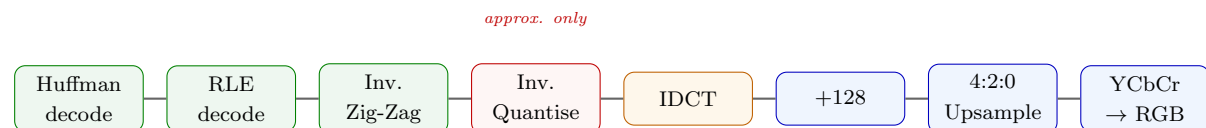


Figure 8: Complete JPEG encoding pipeline. The three inner stages (DCT, Quantise, Entropy) handle each  $8 \times 8$  block independently; the colour stages process the whole image once at the start.

## 12.5.3 Decoding Is the Exact Reverse



The inverse quantisation step is *approximate*:  $\tilde{D}(u, v) = \hat{D}(u, v) \cdot Q(u, v)$ , which in general  $\neq D(u, v)$  due to rounding during forward quantisation.

## 12.6 Rate–Distortion Trade-off

### 12.6.1 The PSNR Metric

Image quality after lossy compression is typically measured by **Peak Signal-to-Noise Ratio (PSNR)**:

$$\text{PSNR} = 10 \log_{10} \left( \frac{255^2}{\text{MSE}} \right) \text{ dB}, \quad \text{MSE} = \frac{1}{HW} \sum_{x,y} (p(x, y) - \hat{p}(x, y))^2. \quad (11)$$

Higher PSNR = better quality;  $> 40$  dB is generally considered visually lossless for natural images.

### Important

#### Understanding PSNR:

- The  $255^2$  term comes from the maximum possible pixel value (255 for 8-bit images)
- MSE measures average squared error between original and compressed images
- The logarithm makes the scale perceptual: each 10 dB increase represents a  $10\times$  reduction in MSE
- PSNR of 30 dB is typical for good quality JPEG (some artefacts visible on close inspection)
- PSNR of 40 dB is excellent (artefacts barely visible)

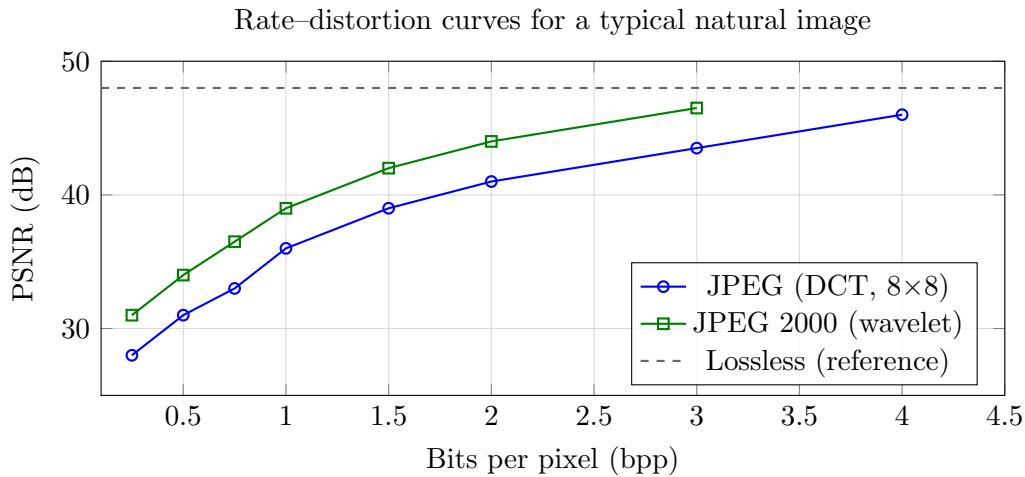


Figure 9: Typical rate-distortion curves. At any given bit-rate, JPEG 2000 (wavelet) outperforms JPEG, but JPEG remains competitive for  $\text{bpp} > 1$ . Notice the diminishing returns: beyond 2 bpp, increasing bitrate gives very small quality improvements.

## 12.7 Compression Artefacts

### Definition

**Compression Artefact:** A visual distortion introduced by lossy compression that was not present in the original image and is not present in the scene being depicted.

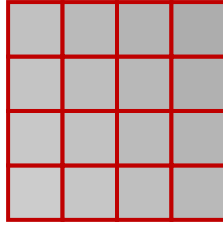
Understanding artefacts helps you choose appropriate quality settings and diagnose compression problems.

### 12.7.1 Blocking Artefacts

**Cause:** Each  $8 \times 8$  block is quantised independently. At low quality, large quantisation errors produce abrupt changes in reconstructed pixel values at block boundaries.

**Why this happens mathematically:** The quantisation error for the DC coefficient (average brightness) can differ between adjacent blocks. If block A's DC is quantised to 59 ( $\times 16 = 944$ ) and block B's DC to 60 ( $\times 16 = 960$ ), the boundary shows a sudden jump of 16 in pixel values where the original image had a smooth gradient.

**Visual appearance:** A visible rectangular grid superimposed on smooth image regions. Most noticeable in flat areas such as skies and skin tones.



Visible block boundaries

Figure 10: Blocking artefact : visible grid lines at  $8 \times 8$  block boundaries.

### 12.7.2 Ringing Artefacts (Gibbs Phenomenon)

**Cause:** Truncating high-frequency DCT coefficients is equivalent to multiplying the frequency-domain representation by a rectangular window. By the convolution theorem, this convolves the spatial image with a sinc-like kernel, producing oscillatory ringing near sharp edges.

**Mathematical intuition:** An ideal step edge contains all frequencies. When we discard high frequencies, the reconstructed edge "rings" : it overshoots on one side and undershoots on the other, creating halos.

**Visual appearance:** Faint light/dark halos around high-contrast edges (e.g., text on a coloured background, tree silhouettes against sky).



Ringing halos near edge

Figure 11: Ringing artefact : oscillatory halos near a sharp black/white edge.

### 12.7.3 Banding / Posterisation

**Cause:** Coarse quantisation of the DC coefficient and the lowest AC coefficients cannot represent the full tonal range of smooth gradients. The reconstructed image can only take on discrete levels spaced  $Q(u, v)$  apart.

**Why this happens:** A smooth gradient requires many subtle intermediate shades. When quantisation is too coarse, the encoder cannot represent these subtle differences, so the decoder outputs the same value for a range of input shades, creating visible bands.

**Visual appearance:** Smooth gradients (blue sky, skin) appear as distinct bands of uniform tone rather than continuous variation.

### 12.7.4 Colour Bleeding

**Cause:** 4:2:0 chroma subsampling assigns a single  $C_b/C_r$  value to each  $2 \times 2$  luma block. At luminance edges, the chroma value may belong to the wrong side of the edge.

**Visual appearance:** Colour "bleeds" across sharp edges; particularly visible on coloured text or saturated objects.

### 12.8 JPEG vs. Modern Formats

Table 7: Comparison of major lossy image compression standards (2024).

| Format    | Year | Transform | Max block | Transparency | Animation | Royalty-free |
|-----------|------|-----------|-----------|--------------|-----------|--------------|
| JPEG      | 1992 | DCT       | 8 × 8     | No           | No        | Yes          |
| JPEG 2000 | 2000 | Wavelet   | Variable  | Yes          | No        | Yes          |
| WebP      | 2010 | DCT (VP8) | 16 × 16   | Yes          | Yes       | Yes          |
| HEIC/HEIF | 2017 | H.265 DCT | 64 × 64   | Yes          | Yes       | Contested    |
| AVIF      | 2019 | AV1 DCT   | 64 × 64   | Yes          | Yes       | Yes          |

Despite modern formats offering better compression at equal quality, JPEG retains dominant market share because of three decades of universal toolchain support.

### 12.9 When to Use (and Not Use) JPEG

**Use JPEG for**

- Photographs and natural images
- Web delivery (universal support)
- Camera output / archival when space is limited
- Any image where approximately 5% quality loss is acceptable

**Avoid JPEG for**

- Text, diagrams, line art (use PNG)
- Images needing transparency (use PNG/WebP)
- Medical / scientific images (use lossless)
- Files that will be re-edited and re-saved (generation loss!)
- Screenshots and UI graphics (use PNG)

**Generation Loss:** Each JPEG save re-quantises the DCT coefficients. Saving a JPEG as JPEG ten times compounds quantisation error. After ten re-saves at quality 80, blocking and ringing artefacts that were invisible after one save become clearly visible. **Always edit in a lossless format; export to JPEG only as the final step.**

## 12.10 Summary

### Important

#### Five Key Insights of Transform Coding:

1. **Decorrelation:** The DCT transforms spatially correlated pixels into (approximately) uncorrelated frequency coefficients.
2. **Energy compaction:** Natural image energy concentrates in a small fraction of DCT coefficients (typically the top-left corner of the  $8 \times 8$  matrix), enabling aggressive but selective discarding.
3. **Single lossy step:** Quantisation is the *only* irreversible operation. Every other step : DCT, zig-zag, RLE, Huffman : is lossless and reversible.
4. **Perceptual exploitation:** The non-uniform Q-table encodes the human visual system's reduced sensitivity to high-frequency detail. We discard exactly what we cannot see.
5. **Universality:** The transform  $\rightarrow$  quantise  $\rightarrow$  entropy-code paradigm extends directly to audio (MDCT in MP3/AAC), video (block DCT in H.264), and even speech coding.

## 12.11 Exercises

### Exercise 12.0

**(DC coefficient)** Compute  $D(0,0)$  for a uniform  $8 \times 8$  block where every pixel equals 200. Show that  $D(0,0) = 8\bar{p}'$  where  $\bar{p}'$  is the mean of the level-shifted block. What does  $D(0,0) = 0$  imply about a block whose average brightness is exactly 128?

### Exercise 12.1

**(Quantisation error)** A DCT coefficient  $D(1,0) = -15$  is quantised with step  $Q(1,0) = 12$ .

- (a) Compute the quantised value  $\hat{D}(1,0)$ .
- (b) Compute the reconstructed value  $\tilde{D}(1,0) = \hat{D}(1,0) \times Q(1,0)$ .
- (c) Compute the absolute quantisation error.

### Exercise 12.2

**(Zig-zag and RLE)** A quantised  $8 \times 8$  block has non-zero coefficients *only* at positions  $(0,0) = 10$ ,  $(2,0) = 3$ ,  $(0,2) = -2$ . All others are zero.

- (a) Write out the first 10 elements of the zig-zag sequence.
- (b) Write the JPEG RLE encoding (DC treated separately).

### Exercise 12.3

**(Medical imaging)** A hospital proposes saving CT scans as JPEG at quality 80 to reduce storage costs. Identify two specific diagnostic risks, explain their technical cause using concepts from this lecture, and propose an alternative compression strategy that reduces storage without those risks.

# 13: Lecture 13: Perceptual Coding and MP3/AAC: Compression by Exploiting Human Hearing

## 13.1 Learning Objectives

By the end of this lecture, you will be able to:

1. Explain why time-domain waveform coding is inefficient and why the frequency domain is the natural representation for perceptual audio coding.
2. Describe the three pillars of the psychoacoustic model: the absolute threshold of hearing, simultaneous masking, and temporal masking.
3. Define the Bark scale and critical bands, and explain their role in masking calculations.
4. Explain the Modified Discrete Cosine Transform (MDCT) and why it is preferred over a plain DCT for audio signals.
5. Trace the complete MP3 encoding pipeline: filterbank  $\rightarrow$  MDCT  $\rightarrow$  psychoacoustic model  $\rightarrow$  bit allocation  $\rightarrow$  quantisation  $\rightarrow$  Huffman coding.
6. Calculate the Signal-to-Mask Ratio (SMR) and determine which frequency bands can tolerate coarser quantisation.
7. Identify and explain pre-echo, and describe how temporal masking and window switching mitigate it.
8. Compare MP3 and AAC in terms of filterbank design, efficiency, and quality at low bit-rates.
9. Connect the perceptual coding paradigm to the JPEG quality factor from the previous lecture.
10. Apply the 6 dB/bit rule and bit-budget allocation in complete multi-band scenarios.

## 13.2 Motivation: Why Is Audio Different from Images?

### 13.2.1 The Raw Cost of Digital Audio

A single CD-quality audio signal requires:

- Sampling rate: 44,100 samples per second (slightly above the Nyquist requirement for the 20 Hz–20 kHz audible range)
- Bit depth: 16 bits per sample
- Channels: 2 (stereo)

The raw bit rate is:

Bit rate = 44,100 samples/s  $\times$  16 bits/sample  $\times$  2 channels = **1,411,200** bits/s  $\approx$  1.4 Mbit/s.

One minute of CD-quality stereo audio occupies approximately 10 MB.

### 13.2.2 Why Not Just Use Lossless Compression?

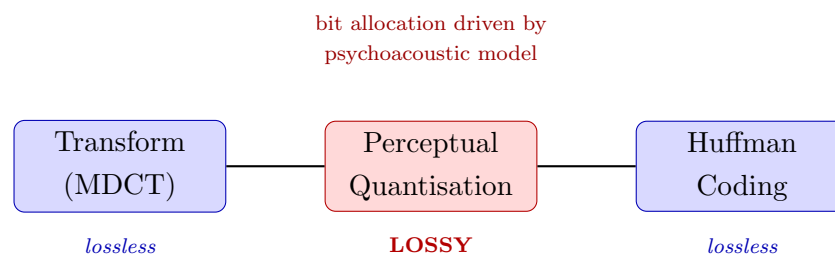
Lossless audio codecs (FLAC, Apple Lossless) achieve roughly 2:1 compression on typical music, reducing 1.4 Mbit/s to about 700 kbit/s. MP3 achieves 10:1 to 14:1 compression (128–320 kbit/s) *while being perceptually transparent for many listeners under typical conditions*.

#### Important

**The Central Insight of Perceptual Coding:** The human auditory system does not perceive all information in a sound signal equally. Large amounts of the signal can be discarded entirely without the listener being able to detect the loss. The goal of MP3 and AAC is not to minimise mathematical error (as in lossless coding) but to minimise *perceptible* error.

### 13.2.3 The Contrast with JPEG

In the JPEG chapter we saw that the human visual system is less sensitive to high-frequency spatial detail. The psychoacoustic model in MP3 is the *audio counterpart*: it determines, dynamically and per-frequency-band, how much quantisation noise the ear will tolerate at each instant in time. Both systems share the same three-stage skeleton:



## 13.3 The Human Auditory System: Three Pillars of Perception

### 13.3.1 Pillar 1 — The Absolute Threshold of Hearing

#### Definition

**Absolute Threshold of Hearing (ATH):** The minimum sound pressure level (SPL) at each frequency that an average human can detect in a perfectly quiet environment. Any sound *below* the ATH at a given frequency is completely inaudible and can be discarded at effectively zero perceptual cost.

The ATH is highly non-uniform across the audible spectrum. A useful approximation (Terhardt’s formula) in dB SPL is:

$$T_{\text{ATH}}(f) = 3.64 \left( \frac{f}{1000} \right)^{-0.8} - 6.5 e^{-0.6(f/1000-3.3)^2} + 10^{-3} \left( \frac{f}{1000} \right)^4, \quad (12)$$

where  $f$  is in Hz and the result is in dB SPL.

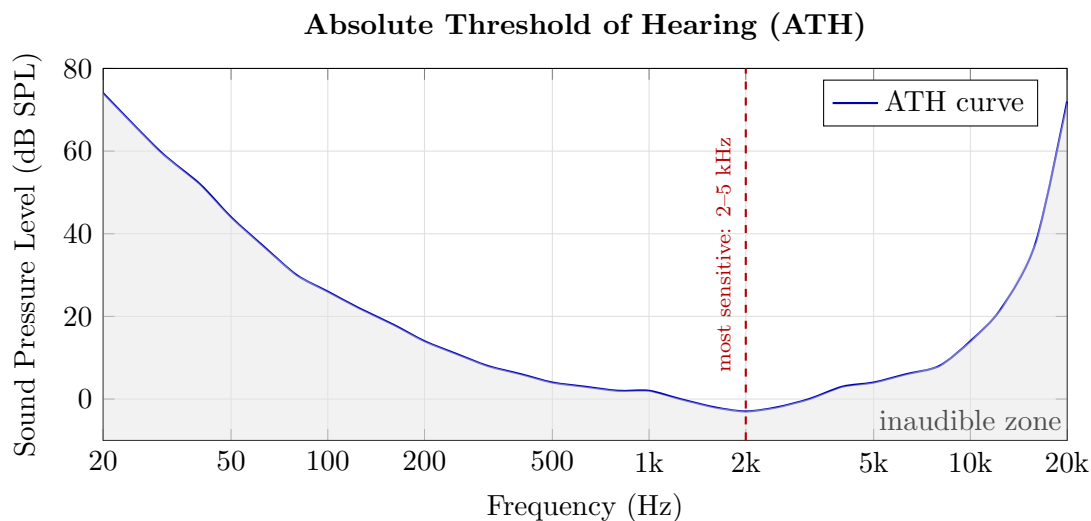


Figure 12: The Absolute Threshold of Hearing. The shaded region below the curve is completely inaudible. The ear is most sensitive around 3–4 kHz and rapidly loses sensitivity below 100 Hz and above 10 kHz.

### Example

#### Example 1: Applying the ATH formula step by step.

Evaluate  $T_{\text{ATH}}$  at four frequencies using Eq. (12) and decide which bands can be discarded for free.

| Band | $f$ (Hz) | Signal (dB SPL) | $T_{\text{ATH}}$ (dB SPL) | Decision |
|------|----------|-----------------|---------------------------|----------|
| A    | 200      | 18              | ?                         | ?        |
| B    | 1000     | 25              | ?                         | ?        |
| C    | 8000     | 10              | ?                         | ?        |
| D    | 16000    | 5               | ?                         | ?        |

Let  $u = f/1000$  for brevity. We compute the three terms  $T_1$ ,  $T_2$ ,  $T_3$  separately.

*Band A:*  $f = 200$  Hz,  $u = 0.2$ .

$$T_1 = 3.64 \times (0.2)^{-0.8} = 3.64 \times 6.899 = 25.11 \text{ dB},$$

$$T_2 = 6.5 e^{-0.6(0.2-3.3)^2} = 6.5 e^{-5.766} \approx 0.020 \text{ dB},$$

$$T_3 = 10^{-3}(0.2)^4 \approx 0 \text{ dB},$$

$$T_{\text{ATH}}(200) \approx 25.1 \text{ dB SPL}.$$

Signal (18 dB) < ATH (25.1 dB): **inaudible – 0 bits needed.**

Band B:  $f = 1000 \text{ Hz}$ ,  $u = 1.0$ .

$$T_1 = 3.64 \times 1^{-0.8} = 3.64 \text{ dB},$$

$$T_2 = 6.5 e^{-0.6(1.0-3.3)^2} = 6.5 \times 0.0418 = 0.27 \text{ dB},$$

$$T_3 = 10^{-3} \times 1 = 0.001 \text{ dB},$$

$$T_{\text{ATH}}(1000) \approx 3.37 \text{ dB SPL}.$$

Signal (25 dB) > ATH (3.4 dB): **audible – allocate bits**.  $\text{SMR} = 25 - 3.4 = 21.6 \text{ dB}$ , so  $\lceil 21.6/6 \rceil = 4$  bits minimum.

Band C:  $f = 8000 \text{ Hz}$ ,  $u = 8.0$ .

$$T_1 = 3.64 \times (8)^{-0.8} = 3.64 \times 0.1768 = 0.643 \text{ dB},$$

$$T_2 \approx 0 \text{ dB} \quad (e^{-13.25} \approx 0),$$

$$T_3 = 10^{-3}(8)^4 = 4.096 \text{ dB},$$

$$T_{\text{ATH}}(8000) \approx 4.74 \text{ dB SPL}.$$

Signal (10 dB) > ATH (4.7 dB): **audible**.  $\text{SMR} = 10 - 4.7 = 5.3 \text{ dB}$ , so  $\lceil 5.3/6 \rceil = 1$  bit.

Band D:  $f = 16000 \text{ Hz}$ ,  $u = 16.0$ .

$$T_1 = 3.64 \times (16)^{-0.8} = 0.347 \text{ dB},$$

$$T_2 \approx 0 \text{ dB},$$

$$T_3 = 10^{-3}(16)^4 = 65.5 \text{ dB},$$

$$T_{\text{ATH}}(16000) \approx 65.8 \text{ dB SPL}.$$

Signal (5 dB)  $\ll$  ATH (65.8 dB): **inaudible by 60.8 dB – 0 bits needed**.

**Summary:** Before any masking is even considered, the ATH eliminates Bands A and D entirely. Band B requires  $\geq 4$  bits; Band C requires only 1 bit. This demonstrates the ATH alone contributing free bit savings.

### 13.3.2 Pillar 2 — Simultaneous Masking (Frequency Masking)

#### Definition

**Simultaneous Masking:** When a loud *masker* tone is present, it raises the hearing threshold in nearby frequency bands, making softer *maskee* sounds inaudible even though they are above the ATH. Masking spreads asymmetrically: more strongly *above* the masker frequency (upward spread) than below it.

## Tonal vs. Noise Maskers.

- **Tonal maskers:** Narrow spectral peaks (e.g., a pure tone). Strong, localised masking with a sharp spreading function.
- **Noise maskers:** Broad spectral humps (e.g., a drum hit). Weaker, broader masking.

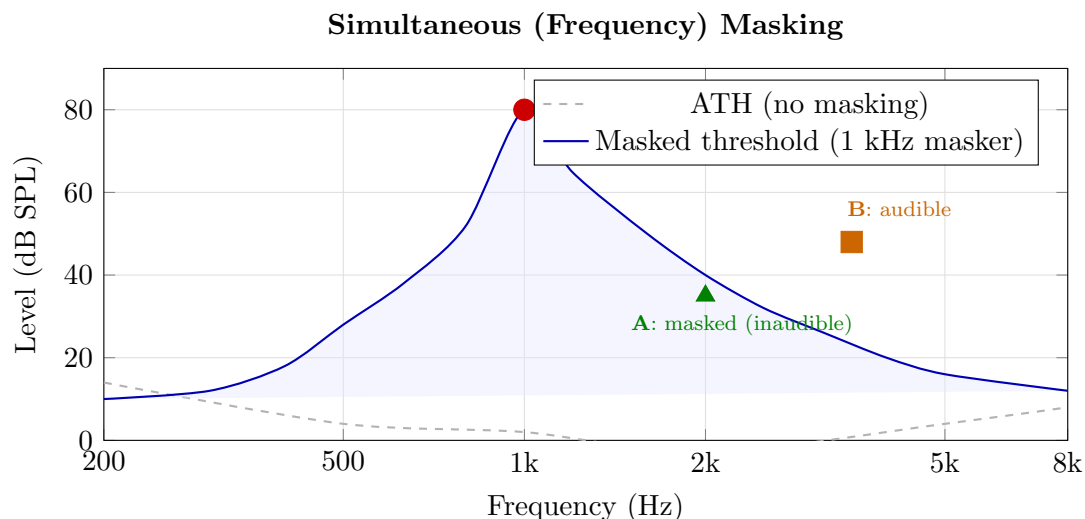


Figure 13: Simultaneous (frequency) masking. A 1 kHz masker at 80 dB SPL raises the threshold over a wide range. Maskee **A** at 2 kHz / 35 dB is inaudible; Maskee **B** at 3.5 kHz / 48 dB remains audible.

## Example

### Example 2: Full masking calculation — two maskers, three maskees.

**Setup.** Two masker tones are present simultaneously:

- **Masker 1:** 500 Hz at 70 dB SPL (tonal).
- **Masker 2:** 4000 Hz at 65 dB SPL (tonal).

Three candidate maskee tones (with ATH approximations):

| Maskee | Frequency | Level (dB SPL) | ATH (approx.)          |
|--------|-----------|----------------|------------------------|
| P      | 1000 Hz   | 35 dB          | 3.4 dB                 |
| Q      | 2500 Hz   | 42 dB          | −2 dB (use 0 dB floor) |
| R      | 6000 Hz   | 20 dB          | 5.5 dB                 |

**Spreading functions (simplified didactic model):**

$$\Delta S_{\text{up}} = 10 + 2.5 \Delta z \quad (\Delta z > 0), \quad \Delta S_{\text{down}} = 25 + 10 |\Delta z| \quad (\Delta z < 0).$$

The masked threshold from masker  $i$  is  $T_{mi} = L_{mi} - \Delta S_i$  (floored at 0 dB).

**Step 1 — Convert all frequencies to Bark.**

| Frequency | Bark (approx.) | Role     |
|-----------|----------------|----------|
| 500 Hz    | 4.9            | Masker 1 |
| 1000 Hz   | 8.5            | Maskee P |
| 2500 Hz   | 15.0           | Maskee Q |
| 4000 Hz   | 17.5           | Masker 2 |
| 6000 Hz   | 19.5           | Maskee R |

**Step 2 — Compute masked thresholds.** The effective mask is  $\max(\mathbf{ATH}, T_{m1}, T_{m2})$ .

*Maskee P (1000 Hz, 8.5 Bark):*

From Masker 1 (500 Hz, 4.9 Bark, 70 dB, upward  $\Delta z = +3.6$ ):

$$\Delta S = 10 + 2.5 \times 3.6 = 19.0 \text{ dB}, \quad T_{m1}(P) = 70 - 19.0 = 51.0 \text{ dB}.$$

From Masker 2 (4000 Hz, 17.5 Bark, 65 dB, downward  $|\Delta z| = 9.0$ ):

$$\Delta S = 25 + 10 \times 9.0 = 115 \text{ dB}, \quad T_{m2}(P) = \max(0, 65 - 115) = 0 \text{ dB}.$$

Effective:  $T_{\text{mask}}(P) = \max(3.4, 51.0, 0) = \mathbf{51.0}$  dB SPL.

*Maskee Q (2500 Hz, 15.0 Bark):*

From Masker 1 (upward  $\Delta z = +10.1$ ):

$$\Delta S = 10 + 2.5 \times 10.1 = 35.25 \text{ dB}, \quad T_{m1}(Q) = 70 - 35.25 = 34.75 \text{ dB}.$$

From Masker 2 (downward  $|\Delta z| = 2.5$ ):

$$\Delta S = 25 + 10 \times 2.5 = 50 \text{ dB}, \quad T_{m2}(Q) = \max(0, 65 - 50) = 15 \text{ dB}.$$

Effective:  $T_{\text{mask}}(Q) = \max(0, 34.75, 15) = \mathbf{34.75}$  dB SPL.

*Maskee R (6000 Hz, 19.5 Bark):*

From Masker 1 (upward  $\Delta z = +14.6$ ):

$$\Delta S = 10 + 2.5 \times 14.6 = 46.5 \text{ dB}, \quad T_{m1}(R) = 70 - 46.5 = 23.5 \text{ dB}.$$

From Masker 2 (upward  $\Delta z = +2.0$ ):

$$\Delta S = 10 + 2.5 \times 2.0 = 15.0 \text{ dB}, \quad T_{m2}(R) = 65 - 15.0 = 50.0 \text{ dB}.$$

Effective:  $T_{\text{mask}}(R) = \max(5.5, 23.5, 50.0) = \mathbf{50.0}$  dB SPL.

**Step 3 — SMR and bit decision.**

| Maskee      | Signal (dB) | $T_{\text{mask}}$ (dB) | SMR (dB) | Audible? | Bits                            |
|-------------|-------------|------------------------|----------|----------|---------------------------------|
| P (1 kHz)   | 35          | 51.0                   | −16.0    | No       | <b>0 bits</b> (masked)          |
| Q (2.5 kHz) | 42          | 34.75                  | +7.25    | Yes      | $\lceil 7.25/6 \rceil = 2$ bits |
| R (6 kHz)   | 20          | 50.0                   | −30.0    | No       | <b>0 bits</b> (masked)          |

**Key observation:** Maskee Q at 2500 Hz is *not* fully masked – it sits 7.25 dB above the combined masking threshold and requires 2 bits. This is a common exam trap: always compute the effective threshold from *all* maskers before deciding.

### 13.3.3 The Bark Scale and Critical Bands

#### Definition

**Critical Band:** The range of frequencies that activate the same region of the basilar membrane. There are approximately 25 critical bands spanning 20 Hz–20 kHz.

**Bark Frequency:** A frequency scale calibrated to critical-band width:

$$z(f) = 13 \arctan(0.00076 f) + 3.5 \arctan\left(\frac{f}{7500}\right)^2, \quad (13)$$

where  $f$  is in Hz and  $z$  is in Bark.

Table 8: The 25 critical bands (Bark bands). Bandwidth in Hz grows with frequency; each band spans approximately 1 Bark.

| Band | Lower (Hz) | Upper (Hz) | Width (Hz) | Band | Lower (Hz) | Upper (Hz) | Width (Hz) |
|------|------------|------------|------------|------|------------|------------|------------|
| 1    | 20         | 100        | 80         | 14   | 2000       | 2320       | 320        |
| 2    | 100        | 200        | 100        | 15   | 2320       | 2700       | 380        |
| 3    | 200        | 300        | 100        | 16   | 2700       | 3150       | 450        |
| 4    | 300        | 400        | 100        | 17   | 3150       | 3700       | 550        |
| 5    | 400        | 510        | 110        | 18   | 3700       | 4400       | 700        |
| 6    | 510        | 630        | 120        | 19   | 4400       | 5300       | 900        |
| 7    | 630        | 770        | 140        | 20   | 5300       | 6400       | 1100       |
| 8    | 770        | 920        | 150        | 21   | 6400       | 7700       | 1300       |
| 9    | 920        | 1080       | 160        | 22   | 7700       | 9500       | 1800       |
| 10   | 1080       | 1270       | 190        | 23   | 9500       | 12000      | 2500       |
| 11   | 1270       | 1480       | 210        | 24   | 12000      | 15500      | 3500       |
| 12   | 1480       | 1720       | 240        | 25   | 15500      | 22050      | 6550       |
| 13   | 1720       | 2000       | 280        |      |            |            |            |

### 13.3.4 Pillar 3 — Temporal Masking

#### Definition

**Temporal Masking:** The masking effect extends *in time* around a loud sound event.

- **Post-masking (forward masking):** A loud sound continues to mask quieter sounds for up to 100–200 ms *after* it ends.
- **Pre-masking (backward masking):** A loud sound masks quieter sounds up to 5–20 ms *before* it begins, because the brain integrates sound over a short window before making a perceptual decision.

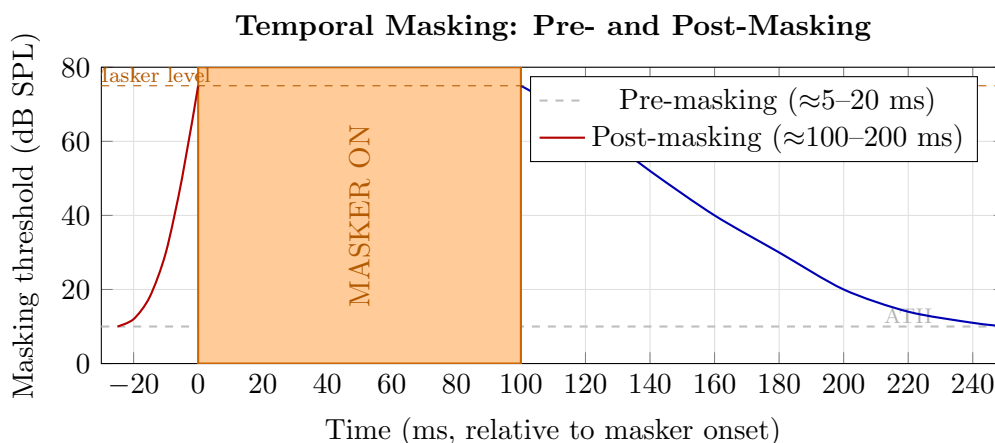


Figure 14: Temporal masking. Before the masker starts (pre-masking, red curve), the threshold rises steeply over  $\approx 20$  ms. After the masker ends (post-masking, blue curve), the threshold decays gradually over  $\approx 200$  ms.

### Important

#### The Three Pillars Working Together:

1. **ATH** sets a global noise floor below which everything is free.
2. **Simultaneous masking** dynamically raises the threshold based on what loud signals are present right now, often by 20–60 dB.
3. **Temporal masking** allows coarser coding in frames just before and after loud transients.

Together these effects mean the perceptual noise floor is often 40–60 dB above the ATH — this is where the 10:1 compression ratio comes from.

## 13.4 The Modified Discrete Cosine Transform (MDCT)

### 13.4.1 Why Not the DCT?

The plain DCT processes blocks independently, creating discontinuities at block boundaries that manifest as a “flanging” or “warbling” sound. A second problem is the trade-off between frequency resolution and time resolution.

### 13.4.2 The MDCT: Overlapping Blocks with Perfect Reconstruction

#### Definition

**Modified Discrete Cosine Transform (MDCT):** An overlapping transform where each block of  $2N$  input samples produces  $N$  output coefficients. Adjacent blocks overlap by 50%. Despite the overlap, the MDCT satisfies **perfect reconstruction**: the overlapping IMDCT blocks, when added using overlap-add, exactly recover the original signal (before quantisation).

## MDCT formula.

$$X(k) = \sum_{n=0}^{2N-1} x(n) w(n) \cos\left(\frac{\pi}{N}\left(n + \frac{1}{2} + \frac{N}{2}\right)\left(k + \frac{1}{2}\right)\right), \quad k = 0, \dots, N-1, \quad (14)$$

where  $w(n)$  is the analysis window.

## Key properties.

1. **50% overlap:** Each frame of  $N$  coefficients describes  $2N$  input samples.
2. **Critical sampling:** Output coefficients ( $N$ ) equals new input samples per frame — no data expansion.
3. **TDAC property** (Time-Domain Aliasing Cancellation): Aliasing introduced by the overlap exactly cancels during overlap-add, giving perfect reconstruction.

## Example

**Example 3: MDCT computation on a minimal block ( $N = 2$ ).**

**Setup.**  $N = 2$ , block size = 4 samples, rectangular window  $w(n) = 1$ .

$$\mathbf{x} = [x(0), x(1), x(2), x(3)] = [1, 2, 3, 4].$$

**The MDCT formula for  $N = 2$ :**

$$X(k) = \sum_{n=0}^3 x(n) \cos\left(\frac{\pi}{2}\left(n + \frac{3}{2}\right)\left(k + \frac{1}{2}\right)\right).$$

**Computing  $X(0)$**  ( $k = 0$ , argument =  $\frac{\pi}{2}(n + \frac{3}{2}) \cdot \frac{1}{2} = \frac{\pi(n+1.5)}{4}$ ):

$$\begin{aligned} n = 0 : & \quad 1 \times \cos(67.5^\circ) = 1 \times 0.3827 = 0.3827, \\ n = 1 : & \quad 2 \times \cos(112.5^\circ) = 2 \times (-0.9239) = -1.8478, \\ n = 2 : & \quad 3 \times \cos(157.5^\circ) = 3 \times (-0.9239) = -2.7717, \\ n = 3 : & \quad 4 \times \cos(202.5^\circ) = 4 \times (-0.3827) = -1.5307. \end{aligned}$$

$$X(0) = 0.3827 - 1.8478 - 2.7717 - 1.5307 = -5.7675.$$

**Computing  $X(1)$**  ( $k = 1$ , argument =  $\frac{3\pi(n+1.5)}{4}$ ):

$$\begin{aligned} n = 0 : & \quad 1 \times \cos(202.5^\circ) = -0.3827, \\ n = 1 : & \quad 2 \times \cos(337.5^\circ) = 2 \times 0.9239 = +1.8478, \\ n = 2 : & \quad 3 \times \cos(112.5^\circ) = 3 \times (-0.3827) = -1.1481, \\ n = 3 : & \quad 4 \times \cos(247.5^\circ) = 4 \times (-0.3827) = -1.5307. \end{aligned}$$

$$X(1) = -0.3827 + 1.8478 - 1.1481 - 1.5307 = -\mathbf{1.2137}.$$

### Interpretation.

- 4 input samples  $\rightarrow$  2 MDCT coefficients: *critical sampling confirmed*.
- $|X(0)| \gg |X(1)|$ : the rising ramp  $[1, 2, 3, 4]$  has strong low-frequency energy — correctly captured in the lowest-frequency MDCT coefficient.
- In real MP3 with  $N = 576$ :  $2N = 1152$  input samples produce 576 MDCT frequency lines per granule.

**What if we used a pure zero input  $\mathbf{x} = [0, 0, 0, 0]$ ?** Then  $X(0) = X(1) = 0$  trivially — an MDCT of silence produces all-zero coefficients, which Huffman coding collapses to a handful of bits. This is why silence is compressed so aggressively in MP3.

### 13.4.3 Window Switching: Handling Transients

For stationary signals, MP3 uses **long blocks** ( $N = 576$ ). When a transient is detected, it switches to **short blocks** ( $N = 192$ ) to avoid pre-echo.

#### Pre-Echo: The Most Audible MP3 Artefact at Low Bit-Rates.

A long block is analysed as a whole. If a loud transient appears mid-block, coarse quantisation spreads noise across the entire block after IMDCT — including the silence *before* the transient. You hear a faint “pre-echo” before the event occurs.

**Solution:** Switch to short blocks (8.7 ms each). Noise cannot spread more than 8.7 ms, and post-masking from the loud transient hides any noise in the preceding short block.

## 13.5 The Complete MP3 Encoding Pipeline

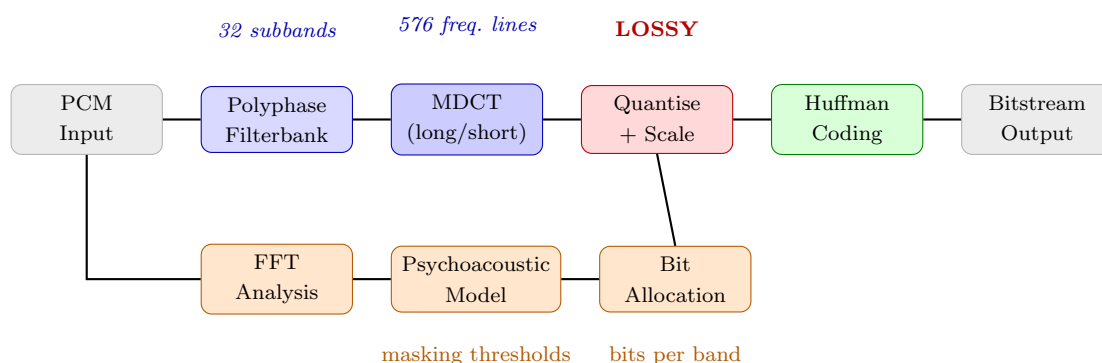


Figure 15: Complete MP3 encoding pipeline. The top path is the main audio processing chain; the bottom path is the psychoacoustic model running in parallel.

### 13.5.1 Stage 1 — Polyphase Quadrature Mirror Filterbank (PQMF)

The MP3 encoder divides audio into 32 equal-width subbands. Each subband produces 18 samples per frame, giving  $32 \times 18 = 576$  values transformed by the MDCT.

### 13.5.2 Stage 2 — Psychoacoustic Analysis (in Parallel)

A separate 1024-point FFT analyses the PCM samples. The psychoacoustic model identifies tonal and noise maskers, computes spreading functions, sums contributions, combines with the ATH, and computes SMR per subband.

### 13.5.3 Stage 3 — Bit Allocation

#### Definition

**Noise Shaping Objective:** Allocate bits such that the quantisation noise in each band is just below the masking threshold — no more (wastes bits), no fewer (makes noise audible).

#### Example

**Example 4: Complete bit allocation across six subbands with a constrained budget.**

**Setup.** The psychoacoustic model has computed the following for six subbands. Total bit budget: **36 bits**.

| Band | Freq. range  | Signal (dB) | Mask thr. (dB) | SMR (dB) | Audible?     |
|------|--------------|-------------|----------------|----------|--------------|
| 1    | 0–689 Hz     | 72          | 20             | +52      | Yes          |
| 2    | 689–1378 Hz  | 85          | 30             | +55      | Yes          |
| 3    | 1378–2067 Hz | 60          | 68             | –8       | No (masked)  |
| 4    | 2067–2756 Hz | 55          | 25             | +30      | Yes          |
| 5    | 2756–3445 Hz | 30          | 42             | –12      | No (masked)  |
| 6    | 3445–4134 Hz | 12          | 5              | +7       | Yes (barely) |

**Step 1 — Compute initial bit requirements.**

Using  $b = \lceil \text{SMR}/6 \rceil$  for audible bands:

| Band                  | SMR (dB) | Calc          | Min bits                                    | Reasoning              |
|-----------------------|----------|---------------|---------------------------------------------|------------------------|
| 1                     | +52      | $52/6 = 8.67$ | 9                                           | Loud; must be accurate |
| 2                     | +55      | $55/6 = 9.17$ | 10                                          | Highest SMR — loudest  |
| 3                     | −8       | masked        | 0                                           | Discarded              |
| 4                     | +30      | $30/6 = 5.00$ | 5                                           | Moderate               |
| 5                     | −12      | masked        | 0                                           | Discarded              |
| 6                     | +7       | $7/6 = 1.17$  | 2                                           | Barely audible         |
| <b>Total required</b> |          |               | $9 + 10 + 0 + 5 + 0 + 2 = \mathbf{26}$ bits |                        |

### Step 2 — Budget check.

Required: 26 bits. Budget: 36 bits. Spare:  $36 - 26 = \mathbf{10}$  bits.

### Step 3 — Distribute spare bits (highest SMR first).

| Band         | Base | Spare | Final          | Achieved SNR (dB)              |
|--------------|------|-------|----------------|--------------------------------|
| 1            | 9    | +4    | 13             | $1.76 + 6.02 \times 13 = 80.0$ |
| 2            | 10   | +4    | 14             | $1.76 + 6.02 \times 14 = 86.0$ |
| 3            | 0    | 0     | 0              | —                              |
| 4            | 5    | +2    | 7              | $1.76 + 6.02 \times 7 = 43.9$  |
| 5            | 0    | 0     | 0              | —                              |
| 6            | 2    | 0     | 2              | $1.76 + 6.02 \times 2 = 13.8$  |
| <b>Total</b> |      |       | <b>36 bits</b> |                                |

### Step 4 — Verify perceptual safety.

| Band | Required SNR | Achieved SNR | Margin   | Safe? |
|------|--------------|--------------|----------|-------|
| 1    | 52 dB        | 80.0 dB      | +28.0 dB | ✓     |
| 2    | 55 dB        | 86.0 dB      | +31.0 dB | ✓     |
| 4    | 30 dB        | 43.9 dB      | +13.9 dB | ✓     |
| 6    | 7 dB         | 13.8 dB      | +6.8 dB  | ✓     |

All bands have quantisation noise below the masking threshold. Budget exactly consumed.

### Step 5 — Compression gain from psychoacoustics.

A naive flat coder allocates 16 bits per band:  $6 \times 16 = 96$  bits. Our model uses 36 bits — a  $96/36 \approx \mathbf{2.7\times}$  additional compression gain from psychoacoustics alone.

### 13.5.4 Stage 4 — Quantisation and Huffman Coding

The MDCT coefficients are quantised using a non-uniform scalar quantiser:

$$\hat{X}(k) = \text{nint}\left(\frac{|X(k)|^{3/4}}{\text{quantstep}}\right) \times \text{sgn}(X(k)), \quad (15)$$

where the  $3/4$  power law approximates perceptual loudness scaling. The quantised values are Huffman coded; zero runs are run-length encoded, analogous to JPEG's End-of-Block (EOB) mechanism.

## 13.6 The 6 dB/Bit Rule and Rate–Distortion in Audio

For a uniform quantiser with step size  $\Delta$  and  $b$  bits:

$$\text{SNR}_{\text{dB}} \approx 1.76 + 6.02b \quad [\text{dB}]. \quad (16)$$

**6 dB per bit rule:** each additional bit improves SNR by approximately 6 dB.

#### Example

**Example 5: Applying the 6 dB/bit rule — bits, step size, and verification.**

A subband has MDCT coefficients with peak amplitude  $A = 1.0$ . The required SNR is **54 dB**.

**Part (a) — Minimum bits.**

$$54 \leq 1.76 + 6.02b \implies b \geq \frac{54 - 1.76}{6.02} = 8.68 \implies \boxed{b = 9 \text{ bits}}.$$

**Part (b) — Quantiser step size.**

$$\Delta = \frac{2A}{2^b} = \frac{2.0}{512} = \mathbf{0.00391}.$$

Every MDCT coefficient is rounded to the nearest multiple of 0.00391.

**Part (c) — Achieved SNR.**

$$\text{SNR} = 1.76 + 6.02 \times 9 = 1.76 + 54.18 = \mathbf{55.94 \text{ dB}}.$$

Exceeds required 54 dB by 1.94 dB. ✓

**Part (d) — Why 8 bits is insufficient.**

$$\text{SNR}|_{b=8} = 1.76 + 6.02 \times 8 = 49.92 \text{ dB} < 54 \text{ dB}.$$

The 49.92 dB SNR falls *above* the masking threshold (we need 54 dB SNR to push noise *below* the threshold). The ninth bit is mandatory.

**Context:** Without psychoacoustics, CD quality demands  $b = 16$  bits. The model reduces this band to 9 bits — a  $16/9 \approx 1.78\times$  reduction in bit usage for this band alone.

## Example

**Example 6: Bit allocation with a tight budget — forced trade-offs and graceful degradation.**

**Setup.** Budget: **24 bits** for five subbands.

| Band      | SMR (dB) | Min bits $\lceil \text{SMR}/6 \rceil$ | Priority                     |
|-----------|----------|---------------------------------------|------------------------------|
| 1         | +42      | 7                                     | High (loud, clearly audible) |
| 2         | +30      | 5                                     | Medium                       |
| 3         | −5       | 0                                     | None (masked)                |
| 4         | +18      | 3                                     | Lower                        |
| 5         | +12      | 2                                     | Lowest                       |
| Total min |          | 17 bits                               |                              |

**Scenario A: Budget = 24 bits (7 spare bits).**

| Band         | Min bits | Spare added | Final   | Achieved SNR (dB)              |
|--------------|----------|-------------|---------|--------------------------------|
| 1 (SMR 42)   | 7        | +4          | 11      | $1.76 + 6.02 \times 11 = 68.0$ |
| 2 (SMR 30)   | 5        | +3          | 8       | $1.76 + 6.02 \times 8 = 49.9$  |
| 3 (masked)   | 0        | 0           | 0       | —                              |
| 4 (SMR 18)   | 3        | 0           | 3       | $1.76 + 6.02 \times 3 = 19.8$  |
| 5 (SMR 12)   | 2        | 0           | 2       | $1.76 + 6.02 \times 2 = 13.8$  |
| <b>Total</b> |          |             | 24 bits |                                |

All margins positive:  $68 - 42 = +26$ ,  $49.9 - 30 = +19.9$ ,  $19.8 - 18 = +1.8$ ,  $13.8 - 12 = +1.8$ . Perceptually transparent.

**Scenario B: Budget = 10 bits (7 bits *fewer* than the minimum).**

Reduce from lowest-priority bands first:

- Remove Band 5 (2 bits):  $10 - 2 = 8$  saved. Band 5 now gets 0 bits; its SMR was +12 dB, so weak audible noise is introduced.
- Remove Band 4 (3 bits):  $8 - 3 = 5$  saved. Band 4 gets 0 bits; SMR +18 dB, more noticeable noise.

- Still need  $17 - 10 = 7$  more cuts. Reduce Band 2: cut from 5 to 3 bits. Achieved SNR: 19.8 dB, but required is 30 dB. Audible noise introduced in Band 2.
- Remaining:  $10 = 7 + 3 + 0 + 0 + 0$ . Band 1 keeps 7 bits (SNR 43.9 dB vs required 42 dB: just sufficient).

**Lesson:** At very low bit-rates, the encoder sacrifices lower-priority bands first (“graceful degradation”). The most perceptually important bands are preserved as long as possible.

## 13.7 Pre-Echo and Window Switching: A Detailed Example

### Example

**Example 7: Pre-echo — quantitative analysis and short-block solution.**

**Scenario.** A drum hit occurs at sample 900 within a 1152-sample long block (at 44.1 kHz: 26.1 ms total). Samples 0–899 are silence at  $-55$  dB; samples 900–1151 contain the percussive attack at 90 dB SPL.

**Step 1 — Masking analysis with a long block.**

The 1152-sample FFT is dominated by the loud attack. The model computes a masking threshold of  $\approx 60$  dB for the entire block.

**Step 2 — Bit allocation (long block).**

$$b = \left\lceil \frac{60}{6.02} \right\rceil = 10 \text{ bits for all bands.}$$

**Step 3 — Quantiser step size and noise level.**

$$\Delta = \frac{2}{2^{10}} = 0.00195, \quad \sigma_q^2 = \frac{\Delta^2}{12} = 3.17 \times 10^{-7}.$$

$$\text{SNR} = 1.76 + 6.02 \times 10 = 61.96 \text{ dB.}$$

With signal at 90 dB SPL, noise floor  $\approx 90 - 61.96 = 28.0$  dB SPL.

**Step 4 — The pre-echo problem.**

In the silent region (samples 0–899), the signal is at  $-55$  dB. The ATH at 1 kHz is  $\approx 3.4$  dB SPL.

$$\text{Noise level (28 dB)} \gg \text{ATH (3.4 dB)}$$

The pre-echo is audible by  $28 - 3.4 = \mathbf{24.6}$  dB above the threshold. This is a severe, clearly perceptible artefact.

**Step 5 — Solution: switch to three short blocks (384 samples = 8.7 ms each).**

| Block | Samples  | Time span    | Content        | Strategy                     |
|-------|----------|--------------|----------------|------------------------------|
| A     | 0–383    | 0–8.7 ms     | silence        | Use many bits (low masking)  |
| B     | 384–767  | 8.7–17.4 ms  | fading silence | Moderate allocation          |
| C     | 768–1151 | 17.4–26.1 ms | full attack    | High masking; tolerate noise |

Post-masking from Block C’s loud attack lasts up to 200 ms. Block B ends 26.1 – 17.4 = 8.7 ms before Block C starts. Block A ends 17.4 ms before Block C starts. Both A and B fall within the pre-masking window ( $\leq 20$  ms before onset), so their noise is temporally masked by the drum.

**Result:** Noise in Block A is now contained within 8.7 ms and hidden by the drum’s pre-masking. Pre-echo is eliminated.

**Trade-off summary:**

| Property                      | Long block | Three short blocks |
|-------------------------------|------------|--------------------|
| Time resolution               | 26.1 ms    | 8.7 ms             |
| Frequency resolution          | 576 lines  | 192 lines          |
| Pre-echo risk                 | High       | Eliminated         |
| Psychoacoustic model accuracy | High       | Reduced            |

## 13.8 AAC: The Successor to MP3

Table 9: MP3 vs. AAC: architectural comparison.

| Feature                       | MP3 (MPEG-1 Layer III)      | AAC (MPEG-4 Part 3)         |
|-------------------------------|-----------------------------|-----------------------------|
| Filterbank                    | Hybrid: 32-band PQMF + MDCT | Pure MDCT only              |
| MDCT size (long)              | 576 lines                   | 1024 or 2048 lines          |
| MDCT size (short)             | 192 lines                   | 128 or 256 lines            |
| Channels                      | Up to 2 (stereo)            | Up to 48                    |
| Huffman tables                | 32 fixed tables             | Adaptive tables (per frame) |
| Joint stereo                  | Simple M/S stereo           | M/S + intensity stereo      |
| Temporal noise shaping        | No                          | Yes (TNS)                   |
| Perceptual noise substitution | No                          | Yes (PNS)                   |
| Max bit-rate                  | 320 kbit/s                  | 576 kbit/s per channel      |
| Typical transparent quality   | 192–256 kbit/s              | 128–160 kbit/s              |

### 13.8.1 The Pure MDCT Filterbank

AAC replaces MP3's hybrid filterbank with a single MDCT, giving 1024 frequency lines ( $\approx 1.78\times$  better frequency resolution) and eliminating PQMF aliasing errors.

### 13.8.2 Temporal Noise Shaping (TNS)

#### Definition

**Temporal Noise Shaping (TNS):** Applies a prediction filter to MDCT coefficients in the frequency domain, shaping quantisation noise in the time domain — concentrating it where masking is strongest. This directly addresses pre-echo without requiring window switching.

### 13.8.3 Joint Stereo Coding

$$M = \frac{L + R}{2} \quad (\text{mid channel}), \quad S = \frac{L - R}{2} \quad (\text{side channel}).$$

#### Example

##### Example 8: Mid/Side stereo — full energy and SNR analysis.

Two stereo channels:

$$L(t) = 0.9 \sin(\omega t), \quad R(t) = 0.6 \sin(\omega t).$$

##### Step 1 — Compute M and S.

$$M = \frac{0.9 + 0.6}{2} = 0.75, \quad S = \frac{0.9 - 0.6}{2} = 0.15.$$

##### Step 2 — Power ratio.

$$P_M \propto 0.75^2 = 0.5625, \quad P_S \propto 0.15^2 = 0.0225. \quad \frac{P_M}{P_S} = 25 \quad (14 \text{ dB advantage for M}).$$

##### Step 3 — Bit allocation (total budget: 12 bits).

| Strategy        | Bits to L or M | Bits to R or S |
|-----------------|----------------|----------------|
| Independent L/R | 6 each         | 6 each         |
| M/S stereo      | 9 to M         | 3 to S         |

M is allocated more bits because it carries  $25\times$  more power.

**Step 4 — Quantiser step sizes.**

$$\Delta_M = \frac{2 \times 0.75}{2^9} = \frac{1.5}{512} = 0.00293, \quad \Delta_S = \frac{2 \times 0.15}{2^3} = \frac{0.30}{8} = 0.0375.$$

**Step 5 — Reconstruction errors.**

$$\Delta L = \Delta M + \Delta S = 0.00293 + 0.0375 = 0.0404,$$

$$\Delta R = \Delta M - \Delta S = 0.00293 - 0.0375 = -0.0346.$$

**SNR for L channel (M/S):**

$$\text{SNR}_L = 20 \log_{10} \left( \frac{0.9}{0.0404} \right) = 20 \log_{10}(22.3) = 26.9 \text{ dB}.$$

**SNR for L channel (independent, 6 bits):**

$$\text{SNR}_L = 1.76 + 6.02 \times 6 = 37.9 \text{ dB}.$$

**Discussion.** Here independent coding gives higher SNR because the S channel, despite carrying little energy, is quantised very coarsely (3 bits, step 0.0375), which is large relative to its amplitude of 0.15. M/S stereo is most beneficial when  $L \approx R$  (i.e., near-mono content, typical in popular music), where  $S \approx 0$  and the 3-bit S quantiser has negligible impact.

**Key formula:**  $\Delta L = \Delta M + \Delta S$ ,  $\Delta R = \Delta M - \Delta S$ . When  $|\Delta S| \ll |\Delta M|$ , both channels are dominated by the high-quality M error. M/S stereo saves  $\approx 10\text{--}25\%$  of the total bit budget for typical music.

## 13.9 Worked End-to-End Example: One MP3 Frame

### Example

**Example 9: Tracing one complete MP3 granule from PCM to bitstream.**

**Signal:** 500 Hz at 75 dB SPL, 1.5 kHz at 40 dB SPL, 3 kHz at 20 dB SPL. Mono, 44.1 kHz, long block.

**Stage 1 — Filterbank.**

| Subband | Freq. range  | Dominant energy |
|---------|--------------|-----------------|
| 0       | 0–689 Hz     | 500 Hz (strong) |
| 2       | 1378–2067 Hz | 1.5 kHz (main)  |
| 4       | 2756–3445 Hz | 3 kHz (weak)    |
| 5–31    | rest         | Negligible      |

### Stage 2 — Psychoacoustic model.

Convert to Bark:  $z(500) = 4.9$ ,  $z(1500) = 11.3$ ,  $z(3000) = 16.7$ .

*Masking at 1.5 kHz from 500 Hz masker:*

$$\Delta z = 11.3 - 4.9 = 6.4, \quad \Delta S = 10 + 2.5 \times 6.4 = 26.0 \text{ dB}, \quad T_m = 75 - 26 = 49.0 \text{ dB}.$$

*Masking at 3 kHz from 500 Hz masker:*

$$\Delta z = 16.7 - 4.9 = 11.8, \quad \Delta S = 10 + 2.5 \times 11.8 = 39.5 \text{ dB}, \quad T_m = 75 - 39.5 = 35.5 \text{ dB}.$$

### Stage 3 — SMR and bit allocation.

| Tone    | Signal (dB) | $T_{\text{eff}}$ (dB)           | SMR (dB) | Bits                      |
|---------|-------------|---------------------------------|----------|---------------------------|
| 500 Hz  | 75          | $T_{\text{ATH}}(500) \approx 4$ | +71      | $\lceil 71/6 \rceil = 12$ |
| 1.5 kHz | 40          | 49.0                            | −9.0     | 0 (masked)                |
| 3.0 kHz | 20          | 35.5                            | −15.5    | 0 (masked)                |

### Stage 4 — Quantisation.

$$\text{SNR}(12 \text{ bits}) = 1.76 + 6.02 \times 12 = 74.0 \text{ dB} \quad (> 71 \text{ required}). \quad \checkmark$$

### Stage 5 — Huffman coding.

Non-zero coefficients:  $\approx 18$  (subband 0). Zeros:  $576 - 18 = 558$ . Run-length encoding collapses 558 trailing zeros to an End-of-Block token ( $\approx 8$  bits).

### Compression estimate:

| Item                                        | Bits                                          |
|---------------------------------------------|-----------------------------------------------|
| Raw PCM (1152 samples $\times$ 16 bits)     | 18,432                                        |
| 500 Hz MDCT coefficients ( $18 \times 12$ ) | 216                                           |
| Huffman overhead + side info                | $\approx 150$                                 |
| <b>Total encoded</b>                        | <b><math>\approx 366</math> bits</b>          |
| <b>Compression ratio</b>                    | <b><math>18,432/366 \approx 50 : 1</math></b> |

*Note:* This is an idealised two-tone signal. Real music at 128 kbit/s achieves  $\approx 11:1$ .

**Decoder.** The 1.5 kHz and 3 kHz tones are absent from the output. The listener cannot detect this: the 75 dB / 500 Hz masker raises the threshold at those frequencies to 49 dB and 35.5 dB respectively — far above the original 40 dB and 20 dB levels.

## 13.10 Audio Coding Artefacts

### 13.10.1 Granular Noise (Birdies)

**Cause:** Coarse quantisation of MDCT coefficients causes them to oscillate between a handful of levels; IMDCT produces tonal noise not in the original.

**Character:** Faint metallic “birdlike” chirping; most audible during quiet passages. Typical threshold: below 80 kbit/s (MP3), 48 kbit/s (AAC).

### 13.10.2 Smearing and Loss of Stereo Imaging

**Cause:** Aggressive M/S stereo coding at low bit-rates collapses the stereo image toward mono.

### 13.10.3 Watery / Phasiness

**Cause:** At very low bit-rates, overlap-add reconstruction creates unnatural phase relationships. Sounds “underwater.”

### 13.10.4 Pre-Echo

**Cause:** Inadequate bit allocation before a transient spreads quantisation noise into preceding silence.

Table 10: MP3 and AAC artefacts: cause and typical bit-rate at which they become audible.

| Artefact        | MP3 threshold | AAC threshold | Technical cause                            |
|-----------------|---------------|---------------|--------------------------------------------|
| Granular noise  | < 80 kbit/s   | < 48 kbit/s   | Coarse quantisation of MDCT coefficients   |
| Pre-echo        | < 96 kbit/s   | < 64 kbit/s   | Noise spreading into pre-transient silence |
| Stereo collapse | < 96 kbit/s   | < 64 kbit/s   | Aggressive M/S / intensity stereo          |
| Watery sound    | < 48 kbit/s   | < 32 kbit/s   | Insufficient MDCT frequency resolution     |

13.11 Connecting to JPEG: A Unified View

The Universal Lossy Compression Paradigm

| Stage             | JPEG (images)                             | MP3/AAC (audio)                                   |
|-------------------|-------------------------------------------|---------------------------------------------------|
| Transform         | 8×8 block DCT                             | MDCT (long: 576/1024, short: 192/128)             |
| Perceptual model  | Fixed Q-table (HVS frequency sensitivity) | Dynamic psychoacoustic model (per frequency band) |
| Quantisation      | Per-coefficient step size from Q-table    | Per-subband step size from bit allocation         |
| Entropy coding    | Huffman + RLE + DPCM for DC               | Huffman (32 tables) + RLE for zero runs           |
| Quality control   | Single quality factor (scales Q-table)    | Target bit-rate (via psychoacoustic model)        |
| Blocking artefact | 8×8 grid at block boundaries              | Granular noise (spectral birdies)                 |
| Ringing artefact  | Gibbs phenomenon at sharp edges           | Pre-echo at transients                            |

**Key conceptual difference:** JPEG’s Q-table is *static*. MP3’s psychoacoustic model is *dynamic*: it recomputes masking thresholds  $\approx 38$  times per second, adapting to the signal in real time.

13.12 Summary

Important

**Six Key Insights of Perceptual Audio Coding:**

- Perception, not mathematics:** The goal is to minimise *audible* distortion, not mathematical MSE.
- Three masking phenomena:** ATH, simultaneous masking, and temporal masking together define a perceptual noise floor typically 40–60 dB above the ATH.
- The Bark scale:** Masking operates in critical-band units (Bark), not in Hz. Frequency resolution of the ear is approximately logarithmic.
- MDCT with overlap:** Reduces blocking artefacts through overlap-add while maintaining critical sampling.
- Window switching:** Long blocks for stationary signals; short blocks for transients.
- The same paradigm:** Transform  $\rightarrow$  perceptual quantisation  $\rightarrow$  entropy coding is universal. JPEG, MP3, AAC, H.264, H.265, and AV1 all follow it.

## 13.13 Exercises

### Exercise 13.0

**(ATH formula application)** Using Eq. (12), compute the ATH at  $f = 500$  Hz and  $f = 4000$  Hz. For each frequency, a tone is present at 15 dB SPL. Determine whether each tone is audible. For audible tones, compute the minimum bits required (use the 6 dB/bit rule with  $\text{SMR} = \text{signal level} - \text{ATH}$ ).

### Exercise 13.1

**(Bark conversion and single-masker SMR)** A 800 Hz tonal masker is present at 78 dB SPL. A candidate maskee is at 2000 Hz with a level of 45 dB SPL.

- Convert 800 Hz and 2000 Hz to Bark using Eq. (13).
- Compute the upward spreading attenuation  $\Delta S = 10 + 2.5 \Delta z$  in dB.
- Compute the masked threshold at 2000 Hz.
- Compute the SMR. Is the 2000 Hz tone audible?
- If audible, how many bits does the encoder allocate? (6 dB/bit rule.)

### Exercise 13.2

**(Two-masker scenario)** Two tonal maskers: Masker A at 1000 Hz, 80 dB SPL; Masker B at 3000 Hz, 70 dB SPL. A maskee at 2000 Hz, 50 dB SPL. ATH at 2000 Hz  $\approx -3$  dB (use 0 dB as floor).

Downward spread:  $\Delta S_{\text{down}} = 25 + 10 |\Delta z|$  dB.

- Convert 1000, 2000, and 3000 Hz to Bark.
- Compute  $T_{mA}$  (upward spread from Masker A) and  $T_{mB}$  (downward spread from Masker B).
- Compute  $T_{\text{mask}} = \max(\text{ATH}, T_{mA}, T_{mB})$ .
- Is the maskee audible? Compute the SMR. How many bits are allocated?

### Exercise 13.3

**(Bit allocation with a constrained budget)** Budget: **40 bits** for one frame across six subbands.

| Band | SMR (dB)     |
|------|--------------|
| 1    | +48          |
| 2    | +36          |
| 3    | -10 (masked) |
| 4    | +24          |
| 5    | +12          |
| 6    | +6           |

- Compute minimum bits per band. Total required?

- (b) Distribute any spare bits to the highest-SMR bands.
- (c) Verify each audible band achieves at least the required SNR.
- (d) If budget drops to 20 bits, which band(s) would you sacrifice first and why? Specify the resulting allocation.

### Exercise 13.4

**(MDCT computation)** An MDCT with  $N = 2$  (block size 4) uses a rectangular window. Input block:  $\mathbf{x} = [3, 1, -1, -3]$  (a falling ramp).

- (a) Compute  $X(0)$  and  $X(1)$  using Eq. (14).
- (b) Compare  $|X(0)|$  and  $|X(1)|$ . Which frequency dominates?
- (c) If the amplitude of  $X(0)$  were doubled (by scaling the input), by how many dB would the SNR need to increase to maintain the same perceptual quality? (Hint: use the 6 dB/bit rule in reverse.)
- (d) Verify that critical sampling holds: how many new samples per frame, and how many output coefficients?

### Exercise 13.5

**(Pre-echo quantitative analysis)** A long block (1152 samples, 44.1 kHz) contains 800 samples of silence at  $-60$  dB, followed by 352 samples of a 2 kHz tone at 85 dB SPL. The bit allocator assigns 8 bits to all subbands.

- (a) Compute the SNR achieved by an 8-bit quantiser.
- (b) With the signal at 85 dB SPL and  $\text{SNR} = 8\text{-bit SNR}$ , what is the absolute noise level in dB SPL? Is it above the ATH at 2 kHz ( $\approx -3$  dB; use 0 dB)?
- (c) By how many dB does the noise in the silent portion exceed the ATH? Is this a severe artefact?
- (d) The encoder switches to 3 short blocks (each 384 samples = 8.7 ms). Post-masking from the 85 dB attack lasts 200 ms, and pre-masking covers 20 ms before the onset (sample 800, i.e., at  $t = 18.1$  ms). Which short block(s) fall within the pre-masking window of the attack, and therefore tolerate coarser quantisation?

### Exercise 13.6

**(Mid/Side stereo analysis)** Stereo channels:  $L(t) = 1.0 \sin(\omega t)$ ,  $R(t) = 0.4 \sin(\omega t)$ .

- (a) Compute  $M$  and  $S$  amplitudes.
- (b) Compute power ratio  $P_M/P_S$  and express in dB.
- (c) Total budget: 12 bits. Propose an independent L/R allocation and an M/S allocation. Justify the M/S split.
- (d) Compute  $\Delta_M$  and  $\Delta_S$  for your M/S allocation (assume  $A_M$  and  $A_S$  from part (a)).

- (e) Using  $\Delta L = \Delta_M + \Delta_S$ , compute the reconstruction error in L. Compute the resulting SNR for the L channel.

### Exercise 13.7

**(Design question: perceptual coding for images vs. audio)** JPEG uses a static quantisation table; MP3 uses a dynamic psychoacoustic model recomputed per frame.

- (a) Why is a static table sufficient for images but not for audio? Consider how content varies across image blocks vs. audio frames, and how rapidly masking properties change.
- (b) Could a “dynamic JPEG” be designed that recomputes the Q-table per  $8 \times 8$  block based on local visual masking? What information would be needed? How would the decoder know which Q-table was used?
- (c) If visual masking similarly raised the effective noise floor by 40 dB in images, how many bits per pixel could be saved using the 6 dB/bit rule? Assuming standard JPEG uses 8 bits per pixel channel, what would the approximate compression ratio become?